



Participant Handbook

Sector
Green Jobs

Sub Sector
Green Hydrogen

Occupation
Entrepreneur

Reference Id
SGJ/Q0121, Version 2.0 NSQF Level 5



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Green Hydrogen plant Entrepreneur

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This “Participant Handbook for Green Hydrogen Plant Entrepreneur” is dedicated to all the aspiring youth who desire to gain special skills and gain a meaningful livelihood in the Green Hydrogen sector. SCGJ will like to express sincere gratitude to all experts and other stakeholders for their invaluable contribution to the development of the Green Hydrogen Plant Entrepreneur Handbook. Their expertise and insights have been instrumental in crafting a comprehensive resource that will undoubtedly empower aspiring entrepreneurs and industry professionals in navigating the complexities of the green hydrogen sector.

About this Book

This Participant Handbook is designed to enable training for the specific Qualification SGJ/Q0121: Green Hydrogen Plant Entrepreneur. The Government of India has launched National Green Hydrogen Mission (NGHM) through which it aims to position the country as the global center for the production, utilization, and export of Green Hydrogen and its derivatives. This endeavor aligns with India's vision of self-reliance through clean energy, serving as a model for the global transition to clean energy. The mission will play a pivotal role in significantly reducing carbon emissions, decreasing reliance on fossil fuel imports, and positioning India as a leader in Green Hydrogen technology and markets.

In pursuit of this goal, the Government aims to develop the capacity and infrastructure to produce 5 million metric tons (MMT) of Green Hydrogen by 2030, along with installing 60-100 GW of electrolyzer capacity and generating 125 GW of renewable energy for Green Hydrogen production. Additionally, the mission targets achieving 50 MMT of cumulative carbon abatement and creating 600,000 new green jobs, across the green hydrogen value chain. This handbook is designed to provide practical guidance on Green Hydrogen plant entrepreneurship in simple language, while avoiding excessive theoretical details and calculations. It is intended to equip participants with the necessary knowledge and skills to establish and operate Green Hydrogen plants while ensuring compliance with applicable codes, standards, and safety regulations. Skill Council for Green Jobs has developed and implemented various skilling programs focusing on project construction, system installation, and operations & maintenance roles related to Green Hydrogen. This updated participant book, titled SGJ/Q0121 Green Hydrogen Plant Entrepreneur, is specifically designed to facilitate theoretical and practical training for Green Hydrogen plant entrepreneurs. It is available for free download and aims to empower individuals with the knowledge and skills required to excel in this nascent but fast-emerging field.

Symbols Used



**Key Learning
Outcomes**



Steps



Exercise



Activity



Notes



**Unit
Objectives**

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1. Introduction to Green Hydrogen

Unit 1.1 Necessity of Green Hydrogen In Sustainable Energy Transition

Unit 1.2 Properties, Characteristics and Nomenclature of Hydrogen

Unit 1.3 Value Chain for Green Hydrogen

Unit 1.4 Key Aspects and Challenges Related To Production, Storage, Transportation, and Distribution of Green Hydrogen

Unit 1.5 End-Use Applications of Green Hydrogen in Industry, Transport and Power Production

Unit 1.6 Technology Options for Production of Green Hydrogen

Unit 1.7 Renewable Energy Sources for Large Scale Production of Green Hydrogen Through Electrolysis of Water

Unit 1.8 Green Hydrogen Economy in Indian Context

Unit 1.9 Role and Responsibilities of a Green Hydrogen Plant Entrepreneur



Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the properties and characteristics of Hydrogen
2. Discuss various colour code nomenclature of Hydrogen and role of Green Hydrogen in sustainable energy transition.
3. Discuss key aspects related to production, storage and transportation of Green Hydrogen.
4. Discuss key aspects of the hydrogen economy
5. Discuss the applications of Green hydrogen in industry, transport and power production.
6. Discuss in brief the emerging entrepreneurial opportunities across Green Hydrogen Value chain
7. Identify opportunities to various renewable energy sources
8. Discuss the role and responsibilities of a Green Hydrogen Plant Entrepreneur.

UNIT 1.1: Necessity Of Green Hydrogen in Sustainable Energy Transition

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the concept of sustainable energy
2. Explain the necessity of green hydrogen in sustainable energy transition.

1.1.1. Introductions

Energy sources and technologies that can be produced and used in a way that satisfies the current demands without compromising the ability of future generations to meet their own energy needs are referred to as sustainable energy, also known as renewable energy or green energy. It is a form of energy generation that has very minimum impact on the environment, does not deplete finite resources, and helps reduce climate change by cutting down the greenhouse gas emissions.

Sustainable energy transition refers to the process of shifting from conventional, fossil fuel-based energy systems to cleaner and more environmentally friendly sources of energy. It involves a comprehensive transformation of the energy sector, with the primary goal of reducing greenhouse gas emissions, mitigating climate change, and achieving a more sustainable energy future.

The sustainable energy transition is necessary to protect the environment, combat climate change, promote economic growth, and create a more sustainable and equitable future for current and future generations.

1.1.2. Green Hydrogen in Sustainable

Green Hydrogen is an important technology for moving towards a cleaner energy system with net zero emissions. It can replace the use of grey hydrogen (derived from fossil fuels) in various industries like oil refineries, fertilizer plants, and chemical plants that produce methanol. Green hydrogen is hydrogen gas that is produced using renewable energy sources, such as wind, solar, or hydropower.

Note: Grey hydrogen is utilised in various industrial applications as a feedstock or process gas. It is derived from fossil fuels by using the steam methane reforming (SMR) or coal gasification process. In these procedures, steam is reacted with fossil fuels, such as coal or natural gas, to create hydrogen gas and carbon dioxide (CO₂) as a byproduct. This process typically results in the release of carbon dioxide into the atmosphere, which increases greenhouse gas emissions and contributes to climate change.

Green Hydrogen can also be used for blending with piped natural gas (PNG) used in domestic, commercial, and industrial applications. It can also be used to power heavy-duty long-haul transportation like trucks and ships through hydrogen fuel cells. Furthermore, it can store energy and be used in combination with renewable energy sources to ensure a stable power supply.

Electricity is a convenient form of energy that became widely used after we found ways to generate it from water and fossil fuels. However, the rapid growth of fossil fuel-based electricity generation has led to a significant increase in greenhouse gas emissions, contributing to global warming and climate change. Similarly, vehicles powered by internal combustion engines using fossil fuels have also contributed to higher fossil fuel consumption and global warming.

Currently, hydrogen is mainly used in the chemical industry for fertilizers and refineries. Unfortunately, more than 95% of global hydrogen production comes from fossil fuels, which emit greenhouse gases and contribute to global warming and climate change.

India aims to achieve Net Zero emissions by 2070. With India's economic growth and development, the demand for energy and resources is expected to increase. Energy usage has already doubled in the past twenty years and is expected to grow by 25% more by 2030. India currently imports over 40% of its primary energy, costing more than USD 90 billion annually. Major sectors like transportation and manufacturing are completely dependent on imported fossil fuels. To address this, there is a need to shift towards technologies that increase the use of renewable energy sources and decrease dependence on fossil fuels. Thus, Green Hydrogen, produced from renewable sources, can play a crucial role in creating a low-carbon and self-sufficient economy.

Green Hydrogen enables us to use the abundant renewable energy available in different regions and seasons. It can be used in various ways, both as a fuel and as a material in industries. For example, it can replace fossil fuels in petroleum refining, fertilizer production, and steel manufacturing. By using Green Hydrogen, we can also make long-distance vehicles and ships more environmentally friendly. In addition, to fulfil the energy needs of remote regions, such as islands, green hydrogen can serve as a flexible and sustainable energy carrier.

Thus, Green hydrogen is essential in achieving a sustainable energy transition for some of the following reasons:

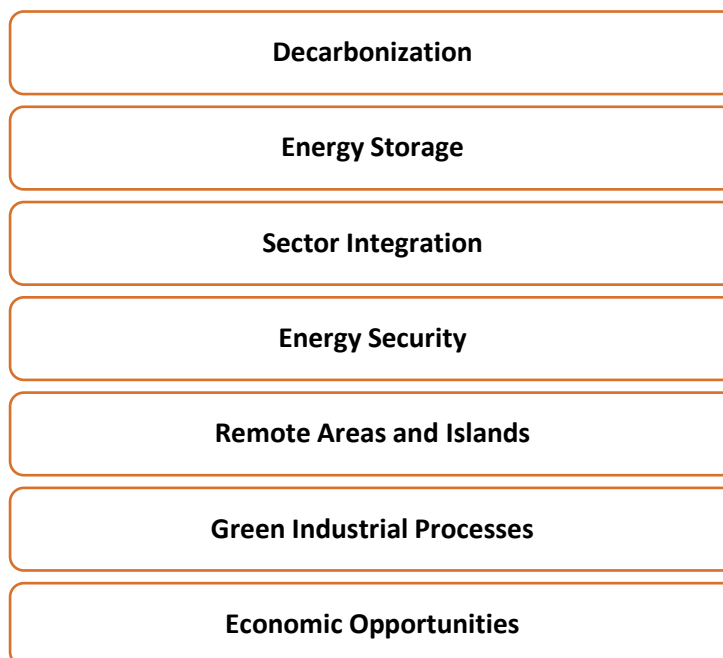


Fig. 1.1.1: Reasons for Considering Green Hydrogen for Substantial Energy Transition

- **Decarbonization:** Green hydrogen is produced using renewable energy sources like wind, solar, or hydropower, and its production process emits no greenhouse gases. By using green hydrogen as a clean fuel and feedstock, we can significantly reduce carbon emissions and combat climate change.
- **Energy Storage:** Green hydrogen can store excess energy from renewable sources during times of high generation and release it when demand is high or renewable energy production is low. This helps to stabilize energy supply and balance the intermittent nature of renewables.
- **Sector Integration:** Green hydrogen can be utilized in various sectors, such as transportation, industry, heating, and power generation. Its versatility makes it an important bridge between different sectors, enabling greater integration and flexibility in the energy system.
- **Energy Security:** By producing green hydrogen domestically, countries can reduce their dependence on imported fossil fuels, enhancing energy security and self-reliance.
- **Remote Areas and Islands:** Green hydrogen is particularly valuable for meeting the energy needs of remote regions and islands that may have limited access to traditional energy infrastructure. It can offer a sustainable and reliable energy solution in such areas.
- **Green Industrial Processes:** Green hydrogen can replace fossil fuels in industrial processes like steel production, ammonia synthesis, and petrochemicals, contributing to cleaner and more sustainable manufacturing.
- **Economic Opportunities:** The development and deployment of green hydrogen technologies create new economic opportunities, driving innovation, creating jobs, and promoting sustainable economic growth.

Note:

To know more about Green Hydrogen, you may refer to the following link:

- <https://www.youtube.com/watch?v=GqVxE2cVGk>

Title of Video: What is Green hydrogen (Hindi)

By: Darvesh

Duration: 6 m 46 sec

<https://www.youtube.com/watch?v=qu1W-6gc36c>

<https://www.youtube.com/watch?v=gEByrL5a27c>

UNIT 1.2: Properties, Characteristics and Nomenclature of Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the properties and characteristics of Hydrogen.
2. Describe basic concepts of Hydrogen as an energy carrier.
3. Discuss various colour code nomenclature of Hydrogen.

1.2.1. Atomic structure

Every matter (anything that occupies space and has mass) is made of tiny particles called atoms. All atoms of the same element are identical in size, mass, and properties, whereas atoms of different elements have different size and mass.

Atomic structure refers to the arrangement of components within an atom. It consists of a nucleus at the centre that is composed of positively charged protons and neutral neutrons. The negatively charged 'electrons' revolve around the nucleus.

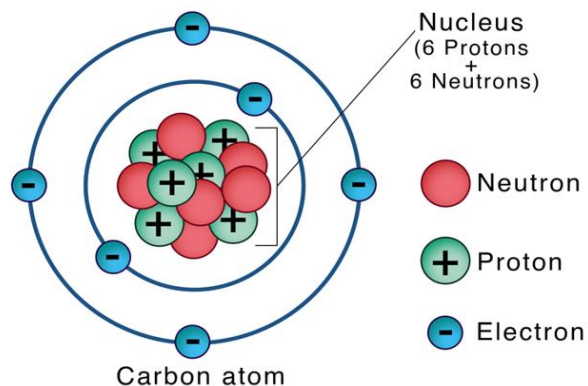


Fig. 1.2.1 Atomic structure of carbon

Source: Science facts. Atom. <https://www.sciencefacts.net/atom-2.html>

Atomic structure of hydrogen

Hydrogen is a colourless, odourless, tasteless, light, and non-toxic gas. It is a combustible gas but it does not support combustion. It is insoluble in water. The atomic structure of hydrogen typically includes one proton and one electron. The number of protons depends upon the Isotopes of hydrogen. Isotopes of hydrogen means different forms of hydrogen atoms that have the same number of protons (which defines the element as hydrogen) but varying numbers of neutrons. Following are the different isotypes of hydrogen:

- Protium(H): It consists of one proton and one electron in its atomic structure and no neutrons.
- Deuterium (H₂): It contains one proton and one neutron in its nucleus, along with one electron.
- Tritium (H₃): It contains one proton and two neutrons in its nucleus, along with one electron.

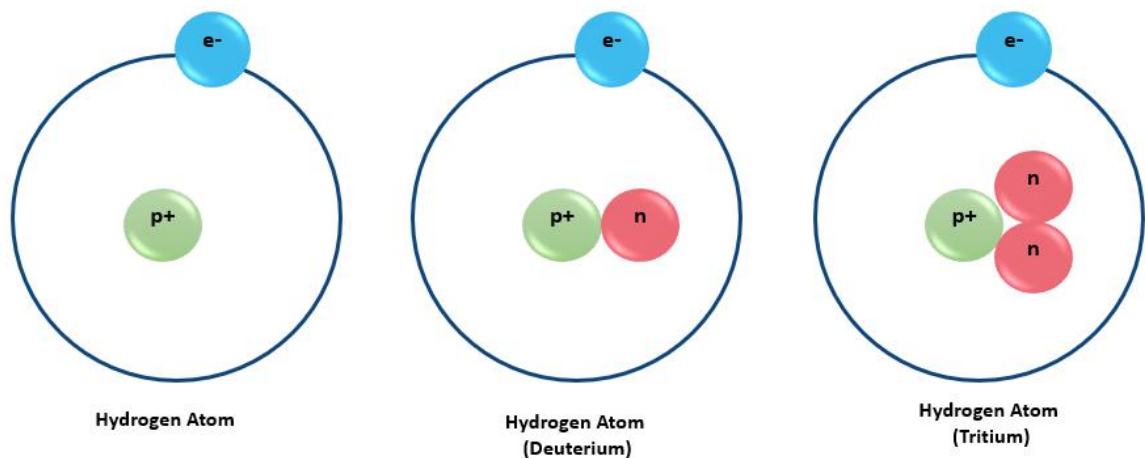


Fig. 1.2.2 Atomic Structure of Isotopes of Hydrogen

Comparison of Hydrogen Volumes

Hydrogen is the lightest gas. One kilogram of hydrogen takes up 11,000 litres of space at normal temperature (20°C) and regular air pressure. However, if hydrogen gas is compressed to 700 bars (bar is the unit to measure air pressure), it will occupy only 24 litres of space.

Whereas, the liquified hydrogen will occupy only 14 litres of space at the temperature -252.87° C and air pressure at 1 bar. But, liquified hydrogen has its challenges/problems and therefore, currently the compressed hydrogen appears to be the best cost-effective storage option due to the challenges associated with liquid hydrogen and until better alternatives become available.

The following figure represents the Storage volume for one kilogram of hydrogen in its different forms:

Storage volume for 1kg of hydrogen

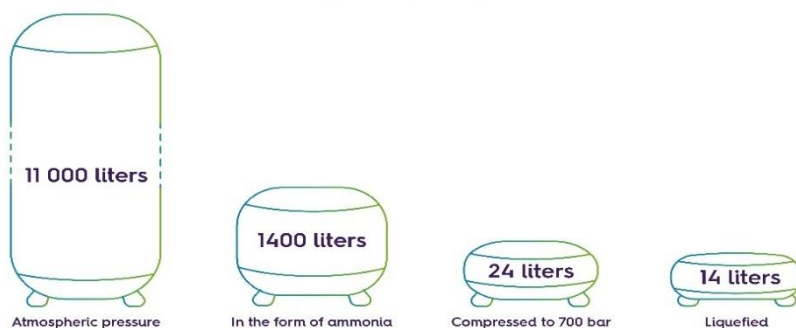


Fig. 1.2.3 Storage volume of 1 kg hydrogen at NTP

Source: Terega: Hydrogen storage: A challenge for the development of the industry

1.2.2. Properties of Hydrogen

Hydrogen is a chemical element with several notable properties. The following are some of the properties of hydrogen that makes it versatile element with potential applications in energy, industry, and various scientific fields:

- Hydrogen is colourless and odourless in its natural state.
- It is the lightest element/gas with atomic number 1 and is most abundant element/gas.
- It is one of the highly flammable gases that can easily catch fire if exposed to any flame or spark. In certain situations where there is enough hydrogen present and an ignition source is added, hydrogen can combine to generate explosive mixtures in the air. Thus, when the hydrogen content in this mixture falls within a certain range (often between 4% and 75% hydrogen in air) it becomes potentially explosive.
- It has low density when it is in gaseous form that can be useful for applications such as in airships and balloons. But due to this property, one of the challenges with hydrogen is its storage and transportation and thus need specialized equipment to prevent leakage.
- It has a high energy content per unit mass that makes it a potential fuel source for various applications that includes powering vehicles and generating electricity in fuel cells.
- It can be used as an energy carrier to store and transport energy generated from renewable sources such as solar and wind. It can be converted back into electricity or other forms of energy when needed.

The following table lists the properties of hydrogen:

Property	Value
Density	0.089 g/litre (gas) 0.07 g/cm ³ (liquid)
Ionization energy	13.5989 eV (electron Volt)
Liquid to gas expansion ration	1:848 (atmospheric conditions)
Melting point	-259.14° C
Boiling point	-252.87° C
Heating value	118.8 MJ/kg (Lower) 142 MJ/kg (Upper)
Flammability range in air	4 to 75%
Laminar flame velocity	3.06 m/s
Flash point	-253° C
Auto ignition temperature	585° C

Table 1.2.2 Chemical and Physical Properties of hydrogen

Before we compare hydrogen with other fuels let us understand the concept of diesel engine and fuel cells.

Diesel Engine and Fuel Cell

A fuel cell is a device similar to a battery. It converts chemical potential energy (energy stored in molecular bonds) into electrical energy. It has two electrodes where the chemical reaction takes place. An electrolyte carries the charged particles from one electrode to the other. It

generates electrical energy as long as hydrogen is supplied to it. The operation of a fuel cell is just the reverse of electrolyzer i.e. a PEM electrolyzer can operate as a fuel cell also.

A fuel cell works like a battery. It turns stored chemical energy into electricity. The chemical reaction happens between two electrodes and the electrolyte transports charged particles between these electrodes. It produces electricity when hydrogen is supplied. A fuel cell is the opposite of an electrolyzer - a PEM electrolyzer can also function as a fuel cell.

The following table represents the comparison between diesel driven internal combustion engine and the fuel cell:

Description	Thermal engine	Fuel cell	Remarks
	Diesel	H ₂ 700 bars	
Raw specific energy (kWh/kg)	12.00	33.33	Energy per unit mass
Specific mass (kg/m ³)	840	42	Mass per unit volume
Energy density (kWh/m ³)	10,080	1400	Energy per unit volume
Mass of fuel/mass of tank or system ratio	0.67	0.056	Ratio = (mass of fuel/ mass of tank)
Volume of fuel/volume of tank or system ratio	0.80	0.65	Ratio = (volume of fuel/ volume of tank)
Overall conversion (%)	33	45	
Net specific energy on-board (kWh/kg)	2.640	0.84	Net usable specific energy per unit mass
Net energy density on board (kWh/m ³)	2661	409	Net energy density per unit volume

Table 1.2.1 Diesel engine and fuel cell systems of electrical energy supply

A comparison of hydrogen with other fuels is given in Table 1.2.2.

Property	Hydrogen	CNG	Petrol	Diesel
Carbon content (mass%)	0	75 ^e	84	86
Lower heating value (MJ/kg)	119.7	45.8	44.8	42.5
Density ^{a,b} (kg/m ³)	0.089	0.72	730 to 780	830
Volumetric energy content ^{a,b} (MJ/m ³)	10.7	33.0	33 × 10 ³	35 × 10 ³
Molecular weight	2.016	16.043 ^e	≈110	≈170
Boiling point ^a (K)	20	111 ^e	298 to 488	453 to 633
Auto-ignition temperature (K)	858	813 ^e	≈623	≈523
Flammability limits in air (vol%)	4 to 76	5.3 to 15	1 to 7.6	0.6 to 5.5.

^a at 1 bar, ^b at 273 K, ^c at 298 K, ^d at stoichiometry, ^e methane, ^f vapour and ^g n-heptane

Table 1.2.2 Comparison of hydrogen with other fuels

Source: *Yip H.L., Srna A., Chun A., and Yuen A.C.Y. 2019. A Review of Hydrogen Direct Injection for Internal Combustion Engines: Towards Carbon-Free Combustion*

The tables above show that hydrogen carries over twice the energy per unit of weight compared to natural gas and diesel. Additionally, using hydrogen for energy does not produce any greenhouse gas emissions. This characteristic of hydrogen is the primary factor driving its selection for the energy transition.

1.2.3 Hydrogen as an Energy Carrier

Similar to electricity, Hydrogen is considered an energy carrier because it doesn't exist in its pure form in nature. Instead, it is primarily derived from water and hydrocarbons through the utilization of external energy sources.

Hydrogen is considered an energy carrier because it can store and transport energy in a form that can be easily converted into usable energy when needed. It can store and carry 33.3 kWh/kg energy which is higher than CNG (13 kWh/kg), diesel (12 kWh/kg) and ammonia (5.4 kWh/kg).

Hydrogen has a fuel conversion efficiency of over 50% (i.e. >50%), far surpassing CNG, diesel, and ammonia, which have efficiencies below 40%. Additionally, hydrogen emits fewer pollutants as compared to these fossil fuels. While hydrogen serves as a superior energy carrier, its primary challenge lies in either compressing it at high pressure or liquefying it for practical and cost-effective bulk transportation.

The following figure represents the production of hydrogen using electricity and its uses as an energy carrier:

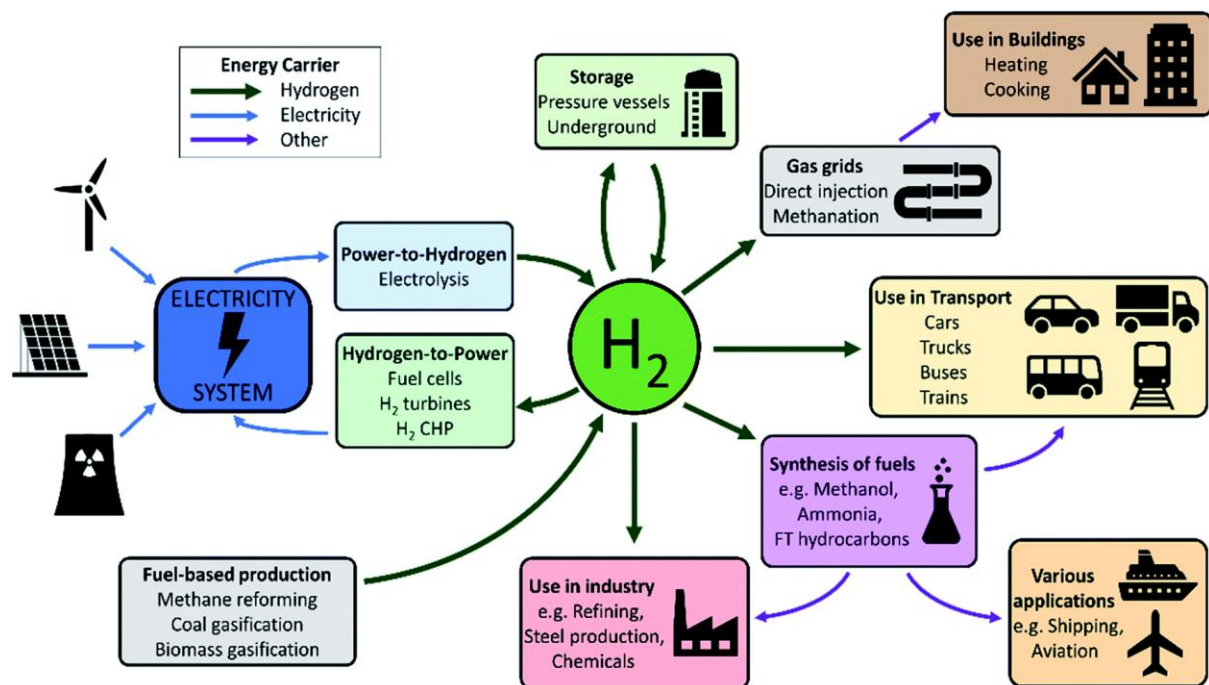


Fig. 1.2.4 Green Hydrogen as a Carrier of Electrical Energy and Its Uses

Source: Christopher J. Quarton C.J. et al. 2020. The curious case of the conflicting roles of hydrogen in global energy scenarios. Sustainable Energy Fuels

1.2.4 Colour Code Nomenclature of Hydrogen

The amount of hydrogen in the atmospheric air varies, it is around 0.6 Parts Per Million (PPM), which is almost negligible. Hydrogen is found in water and hydrocarbons such as natural gas (about 95% methane, CH₄) and biomass (cellulose).

Naturally Hydrogen is a colourless gas and can be produced from different sources of energy using different processes. Based on the source of energy used to produce hydrogen, different colour codes have been assigned to it. Note, that there is no difference in hydrogen produced by any of the sources. The following figure represents the colour codes, source of input energy, and the processes of production of hydrogen:

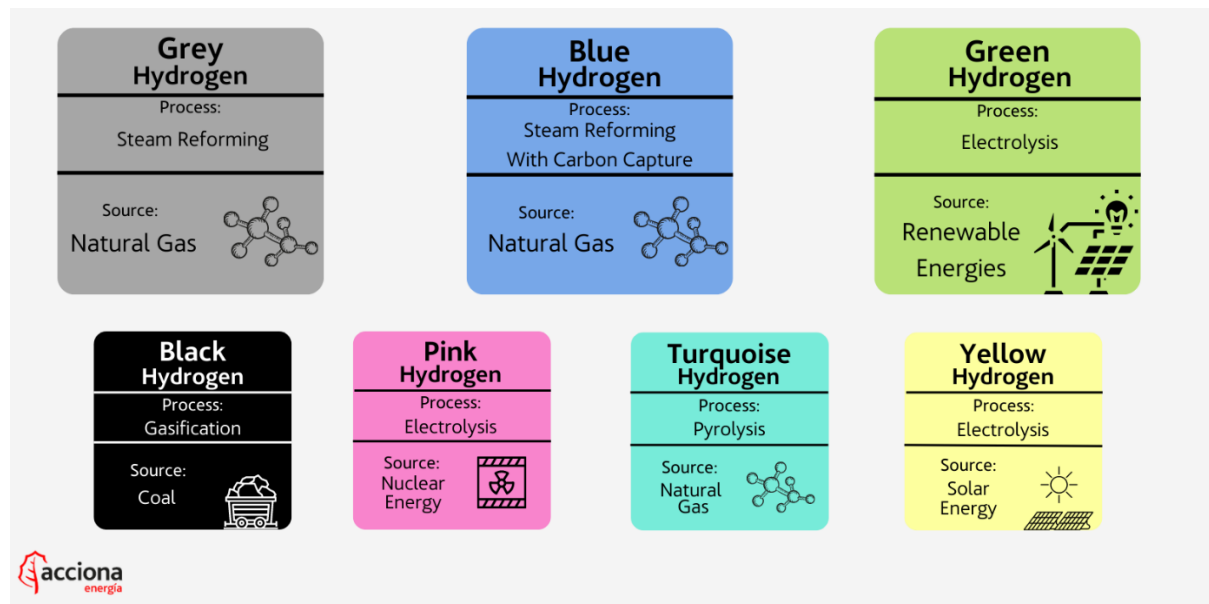


Fig. 1.2.5 Colour code nomenclature of hydrogen

Source: Acciona.2022. What Are the Colours of Hydrogen and What Do They Mean?

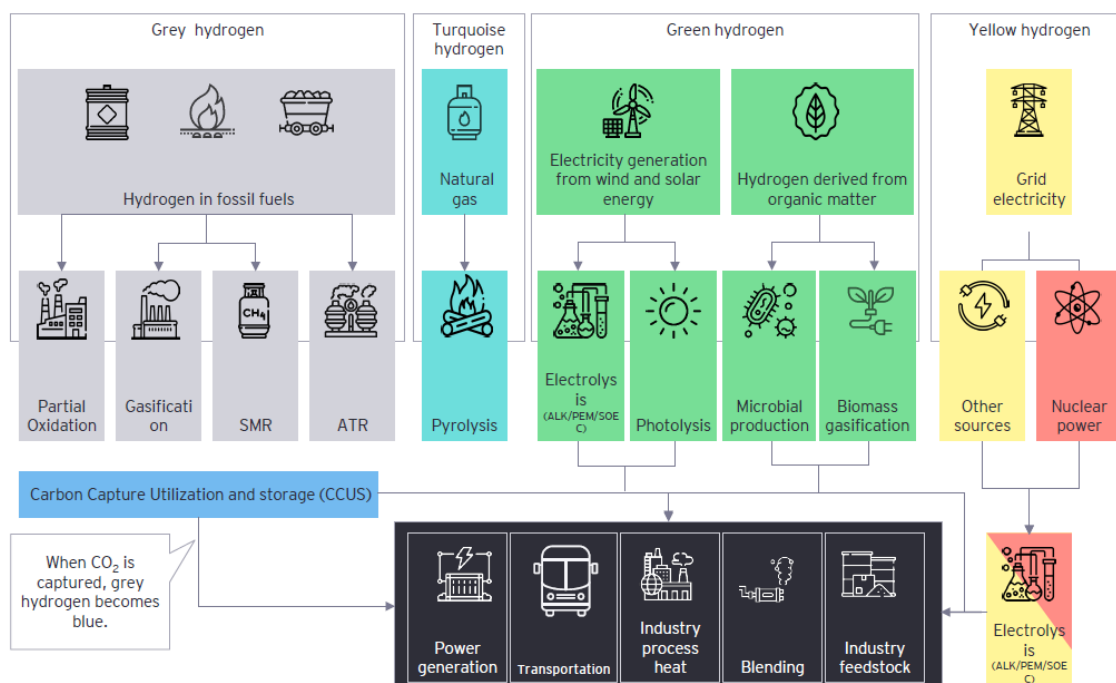
The colour coding of hydrogen is explained below:

- **Grey Hydrogen:** It is produced using the steam methane reformation (SMR) process, which relies on fossil fuels. During this process, carbon dioxide is released into the atmosphere. Consequently, the worldwide production of grey hydrogen is responsible for emitting around 830 million tonnes of carbon dioxide each year. As per the calculations, the production of one kilogram of hydrogen results in the emission of 8 to 9 kilograms of carbon dioxide (CO₂).
- **Blue Hydrogen:** Its production is similar to the production of grey hydrogen; both are produced from fossil fuels. However, with blue hydrogen, carbon dioxide is trapped and stored through a method called carbon sequestration. When the collected hydrogen is used, it's called 'Carbon Capture, Storage, and Utilization (CCSU)'. This process is environmentally friendly and is categorized as carbon neutral. According to International

Renewable Energy Agency (IRENA), the capturing of carbon dioxide is 85% to 95% efficient, i.e. 5% to 15% of carbon dioxide will be leaked out during the process.

- **Green Hydrogen:** It is produced using renewable energy such as wind or solar power. It is produced through electrolysis process, a process in which water is divided into hydrogen and oxygen without releasing greenhouse gases (GHGs). Green Hydrogen is the cleanest and most sustainable form of Hydrogen. However, globally only 1% of the total hydrogen produced is the green hydrogen.
- **Black Hydrogen:** It is produced from the gasification of coal. However, this process is heavily polluting and leads to the emission of carbon dioxide (CO₂) and carbon monoxide (CO) into the atmosphere.
- **Pink Hydrogen:** It is produced by using the electricity produced by a nuclear power plant for electrolysis. It indicates a hopeful advancement in eco-friendly hydrogen production, holding potential for lowering emissions and fostering sustainability.
- **Turquoise Hydrogen:** It can be extracted through methane thermal splitting using pyrolysis. Although this method is still in the experimental phase, it extracts carbon in solid/powder form rather than as carbon dioxide gas.
- **Yellow Hydrogen:** It is produced through the electrolysis of natural gas using solar power. Yellow Hydrogen is considered a transitional option with lower emissions compared to Grey Hydrogen (produced from fossil fuels) but higher than Green Hydrogen, which is considered the most environmentally friendly option.

The following figure represents the process of producing Hydrogen based on the colour code.



Source: EY, Accelerating Green Hydrogen Economy, June 2022

Activity



- Match the process and source of production of hydrogen with the colour codes of hydrogen.

Process		Source of production	
A	SMR	E	Renewable electricity
B	Gasification	F	Biomass
C	Electrolysis	G	Coal
D	Anarobic digestion	H	Methane

- (a) A-E, B-F, C-H, D-G
 (b) A-H, B-G, C-E, D-F
 (c) A-F, B-G, C-E, D-H
 (d) A-G, B-H, C-F, D-E

- Based on the Hydrogen colour code, write the process and the source of input energy respectively.

Hydrogen colour Code	Process	Source of input energy
Grey		
Blue		
Green		
Black		
Pink		
Turquoise		
Yellow		

Equipment:

- Chart of the colour-code nomenclature of Hydrogen.
- Chart of colour-code Hydrogen production processes.

UNIT 1.3: Value Chain for Green Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the concept of value chain for Green Hydrogen.
2. Explain the elements of a value chain of Green hydrogen.
3. Discuss benefits and drawbacks of existing methods of Hydrogen production.

1.3.1 Value Chain

A value chain refers to the lifecycle of a product or service, starting with sourcing of materials and may include production, distribution, after-sales services and disposal/recycling processes. The company/industry analyses the value chain with an objective to increase efficiency to obtain maximum value for the least possible cost.

By analysing each stage of the value chain, companies can identify opportunities to improve their processes, reduce costs, and increase the overall value delivered to customers. Value chain analysis also helps in understanding the competitive landscape and the roles of different stakeholders within an industry.

The hydrogen value chain is typically split into the three stages i.e. production, storage and distribution, and consumption/ application. Each of these fields requires expert know-how to ensure safety, decrease costs, and guarantee compliance with regulations, codes, and standards. Thus, the value chain shows the values to be considered to work out the hydrogen cost to the end user. It is the basic requirement to compete with the existing fossil sources of energy. Further, safety features required for existing fossil fuels are of lower level and have been taken care of very well over a century. As compared to this, safety features required for hydrogen are of higher level and are backed with the limited experience of its use in some industries. Hydrogen value chain has to compete considering this factor also.

In this context of hydrogen value chain, India is in an advantageous position due to availability of low-cost renewable energy in abundance and great potential for export of green hydrogen and green products. In addition, it will minimize the import of fossil fuels which will also reduce the emission of GHGs.

The hydrogen value chain comprising of its production, storage, transportation and utilization is shown in Figure 1.3.1.

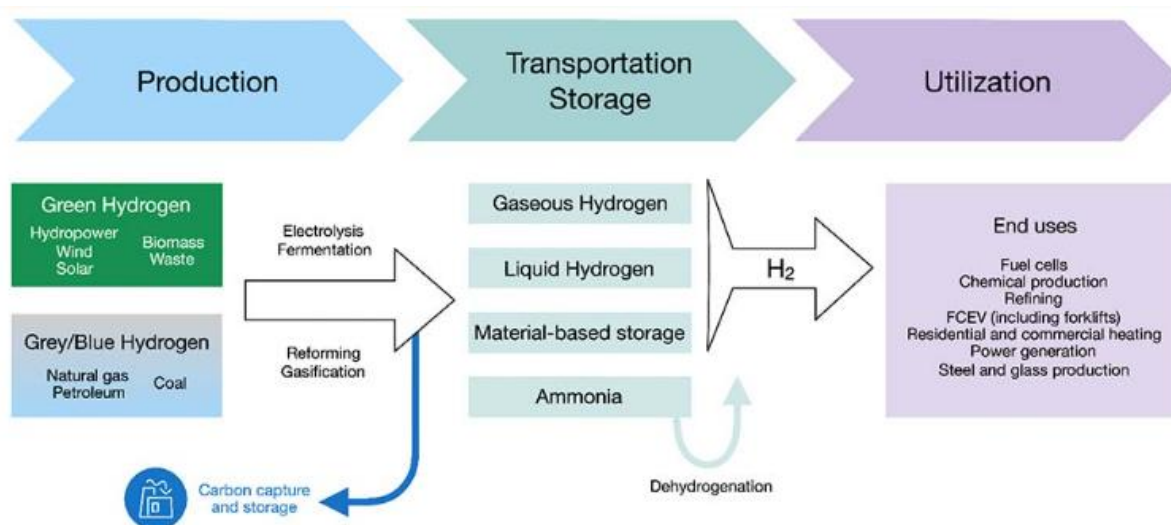


Fig. 1.3.1 Value chain of hydrogen

Source: Hydrogen Energy Global Road Map 2020-2050 / Value Chain, Technology, Road Map, Prevision – D-News February 2021

1.3.2 Benefits and Drawbacks of Existing Methods of Hydrogen

Production

The following table represents the benefits and drawbacks of hydrogen production method:

Hydrogen Production Methods	Benefits	Drawbacks
Steam methane reformation Input: natural gas (methane) Output: Grey hydrogen	-Well established technology -High yield -Low cost	Emission of greenhouse gases (8 to 9 kg carbon dioxide emission during 1 kg production of 1 kg hydrogen)
Coal gasification Input: Coal Output: Black hydrogen	-Syngas also known as synthesis gas can be treated to produce petrol and diesel. -Residual Coal tar for road construction	Highly polluting CO ₂ , CO
Pyrolysis Input: Methane Output: Turquoise hydrogen	-No emissions -Solid carbon is the byproduct	-Process is complex and under development -High operating temperature above 1000° C -Purification of air is necessary before its release due to high CO content
Biomass gasification Input: Biomass Output: Green hydrogen	-Biomass is renewable and easily available	-Some emissions of CO ₂ , CO -Process of biomass gas production is natural and therefore it is quite slow
Electrolysis Input: Wind, Solar, Hydro Output: Green hydrogen	No emissions	-Costly -Dedicated infrastructure for hydrogen storage,

		transportation/distribution are costly. -Higher level safety requirement as compared to existing fuel natural gas
--	--	--

Table 1.3.1: Benefits and drawback of hydrogen production methods

The following table represents the benefits and drawbacks of Alkaline, PEM, and solid oxide electrolysis:

	AEL	PEM	SOEL
Pros	- Established technology - Plants with the highest nominal output (>100MW) - Lowest investment costs (1000€/kW) and long lifetime - Does not require critical raw materials	- Hydrogen with highest purity - Good dynamic properties and high load gradient: good for fluctuating RES - Cold start in only 15 min	- Highest efficiency (80%) - Suitable for co-electrolysis : direct synthesis gas generation
Cons	- Vulnerability to impurities in the product gases - Relatively long cold start time (50 min)	- Electrodes made of precious metals - Higher investment costs (2000 €/kW)	- Currently still under development (only pilot plants so far) - Highest investment costs (2500 €/kW) - Highest cold start time (several hours)

Table 1.3.2 Pros and cons of the electrolyzers

Source: Hussy C. Water electrolysis explained – the basis for most Power-to-X processes. International PtX Hub

VIDEO

Topic: How to produce hydrogen gas at home (Hindi)
 By: Indian Hacker
 Link: https://www.youtube.com/watch?v=RAbjgk1Zlwc&ab_channel=MR.INDIANHACKER
 Duration: 3 m 45 sec

UNIT 1.4: Key Aspects and Challenges related to Production, Storage, Transportation, and Distribution of Green Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the challenges in the production and storage of green hydrogen
2. Explain the challenges in the transportation and distribution of green hydrogen

1.4.1 Challenges in production of green

The primary challenge in the production of green hydrogen is the cost of production technologies. The aim is to reduce the cost of production technologies to match the cost with the production of traditional fuels. To mitigate this challenge, the Government of India has introduced policy that offers incentives for manufacturing of the electrolyzers and production of hydrogen.

1.4.2 Challenges in storage of green hydrogen

While storing green hydrogen, the primary challenge is the cost. Storing green hydrogen comes with higher costs due to its low energy density, requiring larger storage volumes. Ensuring safety against hydrogen's flammability requires specialized equipment. Moreover, advanced storage methods like high-pressure or low-pressure (cryogenic storage) involve technical intricacies and increased infrastructure expenses.

Whereas the storage of fossil fuels is cheaper for several reasons. First, they are stored at regular atmospheric pressure, which reduces storage costs. Second, there's less risk because they have a relatively small range where they can catch fire. Additionally, fossil fuels have a low ratio of their volume or weight to the volume or weight of the storage container. This means that the space they occupy in storage is relatively small compared to the space taken up by the storage containers themselves.

In contrast, for hydrogen, the ratio of its volume or weight to the volume or weight of the container is much higher. (i.e. H_2 volume/weight $\div$ container vol. or wt = $\ll 1$. It is around 0.7 to 0.8 for fossil fuels and their containers).

In a project by NTPC in Leh, there's a plan to store 1 kilogram of hydrogen at a pressure of 100 bars within a cylinder weighing 149 kilograms.

Thus,

- $1 \div 149 = 0.0066$

It means that the weight of the hydrogen is only 0.66% of the total cylinder weight.

In comparison, a typical household LPG cylinder weighing 16 kilograms can store 14 kilograms of LPG gas, resulting in a ratio of 0.9, where the gas is 90% of the cylinder's weight.

When compared with fossil fuels, the cost of hydrogen storage is very high. Hydrogen being the lightest gas needs high compression to store it in its gaseous form. For storing liquid hydrogen, cryogenic vessels are required. It is a challenge to maintain the temperature of -252.8 deg C that may result in boil off losses of about 0.5% per day. Another major challenge is Embrittlement of metals. Due to these phenomena holes and crack are developed in metallic tanks in which hydrogen is stored.

Thus, following are the challenges in storing green hydrogen:

Low Energy Density	•Green hydrogen has a low energy density, necessitating a larger volume for storage compared to other fuels.
Leakage Concerns	•The small molecular size of hydrogen molecules can lead to leakage through materials used for storage.
Embrittlement of Materials:	•Hydrogen's interaction with certain metals can cause embrittlement, potentially compromising storage container integrity.
Storage Methods:	•Choosing appropriate storage methods like high-pressure or cryogenic storage presents technical and cost challenges.
Safety Considerations:	•Green hydrogen is highly flammable and requires careful safety measures to prevent potential explosions.
Infrastructure Development:	•Developing storage infrastructure compatible with green hydrogen's properties and cost-effective poses a significant challenge.

Fig. 1.4.1: Challenges in Storing Hydrogen

1.4.3 Challenges in Transportation of Green Hydrogen

Special tankers are required for high pressure hydrogen gas or cryogenic tankers for liquid hydrogen as described in the case of storage tanks. Following are some of the challenges in transportation of green hydrogen:

Low Energy Density

Green hydrogen's low energy density requires specialized storage and transport methods to accommodate the large volumes needed to transport a significant amount of energy.

Safety

Hydrogen's flammability requires stringent safety measures during transportation to prevent accidents or leaks.

Distance and Efficiency

Over long distances, energy losses can occur during transportation, especially when using conventional methods like compression or liquefaction.

Cost

Establishing efficient transportation methods, such as high-pressure tube trailers or liquid hydrogen carriers, can be costly.

Material Compatibility

Hydrogen's small molecular size can pose challenges in selecting suitable materials for containers and pipelines to prevent leakage or embrittlement.

Storage During Transit

Storage during Transit: Maintaining proper storage conditions during transit is crucial to prevent hydrogen loss or deterioration.

Fig. 1.4.2 Challenges in Transportation of Green Hydrogen

1.4.4 Challenges in the Distribution of Green Hydrogen

Distribution of green hydrogen has various challenges such as:

- Setting up the necessary infrastructure like pipelines and storage facilities requires investments and planning. Due to embrittlement of metals, the existing infrastructure of gas pipelines cannot be used for hydrogen due to the problem of embrittlement of metals. Until specific infrastructure for hydrogen is established, it might be feasible to blend natural gas with hydrogen up to approximately 15% without requiring significant alterations to the current infrastructure.
- Keeping the hydrogen pure and safe during distribution is important.
- Choosing the right ways to transport it, minimizing energy losses, and ensuring safety add complexity.
- Integrating green hydrogen with existing systems, gaining public trust, and managing supply fluctuations from renewable sources are also hurdles.
- Non-availability of the fueling stations is another constraint. High level security is required for both personnel and plant.

Overcoming these challenges is essential to effectively use green hydrogen for energy.

The following schematic diagram represents the green hydrogen for blended CNG, transport, industry, and power generation.

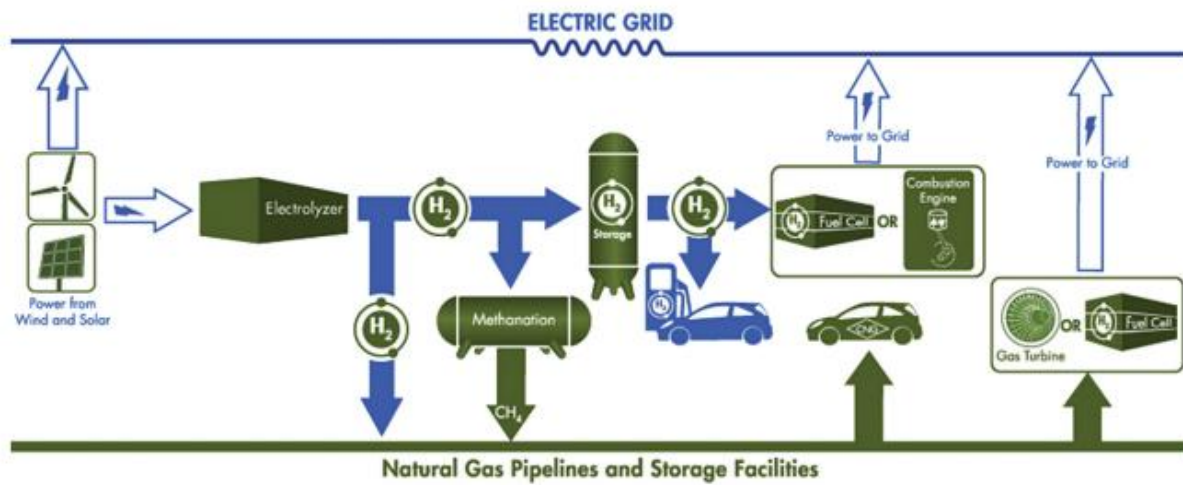


Fig. 1.4.3 Green hydrogen for blended CNG, transport, industry and power generation

Source: Gondal I. A. 2018. Hydrogen integration in power-to-gas networks. *International Journal of Hydrogen Energy*

UNIT 1.5: End Use Applications of Green hydrogen in Industry, Transport and Power Production

Unit Objectives



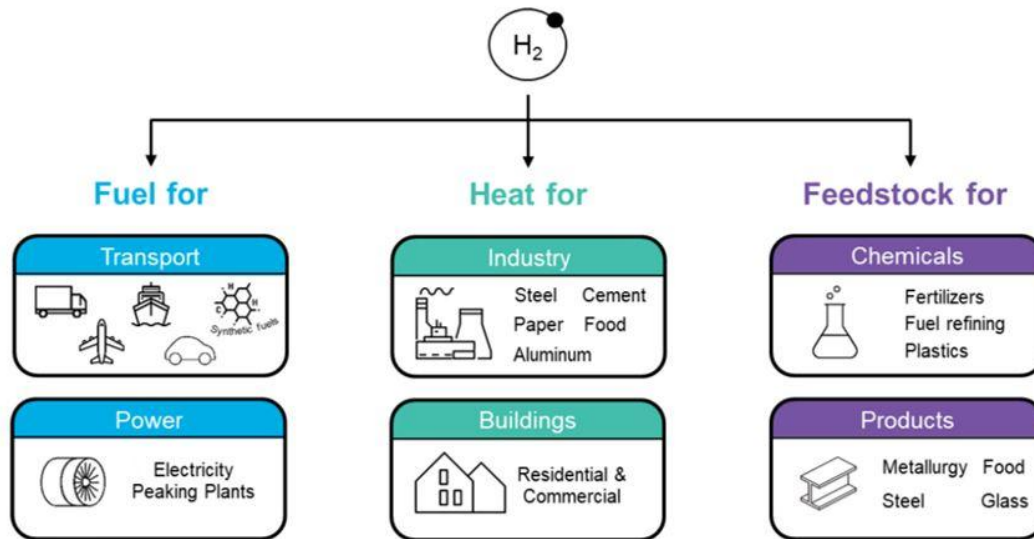
At the end of this unit, you will be able to:

1. Understand the applications of green hydrogen in industry and transport sectors
2. Explain the application of green hydrogen in electric power generation

1.5.1 End Use Application of Green Hydrogen in Industry

Green hydrogen has the potential to be used in various areas like industry, transportation, and producing power. It's a clean and adaptable solution that can replace fossil fuels. Its potential to replace fossil fuels can reduce pollution in industries, make transportation greener, and provide a sustainable energy source for power generation.

For more than a century, hydrogen has been primarily utilized by the chemical and refining industries. The following figure represents the end uses of green hydrogen as a fuel, for heating, and feedstock.



Source: BloombergNEF

Fig. 1.5.1 End uses of green hydrogen

Source: Kane M. K. and Gil S. 2022. Green Hydrogen: A key investment for the energy transition. World Bank Blog

Let us understand the uses of hydrogen across industries.

Agricultural industry

In the agricultural industry, green hydrogen finds practical application as a clean energy source for various processes. It can power machinery and equipment, such as tractors and irrigation systems, reducing carbon emissions. Additionally, green hydrogen can be converted into ammonia, a crucial component of fertilizers, offering a sustainable alternative to traditional production methods that rely on fossil fuels.

Hydrogen is a basic raw material needed to produce ammonia for use in fertilizers such as urea, ammonium nitrate, ammonium sulphate etc.

Petroleum Refining Industry

Hydrogen is generally used in hydrocracking to create petroleum products, including gasoline and diesel. It is also used to produce methanol. The following figure represents the consumption of hydrogen in petroleum industry.

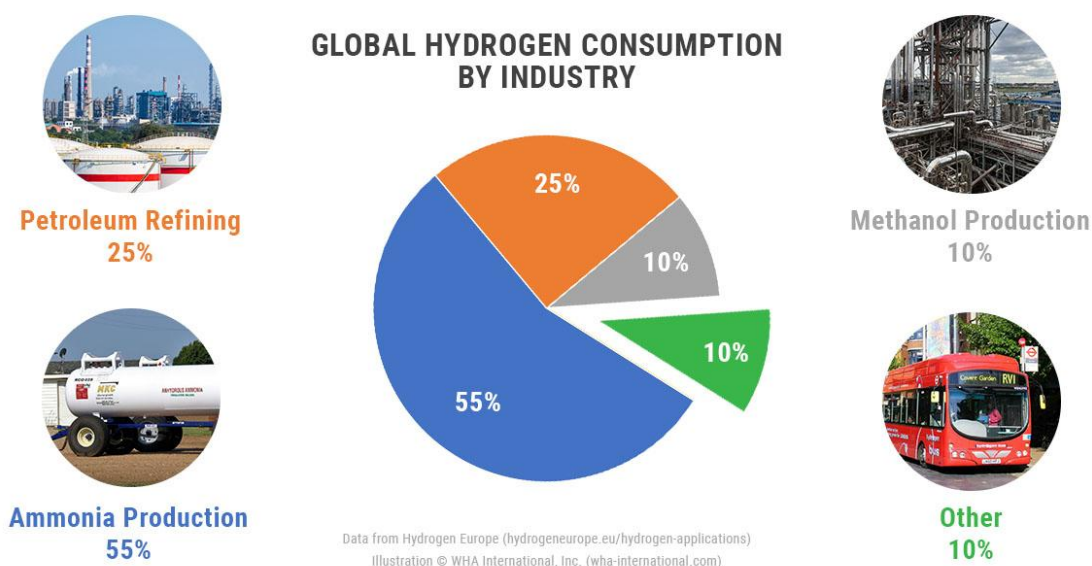


Fig. 1.5.2 Global hydrogen consumption by industries

Source: World Hydrogen Association: Hydrogen industry applications: Past, Present, and Future.
<https://wha-international.com/hydrogen-in-industry/>

Other industries

Following are the uses of hydrogen across different industries:

- **Iron and Steel:** The adoption of green hydrogen aims to substitute natural gas as a fuel source. Acting as a reductant, hydrogen generates direct reduced iron that can be further converted into steel using electric arc furnaces. Moreover, in future there's potential for hydrogen utilization in the cement industry.
- **Food:** Hydrogen is used to convert unsaturated fats into saturated oils and fats, including hydrogenated vegetable oils. It is also used in food packaging to extend shelf life and prevent oxidation. Additionally, hydrogen is employed in producing certain food additives and processing ingredients.
- **Metal works:** Hydrogen is used in multiple applications including metal alloying and iron flash making. It is used for heat treating metals to improve strength, in welding processes for shielding

and efficient heat generation, and as a reducing agent for producing metals like titanium and silicon. The atomic hydrogen welding is done in hydrogen environment.

- **Flat Glass Production:** During flat glass production, a mixture of hydrogen and nitrogen prevents oxidation and thus the manufacturing defects.
- **Electronics Manufacturing:** During the manufacturing phase of electronic components such as semiconductors, LEDs, displays etc. the hydrogen is used as a reducing agent and for etching.
- **Medical:** Hydrogen is used in the production of hydrogen peroxide and also its use in number of different diseases. Studies suggest that hydrogen may act as an antioxidant, protecting cells from oxidative stress and inflammation, both of which play roles in various diseases. These diseases range from cardiovascular issues and neurodegenerative conditions like Alzheimer's and Parkinson's diseases to inflammatory disorders. Also, it is used in cooling systems for MRI machines, as a carrier gas in gas chromatography, and as a fuel for fuel cells powering medical devices. Additionally, hydrogen is researched for potential therapeutic uses due to its antioxidant properties.

1.5.2 End Use Application of Green Hydrogen in Transport

Hydrogen is utilized as a fuel in different ways. It is used as a fuel for burning space exploration, internal combustion engines, and gas turbines to produce energy. Moreover, hydrogen can also be used to generate electricity using fuel cells. These fuel cells can create electric power, which is then used to drive the motors of electric vehicles. This dual usage of hydrogen showcases its versatility in providing energy for various applications such as:

- **Space Exploration:** Liquid hydrogen fuel has played a significant role in space exploration since NASA's Apollo mission. It has been utilized as a propellant to power rockets, enabling spacecraft to travel vast distances.
- **Aviation:** Hydrogen fuel cells are used in several experimental flights which were part of the projects, Pathfinder and Helios unmanned aircrafts. Airbus, a leading aerospace company, has a future strategy to use liquid hydrogen as a fuel to operate aircraft. They intend to do this by modifying traditional gas turbine engines to run on liquid hydrogen. In this approach, the liquid hydrogen would be used as a clean and sustainable energy source to power the engines of the aircraft.
- **Logistics:** Several logistics companies with large warehouses and distribution network around the world are exploring the use of hydrogen fuel cells to power trucks, forklifts, etc.
- **Public Transportation:** In certain countries, there is a growing trend of using hydrogen fuel cells to operate large buses and trucks. Moreover, hydrogen-fuelled trains have been introduced in Germany, and will start in India in 2023-24 besides Great Britain, France, Italy, Japan, South Korea, And The United States. Some of the major auto manufacturers have developed hydrogen fuel cell cars for personal use.

1.5.3 End Use Application of Green Hydrogen in Power Sector

Hydrogen is already used for cooling the hydrogen cooled stator winding of thermal power generators. In thermal power generators, there are parts called stator windings that can get very hot during operation. To prevent overheating and ensure safe operation, hydrogen gas is used to cool these stator windings. This is because hydrogen is an excellent conductor of heat and can efficiently dissipate the heat generated, helping to keep the generators running smoothly.

Hydrogen is also being used to generate electricity in a clean and reliable way. This electricity generation can serve as a stable source of power to help balance and stabilize the electricity grid. When there's high demand for electricity or when there are fluctuations in supply, hydrogen-based power generation systems can quickly provide the needed electricity, contributing to a more consistent and dependable electricity supply.

The following figure represents the Grid connected fuel cell power generation

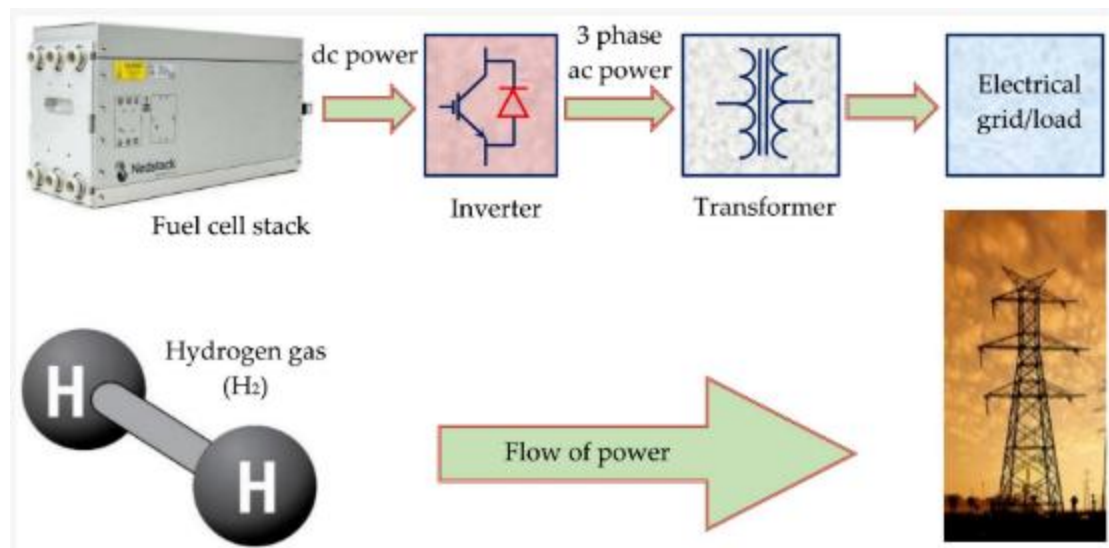


Fig. 1.5.3 Grid connected fuel cell power generation

Source: Koushik Ahmed et al. 2020. Proton Exchange Membrane Hydrogen Fuel Cell as the Grid Connected Power Generator. Energies

In reference the preceding figure, a fuel cell is an electrochemical power generator which combines hydrogen and oxygen to produce electricity. The byproducts are water and heat without emission of Green House Gases (GHGs). The Proton Exchange Membrane fuel cell (PEMFC) stack generates direct current (DC) power, which is then converted into three-phase alternating current (AC) power using an inverter. The inverter's settings are adjusted to align with the characteristics of the grid-connected step-up transformer.

Backup Power Generation

Stationary fuel cells have practical applications as uninterruptible power supply (UPS) systems. These systems are valuable for fulfilling the uninterrupted power needs of places like hospitals and data centres. In 2020, Microsoft achieve a success in using Hydrogen as the backup source to run a part of their data centre for almost two days.

The stationary fuel cells useful as uninterruptible power supply (UPS) systems for applications such as hospitals and data centres to fulfil their need of uninterruptible power supply. Similarly, in small residential areas, schools, hotels etc., fuel cells can provide backup power if the main grid fails. Alternatively, they can be operated as independent electric power systems.

UNIT 1.6: Technology Options for Production of Green Hydrogen

Unit Objectives



At the end of this unit, you will be able to:

1. Explain the electrolysis process
2. Identify the different types of electrolyzers

1.6.1 Electrolysis Process

Electrolysis is one the options to produce green hydrogen. The electrolysis process splits water into oxygen and hydrogen with the help of renewable electricity. The splitting occurs at two electrodes; cathode (-) and anode (+) in the electrolyzer, as shown in the following figure.

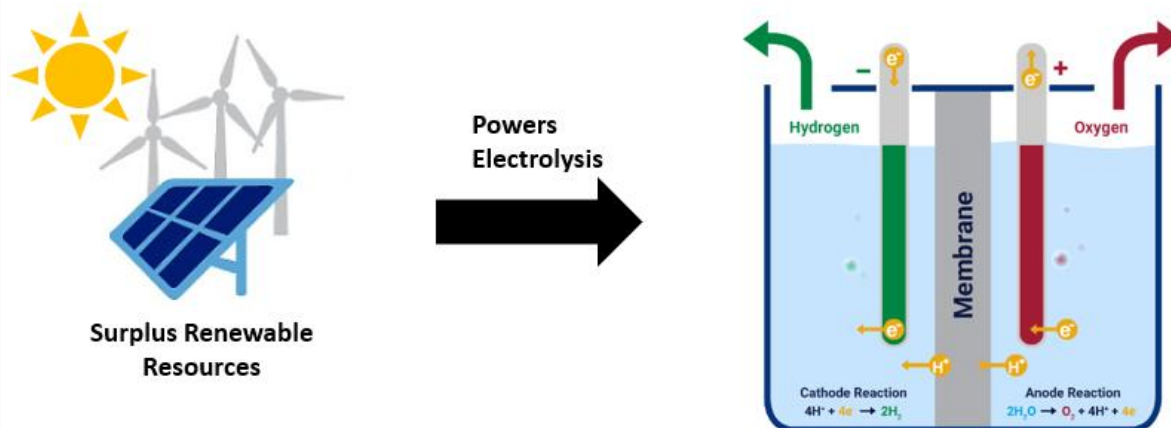


Fig. 1.6.1 Electrolysis

An electrolyzer is composed of a stack created by connecting multiple electrolysis cells. When direct current (DC) voltage is applied to the electrodes, hydrogen is generated at the cathode and oxygen is produced at the anode. This charge balance happens through ion movement facilitated by an electrically conductive electrolyte. Additionally, a membrane is used to keep the two reactions separate and prevent the gases produced from mixing. The nature of ions and the type of electrolyte used vary in different electrolysis technologies.

The following figure represents the renewable electrical energy and water as inputs to produce green hydrogen.



Fig. 1.6.2 Inputs and outputs of an electrolyzer

Source: Hussy C. Water electrolysis explained – the basis for most Power-to-X processes. International PtX Hub

1.6.2 Types of Electrolysis

Following are the different types of electrolysis:

Alkaline Electrolysis (AEL)

Proton Exchange Membrane (PEM) Electrolysis

Solid Oxide Electrolysis (SOE)

Anion Exchange Membrane Electrolysis (AEM)

Fig. 1.6.2: Different Types of Electrolysis Processes

Alkaline electrolysis (AEL)

Alkaline Electrolysis (AEL) processes uses a liquid electrolyte, which is generally potassium hydroxide (KOH) or sodium hydroxide (NaOH) and metallic electrodes. A non-permeable diaphragm separates the anode and cathode, preventing the passage of hydrogen and oxygen.

AEL is the longest-proven process as it has been deployed on a large scale since 1927. It comes with reduced initial expenses, a longer operational lifespan compared to other technologies, and is currently capable of producing the highest amounts of power (over 100MW) without requiring essential raw materials. However, AELs are not as robust as PEM electrolyzers. They

are more sensitive to impurities in the product gas as the gases dissolved in the electrolyte remain in the cycle. In addition, the AEL has a long cold start time of 50 minutes.

Proton Exchange Membrane (PEM) electrolysis

PEM electrolysis takes place in an acidic set-up. Therefore, precious metals such as iridium or platinum are used to protect the electrodes from corrosion. In order to produce hydrogen with higher level of purity, a gas-proof, proton-exchange plastic membrane is used. It is available on an industrial scale with outputs of over 10 MW i.e. producing more than 10 megawatts of power.

The PEM electrolysis is the recommended for operation with fluctuating electric supply from renewable sources of energy. Its cold start time is approximately 15 minutes.

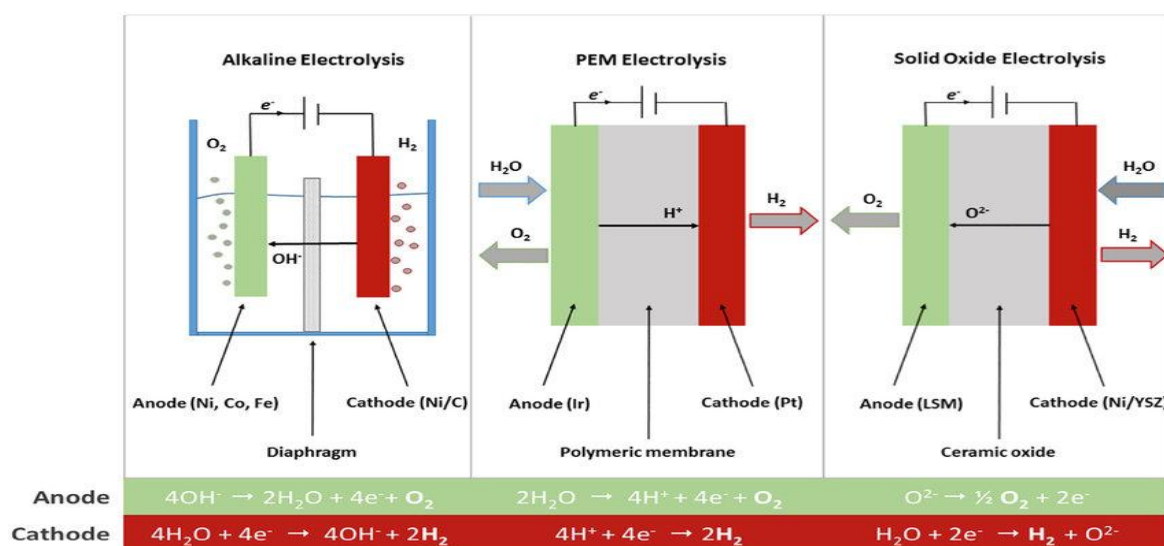


Fig. 1.6.3 Alkaline, PEM and solid oxide electrolyzers

Source: Sapountzi, F.M., J.M. Gracia, C.J. Weststrate, H.O.A. Fredriksson, and J.W. Niemantsverdriet. 2017. *Electrocatalysts for the generation of hydrogen, oxygen and synthesis gas. Progress in Energy and Combustion Science (Elsevier). vol. 58*

Note: Cold start time refers to the amount of time it takes for a system or device to go from a completely powered-off or idle state to full operational functionality.

Solid Oxide Electrolysis (SOE)

In this electrolyzer, the electrodes are separated by a solid oxide that conducts oxygen ions. It operates at high temperatures up to 1000° C and uses the superheated water vapour. The highest efficiency of 80% can be achieved with the help of an external heat source, such as industrial waste heat. However, the high temperature results in high material stresses. It is still in the development stage though it is being used as a pilot project. Its capital cost is higher in comparison to AEL and PEM. In addition, the cold start times are of several hours.

Anion Exchange membrane electrolysis (AEM)

Anion exchange membrane electrolyzers are being developed with the aim to club the advantages of alkaline electrolysis, such as stability of the cheap catalyst and proton-exchange membrane systems, like the ability to operate at differential pressure, fast dynamic response, low energy losses, and higher current density.

VIDEO-1

Topic: Hydrogen Electrolyzer

By: Phi Suea House Project

Link:

https://www.youtube.com/watch?v=WfkNf7kMZPA&ab_channel=PhiSueaHouseProject

Duration: 4 m 07 sec

VIDEO-2

Topic: What is Hydrogen Fuel with Full Information? (Hindi)

By: Quick Support

Link: https://www.youtube.com/watch?v=Bn6_fQGqjIQ&ab_channel=QuickSupport

Duration: 8 m 4 sec

UNIT 1.7: Renewable Energy Sources for Large Scale Production of Green Hydrogen through water electrolysis

Unit Objectives



At the end of this unit, you will be able to:

1. Explain the concept of renewable energy.
2. Explain various sources of renewable energy

1.7.1 Renewable Energy

Renewable energy sources are sustainable and naturally replenished forms of power, such as sunlight, wind, water flow, geothermal heat, and biomass. These sources have minimal environmental impact and can be continuously harnessed to generate clean electricity, making them vital contributors to a sustainable energy future.

These sources can provide a sustainable and clean power supply for water electrolysis, contributing to the large-scale production of green hydrogen.

Main Sources of Renewable Energy

The following figure represents the main sources of renewable energy are:

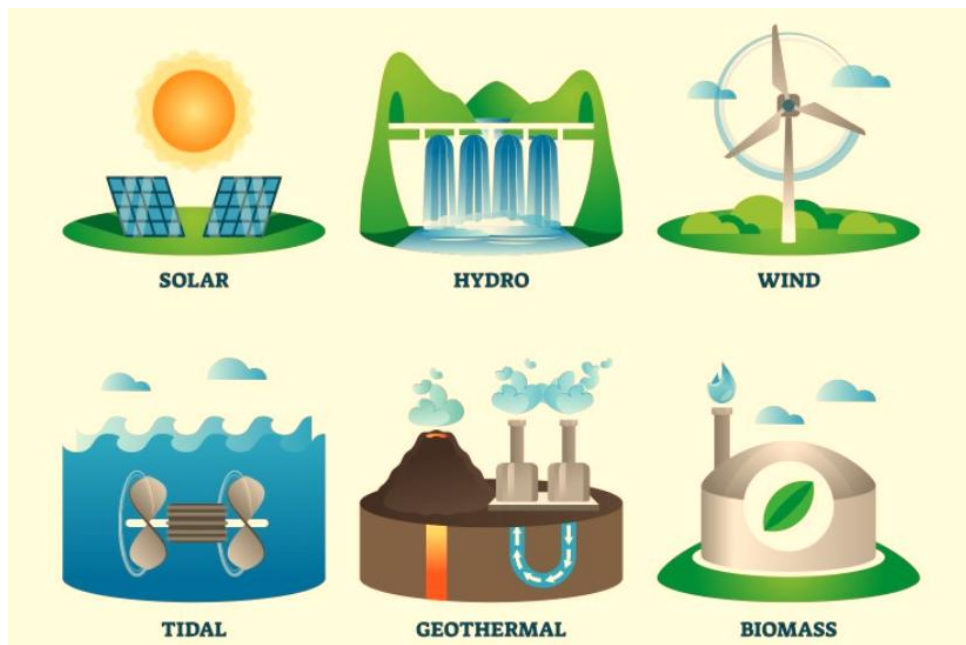


Fig. 1.7.1(a): Main Sources of Renewable Sources of Renewable Energy

The following figure describes the main sources of renewable energy are:

Solar Energy	•Captures sunlight to generate electricity, offering clean and abundant power.
Wind Energy	•Converts wind's kinetic energy into electricity, providing sustainable and widely available source.
Hydro Energy	•Utilizes flowing water to produce power, offering reliable and established renewable option.
Ocean Energy	•Uses the power of tides, waves, and temperature differences in the water to generate electricity, making the most of the ocean's big energy potential.
Geothermal Energy	•Extracts Earth's heat for power, offering continuous and dependable energy supply.
Biomass Energy	•Converts organic matter into energy, offering versatile and renewable fuel source.

Fig. 1.7.1(b): Main Sources of Renewable Sources of Renewable Energy

Working of Main Sources of Renewable Sources of Renewable Energy

- **Solar energy:** It is the most abundant resources and is used to generate power for domestic and commercial applications. Photovoltaic (PV) systems captures by solar light energy and then used for electric power generation or for heating. The solar energy depends largely on geographical location, the season of the year, and the time of day.
- **Wind energy:** Wind energy is harnessed by using turbines. When wind blows, it causes the blades to rotate and produce electricity through generator which is connected to turbine. The wind speed and direction are two important parameters which are measured for a long time to assess the wind resource. The wind speed increase with the height above ground. The power generated by wind is directly proportional to the cube (third power) of the wind speed.
- **Hydro energy:** Hydro energy functions by capturing the energy inherent in moving water to generate electricity. It employs hydroelectric power plants, often using a river's natural flow or, in some cases, creating a reservoir by constructing a dam. The gravitational pull propels water downhill, building up kinetic energy as it moves. The water is guided through an intake structure and directed onto the blades of a turbine inside the power plant. As the turbine spins, it transforms the water's kinetic energy into mechanical energy. This rotation activates a generator connected to the turbine, producing an electric current through electromagnetic induction. The generated electricity is then processed, transformed, and transmitted via power lines to supply homes, industries, and more. Hydro energy's sustainability is rooted in the continuous water cycle, ensuring a renewable and dependable energy source.
- **Tidal energy:** Tidal energy of ocean is another form of hydro energy to generate power. The tidal currents are used to drive turbines connected to generators. The amount of energy

generated during each tidal current can be estimated. Tidal energy is very useful for its use in the nearby coastal areas.

- **Geothermal energy:** As we go deep inside the earth, the temperature increases and the energy is called geothermal energy. Thermal energy is there since the formation of earth or due to radioactive decay. Some of the heat escapes naturally in the form of a volcanic or geyser, but it can also be captured to produce electric power by driving steam turbine-generator. The heated water within a geothermal reservoir is pumped up to a geothermal power plant on the surface where it operates a steam turbine which is coupled to a generator. The electricity is supplied to the grid and the steam is cooled back into water which flows back into the geothermal reservoir for reuse. This heat energy can also be used to heat homes.
- **Biomass energy:** Biomass is an organic matter that is a by-product of recently living organisms and plants, and can be converted into a solid fuel or a methane gas. The solid fuel is burned to produce electricity. Methane gas is typically produced through anaerobic processes (chemical reactions or biological activities that occur in the absence of oxygen) during the natural decomposition of organic matter. It occurs particularly in bodies of water and landfills. Biomass energy can be produced at low costs in an environmentally friendly manner. In fact, it is a waste management because it uses agricultural, domestic, and industrial waste as a fuel.

Benefits of Renewable Energy

As compared to traditional fossil fuels, following are some of the benefits of renewable energy sources:

- **Reduced Greenhouse Gas Emissions:** Renewable energy sources such as solar, wind, hydro, and geothermal power generate electricity without emitting carbon dioxide or other harmful greenhouse gases. This contributes to the global effort to reduce the overall carbon footprint and combat climate change.
- **Inexhaustible Resources:** Renewable energy sources are essentially inexhaustible, as they are derived from natural processes that continually replenish themselves. For example, sunlight, wind, and water are continuously available resources.
- **Reduces dependency on imported fuels:** Renewable energy sources are abundant and widely distributed. This helps enhance energy security and reduces dependency on imported fuels.
- **Rural Development:** Renewable energy projects, such as wind farms and solar installations, can bring economic development to rural areas by providing revenue to landowners and local governments and creating job opportunities.
- **Reduced Water Usage:** Unlike fossil fuel power plants, many renewable energy sources, such as wind and solar, require little to no water for operation. This helps conserve water resources, especially in water-scarce regions.
- **Community Empowerment:** Distributed renewable energy systems, such as rooftop solar panels, allow individuals and communities to generate their own electricity, reducing reliance on centralized power grids and giving people more control over their energy consumption.
- **Long-Term Sustainability:** Investing in renewable energy promotes a sustainable energy future by reducing the depletion of finite fossil fuel resources and mitigating the environmental impacts associated with their extraction and use.
- **Low Operating and Maintenance Costs:** Once renewable energy systems are installed, their operating and maintenance costs are typically lower compared to fossil fuel power plants. This can lead to stable and predictable energy prices over the long term.

- **Job Opportunities:** The renewable energy sector provides substantial job opportunities in various fields, including manufacturing, installation, maintenance, and research. This can contribute to local economic development and job growth.

VIDEO

Topic: Renewable Energy Sources Hindi

By: Sampurna Engineering: 2020

Link: https://www.youtube.com/watch?v=wl6eKPZ_qkc

Duration: 15 m 31 sec

UNIT 1.8: Emerging Entrepreneurial Opportunities Across Green Hydrogen Value chain

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss in brief the emerging entrepreneurial opportunities across Green Hydrogen Value chain
2. Identify opportunities to various renewable energy sources.

1.8.1 Emerging Entrepreneurial

Emerging entrepreneurial opportunities across the Green Hydrogen value chain in India are driven by the country's commitment to clean energy and sustainability. Entrepreneurs in India have a significant role to play in advancing the Green Hydrogen ecosystem. Collaborations, partnerships, and innovative solutions across the value chain are essential for harnessing the potential of Green Hydrogen as a clean energy source and contributing to a sustainable future.

Following are some of the emerging entrepreneurial opportunities in India:

- **Production:** Entrepreneurs can venture into the production of Green Hydrogen through various methods such as water electrolysis powered by renewable energy sources (solar and wind). Establishing hydrogen production facilities and optimizing electrolysis technologies is a growing opportunity.
- **Renewable Energy:** Investing in renewable energy infrastructure, particularly solar and wind farms, is essential for providing the energy needed for hydrogen production. Entrepreneurs can focus on developing and expanding renewable energy projects.
- **Storage:** Green Hydrogen storage is a critical aspect of the value chain. Entrepreneurs can explore opportunities in advanced hydrogen storage technologies, including high-pressure tanks, underground storage, and chemical storage solutions.
- **Distribution:** Entrepreneurs can invest in the distribution infrastructure, including pipelines and transportation networks, to ensure efficient delivery of Green Hydrogen to end-users, industries, and refuelling stations.
- **Applications:** Developing innovative applications for Green Hydrogen is a promising area. This includes using hydrogen in industries such as steel, chemicals, and transportation (hydrogen fuel cell vehicles) and even as a clean cooking fuel.
- **Research and Development:** Entrepreneurs can focus on research and development to improve the efficiency and cost-effectiveness of Green Hydrogen production, storage, and utilization technologies. Collaborations with research institutions and universities are valuable.
- **Policy and Consulting:** Providing consulting services to navigate the evolving regulatory landscape and policy frameworks related to Green Hydrogen can be a viable entrepreneurial opportunity.
- **Financing and Investment:** Facilitating investment and financing for Green Hydrogen projects and startups can be a lucrative niche, especially as interest in sustainable investments grows.
- **Export Opportunities:** India can become a hub for Green Hydrogen production and export. Entrepreneurs can explore opportunities for exporting Green Hydrogen to countries looking to reduce their carbon footprint.

- **Hydrogen Infrastructure:** Building infrastructure such as refuelling stations for hydrogen fuel cell vehicles is an emerging entrepreneurial area, particularly in urban centers.

1.8.2 Opportunities to Various

The following table describes some opportunities for various renewable energy sources in India that can contribute to India's energy security, reduce greenhouse gas emissions, and promote sustainable development.

Renewable Energy	Opportunities
Solar Energy	India has abundant sunlight throughout the year, making it a great place for solar power generation. Opportunities include solar panels on rooftops, solar farms in open areas, and solar-powered appliances.
Wind Energy	Coastal regions and certain inland areas in India have strong wind resources. Opportunities involve setting up wind farms and harnessing wind energy for electricity generation.
Hydropower	India has rivers and hilly terrain suitable for hydropower projects. Opportunities include building small and large-scale hydropower plants to generate electricity.
Biomass Energy	India has a vast agricultural sector, which can provide biomass sources like crop residues and organic waste. Opportunities involve converting biomass into biofuels and using it for power generation.
Geothermal Energy	Certain regions in India have geothermal potential. Opportunities include exploring and utilizing geothermal energy for heating and electricity.
Tidal and Wave Energy	India has a long coastline with potential for tidal and wave energy generation. Opportunities involve deploying technologies to harness energy from tides and waves.
Waste-to-Energy	India's urban areas produce significant amounts of waste that can be converted into energy through waste-to-energy plants. Opportunities include waste management and energy generation.
Microgrids	Setting up microgrids in remote areas can provide clean and reliable energy access. Opportunities involve decentralized renewable energy solutions.
Energy Efficiency	Opportunities also lie in improving energy efficiency across various sectors, reducing energy wastage, and promoting energy-efficient technologies.

Table 1.8.1: *Some Opportunities for Various Renewable Energy Sources in India*

UNIT 1.9: Role and Responsibilities of a Green Hydrogen Plant Entrepreneur

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the role and responsibilities of a Green Hydrogen Plant Entrepreneur.
2. Learn about Career Progression for Green Hydrogen Plant Entrepreneur

1.9.1 Role and Responsibilities

The Green Hydrogen Plant Entrepreneur role involves identifying the potential market and the client needs/ requirements to propose the right kind of technically and economically feasible solution to set up green hydrogen plant. He/she is also expected to undertake specific works/sub component installations of a green hydrogen plant. The role of entrepreneur in this nascent but fast evolving field is crucial as they undertake various small projects within the green hydrogen value chain while managing various resources and monitor project progress, and prioritizing environmental sustainability. By combining technical expertise with project management skills, they drive the growth of green hydrogen production in India, contributing significantly to the nation's clean energy objectives and the global shift towards sustainable energy sources. The following figure represents the various roles and responsibilities of a Green Hydrogen Plant Entrepreneur.



Fig 1.9.1: Roles and Responsibilities of a Green Hydrogen Plant Entrepreneur

- **Market Analysis:** Conduct comprehensive market research to identify potential opportunities for green hydrogen plant projects in India. Analyze market trends, competition, and regulatory developments related to green hydrogen production.
- **Client Needs Assessment:** Engage with prospective clients and stakeholders to understand their specific requirements and objectives for green hydrogen production. Collaborate closely with clients to tailor solutions that align with their sustainability goals and economic considerations.
- **Client Relationship Management:** Build a strong relationship with clients and stakeholders, addressing their concerns and ensuring their satisfaction throughout the project lifecycle.
- **Feasibility Assessment:** Evaluate the technical and economic feasibility of green hydrogen plant projects, considering factors such as resource availability, infrastructure, and investment requirements. Perform techno-economic analysis to determine project viability and potential returns on investment.
- **Proposal Development:** Develop comprehensive proposals outlining project details, including technology choices, project scope, timelines, and cost estimates. Present proposals to clients, highlighting the advantages and benefits of green hydrogen production.
- **Project Planning:** Create detailed project plans, including engineering designs, procurement strategies, and construction schedules. Collaborate with engineering and technical teams to ensure project readiness and adherence to industry standards.
- **Component Selection and Installation:** Identify and procure essential components and equipment required for the green hydrogen plant. Oversee the installation and integration of key components, such as electrolyzers, renewable energy sources, and hydrogen storage solutions.
- **Regulatory Compliance:** Stay updated on relevant regulations, permits, and environmental requirements related to green hydrogen production. Ensure that all aspects of the project adhere to local, state, and national regulations.
- **Risk Management:** Identify potential risks and challenges associated with project implementation and take proactive measures to mitigate them. Develop contingency plans to address unexpected issues that may arise during construction and operation.
- **Resource Management:** Efficiently allocate resources, including manpower, materials, and finances, to ensure the successful execution of green hydrogen projects.
- **Monitoring and Reporting:** Implement monitoring systems to track project progress, energy production, and hydrogen quality. Generate regular reports for clients and stakeholders, providing updates on project milestones and performance.
- **Environmental Sustainability:** Emphasize sustainability and environmental responsibility in project design and operation, minimizing the carbon footprint of green hydrogen production.

1.9.2 Career Progression for Green

The role of a green hydrogen plant entrepreneur entails comprehensive knowledge of the key process involved in producing green hydrogen. This includes understanding the key aspects of various subsections of the plant and their operations, as well as possessing an overview of key technical and business skills necessary for installing & procuring critical components. The entrepreneur must be multidisciplinary, with expertise in hydrogen production plant EPC work, installation etc. Key responsibilities include conducting market research, identifying customer needs, performing cost-benefit analysis for components, sourcing manufacturers, overseeing

procurement, ensuring proper installation, and managing a team of technicians and other workers.

The following image represents the career progression for Green Hydrogen Plant Entrepreneur:

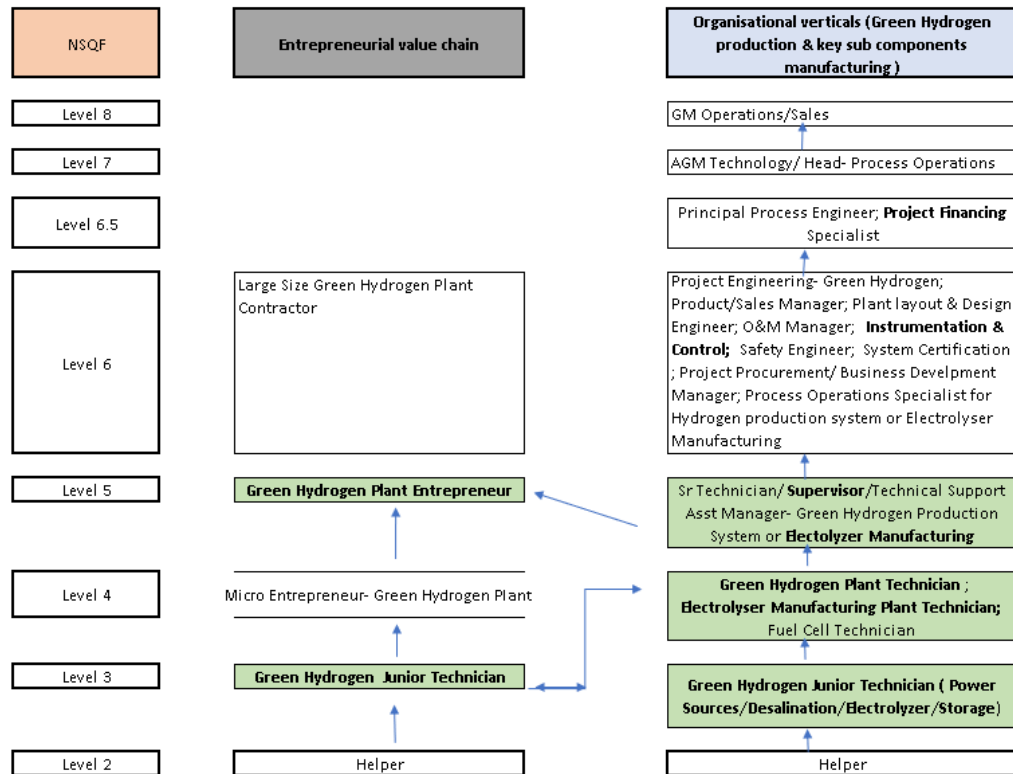


Fig. 1.9.2: career progression for Green Hydrogen Plant Entrepreneur

Exercise



1. Why green hydrogen is necessary in energy transition?
2. Name three unique properties of hydrogen?
3. What do you understand by energy carrier and why hydrogen is an energy carrier?
4. What is the basis of colour codes of hydrogen?
5. What inputs are given to an electrolyzer to produce green hydrogen?
6. What is value chain and which elements forming the value chain of hydrogen?
7. Write the inputs (feedstock, water, electricity etc.) to be provided in various methods of hydrogen production?
8. What are the challenges associated with the hydrogen production and storage?
9. What are the challenges in the transportation and distribution of hydrogen?
10. List out the end use applications of hydrogen?
11. Name 4 types of electrolyzers for the production of green hydrogen and name the electrolyte used in each of them?
12. What do you understand by the renewable source of energy?
13. Name 3 most common sources of renewable energy in India?
14. What do you understand by green hydrogen economy?
15. What precautions are necessary for safe working with hydrogen?





2. Components of Green Hydrogen Plant and its Layout

Unit 2.1: Key Components of the Green Hydrogen Plant

Unit 2.2 Functions of Key Components and Their Fundamental Principles

Unit 2.3 Types of Electrolyzer: Brief Description

Unit 2.4 Overall Layout of the Plant and Sparking items

Unit 2.5 Read, Interpret and Understand the Relevant Drawings and O&M Manuals of the Plant Equipment

Unit 2.6 Key Material and Safety Codes, Technology Protocols and Standards Applicable in Green Industry

Unit 2.7– Key Technical Insights for Setting Up and Running Successful Enterprise



Key Learning Outcomes

At the end of this module, you will be able to:

1. Identify all key components of the Green Hydrogen plant including electrical, mechanical and civil components
2. Discuss functions of the key components of electrolyzer stack, renewable power plant, feed water supply unit, gas separator, transformer and rectifier, gas compression unit, etc.
3. Explain overall layout of the plant
4. Read and interpret various electrical codes, data sheet, safety features etc relevant to the plant.
5. Discuss working principles of main components including electrolyzer stack, gas separator, power source, etc.,
6. Discuss key technical insights of Green hydrogen system for setting up and running successful enterprises
7. Discuss how to perform SWOT analysis
8. Discuss safety aspects to be considered while preparing the layout of hydrogen production plant for prevention from fire caused by sparking, e.g. to ensure safe distance of hydrogen from electrical devices like MCBs, switches.

UNIT 2.1: Key Components of the Green Hydrogen Plant

Unit Objectives

At the end of this unit, you will be able to:

1. Identify key components of green hydrogen plant
2. Categorise electrical, mechanical and civil components

2.1.1 Green Hydrogen Production Plant

A green hydrogen production plant is a facility that generates hydrogen gas using renewable energy sources, primarily through the process of water electrolysis. Unlike conventional hydrogen production methods that rely on fossil fuels, such as natural gas, green hydrogen production involves using renewable electricity to split water molecules into hydrogen and oxygen. This process produces hydrogen without emitting greenhouse gases or contributing to carbon emissions, making it environmentally friendly and sustainable.

The following figure represents a Green hydrogen production plant equipped with alkaline electrolyzers.

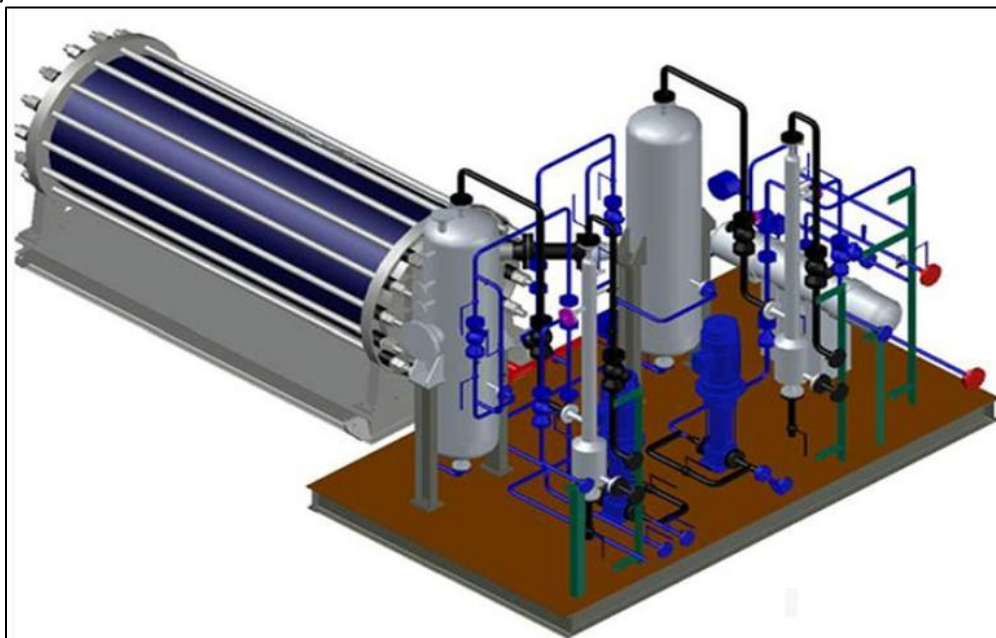


Fig. 2.1.1 A green hydrogen production plant equipped with alkaline electrolyzer

2.1.1 Key Components and Their Categorisation

The following figure represents the components of a Green Hydrogen production plant equipped with alkaline electrolyzers.

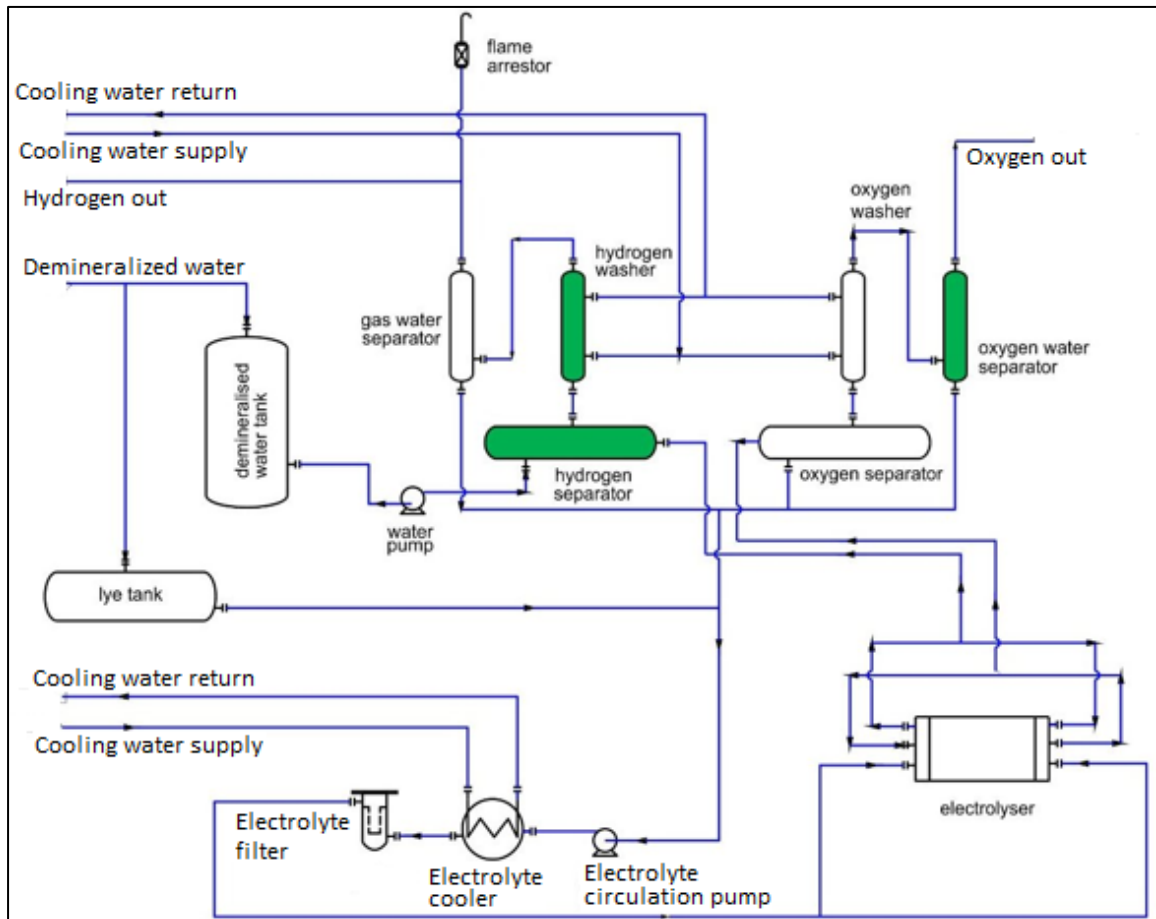


Fig. 2.1.2 Components of a green hydrogen production plant with alkaline electrolyzer
Source: Brise Chemicals. Hydrogen generation plant

Components

The key components of green hydrogen production plant are shown in Figure 2.1.2. The function, inputs, processes, and outputs of these components are explained.

(i) Electric power supply

- Transformer
- Rectifier

NOTE: In case of a dedicated solar PV power (DC power), A.C. switchgear, transformer etc. are not required.

- Electrolyzer: It is a stack consisting of number of cells connected in series as shown in Figure 2.1.3.

- Electrolyte

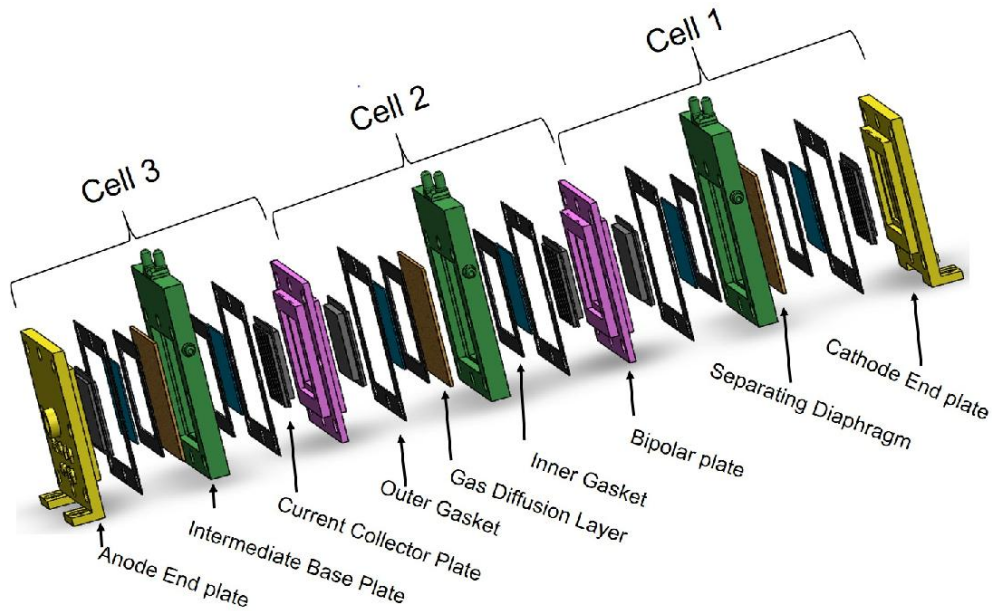


Fig. 2.1.3 Cells of an alkaline electrolyzer connected in series

Source: Kumara S. and Yakdehige D.S. 2017. Design of an Alkaline Electrolysis Stack. Semantic Scholar

2.1.3 Mechanical

Demineralization (DM) water plant

Piping, pumps, valves

Compressors

Hydrogen separator

Oxygen separator

2.1.4 Civil

Plant buildings, Administrative building

Foundation of equipment, panels, feed water supply

Earthing pits, trenches etc. in the plant including substation

Some components of a hydrogen production electrolyzer plant installed in a thermal power station are shown in Figure 2.1.4. The stator winding of generator is hydrogen cooled.



Streams of electrolyzers



An electrolyzer

Fig. 2.1.4 Electrolyzers in a thermal power plant

The key components of the Green Hydrogen and Their categorization is as following:

- **Electrical Components:**
 - **Electrolyzer Stacks:** Electrolyzer stacks split water into hydrogen and oxygen using electrical current.
 - **Renewable Power Supply System:** This provides the required electrical power from renewable sources like solar or wind for electrolysis.
 - **Transformer and Rectifier:** The transformer adjusts voltage, and the rectifier converts alternating current (AC) to direct current (DC) for electrolysis.
- **Mechanical Components:**
 - **Feed Water Supply and Demineralization System:** Provides purified water to the electrolyzer, preventing mineral deposits that can affect its efficiency.
 - **Gas Separator:** Separates hydrogen and oxygen gases produced during electrolysis.
 - **Gas Compression Unit:** Compresses the produced hydrogen gas to higher pressures suitable for storage or further use.
- **Civil Components:**
 - **Plant Structure:** Provides the physical framework for all components, ensuring safety and stability.
 - **Piping and Tubing:** Connects various components for the flow of water, gases, and other substances.
 - **Storage Tanks:** Stores both hydrogen and oxygen gases before they are used or processed further.

Note: In case of a dedicated solar PV power, A.C. switchgear, transformer etc. are not required.

As per the complete layout of the green hydrogen plant, it comprises of several components that work together to generate hydrogen through the electrolysis of water using renewable energy sources. Following are the essential components of a green hydrogen plant:

- **Renewable Energy Source:** Solar panels or wind turbines provide the necessary renewable electricity to power the electrolysis process. These sources ensure that the hydrogen production is environmentally friendly and sustainable.
- **Electrolyzer:** The heart of the plant, the electrolyzer, splits water into hydrogen and oxygen gases using electricity. Different types of electrolyzer technologies exist, such as alkaline, PEM (Proton Exchange Membrane), and solid oxide electrolysis.

- **Power Conversion System:** This system adapts and converts the electricity generated by renewable sources to the required voltage and current for the electrolyzer. It optimizes power efficiency and control.
- **Water Supply and Treatment:** High-quality water supply is essential for efficient electrolysis. Water treatment systems remove impurities and ensure that the water is suitable for the electrolysis process.
- **Gas Separation System:** Once hydrogen and oxygen gases are produced, a gas separation system is employed to isolate and collect the hydrogen gas, which will be further processed for various applications.
- **Compressor:** Hydrogen gas is often compressed for storage and transportation. A compressor increases the gas pressure for storage in tanks or for use in other applications.
- **Storage Tanks:** Hydrogen is stored in high-pressure tanks or, in some cases, in liquid form. These tanks ensure that hydrogen is available when needed and can be transported if required.
- **Safety Systems:** Safety measures are vital in handling hydrogen gas. Explosion-proof systems, leak detection, and ventilation systems are incorporated to ensure safe operation.
- **Control and Monitoring System:** Automation and control systems manage the operation of various components, optimizing efficiency, and maintaining safe and stable conditions. Monitoring systems oversee the process and provide real-time data to operators.
- **Heat Recovery System:** Electrolysis can generate excess heat, which can be captured and utilized for other processes, enhancing overall plant efficiency.
- **Hydrogen Purification System:** If high-purity hydrogen is required for specific applications, a purification system may be added to remove impurities from the generated hydrogen gas.
- **Distribution and Transportation:** Depending on the intended use, hydrogen can be transported through pipelines or in specialized containers for various industrial, commercial, or transportation applications.
- **Energy Management System:** For optimal operation, an energy management system coordinates the production of hydrogen with the availability of renewable energy sources and the demand for hydrogen.
- **Waste Management:** Byproducts from the electrolysis process, such as oxygen gas and residual water, need to be properly managed and disposed of or reused.

These components of the Green Hydrogen Plan can be categorised into electrical, mechanical and civil, as shown in the following figure:

Electrical Components	Mechanical Components	Civil Components
•Renewable Energy Source (Solar Panels or Wind Turbines) •Electrolyser •Power Conversion System •Control and Monitoring System •Gas Separation System •Compressor •Safety Systems •Energy Management System	•Electrolyser •Gas Separation System •Compressor •Storage Tanks •Hydrogen Purification System •Distribution and Transportation Infrastructure •Heat Recovery System	•Foundation and Infrastructure for Solar Panels or Wind Turbines •Building Structures to House Equipment •Storage Tank Foundations •Pipeline and Transportation Infrastructure •Waste Management Facilities •Safety Infrastructure (Ventilation Systems, Fire Suppression Systems) •Access Roads and Pathways

Fig 2.1.5. Components categorised into electrical, mechanical and civil

VIDEO

Topic: Demineralization process, DM plant, RO plant (Hindi)

By: Chemical Pedia

Link:

https://www.youtube.com/watch?v=gCMxFmGkG8s&ab_channel=ChemicalPedia

Duration: 9 m 13 sec

Activity



- Illustrate the schematic of Green hydrogen production plant.
- Draw a block diagram for PEM electrolyzer based hydrogen production plant.

REFERENCE: Block diagram of alkaline electrolyzer based green hydrogen production plant given under following link.

Link: https://www.researchgate.net/figure/Schematic-for-alkaline-electrolysis-of-water-for-hydrogen-production-with-compression_fig1_282746267

UNIT 2.2: Functions of Key Components and Their Fundamental Principles

Unit Objectives

At the end of this unit, you will be able to:

- Discuss functions of the key components including electrolyzer stacks, renewable power supply system, feed water supply and demineralisation system, gas separator, transformer and rectifier, gas compression unit, etc.
- Discuss fundamental principles of main components on which they operate (e.g. electrolyzer stack, gas collector, power source, etc.)

2.2.1. Function of Electrical Components

The incoming power supply to the green hydrogen plant is at a voltage level of 33/11 kV. It is protected using a circuit breaker (CB) or disconnecting (DO) fuse. This high-voltage supply is then transformed to a lower voltage of either 33 kV or 11 kV, depending on the plant's size and requirements, using a transformer. The capacity of this transformer is determined by the plant's capacity and the voltage of the power transmission system.

From the transformer, the voltage is further reduced to 415 V using a distribution box. This lower voltage is used for general purposes and to supply power to components like the rectifier. The control and protection panels are an integral part of the system, ensuring safe and efficient operation.

To handle the electrical connections, cables of 33/11 kV are used. Proper grounding is essential for safety, and both earthing rods/mat and separate electronic earthing are employed. The separation of electronic earthing from electrical earthing prevents AC voltage spikes from affecting electronic systems.

Transformer

The function of a transformer is to adjust the voltage as per the requirement, such as:

- High voltage on primary side to low voltage on secondary side (step-down transformer)
- Low voltage on primary side to high voltage on secondary side (step-up transformer)

The following figure represents the core and coils of a transformer and its symbols.

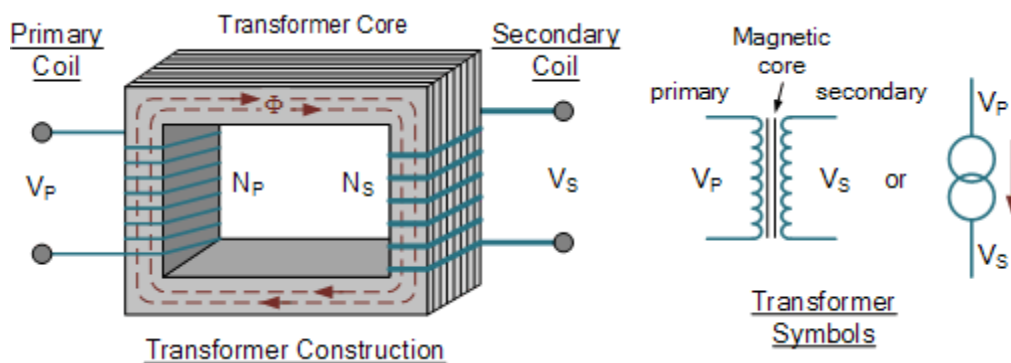


Fig. 2.2.1 Construction of transformer
Source: Electronics tutorials. Transformer basics.

With the change in voltage, corresponding currents also change but the frequency remains unchanged.

The transformer is installed in a switchyard along with the switchgear, terminal box (for connecting the wiring of measuring instruments), controls & protection systems etc.

Rectifier

Rectifier converts input AC power from the transformer into DC power for feeding to the electrolyzers. Full wave circuit and voltage waveforms, as shown in the following figure.

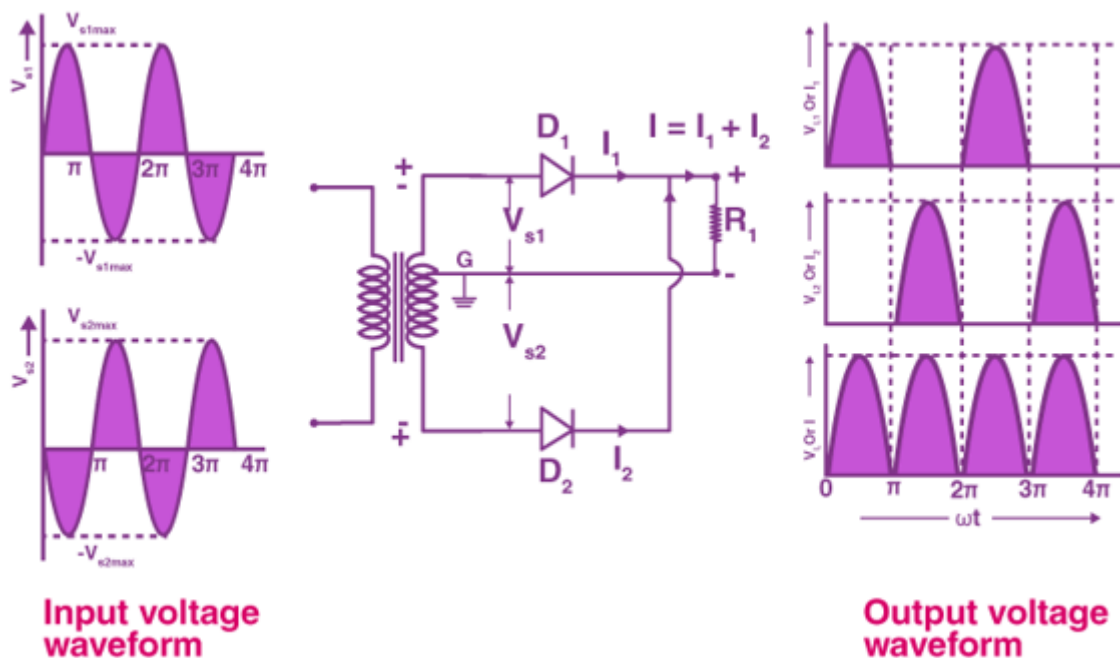


Fig. 2.2.2 Full wave rectifier

Renewable Power

The world is passing through the energy transition based on green hydrogen. The production of green hydrogen needs electrical energy obtained from renewable sources of energy such as wind, solar, hydro, biomass, geothermal, ocean etc. Thus, renewable energy has to be crucial role in the large-scale production of green hydrogen.

The function of renewable power in a green hydrogen plant is to provide a sustainable and environmentally friendly source of energy for the electrolysis process that produces hydrogen. Renewable power sources, such as solar and wind energy, generate electricity without producing greenhouse gas emissions or depleting finite resources.

Electrolyzer Stack

The function of an electrolyzer is to use electrical energy for splitting water into hydrogen and oxygen. The electrolyzer stack is built of number of cells connected in series. Each cell has an anode and a cathode which are separated from each other.

Following the key functions of electrolyzer stack:

- **Electrolysis Reaction:** The primary function of the electrolyzer stack is to facilitate the electrolysis process. It consists of multiple electrolysis cells connected in series or parallel. Each cell contains an anode and a cathode, separated by an electrolyte. When an electric current is passed through the cells, water molecules are split into hydrogen ions (protons) and oxygen ions (hydroxide) at the anode and cathode, respectively.
- **Gas Separation:** The electrolyzer stack is designed to efficiently separate the generated hydrogen gas from the oxygen gas. This is crucial to ensure the purity of the produced hydrogen.
- **Proton Exchange Membrane (PEM) or Alkaline Electrolysis:** Depending on the type of electrolyzer stack (PEM or alkaline), the stack facilitates the appropriate chemical reaction that corresponds to the specific technology. In PEM electrolysis, a proton-conductive membrane facilitates the movement of protons, while in alkaline electrolysis, an alkaline electrolyte enables the reaction.
- **High Energy Conversion Efficiency:** The electrolyzer stack is designed to achieve high energy conversion efficiency. It minimizes energy losses during the electrolysis process, ensuring that a significant portion of the input electrical energy is used to produce hydrogen.
- **Stack Cooling:** During the electrolysis process, heat is generated as a byproduct. The electrolyzer stack is equipped with cooling mechanisms to manage temperature and prevent overheating, optimizing the efficiency of the electrolysis reaction.
- **Hydrogen Production Rate:** The size and design of the electrolyzer stack determine the rate of hydrogen production. Larger stacks with more cells can produce higher volumes of hydrogen.
- **Controllability:** The stack's design allows for precise control of the electrolysis process, including adjusting current and voltage to optimize production based on demand and energy availability.

Electrolyte

The function of an electrolyte is to provide electrically conducting media. An electric current is a flow of electric charge. In an electrolyzer, the charge is carried by moving electrons in an external circuit and by ions in an electrolyte.

Thus, primary function of an electrolyte is to conduct ions, which are electrically charged particles. In an electrolysis setup, the electrolyte allows ions to move between the electrodes. For example, in the electrolysis of water to produce hydrogen and oxygen, the electrolyte enables the movement of hydrogen ions (protons) towards the cathode and oxygen ions towards the anode.

2.2.2 Functions of Mechanical Components

It comprises of compressors, pumps, valves, ventilation, piping, firefighting system, DM water plant etc. In case of a large hydrogen production plan, the source of feed water may be the river, dam, lake etc. The functions of mechanical components are as follows:

- **Feed water supply**
 - Industrial water, which can include municipal water or groundwater, is the most common source of water used in electrolysis. The Feed water supply system in a green hydrogen plant ensures a steady source of pure water for electrolysis. It sources, purifies, and regulates water flow to maintain consistent electrolysis performance. This system also manages temperature, pressure, and cooling, contributing to efficient and safe hydrogen production. Reliable water quality and flow directly impact electrolysis efficiency and the overall production process.
- **DM water plant**
 - Demineralized (DM) i.e. deionised water is used for electrolysis of water to prevent the deactivation of electrolyte. DM water is split into hydrogen and oxygen. The Demineralization (DM) water plant plays a crucial role in a green hydrogen facility by purifying water to an extremely high level of purity. This highly treated water, also known as demineralized water, is essential for preventing mineral deposits and impurities from accumulating in the electrolyzer stack. By maintaining the electrolyzer integrity and performance, the DM water plant ensures consistent hydrogen production and prolongs the lifespan of the electrolysis equipment.
- **Hydrogen separator**
 - Hydrogen with some level of purity is needed depending on its end use/application. Hydrogen produced by the electrolysis process contains water. A hydrogen separator unit is provided for separation of hydrogen from water.
- **Compressor**
 - Hydrogen is typically produced at relatively low pressures (20–30 bar) and must be compressed for storage or transport. Hence, the hydrogen gas flows to the compressor for high pressure compression.
- **Oxygen Separation**
 - Oxygen is produced inside the cell stack on the same side of the membrane where the feed/cooling water flows. The oxygen is separated from the water so that water can be reused in the electrolysis process.
- **Pipelines, pumps, valves**
 - The pipelines are provided to facilitate flow of liquid and gases between various components. The pumps are provided for circulation whereas valves are provided to stop/regulate the flow. high-purity stainless steel is the common materials used for hydrogen piping systems. Composite pipes (made from different thermoplastics of PTFE, FRP etc.) are also used. The following figure represents the piping network of hydrogen.



Fig. 2.2.3 Piping network of hydrogen

Source: PCN Europe.2022. Green Hydrogen for Supply Trucks

Activity



- Match the components of an electrolyzer with their functions in the following Table.

	Component		Function
A	Electrolyte	E	Provides electrical connections between one cell to next cell of the stack.
B	Membrane	F	Speeds-up a chemical reaction without itself getting consumed
C	Catayst	G	Provides an electrically conducting media
D	Biopolar plate	H	Separates anode and cathode. It blocks flow of electrons through it.

- (a) A-E, B-F, C-H, D-G
 (b) A-H, B-G, C-E, D-F
 (c) A-G, B-H, C-F, D-E
 (d) A-F, B-G, C-E, D-H

2.2.3 Fundamental Principles of Main Components

The fundamental principles of key components in a green hydrogen plant play a crucial role in its operation. The electrolyzer stack, for instance, employs the process of electrolysis, where water is split into hydrogen and oxygen using an electric current. The gas collector gathers and stores the produced hydrogen, ensuring its safe containment. The power source, typically a renewable energy system, supplies the necessary electricity for the electrolysis process. Each of these components operates based on scientific principles that enable the production, storage, and utilization of green hydrogen, contributing to sustainable energy solutions.

Electrolyzers

There are several types of electrolyzers available, each with specific characteristics based on the materials used in their construction. These materials include:

- Electrodes, which facilitate the chemical reactions
- Electrolyte, which conducts ions within the system
- Membrane that separates the gases
- Catalysts that expedite the reactions

For example, the membrane used in PEM electrolyzer separates oxygen and hydrogen, while allowing ions to pass through and also provide electrical insulation between the electrodes. The performance of an electrolyzer is assessed by the efficiency of electrolysis, the range of current densities and operating temperatures, and the purity of the hydrogen produced.

Inputs to an electrolyzer (which may vary widely)

- Demineralized water: 10 litres per kg of hydrogen
- Electrical energy: 42 kWh per kg of hydrogen (may vary widely, say up to 60 kWh/kg)

The electrolysis process can be represented by the following equation:

- Electricity + $2\text{H}_2\text{O} = \text{O}_2 + 2\text{H}_2$

The following figure represents the working principle of electrolyzer in terms of chemical reactions.

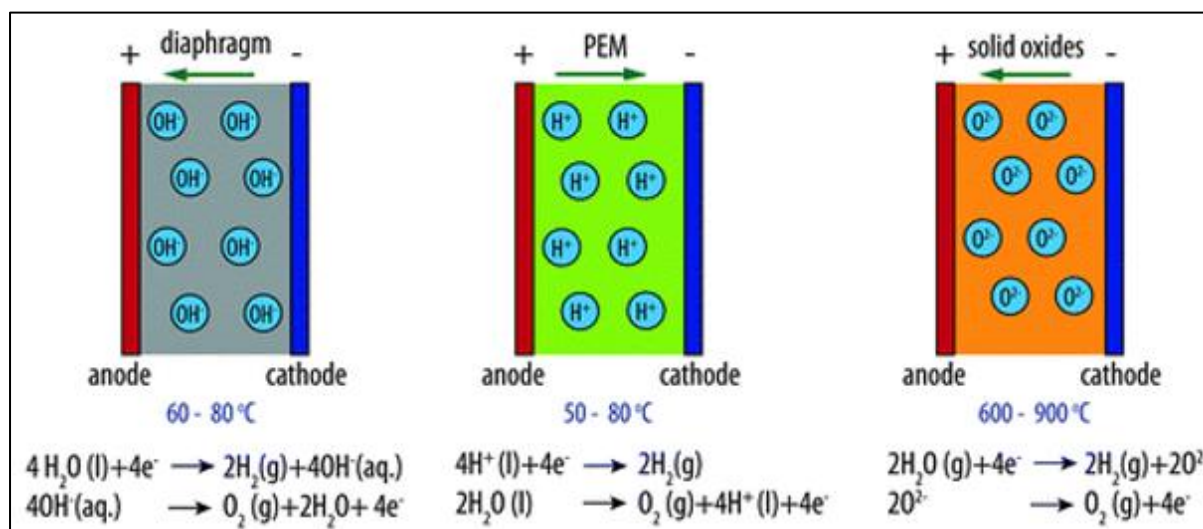


Fig. 2.2.4 Principle of working of electrolyzers

Source: Xiang C., Papadantonakis K. M. and Nathan S. Lewis N.S. 2016. Principles and implementations of electrolysis systems for water splitting. Royal Society of Chemistry

The technologies of alkaline and PEM electrolyzers are well established and are being used commercially. The following figure represents the parts of an alkaline and PEM electrolyzers respectively.

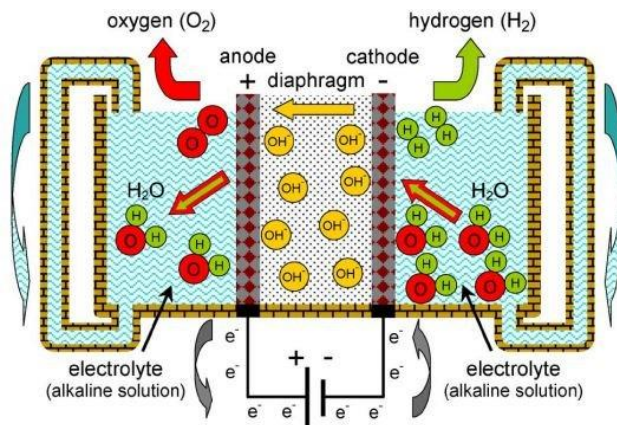


Fig. 2.2.5 Parts of alkaline electrolyzer [1]

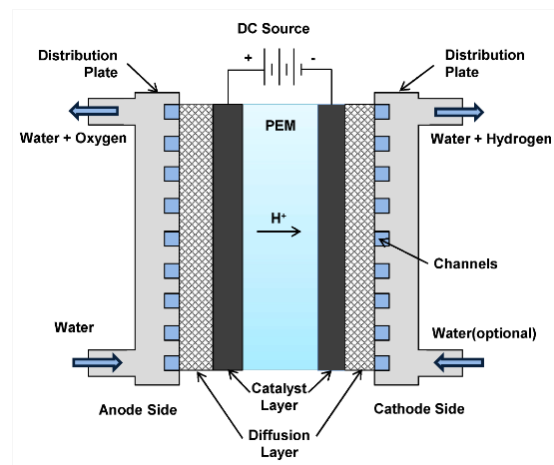


Fig. 2.2.5 Parts of PEM electrolyzer [2]

Sources: [1] De Silva Y.S.K. and Middleton P.H. 2017. *Design of an Alkaline Electrolysis Stack*. M.Sc. Thesis. University of Egder, Norway

[2] Sumit Sood et al. 2020. *Generic Dynamical Model of PEM Electrolyzer under Intermittent* Sources

Electrolyzer stacks

The stacks are the assembly of number of cells. The following figure represents the alkaline electrolyzer stack.



Fig. 2.2.6 Alkaline electrolyzer stacks

Sources: Noordende H. and Ripson P. 2022. *A 1-GigaWatt Green hydrogen plant*. Public Report

The following figure represents the cells of an alkaline electrolyzer stack.

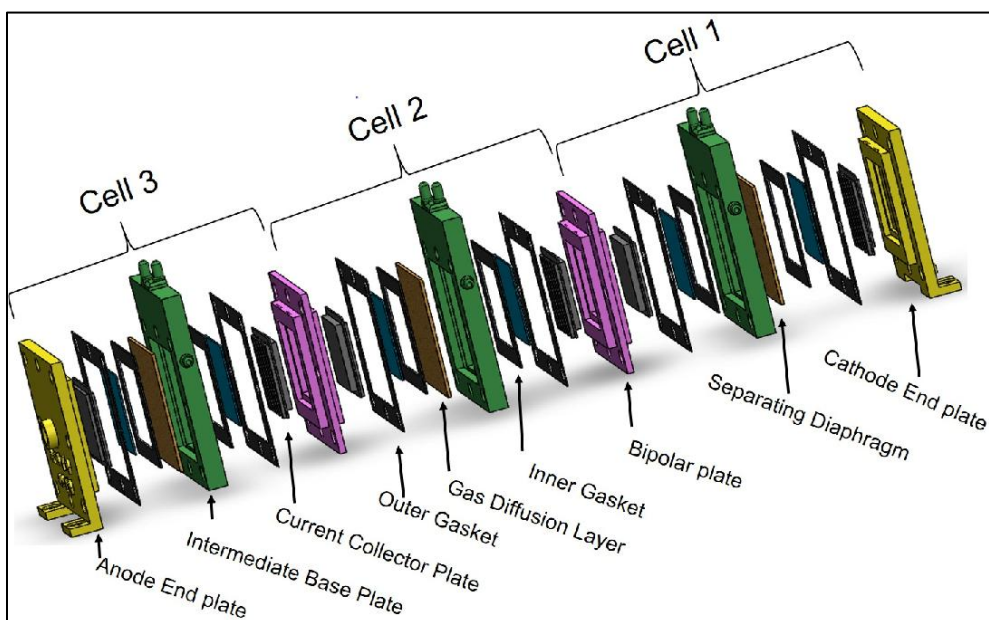


Fig. 2.2.7 Cells of an alkaline electrolyzer connected in series

Source: Kumara S. and Yakdehige D.S. 2017. *Design of an Alkaline Electrolysis Stack*. Semantic Scholar

The following figure represents the stacks of PEM electrolyzers are shown in Figure 2.2.8.



Fig. 2.2.8 PEM electrolyzer stacks

Source: Noordende H. and Ripson P., 2022. *A 1 GigaWatt Green hydrogen plant*. Public Report

The following figure represents a single cell of PEM electrolyzer.

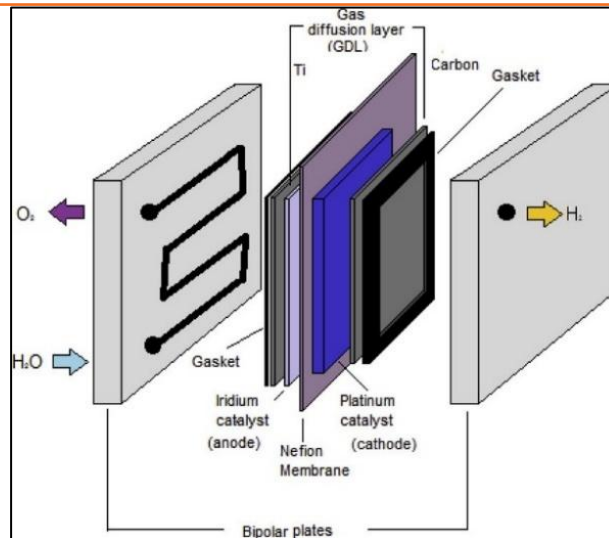


Fig. 2.2.9A cell of PEM electrolyse

Source: Rivera X.C.S., Topriska E., Kolokotroni M., and Azapagic A. 2018. Environmental sustainability of renewable hydrogen in comparison with conventional cooking fuels. *Journal of Cleaner Production*

The function principle of electrolyzer stacks involves the application of an electric current to split water molecules into hydrogen and oxygen gases through an electrochemical reaction. This process occurs within the stack's multiple cells, where electrodes and electrolytes facilitate the separation of ions, allowing the production of hydrogen at the cathode and oxygen at the anode. The design of the stack optimizes efficiency, scalability, and safety while ensuring reliable hydrogen generation.

Electrolyte

Pure water (pH value = 7) cannot conduct electricity because it has no free ions. Adding an alkali or acid makes the water to conduct electricity. An electrolyte is a substance that provides an electrically conducting media. In the case of an alkaline electrolyzer, an alkaline electrolyzer, potassium hydroxide (KOH) or sodium hydroxide (NaOH) are added to water. The electrolyte separates into cations (+ve ions) and anions (-ve ions) which disperse uniformly. The following figure represents the concept of cation and anion.

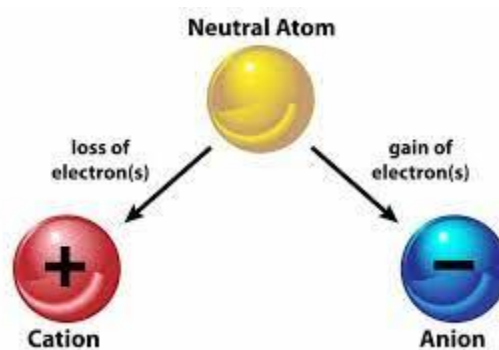


Fig. 2.2.10 Concept of cation and anion

Source: Kinsey N. The anion-cation connection in soil. *Eco Farming Daily*

The following table lists the electrolytes used in respective electrolyzers:

Electrolyzer	Electrolyte
Alkaline electrolyzer	Alkaline solution (e.g. KOH, NaOH etc.)
PEM electrolyzer	Acidic: Solid polymer membrane
Solid oxide electrolyzer	Zirconium dioxide (ZrO_2) ceramic doped with Yttrium oxide (Y_2O_3)

Table 2.2.11: Electrolytes Used in Respective Electrolyzers

Electrodes

Each cell of an electrolyzer cell has an anode and a cathode. In the electrolysis process, positively charged ions move to the cathode (-ve electrode) and negatively charged ions move to the anode (+ve electrode). The positively charged ions receive electrons and negatively charged ions lose electrons. Both the products of the dissociation get collected at the electrodes. The anodes are sacrificed in electrolysis as they get rusted (oxidized) and the cathodes are de-rusted (reduced).

Catalyst

A catalyst is a substance that accelerates a chemical reaction without undergoing any permanent change itself. In reference to electrochemical reactions, catalysts play a crucial role in reducing the energy barrier required for the reaction to happen, known as activation energy. When it comes to water electrolysis, which involves splitting water into hydrogen and oxygen, electrocatalysts are employed to facilitate this process. Noble-metal oxides are a type of electrocatalyst that can speed-up the rate of water electrolysis, but they are both rare and expensive. Currently, precious metals like iridium, ruthenium, and platinum are commonly used as catalysts in industrial electrodes to achieve efficient water electrolysis.

Membrane

A membrane is a thin separator like a plastic used in PEM electrolyzer. It separates the anode loop from the cathode loop, in order to avoid unwanted secondary reactions.

Diaphragm

A diaphragm is a flexible and relatively thick barrier that can partially support itself. The diaphragms are made of a heat-resistant nickel and oxide ceramic particles which are supported by a woven nickel-net. In the context of an alkaline electrolyzer, it serves as a partition that separates different compartments within the electrolyzer. This diaphragm allows the passage of ions while preventing the mixing of gases or solutions in the different compartments. By maintaining separation, the diaphragm facilitates specific electrochemical reactions to occur in each compartment, contributing to the overall efficiency of the electrolysis process.

Thus, it performs the following two functions:

1. Allow the ions to pass through it without hindrance.
2. Keep the products formed during the electrolysis separated from each other in order to prevent undesirable reactions.

Bipolar plates

The bipolar plates provide the electrical connections between one cell and the next cell of the stack. These thin plates have flow fields on both sides. They also distribute the fuel and the oxidant to anode and cathode.

Demineralized (DM) water

Demineralization (DM) is the process of removing all the soluble minerals from water. DM water is supplied to an electrolyzer. In general, a DM water plant contains the following components.

Components	Action
Sand Filter	Pretreatment of water
Cationic bed	Removes cations (positively charged ions such as calcium and magnesium) and replace them with sodium
Anionic bed	- Removes anions (negatively charged ions such as arsenic and - nitrate) and replace them with chloride
Mixed bed	Removes both cations and anions

Table 2.2.12: Components of DM Water Plant

The following figure represents the various components of a DM water plant.

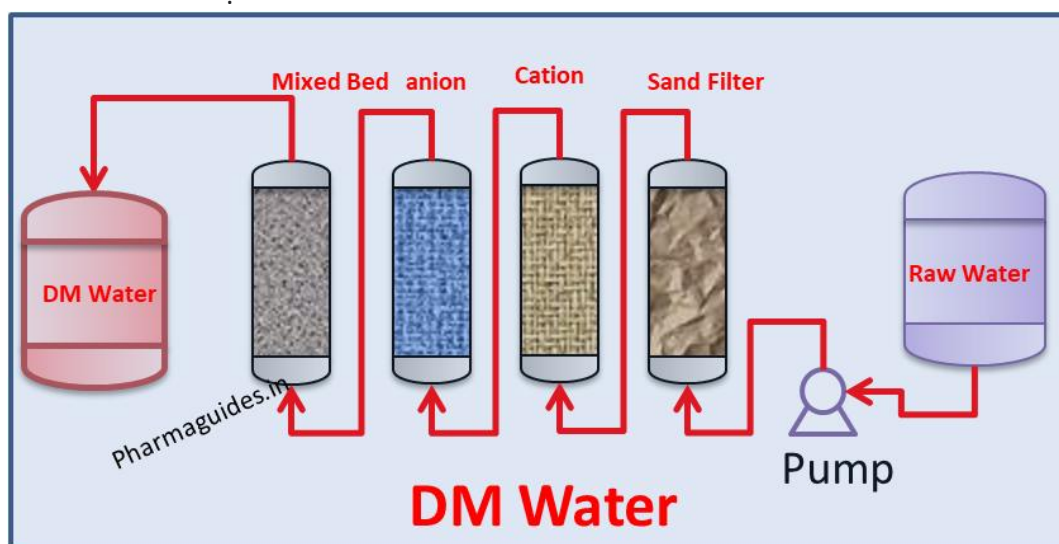


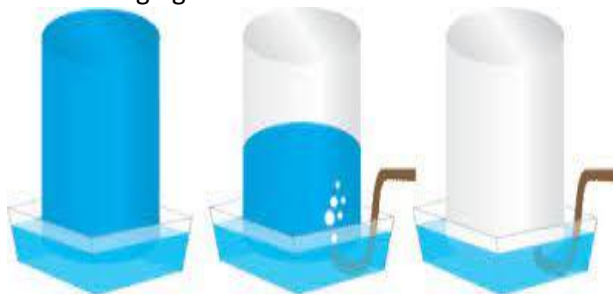
Fig. 2.2.13 Schematic of DM water plant Cells

Source: Pharmaguides.in. DM Water plant. Chemical Engineering Guide

Gas Collector

Hydrogen gas produced by electrolysis of water is often collected by displacement of water. A tank is placed inverted (upside-down) in a large vessel of water. Hydrogen is collected in an inverted tank, floating on water in a larger tank similar to that in gobar gas plant. As the gas is produced, it exits

through the rubber tubing and displaces the water in the hydrogen tank till it gets filled-up with hydrogen, as shown in the following figure.



*Fig. 2.2.14 Hydrogen gas collector**

**Source: Flex books 2.0 (CK-12). Gas collection by water displacement*

The following figure represents the hydrogen gas collector (drum) in a thermal power plant where hydrogen gas is used for cooling of generator stator winding.



Fig. 2.2.15 Hydrogen gas collector at Thermal PP

Hydrogen Separator

The hydrogen produced during electrolysis gets mixed with electrolyte. Hydrogen is separated from the electrolyte in a hydrogen separation unit. The hydrogen collected in the inverted tank pressurizes to some extent (such as 2 to 3 bars), so as to flow into inlet of water ring compressor which is connected to hydrogen gas/water separator. The water separated from hydrogen returns to water ring compressor.

Oxygen separator

Separation of electrolyte from Oxygen which is produced as a by-product during electrolysis. The oxygen may be used in industries, hospitals etc. or may be vented out in the atmosphere.

UNIT 2.3: Types of Electrolyzers: Brief Description

Unit Objectives

At the end of this unit, you will be able to:

1. Describe different types of electrolyzer

2.3.1 Brief Description of Electrolyzers

There are different types of electrolyzers available. Each type of electrolyzer has its own advantages and disadvantages, making them suitable for different applications based on factors like efficiency, operating conditions, scalability, and cost. The following figure represents the four main types of electrolyzers.

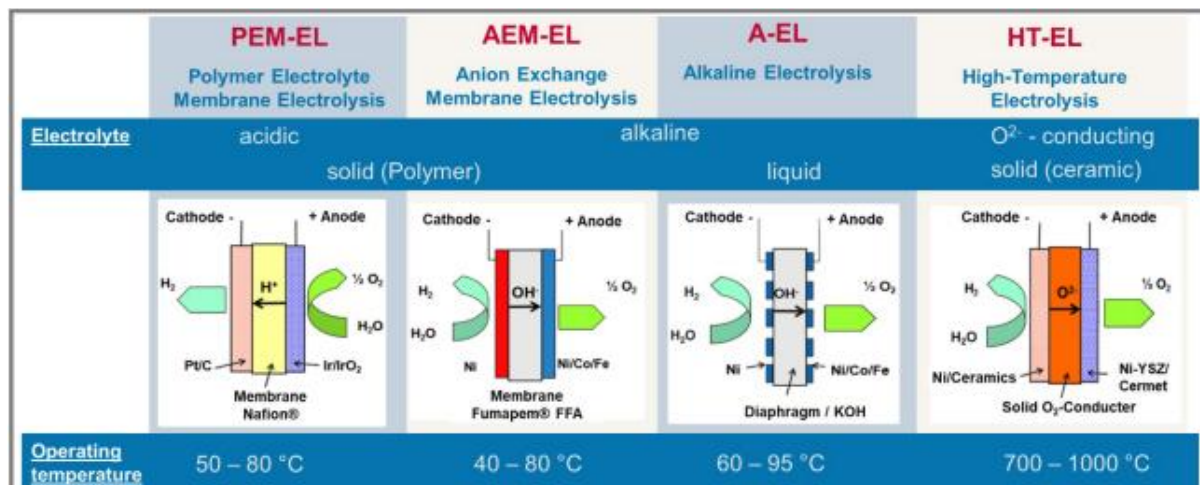


Fig. 2.3.1 Main electrolyzers

Source: Alexander Trattner. 2021. *Renewable Hydrogen: Modular Concepts from Production over Storage to the Consumer*

Polymer Electrolyte Membrane (PEM) Electrolyzers

In a Proton Exchange Membrane (PEM) electrolyzer, a solid polymer membrane serves as the electrolyte:

- At the anode, water molecules undergo a reaction that results in the formation of oxygen molecules and positively charged hydrogen ions, known as protons. Simultaneously, electrons are conducted through an external circuit.
- The PEM allows the selective movement of hydrogen ions to the cathode.
- Once at the cathode, these hydrogen ions join with electrons from the external circuit to produce hydrogen gas. This process effectively splits water into its constituent elements, oxygen and hydrogen, utilizing the solid polymer electrolyte to facilitate the controlled movement of ions and electrons.

The proton exchange membrane (PEM) electrolyzer has one of the most popular technologies due to highly dynamic and fast response. In addition, PEM electrolyzers work on high current densities up to 3 A/cm², high stack efficiencies, and stack size up to 2.5 MW. Moreover, hydrogen can be produced at higher pressure levels for further usage like for refuelling. However, there are certain drawbacks also. They need costly and rare metal catalysts like platinum and iridium. There is still potential for extending life and plant size.

Alkaline Electrolyzers

Alkaline electrolyzers are the most common with well-established technology since early 1900s. It uses a liquid alkaline solution of potassium or sodium hydroxide as the electrolyte. The hydroxide ions (OH⁻) move through the electrolyte from the cathode to the anode while hydrogen is generated on the cathode. The following table describes its the advantages and disadvantages.

Advantages	Disadvantages
• Low-cost technology with long life and use of the low-cost material • Systems up to 100 MW are in operation	• Low current densities up to 0.6 A/cm² and low pressure • Electrolyte, however, is a liquid potassium hydroxide solution, which is corrosive and also is a safety issue • Very high hydrogen purity is hardly reached

Table 2.3.1: Advantages and Disadvantages of Alkaline Electrolyzers

Solid Oxide Electrolyzers

In Solid oxide electrolyzers, a solid ceramic material is used as the electrolyte. It conducts negatively charged oxygen ions (O₂⁻) at high temperatures. It generates hydrogen in the following way:

Steam at the cathode combines with electrons from the external circuit to form hydrogen gas and negatively charged oxygen ions.

The oxygen ions pass through the solid electrolyte and react at the anode to form oxygen gas. The generated electrons travel through the external circuit.

High-temperature electrolysis results in high efficiencies. This technology is still under development stage and further research is needed. At present, up to 150 kW electrolyzers are commercially available but the current densities up to 1 A/cm² are still low. Further, due to the operation at high temperatures, the commonly used ceramics do not withstand high pressures. Another disadvantages include long warm-up times and non-suitability for frequent start-stop.

Anion Exchange Electrolyzer

The anion exchange membrane electrolyzer uses solid alkaline membrane as an electrolyte. The advantage this technology are the low-cost catalysts, which substantially reduces the system costs. The main disadvantage of this electrolyzer is its low current density compared to PEM electrolyzer and short lifetime. The anion exchange electrolyzers are in development stage, the possibilities to increase current densities and lifetime are expected. These are available up to 5 kW capacity with a system efficiency up to, say 70%. Possible current densities go up to 2 A/cm².

UNIT 2.4: Overall Layout of the Plant and Sparking items

Unit Objectives

At the end of this unit, you will be able to:

1. Explain overall layout of the plant
2. Identify the key components in the layout
3. Discuss Sparking items e.g. MCBs, switches to be installed isolated from the hydrogen production area

2.4.1 Green Hydrogen of Plant

The green hydrogen plant layout has been discussed in Unit 2.1 in brief. Some more points regarding plant layout are discussed here.

Electrical: The HV line equipment and transformer are placed outside and at a distance from electrolyzer, hydrogen handling equipment, pipelines to avoid any disaster. Indoor equipment is also kept away in a similar manner.

Mechanical equipment is a part of hydrogen production system and hence these components handle hydrogen e.g. compressors, valves, pipelines, drier, gas holder etc. These need to be inspected frequently to detect hydrogen leakage, if any. The layout takes care of easy and quick approach to various equipment. Hydrogen pipelines are laid at a height above 1.5 m to avoid hydrogen inhaling in case of any leakage of hydrogen.

In the layout of a green hydrogen production plant, the key components of a green hydrogen production plant are to be located with adequate spacing from safety point of view. The main areas and components that require adequate spacing are the hydrogen work areas and electrical spark prone areas. The following figure represents an example of a typical layout of plant.



Fig. 2.4.1 Typical layout of a green hydrogen electrolyzer plant

Source: Atchison J. 2022. New report from ISPT. What does a 1 GW electrolyzer plant look like?
Ammonia Energy Association

The hydrogen work areas are electrolyzer building item 8, compressor building item 9, and hydrogen cooling building item 13. Similarly, the areas prone to electrical sparks are high voltage transformers item 4 in switchyard and rectifier building item 5.

2.4.2 Optimum utilization of the energy and materials

The following figure represents the ways to make optimum utilization of the resources in various processes:

Hydrogen Separation Unit	The hydrogen and water are separated and water may be reused in electrolyte.
Oxygen Separation Unit	The separation of oxygen and water has dual advantage. Oxygen may be used to meet in-house needs, if any or it can be supplied to industries, hospitals etc. As far as water is concerned, it is reused with the electrolyte.
Solid Oxide Electrolyser	The splitting of water takes place at high temperature. The waste heat from the electrolyser can be used for preheating to save the energy. Alternatively, solar heating or heat available from some nearby industry may be used to save the energy to the electrolyser.

Fig 2.4.2. Optimum Utilization of the Resources in Various Processes

Activity

- Draw a block diagram of the green hydrogen production plant and write justification for positioning of various blocks.

2.4.3 Sparking Items

The rate at which hydrogen is produced in an electrolyzer is directly proportional to the amount of electric charge that is engaged in the electrolysis process. This means that the more electric charge is applied, the faster hydrogen gas will be generated. Additionally, the total volume of hydrogen produced over a specific period of time corresponds to the mean value of the electric current flowing through the cell stack of the electrolyzer (i.e. higher the current, the more hydrogen is produced within a given time frame).

Thus, high current and low voltage are required. The per cell voltage is around 2 VDC and it varies with the type of electrolyzer. The D.C. voltage supply to stream of electrolyzer cells depends on series-parallel combination of the cells. The electrolyzer plant generally operates on low D.C. voltage of 100 to 200 Volts and high current of 1000 to 2000 A. For such D.C. supply, suitable rectifier is connected to main A.C. supply line.

However, main supply to the hydrogen plant is provided through a suitable transformer connected to high voltage supply line of RE supplier. In consideration of operation, maintenance, and safety, isolating switches, molded circuit breakers, fuses are provided.

As the hydrogen plant is high current so operation of the electrical switching elements will produce substantial sparks. Therefore, installation of these switchgears should be air tight and away from hydrogen production area.

Generally, they should be installed at a level lower than lowest level of hydrogen pipelines, valves, storage tanks. The main supply cutoff switch should be in other section to operate it safely in case of severe emergency in production plant. As an entrepreneur you should also get broadly acquainted with complete electrical schematic of plant.

With the advancement in technology (such as artificial intelligence), such provisions can be easily made for remote operation of supply switches located in other section of plant. You should keep monitoring the air tightness of such switch gear covers. After switching-off of supply, the plant side conductors should get earthed automatically to earth the residual D.C. charge on the conductors.

UNIT 2.5: Read, Interpret and Understand the Relevant Drawings and O&M Manuals of the Plant Equipment

Unit Objectives

At the end of this unit, you will be able to:

1. Explain how to read and interpret electrical and other applicable codes, standards and protocols relevant to the plant.

2.5.1 Engineering Drawings

Engineering drawings serve as visual tools to convey design concepts and technical specifications, facilitating their application in manufacturing, installation, operation, and maintenance processes. It complies with the contractual requirements and implied needs, applicable standards including those on safety. It represents the specific outline that showcase all the information and requirements needed to manufacture an item or assemble various items. In fact, more than simply a drawing, it is a graphical language that communicates ideas and information. An engineering drawing includes the following information, which is specific to the project:

- Dimensions
- Geometry
- Tolerances
- Material type
- Finish
- Hardware

The following figure represents the sample drawing.

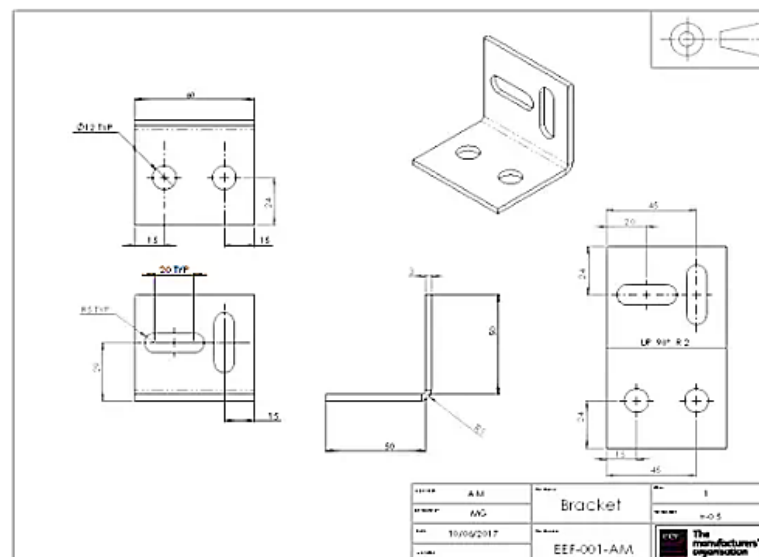


Fig. 2.5.1 A sample drawing

Source: MAKE UK. How to Read Engineering Drawings - A Simple Guide

Information blocks

The information blocks are usually positioned in the lower right-hand corner of the drawing and contains crucial details about the assembly. It provides details about the following:

- For whom
- Part number
- Description
- Material and Finish

The main information blocks are:

Title Block	Revision Block
• It is at the bottom right-hand corner of the drawing. It serves as the context in which drawing should be perceived. It contains the following information: • Name and address of the company or agency who prepared or owns the drawing • Part number and description • Material • Mass • Finish • General tolerances • Projection details • Projection details • Scale used in the drawing • Revision numbers • Status of the drawing (Preliminary, Approved, etc.) • Units used in the drawing	• It is located in the upper right hand corner, shows details about the changes made during the revision. It includes the following information: • Revision number • Description of the changes • Date of Version • Approval of the revision

Fig 2.5.2. Main Information Blocks

Bill of Materials (BOM)

Bill of Materials is a comprehensive list of the items and quantities of all the components, parts, materials that are required for the project or to manufacture or assemble a product.

Lines

It is important to understand what each line type is and what they mean. There are three types of lines, as shown in the following figure:

Visible Lines	• Indicates an edge is visible in relevant view
Hidden Lines	• Indicates the edge is behind a face
Centre Lines	• Drawn to indicate the exact geometric centre of the assembly. They are made from a series of lighter long and short dashes.

Fig 2.5.3. Types of Lines

Reading Engineering Drawings

Try to visualize the assembly in 3D. Sometimes, an isometric view is given to facilitate your understanding. Use the Bill of Materials to find the items in the drawing in order to understand their role in the assembly.

IMPORTANT: Remember that reading an engineering drawing can take a long-dedicated efforts to gain experience. It is in line with the proverb, '**Practice makes a man perfect.**'

2.5.2.O&M manual

An Operation and Maintenance (O&M) manual is a document that provides essential details about the upkeep of equipment and machinery. It also provide the detailed guidelines on extending asset life, minimizing unplanned forced shutdowns, reducing maintenance costs along with the safety of man and materials. These details may vary from product to product. The following figure represents the list of information included in a typical O&M manual (but is not be limited to this).

Details of the project

As built drawings

Material storage requirements

Operation and maintenance instructions

Guarantees and warranties certificates

Tests and commissioning certificates

Completion and handing over certificates

Planned periodic maintenance schedule

Fig 2.5.4. General Details Included in O&M

IMPORTANT:

The performance of the hydrogen production plant will depend on:

- The operation of the plant within permissible operation limits
- The periodic maintenance of the plant as specified in O&M manual

2.5.3 Codes and Standards for Referring in Drawings/Manuals

Codes are generally accepted sets of rules that outline what you need to do. Whereas, standards provide the way of implementing the codes, particularly across various ranges of power supply, pressures, temperatures, etc., in order to safeguard the plant from the severity of accidents.

Codes generally evolved after certain typical accidents in which may be due to the manual errors or negligence, lacuna in design, manufacturing or systems.

Codes and Standards

The codes and standards are mainly referred during design, manufacturing, testing etc. The operation and maintenance manuals supplied with the equipment or plant are prepared considering applicable codes and standards. Usually, technicians/Supervisors/Engineers have to refer these manuals in day-to-day operations, routine maintenance etc. However, entrepreneurs should also have an understanding of standards, codes especially regarding hydrogen handling equipment, safety, fire etc.

Some of the Indian Standards on hydrogen are given in the following table:

Indian Standards	Description
IS 15201 (2002)	Hydrogen – Code of Safety
IS 1090 (2002)	Compressed hydrogen
IS/ISO 11114-4	Transportable Gas Cylinders - Compatibility of Cylinder and Valve Materials with Gas Contents, Part 4: Test Methods for Selecting Metallic Materials Resistant to Hydrogen Embrittlement
IS/ISO 13985:2006	Liquid Hydrogen--Land vehicle fuel tank
IS 16749:2018	Basic considerations for the safety of hydrogen systems
IS 17445: 2020	Fasteners preloading test for detection of hydrogen
IS/ISO 17268:2020	Gaseous Hydrogen Land Vehicle Refuelling Connection Devices

Table 2.5.1: Indian Standards on hydrogen

Note: The Indian Standard 15201: 2002, 'Hydrogen – Code of Safety' [6] was adopted by the Bureau of Indian Standards, after the draft finalized by the Industrial Safety and Chemical Hazards Sectional Committee and approved by the Chemical Division Council.

UNIT 2.6: Key Material and Safety Codes, Technology Protocols and Standards Applicable in Green Industry

Unit Objectives

At the end of this unit, you will be able to:

1. Explain key material and safety codes, Technology protocols and standards as applicable in green hydrogen industry
2. Discuss safety aspects to be considered while preparing the layout of hydrogen production plant for prevention from fire caused by sparking, e.g. to ensure safe distance of hydrogen from electrical devices like MCBs, switches.

2.6.1 Key Materials

The iridium and platinum used in PEM electrolyzers are hard-to-get materials. The iridium is corrosion resistant and is used as a catalyst which is added in thin layers to anodes. Another catalyst for coating the electrodes of PEM electrolyzers is platinum. Often, a thin layer made from platinum covers bipolar plates to lower their surface contact resistance. These bipolar plates separate individual cells in a stack. They not only provide electrical conduction between cells but also supply water used for electrolysis and remove heat as well as reaction products. In several countries, R&D continues in search of suitable materials with a focus to substitute the hard-to-get materials for electrolyzers.

Various materials presently used in electrolyzers are summarized in the following table.

Item	Alkaline	PEM	Solid oxide
Electrolyte	30% KOH + DM water	Solid polymer membrane +DM water	Solid oxide or ceramic
Cathode	Ni with catalytic coating such as Pt	Pt nano particles on carbon	Ni doped YSZ (yttria-stabilized zirconia)
Anode	Ni based	Titanium (Ti) mesh	A cermet of nickel and yttria-stabilized zirconia

Membrane/ Diaphragm	Zirfon / ZrO ₂	Nafian (Du Pont)	Solid oxide or ceramic
Bipolar/sepa rator plate	Stainles s steel	Ti is the base material for bipolar plates	Graphite or metal
Catalyst: anode	Ni/Co/F e	Pt, Iridium oxide	Pt
Catalyst: cathode	Stainless steel mesh	Pt	Pt

Table 2.6.1 Materials used in electrolyzers

2.6.2 Hydrogen Safety Challenges

Currently, the focus of safety codes and standards are specific and limited to industries using hydrogen whereas the safety of general public has not been adequately addressed. Now, the scope of use of hydrogen is widening and will reach to public domain.

So it is important to learn about the nature of hazards associated with hydrogen, preventive measures for controlling the hazards, and essential information on storage, handling, labelling, transportation, waste disposal, training of personnel, personal protective equipment, first aid and firefighting. In addition, you should be aware of the important applicable standards.

Hydrogen is a colourless, odourless and tasteless gas and hence its detection is one of the major challenges. It is stable under normal temperature and pressure; however, violent reaction or ignition takes place with air. The platinum and similar metals containing absorbed oxygen or hydrogen etc. act as catalyst.

Fire and explosion hazards

One of the biggest challenges imposed by hydrogen is the risk of fire. The risk of fire depends on the various factors, as shown in the figure.

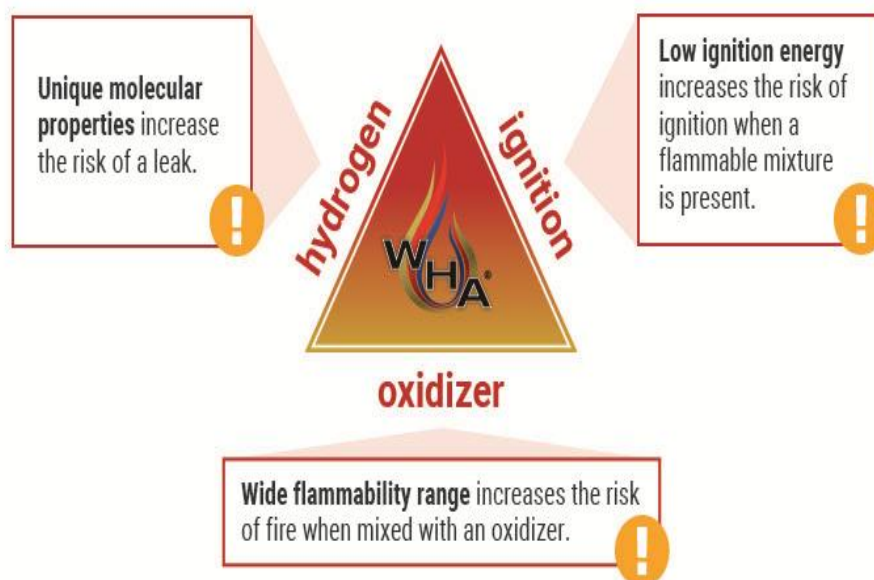


Fig. 2.6.1 Hydrogen fire risk

Source: WHA's hydrogen fire risk management philosophy. WHA International.
<https://wha-international.com/hydrogen-fire-risk-management/>

Hydrogen Leakage

The carbon dioxide is a byproduct when hydrogen is produced by fossil fuel-based steam methane reformation process. Hydrogen itself emits no carbon dioxide when used in a fuel cell or burnt. The size of hydrogen molecule is the smallest and hence many a times, high pressure hydrogen gas continue to leak very slowly into the atmosphere through its containers, pipe line joints etc.

Flammability Range

The flammable range (also called Explosive Range) is the concentration range of a gas or vapor that will burn (or explode). The following figure represents the basic conditions for burning:

Flammable substance - fuel

Oxidizer - oxygen or air

Source of ignition - spark or high heat

Fig 2.6.2. Basic Conditions for Burning

Below the flammable range the mixture is too low or poor to burn and above the upper flammable limit the mixture is too rich to burn. Hydrogen has a very wide flammability range of 4 to 75% by volume in air as compared to other fuels as shown in the following figure.

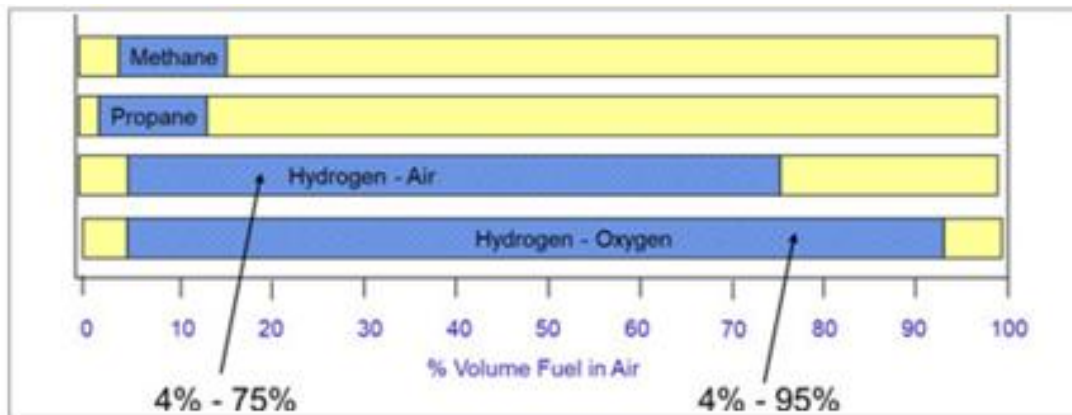


Fig. 2.6.3 Hydrogen flammability range

Similarly, hydrogen has a very broad flammability range of 4 to 95% by volume in oxygen. Hence, prevention of mixing of air or oxygen with hydrogen inside a closed spaces is very important.

Source: WHA's hydrogen fire risk management philosophy. WHA International.
<https://wha-international.com/hydrogen-fire-risk-management/>

Low Ignition Energy

Very small about 1/100th of fossil fuel ignition energy is needed to ignite hydrogen as compared to other common fuels as shown in the following figure.

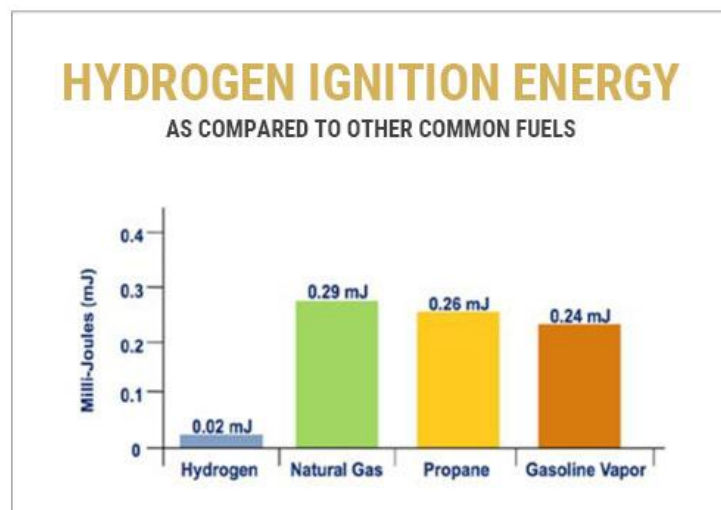


Fig. 2.6.4 Hydrogen ignition energy

Source: WHA's hydrogen fire risk management philosophy. WHA International.
<https://wha-international.com/hydrogen-fire-risk-management/>

Consequently, there is a risk of explosion for hydrogen pipelines, storage tanks and containers.

Toxicity and health effects

Hydrogen is not toxic but may harm in following ways:

- Hydrogen displaces the oxygen and occupies upper space in the closed workplace having no or poor ventilation. If atmosphere does not contain enough oxygen, then inhalation can cause dizziness, unconsciousness or even death. This phenomenon is called asphyxiation.
- Direct contact of body tissue with liquid hydrogen can cause cold burns or frost bite.

2.6.3 Hydrogen Leak Detection

Significant amounts of hydrogen are employed safely in various industries, despite the challenges associated with containing and detecting potential leaks. With its potential use in abate sectors and growing market, Hydrogen would be significantly a larger industrial commodity. The recent commencement of using hydrogen as a consumer fuel, primarily through blending, has brought about significant safety concerns. As a result, there is a growing interest in hydrogen sensors and leak detection technologies.

However, during a visit to an oil refinery it was observed that hydrogen leakage detectors are provided in the areas prone to leakage having capacity to detect 5% of minimum inflammable hydrogen (i.e. 5% of 4% = 0.2%) which is quite safe. Also, the mobile hydrogen detectors are provided to operation and maintenance personal.

By utilising interlock systems that work with hydrogen and fire detectors, hydrogen production plants can increase their level of safety.

The safety of hydrogen production plant can be improved by using Interlock systems that work in conjunction with hydrogen and fire detectors. The control systems should be integrated with the leak and flame detection systems so that when a hydrogen leak or a fire is discovered the control system can do the following tasks:

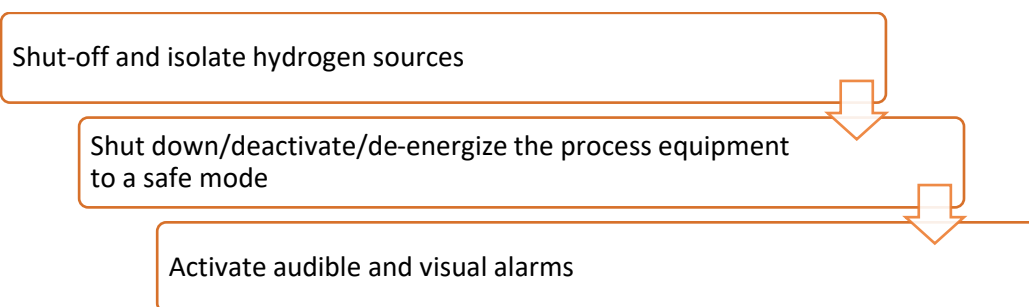


Fig 2.6.5. Task to be Performed on the Leak and Flame Detection

The Hydrogen production plant should also have an active ventilation that can be increased as required. There should be manual reset option to restart the system. In addition, wherever appropriate, remote notification of facility management and first responders should also be taken into account.

2.6.4 Indian Standards on Hydrogen

A regulatory framework, safety codes and relevant quality and performance standards are the prerequisites for hydrogen ecosystem. There is an urgent need to develop regulations and standards in line with industry requirements for emerging technologies. Existing statutory approvals and permissions methods are to be streamlined, and to be replaced by new processes, as required. In view of 'Ease of Doing Business', efforts are required to simplify processes and quick approvals. The efforts needed to align regulations and standards with internationally accepted norms and best practices.

At present, the hydrogen is used in industries generally at low pressure. However, in the recently initiated energy transition phase, hydrogen is being increasingly used at a very high pressure in the transport sector. The hydrogen fuelling stations are being installed to cater the need of transport sector. In view of this, the existing Indian Standards on hydrogen are being reviewed and modified as required.

India already has rich experience of framing standards and codes for use of gaseous fuels e.g. CNG, LPG etc. for domestic as well as transport sectors. These standards or codes need to be reviewed and upgraded, as required to take care of specific properties of hydrogen. The dedicated new standards for hydrogen needs to be developed, as per the requirements.

Hydrogen demands high levels of risk assessment to ensure safety. Stringent codes have been developed by various organization based on their experiences while dealing with hydrogen. The supply chain for hydrogen production, distribution, storage, and utilization has been analysed by various countries in the last decade. The manufacturer of equipment producing hydrogen need to follow standards for construction, safety, and performance requirements of the equipment.

The following table list some of the existing Indian standards on hydrogen.

Indian Standards on Hydrogen	Description
IS 15201 (2002)	Hydrogen – Code of Safety
IS 1090 (2002)	Compressed hydrogen
IS/ISO 11114-4	Transportable Gas Cylinders - Compatibility of Cylinder and Valve Materials with Gas Contents, Part 4: Test Methods for Selecting Metallic Materials Resistant to Hydrogen Embrittlement
IS 16749:2018	Basic considerations for the safety of hydrogen systems
IS 17445: 2020	Fasteners preloading test for detection of hydrogen

Table 2.6.2: Indian Standards on Hydrogen

IMPORTANT:

In line with the National Green Hydrogen Mission-2023, the Ministry of New and Renewable Energy has constituted a Working Group to recommend a national framework for Standards and Regulations required for the Green Hydrogen ecosystem. The standards on hydrogen are being reviewed. The existing standards may be retained as it is for green hydrogen or may be done so after revision. New standards for green hydrogen will be issued, as required.

2.6.5 Personal Protective Equipment

Personal protective equipment (PPE) should not be seen as a complete replacement for essential elements such as safe working conditions, proper ventilation, and responsible employee behaviour. It's crucial to recognize that PPE safeguards only the individual using it, leaving unprotected workers vulnerable to risks. Instead of relying only on PPE, fostering secure work environments and promoting informed conduct among employees are essential safety measures that safeguard both individuals and the workforce as a whole from potential hazards.

Personal protecting equipment should also include non-respiratory equipment such as:

Splash-proof safety goggles

Splash proof face shield when working with liquid

Safety helmets

Boots or high top shoes

Fig 2.6.6. PPE (non-respiratory)

In situations where employees might come into contact with cold equipment, vapours, or liquids, it's essential for them to wear specialized clothing intended to prevent the freezing of body tissues. To ensure consistent safety practices, it's necessary to establish standards and safety codes that outline these specific requirements. Such guidelines are crucial in maintaining the well-being of workers and preventing injuries or health issues that can arise from exposure to extreme cold conditions and substances.

Hydrogen as an Asphyxiant

Hydrogen functions as an asphyxiant by displacing oxygen, potentially causing suffocation. To mitigate this risk in environments with deficient oxygen levels, it is important to utilize an authorized self-contained breathing apparatus equipped with a full facepiece, operating in positive pressure mode. This ensures a safe oxygen supply and guards against the hazards associated with hydrogen's capacity to reduce available oxygen, safeguarding individuals from potential harm.

Activity



- Your supervisor has asked you to check for gas leakage in gas pipeline. Using a leakage detector, you need to check the leakage and if detected, repair it.

Equipment:

- PPE Kit
- Leakage Detector
- Tool Kit

Suggested Steps:

- Wear the PPE.
- Identify the place of leakage
- Repair the leakage
- Clean and keep the equipment used safely

2.6.6 Hazards and Safety Measures

To avoid any casualty, following sign should be displayed at hydrogen work area.



Fig. 2.6.7 Danger Sign That Needs to Be Displayed in hydrogen work area
Source: Hydrogen. ComplianceSigns.com

Storage, Handling, and Labelling

Adhere to the following guidelines while storing, handling and labelling hydrogen:

Storage Guidelines:

- Store hydrogen away from heat, flames, sparks, and oxygen.
- Prohibit smoking and open flames near hydrogen storage areas.
- Bond and ground metal containers and piping for large hydrogen quantities.
- Equip containers with self-closing valves, pressure vacuum bungs, and flame arrestors.

Handling Tools:

- Use non-sparking tools for opening and closing hydrogen containers.
- Employ approved explosion-proof electrical equipment during handling, manufacturing, or storage.

Safety Measures in Storage Areas:

- Ensure functional automatic/manual shutdown processes, gas and heat sensors.
 - Install early warning audio-visual alarms, sprinklers, flooding and water spraying systems, and explosion relief devices.

Ventilation:

- Ensure adequate natural/forced ventilation in the building, especially at higher elevations.

Indoor Storage Precautions:

- Prevent mechanical damage and overheating of storage tanks and cylinders.
- Direct vents and relief valves to safe locations with flame arrestors.

Outdoor Storage Precautions:

- Place outside storage tanks 15 m away from buildings or adjacent to blank masonry walls.
- Shield cylinders from direct sunlight with sheds.
- Maintain distance from flammable liquid storage, and install fire protection systems.

Bulk Storage (Non-refrigerated):

- Use gas-tight containers, with protection by fire hydrants and automatic sprinklers.
- Design storage tanks according to applicable standards with relief valves and flame arrestors.

Handling Precautions:

- Wear approved respirators, chemical-resistant gloves, safety goggles, and protective clothing.
- Prohibit sources of ignition during hydrogen handling.

Safety Adherence:

- Strictly follow safety recommendations and safe operating procedures when handling hydrogen.

Handling Hydrogen Cylinders

Before filling the cylinder with hydrogen, the following checks should be conducted:

Check and confirm for cylinder's validity. Both seller and consumer should verify the validity period.

Cylinder should have a valid test certificate from the concerned authorities.

Cylinder must be hydraulically tested before filling hydrogen gas.

Fig 2.6.8. Checks before filling the cylinder with Hydrogen

Following guidelines should be followed while handling Hydrogen cylinders:

- A. Cylinders must avoid exposure to rough handling or unusual mechanical impacts, such as drops, bumps, drags, or slides. They should always remain in a vertical position.

- B. Refrain from using ropes, slings, hooks, tongs, or similar handling equipment that might generate sparks or static electricity while dealing with hydrogen cylinders.
- C. Utilize a suitable hand truck, fork truck, or analogous equipment capable of securely holding and transporting cylinders.
- D. Employ a fitting platform, cradle, or boat when using a crane or derrick for handling. Secure cylinders with a rack or chain during discharge.
- E. Prevent placing or manipulating cylinders in locations that could incorporate them into an electrical circuit.
- F. Only remove the protection cap when ready to extract hydrogen. Employ appropriate standard valves; perform explosive gas checks on valves and fittings before usage.
- G. Adopt the standard colour code (i.e. Red colour without bands) for hydrogen cylinders.
- H. Never use cylinders, whether full or empty, as rollers for moving heavy objects.
- I. It is imperative not to alter numbers, markings, or test dates imprinted on cylinders.
- J. Upon emptying the cylinder, disconnect it, insert the valve plug, and replace the protective cap.
- K. Store empty cylinders separately from filled ones and promptly attach an 'EMPTY' tag. Close the valve, replace the valve outlet plug or nut, and secure the valve protective cap firmly.

Labelling a Hydrogen Container

Any Hydrogen vessel/container must be labelled as per the standard rules and regulating stating it as a "FLAMMABLE GAS, "CLASS 2" as shown in the following figure.



Fig. 2.6.9 Label on compressed hydrogen vessel

Source: Accuform NMC. Safety label: Hydrogen compressed.

<https://www.accuform.com/safety-label/chemical--hazardous-safety-labels-MCSLHYR>

The above label should also be displayed on the vehicle carrying the Hydrogen. The vehicle must also display the information as shown in the following figure:

Compressed Gas	•UN No. 1049 (United Nations number for hazardous hydrogen)
Refrigerated Liquid	•UN No. 1966

Fig 2.6.10. Information to Be Displayed on the Vehicle Carrying Hydrogen

Spillage/ Leakage of Hydrogen

It is recommended to conduct regular inspections on equipment, containers, and vessels containing hydrogen to promptly detect or prevent potential leaks. Also, there should be a well-trained and around-the-clock emergency response team for organizations handling hydrogen. This trained team should manage spills or leaks effectively. The following figure represents the guidelines/preaution that needs to be followed/taken care while handling spillage or leakage of Hydrogen:

Limit access to the area of spillage/leakage to properly protected personnel only, and restrict entry for others.

Address spills or leaks by taking actions that facilitate the swift dissipation of harmful vapors.

During a significant spill or equipment breach, evacuate the exposed area and refrain from re-entry by unprotected personnel until it is deemed safe.

Furnish suitable respiratory protection for individuals entering the area to conduct tests or repair damaged equipment, as elevated hydrogen concentrations might be present.

Halt gas flow or shut off the leak, if feasible without compromising safety.

If a cylinder is the source of the leak and it cannot be stopped in place, relocate the leaking cylinder to a secure outdoor location, either for leak repair or until the cylinder empties.

In cases of both gas and liquid spills, eliminate all potential ignition sources.

Fig2.6.11. Handling Spillage/Leakage of Hydrogen

Evacuation

As we know Hydrogen is highly flammable and can form explosive mixtures with air, therefore when hydrogen is spilled or leaked, it can create a dangerous situation, especially in enclosed spaces where the gas can accumulate. Even a small spark or ignition source can lead to an explosion or fire in the presence of hydrogen gas. Evacuating the area is essential to protect individuals from potential ignition sources and to prevent the accumulation of hydrogen, which could lead to a hazardous incident.

Waste Disposal

The waste generated during Hydrogen plant operations, regardless of its type, must be gathered and appropriately managed in accordance with the relevant standards, regulations, established practices, and guidelines that are specific to areas where hydrogen is handled. This ensures that waste disposal is carried out in a manner that prioritizes safety, environmental protection, and compliance with industry norms. Proper waste management in hydrogen handling areas prevents potential hazards and minimizes the impact on both personnel and the surroundings.

2.6.7 Fire Prevention and Fire

Hydrogen flame are almost invisible and can cause severe explosions. Any source of energy can cause ignition to the Hydrogen. Some of such sources are listing in the following figure.

Open flames	Electrical equipment	Electrostatic sparks	Sparks from striking objects	Friction
Thermite sparks	Air or oxygen, other oxidants	Hot surfaces	Hot hydrogen leaking into ambient air	Catalytic surfaces

Fig 2.6.12. Energy Sources that can cause Ignition to Hydrogen

Fire Prevention

By adhering to the following precaution one can prevent fire:

- Reduce all potential sources of leaks to a minimum.
- Mitigate sources of ignition as much as feasibly achievable.
- Ensure provisions for hydrogen isolation, escape routes, and fire control measures.
- Furnish effective fire detection and early warning systems, including alarms, in the case of a fire event.

Emergency Response Procedures and Systems

To avoid any casualty, the following measures should be taken:

- Establish complete emergency procedures and systems, and conduct regular mock drills.
- Ensure the emergency response team is well-trained in dos and don'ts, standard operating procedures (SOPs), mitigation strategies, and the hazards linked to hydrogen handling

Fire Fighting

During the Hydrogen spills one needs to take care of the unignited spills and ignited spills as mentioned below:

Unignited Spills:

- In case of a spill that hasn't ignited, use water spray to redirect flammable gas-air mixtures away from potential ignition sources.

Ignited Spills:

- Keep in mind that hydrogen flames are nearly invisible. Therefore, exercise caution when approaching flames for firefighting purposes.
- Attempt to halt gas flow if feasible.
- Due to re-ignition risks, hydrogen fires should generally not be extinguished until the hydrogen supply is cut off. Beware of possible flashback along the trail of the spill.
- If gas flow can't be stopped, allow cylinder contents to burn under controlled conditions.
- For ignited liquid hydrogen, use water to cool exposed fire-affected containers and shield individuals attempting to stop the spill source.

Best Practices

The following figure lists the best practices to deal with the Hydrogen fire:

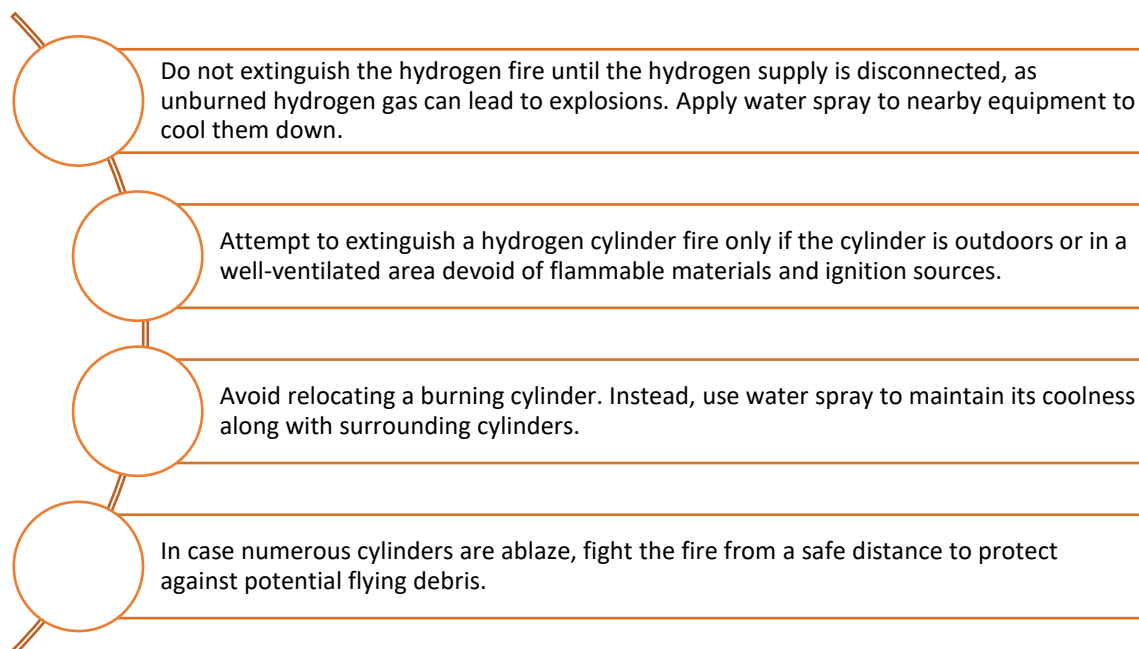


Fig 2.6.13.: Best Practices to Deal with the Hydrogen Fire

Fire Extinguishers

The following figure represents the classification of fire, their symbols, and extinguishing agents are used in extinguishing the fire.

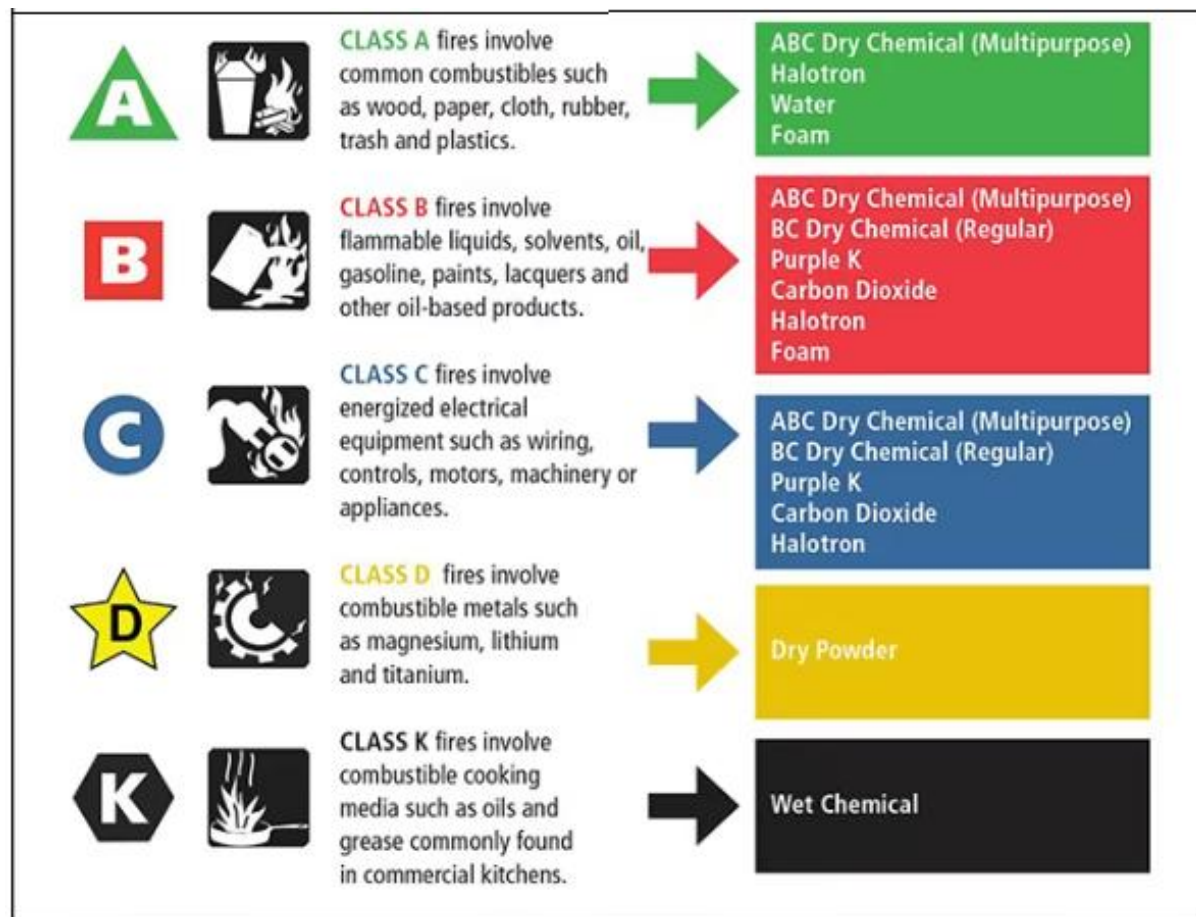


Fig. 2.6.14 Fire classification, symbols and fire extinguishing agents

Source: Kidde. 2019. Types of fires. https://www.kidde.com/home-safety/en/us/support/help-center/browse-articles/articles/types_of_fires.html

- ABC dry chemical is a versatile extinguishing agent effective against fires in classes A, B, and C.
- BC agent, based on sodium bicarbonate and chemical additives, is tailored for class B and C fire extinguishing.
- Typically, BC agent comprises a combination of mono-ammonium phosphate and ammonium sulphate.

Note that hydrogen fires are classified as class C fires.

The following figure represents the typical firefighting equipment provided at a hydrogen production plant of a thermal power station.



Fig. 2.6.15 Typical firefighting equipment for hydrogen plant

2.6.8 Safe Distance from Sparkling Devices

In order to avoid the risk of fires, explosions, and harm to personnel and property, it is important to maintain a safe distance between hydrogen and electrical devices is a fundamental safety practice. The layout design should align with safety standards and regulations, contributing to the overall safety and reliability of hydrogen-related operations. Therefore, it is crucial to maintain a safe distance between hydrogen and electrical devices like MCBs (Miniature Circuit Breakers), switches, and outlets for the following reasons:

- **Fire and Explosion Risk:** Hydrogen is highly flammable and can ignite easily in the presence of an ignition source, such as sparks or electrical arcs. Placing electrical devices too close to hydrogen storage or production areas increases the risk of these devices generating sparks, potentially leading to fires or explosions.
- **Personnel Safety:** Ensuring a safe distance helps protect the safety of personnel working in or around hydrogen facilities. In the event of an accidental hydrogen release, a safe separation reduces the chances of harm to workers.
- **Asset Protection:** Hydrogen-related facilities often involve significant investments in equipment and infrastructure. Maintaining a safe distance helps protect these assets from damage or destruction in the event of a fire or explosion.
- **Compliance with Safety Standards:** Safety standards and regulations, specific to hydrogen installations, often mandate minimum separation distances between hydrogen sources and electrical equipment. Compliance with these standards is necessary to meet safety requirements and avoid legal and regulatory issues.

Additionally, ensuring safe distances contributes to operational continuity. By preventing fires and equipment damage resulting from hydrogen-related incidents, it helps maintain uninterrupted operations, preventing costly downtime. In essence, this safety measure not only protects lives and property but also supports the reliability and sustainability of hydrogen-related processes.

2.6.9 Training and Health Monitoring

Training

All individuals involved in the operation, maintenance, and response to emergencies involving hydrogen and associated systems must undergo training relevant to their specific roles. The training program should comprehensively address the various aspects and potential risks pertinent to the tasks a given operator might encounter. The following figure lists the topics that must be covered in the training:

Hazards of hydrogen work areas

Emergency response procedures (leaks/spills/evacuation)

Use of fire fighting equipment

Use of protective clothing/apparatus including breathing sets

First aid and Rescue operations

Fig 2.6.15. Topics of Training

Furthermore, individuals must undergo specialized training as per their assigned tasks. Regular refresher courses should be scheduled, particularly for senior staff. Experienced employees should also undergo retraining.

Each staff member should be familiar with the precise location and proper operation of gas and fire alarm systems, sprinklers, monitors, and other firefighting apparatus. Awareness of the whereabouts of safety showers, eye wash stations, and first aid kits is also essential.

Health Monitoring

Health monitoring for hydrogen plant workers is essential as it helps identifying any potential exposure risks, ensuring their well-being, and mitigating any health issues related to hydrogen handling activities. It is recommended that all the employees must take bath/shower daily after work. Consuming or storing food near hydrogen handling areas should be strictly forbidden to prevent potential safety hazards.

Physical Examination

Physical examination of all the employees should include the following:

- Thorough pre-placement medical assessments must be conducted to evaluate the physical suitability of personnel before job assignment.
- Annual medical examinations, as mandated by relevant laws/regulations (such as the Factories Act/State Factories Rules), should be conducted.

First Aid and Medical Treatment

If there's contact with hydrogen or any injury, first aid should be provided immediately. Even for minor injuries, consult a doctor and explain the accident in detail.

If in case, there is excessive inhalation of Hydrogen gas (that leads to asphyxiation), following first aid should be provided to the affected person:

Move the affected person away from the exposed area.

Consider placing them in the recovery position.

Start rescue breathing if breathing stops.

Initiate CPR if heart action ceases.

If medical oxygen and skilled personnel are present, provide oxygen.

Don't offer liquids to an unconscious person.

If conscious, guide the person to sit or lie down calmly.

Fig 2.6.16. First Aid in Case of Excessive Inhalation of Hydrogen Gas

Skin Contact with Liquid Hydrogen

Exposure of skin to liquid hydrogen can cause frost bite. Frost bite is a condition that occurs when skin and underlying tissues freeze due to exposure to cold temperatures. The skin becomes cold, numb, and pale, and in severe cases, ice crystals can form in the affected areas. Frostbite requires prompt medical attention and can lead to tissue damage or even amputation if not treated properly.

In case of frost bite, soak the affected part in lukewarm water and immediately Seek for medical help.

Emergency Action Plan

An emergency action plan is a set of guidelines outlining procedures to follow during crises, ensuring safety and effective response in various emergency situations. A detailed emergency action plan is established for addressing potential emergencies within the thermal power plant. An emergency action plan should adhere to the following guidelines:

- It should include a comprehensive list of potential emergency situations.

- It should be thoughtfully presented in the local language to guarantee understanding by the local workforce.
- Emergency contact numbers should be visibly displayed throughout the power plant for quick access.

The following figures represent the emergency action plan along with emergency numbers at the thermal power plant in local language.





Fig. 2.6.17 Emergency action plant in a thermal power plant and contact numbers

Safety Instructions for Hazardous Areas

Chemicals are used across industries all over the world which pose certain chemical risks. Hence, it is essential to establish a systematic approach to safety in the use of chemicals.

For example, oil refineries employ a range of chemicals, also there is heightening risk due to hydrogen usage in desulphurization of feedstock such as natural gas, methane, and naphtha. In addition to safety instructions, mandatory personal protective equipment and emergency siren codes are issued. The behaviour-based safety instructions for every one including visitors in one of the oil refineries are shown in the following figures.

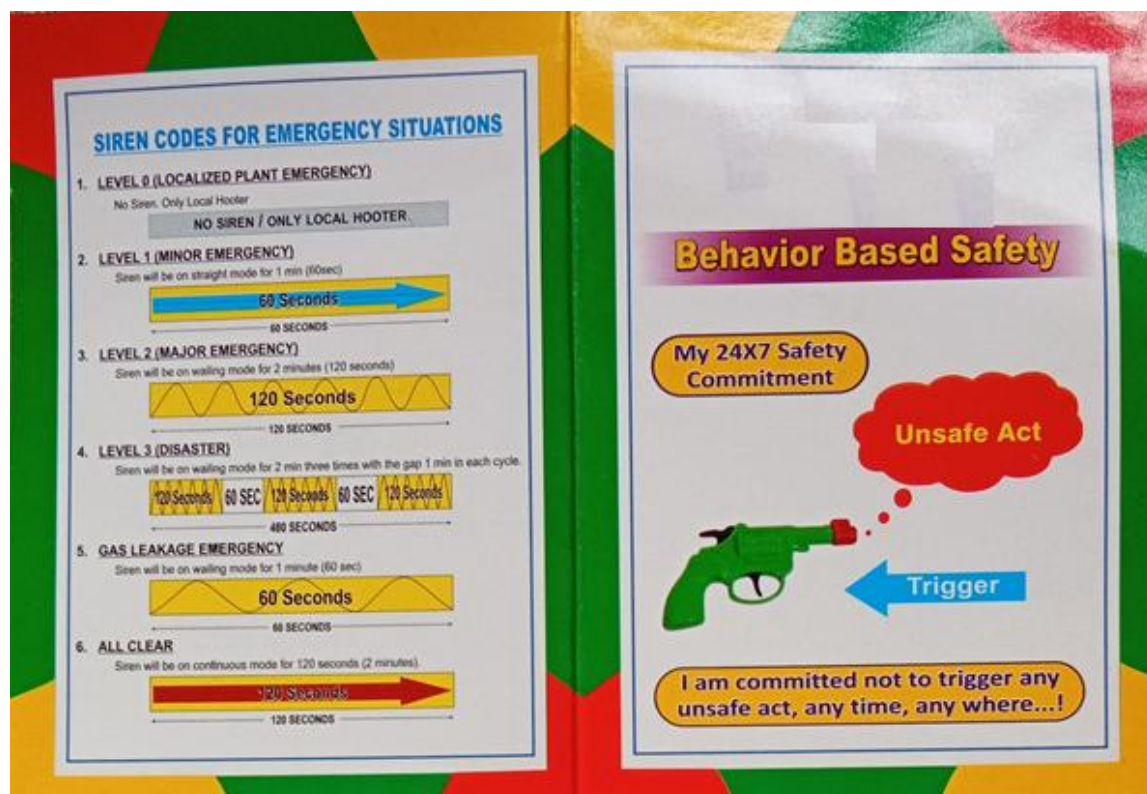


Fig. 2.6.18(a) Safety instructions in an Oil Refinery

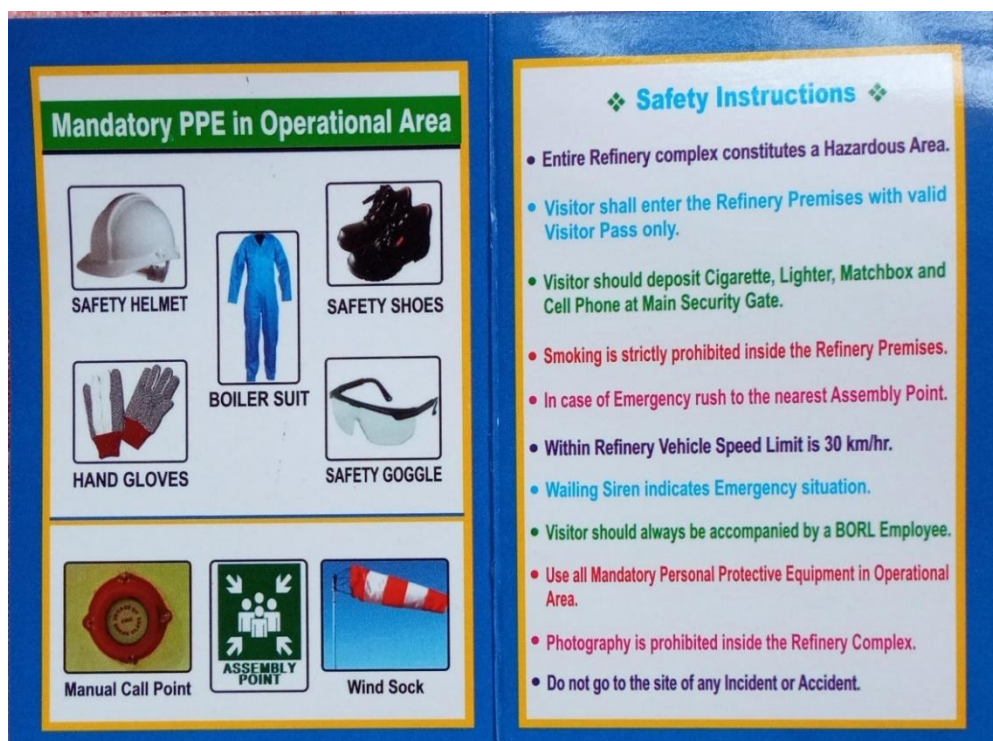


Fig. 2.6.18(b) Safety instructions in an Oil Refinery

Note:

The **National Green Hydrogen Mission** was approved by the Union Cabinet on 4th January 2023 and released by The Ministry of New and Renewable Energy who have constituted a Working Group to recommend a national framework for Standards and Regulations required for the Green Hydrogen ecosystem. The working group comprises of relevant Ministries, government agencies, standardization and certification bodies, and stakeholders from concerned industries.

UNIT 2.7: Key technical insights for setting up and running Successful Enterprise

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss key technical insights of green hydrogen system for setting up and running successful enterprises
2. Discuss how to perform SWOT analysis.

2.7.1 Key Technical Insights

Before setting up and running successful enterprises, it is important to gain its key technical insights. These insights provide a solid foundation for informed decision-making, enhancing the likelihood of business success. In the context of Green Hydrogen, understanding the details of production methods, renewable energy integration, safety protocols, and regulatory compliance is pivotal. It allows entrepreneurs to adapt various strategies to harness India's abundant renewable energy potential, plan and prioritize safety risks etc.

Thus, by considering the technical insights and aligning them with India's unique energy landscape, you can establish and run prosperous Green Hydrogen enterprises that contribute to the nation's clean energy goals while achieving business success.

Following are some of the key technical insights for setting up and running successful Green Hydrogen enterprises in India:

- **Renewable Energy Integration:** India boasts abundant solar and wind resources. Leveraging these renewables for electrolysis, the heart of Green Hydrogen production, is cost-effective and sustainable. Understanding how to efficiently integrate renewable energy sources into the hydrogen production process is essential.
- **Electrolysis Technology:** India's climate and energy infrastructure can influence the choice of electrolysis technology. Proton Exchange Membrane (PEM) electrolyzers may be preferable for regions with variable renewable energy, while Alkaline electrolyzers can be cost-effective in areas with stable energy supply.
- **Efficiency Optimization:** Given India's diversity, optimizing electrolyzer efficiency is crucial. Indian entrepreneurs should explore advanced catalysts and materials to enhance efficiency, reduce energy consumption, and lower production costs.
- **Hydrogen Storage:** Efficient hydrogen storage is key, especially for remote regions with intermittent energy supply. Exploring high-pressure tanks, liquid organic hydrogen carriers (LOHCs), or metal hydrides can address India's diverse storage needs.
- **Distribution Networks:** Building a reliable hydrogen distribution network is essential for reaching various sectors. Developing transport and logistics solutions that align with India's infrastructure and market demands is a priority.

- **Safety Protocols:** India's regulatory environment demands rigorous safety standards. Understanding and implementing comprehensive safety protocols is vital due to hydrogen's flammability.
- **Regulatory Compliance:** Navigating India's evolving regulatory landscape, which promotes clean energy, requires a clear understanding of compliance requirements. Staying abreast of incentives, permits, and environmental standards is essential.
- **Scale-Up Strategies:** India's growing demand for clean energy necessitates scalable solutions. Entrepreneurs should plan for expansion to meet increasing hydrogen requirements across sectors.
- **Environmental Sustainability:** Aligning with India's commitment to environmental sustainability, Green Hydrogen enterprises should focus on reducing carbon footprints and minimizing environmental impact.
- **Collaborations and Partnerships:** Collaborating with research institutions, government agencies, and industry partners in India can facilitate knowledge sharing, access to resources, and funding opportunities.
- **Market Diversification:** India's diverse market offers opportunities in industries, transportation, and energy storage. Diversifying the client base can mitigate risks associated with dependence on a single sector.
- **Workforce Development:** India's skilled labour force is an asset. Training employees in hydrogen technologies, safety practices, and maintenance is essential for long-term success.

SWOT analysis is a strategic tool that is used to assess internal Strengths and Weaknesses, along with external Opportunities and Threats. It helps organizations make informed decisions by identifying areas of advantage, areas needing improvement, potential avenues for growth, and external challenges that may impact their success. The following figure represents the full form of SWOT analysis.

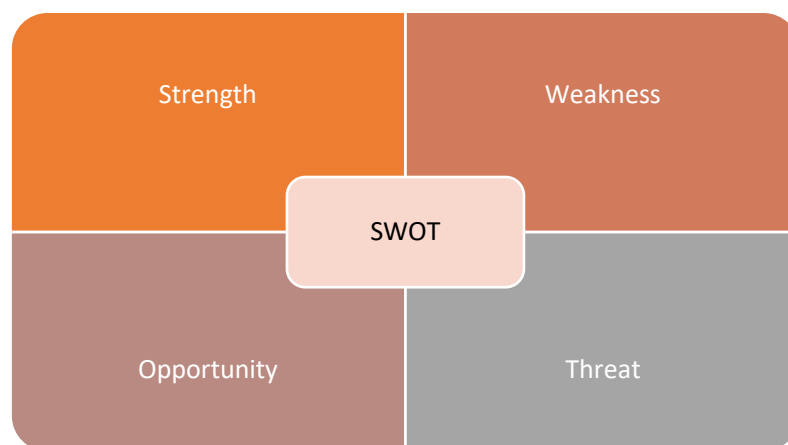


Fig 2.7.1. SWOT Analysis

By conducting a SWOT analysis, Green Hydrogen entrepreneurs can identify strategies to leverage their strengths, mitigate weaknesses, capitalize on opportunities, and

prepare for potential threats. This analysis serves as a valuable roadmap for informed decision-making and business planning in a dynamic and promising sector like Green Hydrogen.

Now, let us understand how to perform a SWOT analysis for setting up a Green Hydrogen enterprise involves evaluating its Strengths, Weaknesses, Opportunities, and Threats.

Strengths

- **Abundant Renewable Resources:** India has vast solar and wind potential, offering a consistent source of renewable energy for hydrogen production.
 - **Example:** A Green Hydrogen startup in Rajasthan benefits from ample sunlight, ensuring a stable and cost-effective energy source for electrolysis.
- **Growing Clean Energy Demand:** India's increasing emphasis on clean energy and environmental sustainability creates a favourable market for Green Hydrogen.
 - **Example:** The Indian government's push for hydrogen-powered vehicles creates a ready market for a Green Hydrogen producer in Pune.

Weaknesses:

- **High Initial Capital:** Establishing a Green Hydrogen production facility requires significant upfront investment, which might strain financial resources.
 - **Example:** Limited access to funding might hinder a startup's ability to procure advanced electrolyzers.
- **Technical Expertise:** The Green Hydrogen sector demands specialized knowledge and skills, which may pose challenges in recruiting a skilled workforce.
 - **Example:** A startup in Bengaluru faces difficulties in finding experienced hydrogen engineers and technicians.

Opportunities:

- **Government Initiatives:** India's supportive policies and incentives for clean energy can significantly benefit Green Hydrogen enterprises.
 - **Example:** A startup in Gujarat aligns with government schemes, receiving grants and tax benefits.
- **Diverse Applications:** Hydrogen finds applications in transportation, industry, and energy storage, providing a range of market opportunities.
 - **Example:** An enterprise in Maharashtra caters to multiple sectors, diversifying its revenue streams.

Threats:

- **Competition:** The Green Hydrogen sector is competitive, with established players and new entrants vying for market share.
 - **Example:** A startup in Tamil Nadu faces intense competition from well-established hydrogen producers.
- **Regulatory Changes:** Rapidly evolving regulations and standards may pose compliance challenges and affect business operations.
 - **Example:** An enterprise in Haryana grapples with changing emission norms affecting its hydrogen supply contracts.

Exercise



1. List out electrical, mechanical, and Civil components of the green hydrogen production plant
2. Describe the function of key components of an electrolyzer hydrogen production plant.
3. Explain working principle of main components of green hydrogen production plant
4. Describe briefly, the working of an alkaline, PEM and solid oxide electrolyzer?
5. What do you understand by 'proton exchange' in a PEM type electrolyzer?
6. How the positioning of various component of hydrogen production plant is decided?
7. What are the sparking items? Give examples.
8. How to safeguard against sparking items in hydrogen work area?
9. Try to read and understand the drawing of green hydrogen production plant and identify and list out various items/assemblies.
10. Write down the safety guidelines that are to be followed by everyone in hydrogen work zone.
11. List out the items of a personal protection kit in hydrogen work area.?
12. Fill in the following Table by the materials used in different parts of an electrolyzer?

Table 2.7.1 Materials used in electrolyzers

Item	Alkaline	PEM	Solid oxide
Electrolyte			
Cathode			
Anode			
Membrane/ Diaphragm			
Bipolar/separator plate			
Catalyst: anode			
Catalyst: cathode			

Interpret following safety signs in a hydrogen production plant.

Table 2.7.2 Safety signs and their interpretation

Safety sign/Symbol	Your Interpretation
	
	
	



This image shows a blank sheet of white paper designed for taking notes. At the top left, there is a header area with the word "Notes" written in a bold, orange font. To the right of the text is a small, square icon of a notepad with a pencil. Below the header, the rest of the page is filled with horizontal black lines, providing space for writing. The entire page is framed by a thin orange border.





3. Key technical and entrepreneurial aspects for supporting growth and business development green hydrogen production



- UNIT 3.1: Setting Up Hydrogen Production Plant
- UNIT 3.2: Cost Estimation
- UNIT 3.3: Taxation and Related Compliances
- UNIT 3.4: Policies and Corresponding Incentives/Subsidies



Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss process for setting up hydrogen production plant including required registration, availing low-cost financing from Financial Institutions/Venture Capital/Banks, government subsidies, green power, freewheeling/banking of renewable energy etc.
2. Discuss various evolving opportunities for setting up of hydrogen production plants at different sites, technologies, services, finance and market ecosystem
3. Discuss how to identify the key ingredients of business plan for land, water, renewable energy, finance, technology and market mechanism
4. Discuss how to assess the potential and demand of Hydrogen in different sectors i.e. transport, power, building, industry, etc.
5. Discuss all key aspects of a Detailed Project Report (DPR) and explain how to develop a bankable DPR which includes market demand, storage, technology and finance for hydrogen production
6. Discuss various parameters for calculating lifecycle cost of hydrogen production, storage system, along with cash flow, annual savings, payback period, IRR, inflation, increase in fossil fuel cost, sensitivity analysis by varying costs of renewable energy, water, man power, and such other factors etc. and how to include those in the detailed project report (DPR)
7. Discuss key aspects to successfully install, operate, safely store and market Hydrogen as Energy source.
8. Discuss O&M related cost for the system. If required, possibility of Annual Maintenance Contract (AMC) of electrolyser from supplier/ manufacturers, and other systems may be explored.
9. Explain different taxation and related compliances to setting up and operations
10. Discuss National Green Hydrogen policy and corresponding incentives/subsidies available and procedures for availing the same
11. Explain how evolving start-ups and new ventures boost-up hydrogen economy.

UNIT 3.1: Setting Up Green Hydrogen Production Plant

Unit Objectives

At the end of this unit, you will be able to:

- Discuss process for setting up hydrogen production plant including required registration, availing low-cost financing from Financial Institutions/Venture Capital/Banks, government subsidies, green power, freewheeling/banking of renewable energy etc.
- Discuss various evolving opportunities for setting up of hydrogen production plants at different sites, technologies, services, finance and market ecosystem
- Discuss how to identify the key ingredients of business plan for land, water, renewable energy, finance, technology and market mechanism
- Discuss how to assess the potential and demand of Hydrogen in different sectors i.e. transport, power, building, industry, etc.
- Discuss all key aspects of a Detailed Project Report (DPR) and explain how to develop a bankable DPR which includes market demand, storage, technology and finance for hydrogen production

3.1.1 Setting up a Hydrogen Production Plant

Setting up a hydrogen production plant in India involves a comprehensive process. It begins with a feasibility study to assess demand and suitability, followed by business registration and environmental clearances. Securing the right location and permits is crucial, as is securing financing, which can come from banks, investors, or government programs. Government incentives, like tax reductions and lower interest rates, can also be advantageous. Using green energy sources such as solar or wind power helps reduce environmental impact, and energy storage strategies are akin to saving money in a bank. Selecting the right technology, proper construction, and ongoing maintenance are vital for safety and efficiency. Lastly, a sales plan targeting industries like transportation and manufacturing is essential. Collaboration with local authorities, experts, and legal advisors is key throughout, and staying updated on evolving government policies and incentives is advised for project success. Following are the basics steps involved in setting up a hydrogen production plant in India:

- **Feasibility Study:** Assess the viability of your hydrogen production plant by considering factors like location, demand, competition, and resource availability.
- **Business Registration:** Register your business with the appropriate authorities, such as the Registrar of Companies (RoC), and choose a suitable business structure.
- **Environmental Clearances:** Obtain environmental clearances from the State Pollution Control Board and/or the Ministry of Environment, Forest and Climate Change (MoEFCC), depending on your project's scale and potential environmental impact.
- **Land Acquisition:** Acquire suitable land for your plant, ensuring it complies with local zoning and regulatory requirements.

- **Permits and Licenses:** Secure permits and licenses required for hydrogen production, including those from the Petroleum and Explosives Safety Organization (PESO) and other relevant authorities.
- **Financing Options:** Explore various financing sources, including banks, venture capital firms, and government-backed programs like those offered by IREDA and IL&FS.
- **Government Subsidies and Incentives:** Investigate available subsidies, tax benefits, and concessional interest rates offered by government agencies like the Ministry of New and Renewable Energy (MNRE).
- **Green Power Procurement:** Consider sourcing green power from renewable energy sources like solar or wind, either by setting up your own generation facility or through power purchase agreements (PPAs).
- **Investment of Renewable Energy:** Implement strategies for efficient power consumption and storage by transmitting excess energy to the grid and withdrawing it when needed.
- **Technology Selection and Plant Construction:** Choose the appropriate hydrogen production technology and build the plant while adhering to safety and environmental standards.
- **Operation and Maintenance:** Develop a robust operation and maintenance plan to ensure the safe and efficient operation of the hydrogen production plant.
- **Marketing and Sales:** Create a market entry strategy and identify potential customers, distributors, and partners in industries such as transportation, manufacturing, and energy.

Throughout the entire process, it's essential to collaborate with local authorities, industry experts, and legal advisors to navigate specific state-level regulations and requirements effectively. Additionally, staying informed about evolving policies and incentives related to renewable energy and hydrogen production in India is critical for the success of your project.

3.1.2 Registrations Required for Setting Up Hydrogen

Setting up a hydrogen production plant in India involves several registrations and regulatory clearances. These requirements may vary depending on the location. Let us understand the key registrations and the general steps involved in the process.

Business Registration

For business registration, you need to choose a suitable business structure, such as a private limited company, LLP (Limited Liability Partnership), or any other legal entity. You have to register your business with the Registrar of Companies (RoC) under the Companies Act, 2013. After that obtain a Certificate of Incorporation, Permanent Account Number (PAN), GST number for your business. The documents required to keep in hand while running your daily business tasks are as follows:

Current Account Number of the Company

GST Number

PAN Number

Fig. 3.1.1: Documents

It is of utmost importance to fulfil all specified procedures and requirements before starting a business legally.

Environmental Clearance

It is necessary for green hydrogen plants to assess and mitigate potential environmental impacts. It ensures compliance with regulations, identifies and addresses risks, and involves public consultations for transparency. Obtaining clearance is a legal requirement in many countries, promoting long-term sustainability and the protection of sensitive areas. It also aligns with international standards and facilitates access to financing. Overall, it's crucial to demonstrate a commitment to responsible environmental practices, even in environmentally friendly projects like green hydrogen production. Following are the steps to apply for environmental clearance

- Step 1: Identify the appropriate regulatory authority based on your project's location and scale. It could be the State Pollution Control Board and/or the Ministry of Environment, Forest and Climate Change (MoEFCC).
- Step 2: Prepare an Environmental Impact Assessment (EIA) report and submit it to the relevant authority.
- Step 3: Await approval and clearances, which may involve a public hearing and consultation with stakeholders.

Industrial Licenses and Permits

To set up a green hydrogen plant in India, you may require various industrial licenses and permits, depending on the specific location and nature of your project. Following are some of the key licenses and permits that may be necessary include:

- **Factory License:** A Factory License is required for setting up a green hydrogen plant to ensure compliance with the Factories Act, 1948. It regulates safety, health, and working conditions in factories, including those engaged in hydrogen production. This license is essential to establish and operate the plant legally while ensuring the safety and well-being of workers and adherence to environmental norms.
- **Consent to Establish (CTE):** Obtaining a CTE from the State Pollution Control Board is essential to establish that your plant complies with environmental norms and standards.
- **Consent to Operate (CTO):** After successful commissioning and compliance with environmental regulations, a CTO is required to legally operate the plant.
- **Fire License:** Depending on the scale and nature of your hydrogen plant, you might need a fire safety license, including a No Objection Certificate (NOC) from the local fire department.
- **Land Use Conversion:** Ensure that the land acquired for your plant is appropriately zoned for industrial use. If not, you may need to apply for land use conversion or change of land use permits. It ensures compliance with zoning regulations, availability of necessary infrastructure, and environmental compatibility.
- **Water and Air Pollution Control Consents:** Obtain necessary consents for water and air pollution control from the State Pollution Control Board. Compliance with emissions and effluent standards is essential.
- **Hazardous Waste Authorization:** If your plant generates hazardous waste, you may need authorization for its collection, transportation, and disposal.

- **Boiler Operation License:** If your plant uses boilers for hydrogen production, you may need a license to operate them safely and in compliance with boiler safety regulations.
- **Renewable Energy Permits:** If you plan to use renewable energy sources, such as solar or wind power, to generate electricity for your plant, you may need permits and clearances related to these sources.
- **Customs and Excise Duty:** Depending on the equipment and materials you import for your hydrogen production plant; you may need to comply with customs and excise duty regulations.
- **Environmental Impact Assessment (EIA) Clearance:** For larger-scale projects with potential environmental impacts, an EIA clearance from the Ministry of Environment, Forest and Climate Change (MoEFCC) may be required.
- **Petroleum and Explosives Safety Organization (PESO) Licenses:** If your plant handles hydrogen gas, you may need licenses and approvals from PESO to ensure safety and compliance with explosive and safety regulations.

Note: The specific licenses and permits required may vary depending on the state and local regulations in your project's location. It is important to consult with local authorities, environmental experts, and legal advisors to ensure that you obtain all the necessary licenses and permits for your green hydrogen plant while complying with relevant safety and environmental standards.

Other Clearances

Depending on the specific location and project details, you may need additional clearances such as coastal regulation zone (CRZ) clearance, forest clearance, and more.

It's important to note that the process may involve extensive documentation, site inspections, and compliance with safety and environmental standards. Additionally, consulting with legal experts, environmental consultants, and industry professionals can help streamline the registration and clearance process for setting up a hydrogen production plant in India.

3.1.3 Availing Low-Cost Financing

Availing low-cost financing, government subsidies, and implementing green power strategies, as well as freewheeling/banking renewable energy for a hydrogen plant in India involves the following steps:

1. Low-Cost Financing: It can be availed from banks financial institutions or via investment from the venture capital firms:

- **Banks and Financial Institutions:** Approach banks and financial institutions with a well-structured project proposal. Highlight the viability and sustainability of your hydrogen plant. Many banks offer loans, credit lines, and financial assistance for renewable energy projects at competitive rates.
- **Venture Capital:** Seek investment from venture capital firms specializing in clean energy and technology. Prepare a compelling business plan and demonstrate the potential for a high return on investment.

2. Government Subsidies and Incentives: Consult with the Ministry of New and Renewable Energy (MNRE) and relevant state-level agencies to identify available subsidies, grants, and incentives for renewable energy projects. Ensure that your project meets the eligibility criteria and compliances required to access these benefits.

3. **Green Power Procurement:** Set up your own renewable energy generation facility, such as solar or wind power, to power your hydrogen plant. Alternatively, enter into Power Purchase Agreements (PPAs) with existing renewable energy providers. Ensure that the power generated aligns with your plant's energy requirements.

4. **Freewheeling/Banking of Renewable Energy:** Implement a strategy for freewheeling or banking excess renewable energy. This involves transmitting surplus energy generated from your renewable sources to the grid and withdrawing it when your plant requires it. Coordination with the local electricity distribution company is essential for this.

5. **Compliance and Documentation:** Ensure that your project meets all regulatory and compliance requirements to access low-cost financing and government subsidies. Maintain detailed records and documentation of your project's financial aspects to facilitate financing approvals.

6. **Project Viability and Financial Modeling:** Develop a robust financial model that demonstrates the feasibility and profitability of your hydrogen plant. Banks and investors require a clear understanding of the expected returns and risk mitigation strategies.

7. **Legal and Regulatory Compliance:** Comply with all legal and regulatory requirements related to financing, subsidies, and energy procurement. Any non-compliance can delay funding and access to incentives.

8. **Collaboration and Networking:** Engage with industry associations, renewable energy forums, and government agencies to stay informed about financing opportunities, subsidies, and best practices in the sector.

9. **Expert Consultation:** Seek guidance from financial advisors, legal experts, and consultants with experience in renewable energy projects. Their expertise can help streamline the financing and subsidy application processes.

Some initiatives and schemes related to green hydrogen production in India included:

- **Production Linked Incentive (PLI) Scheme:** The government proposed a PLI scheme for green hydrogen to incentivize domestic manufacturing and production of green hydrogen technologies and components.
- **Green Hydrogen Fund:** There were discussions about the establishment of a Green Hydrogen Fund to support research, development, and deployment of green hydrogen technologies.
- **National Hydrogen Mission:** The government was considering the launch of a National Hydrogen Mission to promote hydrogen production, storage, and utilization, with a focus on green hydrogen.
- **Renewable Energy Schemes:** Existing renewable energy schemes like the Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM-KUSUM) and various state-level renewable energy policies were being explored to support green hydrogen projects.
- **Subsidies and Incentives:** Green hydrogen projects were expected to benefit from subsidies, tax incentives, and concessional financing under various government initiatives, including those by the Ministry of New and Renewable Energy (MNRE).

3.1.4 Evolving Opportunities

Entrepreneurs in India are at the forefront of a rapidly evolving hydrogen economy, where numerous opportunities await in the establishment of hydrogen production plants. Geographically, regions with abundant solar and wind resources offer ideal sites, with states like Gujarat and Rajasthan standing out. In terms of technologies, advanced electrolysis methods, such as proton exchange membrane (PEM) and alkaline electrolyzers, are gaining traction for their efficiency and scalability. Services encompass not only hydrogen production but also energy storage and the development of hydrogen infrastructure, including refuelling stations. Securing financing through green bonds, government incentives, and strategic partnerships is key to realizing these opportunities. Moreover, India's market ecosystem is witnessing increased demand for hydrogen in industries such as steel, ammonia, and transportation, making it a promising arena for entrepreneurs to drive innovation and contribute to the country's clean energy transition.

The following figure represents the overview of the of India's Nation Green Hydrogen Mission:

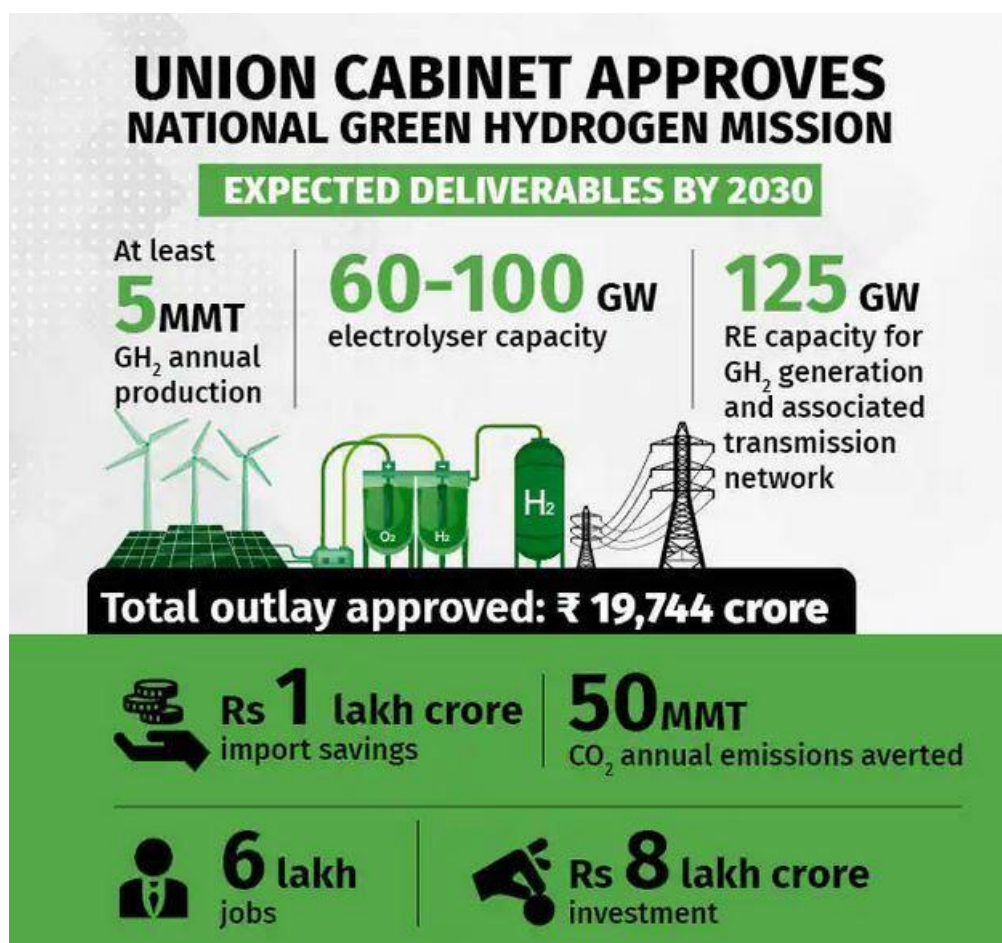


Fig 3.1.2: Overview of India's National Green Hydrogen Mission

Source: <https://fsr.eui.eu/the-green-hydrogen-economy-insights-from-india/>

Following are the key points representing the overview of India's Nation Green Hydrogen Mission:

- Hydrogen is becoming important in the world, and India plans to use more of it. They think that about 6% of all the energy used will be hydrogen by 2050.
- In India, they expect to use four times more hydrogen by 2050. Factories and big trucks will use most of it.
- India wants to make "green hydrogen" using clean energy like wind and solar power. They want to use it for things like cars, ships, making steel without polluting, and making electricity when needed.
- Right now, India spends a lot of money on importing energy, and it might double in 15 years. Green hydrogen can help save money and reduce pollution.
- India has a plan called "Strategic Interventions for Green Hydrogen Transition Programme" to make this happen. They want to make hydrogen machines in India and produce green hydrogen. They also want to do research and training.
- India wants to be energy independent by 2047 and self-reliant, and this plan is part of that. They have already started making green hydrogen in Assam.
- India's plan can help not only the country but also the world by reducing pollution and using clean energy.

Thus, Following are the evolving opportunities in Hydrogen Production in India:

- Solar-Rich Regions: States like Rajasthan, Gujarat, and Karnataka offer ideal conditions for solar-driven green hydrogen production.
- Coastal Locations: Coastal regions present potential for offshore wind-driven hydrogen production and easy access to water for electrolysis.
- Industrial Clusters: Proximity to industrial hubs, including steel, petrochemicals, and ammonia sectors, ensures immediate demand for hydrogen.
- Advanced Technologies: Electrolysis methods like PEM and alkaline electrolyzers are efficient and scalable. High-temperature electrolysis (SOEC) and biomass gasification offer innovation.
- Energy Storage: Hydrogen serves as an energy carrier and aids grid stability, complementing intermittent renewables.
- Hydrogen Refuelling Stations: Developing stations for fuel cell vehicles, especially in urban areas, taps into a growing market.
- Hydrogen Infrastructure: Establishing storage, compression, and transportation infrastructure is vital for a robust supply chain.
- Financial Support: Entrepreneurs can access green finance, government grants, and international funding for project viability.
- Market Expansion: Industries like steel, ammonia, and fuel cell vehicles offer substantial market potential, including export opportunities.
- Regulatory Compliance: Staying informed about evolving regulations and policies is essential for project planning and compliance.
- Research and Development: Investment in R&D and collaboration with institutions and industry players enhance competitiveness.

India's evolving hydrogen landscape provides diverse opportunities, but successful implementation requires navigating regulations, securing financing, and building strategic alliances, all while staying abreast of policy changes and market dynamics.

3.1.5 Key Ingredients of a Business Plan

A business plan is a formal written document that outlines a company's goals and how it plans to achieve them. It serves as a roadmap for the business, providing a detailed description of its products or services, target market, competition, financial projections, and strategies for growth and profitability. A well-structured business plan typically includes the following key ingredients:

Executive Summary

Company Description

Market Research and Analysis

Products or Services

Marketing and Sales Strategy

Operations Plan

Financial Plan

Management and Team

Risk Assessment

Sustainability and Environmental Impact

Fig3.1.3. Key Ingredients of a Business Plan

- **Executive Summary:** A concise overview of the entire business plan, summarizing the company's mission, key objectives, and highlights.
- **Company Description:** Detailed information about the business, including its history, legal structure, ownership, location, and a brief background of the founders.
- **Market Research and Analysis:** A comprehensive analysis of the target market, including size, trends, customer demographics, and competition. This section helps the business understand its market and potential opportunities.
- **Products or Services:** A detailed description of the products or services offered, including their features, benefits, and unique selling points.
- **Marketing and Sales Strategy:** An outline of the marketing and sales tactics the business plans to use to attract and retain customers, including pricing, distribution channels, and promotional strategies.

- **Operations Plan:** Information on how the business will operate, including production processes, supply chain management, facilities, and staffing requirements.
- **Financial Plan:** Financial projections, including income statements, balance sheets, cash flow statements, and funding requirements. This section also includes a break-even analysis and return on investment (ROI) projections.
- **Management and Team:** Details about the management team, their qualifications, roles, and responsibilities. This section highlights the expertise of the team members and their ability to execute the business plan.
- **Risk Assessment:** Identification of potential risks and challenges that the business may face and strategies for mitigating them.
- **Sustainability and Environmental Impact:** Information on the company's commitment to sustainable practices and its impact on the environment, including any relevant certifications or initiatives.

In addition, it also includes supplementary documents, such as resumes of key team members, market research data, legal documents, and additional information that supports the plan.

3.1.6 Assess the Potential and Demand of Hydrogen in Different Sectors

Assessing hydrogen's potential and demand across sectors is crucial for informed decision-making. It enables the identification of growth opportunities, cost-effective applications, and environmental benefits. This assessment guides investments, technology development, and policy formulation, facilitating the transition to a sustainable hydrogen-based economy and reducing dependency on fossil fuels.

Assessing the potential and demand for hydrogen in different sectors, such as transport, power, buildings, and industry, involves a systematic analysis of the following key factors:

Sector-Specific Analysis

- **Transport Sector:** Identify the current and future demand for hydrogen in various modes of transportation, including passenger vehicles, buses, trucks, trains, and ships. Also, consider the availability and infrastructure for hydrogen refuelling stations, as well as the growth potential. In addition, evaluate the cost competitiveness and environmental benefits of hydrogen fuel cells compared to other propulsion technologies like electric batteries or internal combustion engines.
- **Power Sector:** Analyze the potential for using hydrogen in power generation, such as hydrogen-powered turbines or fuel cells. Assess the role of hydrogen in energy storage and grid balancing, especially with intermittent renewable energy sources like wind and solar. Also, Calculate the economic viability and environmental advantages of hydrogen in power generation.
- **Building Sector:** Evaluate the demand for hydrogen as a clean heating fuel for residential, commercial, and industrial buildings. Consider the existing heating infrastructure and the feasibility of retrofitting or incorporating hydrogen-based heating systems. Examine energy efficiency, emissions reductions, and cost-effectiveness.

- **Industrial Sector:** Identify industries with high-temperature heat requirements that can benefit from green hydrogen. Assess the potential for hydrogen to replace fossil fuels in industrial processes such as steel and cement production. Calculate the emissions reductions and cost savings associated with hydrogen adoption in industrial applications.

Market Research

Conduct thorough market research in each sector to understand customer preferences, trends, and regulatory frameworks. Conduct a survey among potential customers, industry associations, and experts to gather insights into the acceptance and demand for hydrogen solutions.

Infrastructure and Supply Chain Analysis

Evaluate the existing and required infrastructure for hydrogen production, storage, transportation, and distribution in each sector. Assess the availability of hydrogen production technologies, such as electrolysis and steam methane reforming. Consider the logistical challenges and costs associated with hydrogen supply chains.

Cost-Benefit Analysis

Conduct a cost-benefit analysis to compare hydrogen-based solutions with existing technologies or alternatives. Do not forget to consider the factor involved in the initial investment, operational costs, maintenance, and expected returns on investment.

Environmental Impact Assessment

Evaluate the environmental benefits of hydrogen adoption in each sector, including greenhouse gas emissions reduction and air quality improvement. Also, consider the life cycle emissions of hydrogen production and use.

Policy and Regulatory Considerations

Analyze the role of government policies, incentives, and regulations in promoting hydrogen adoption and keep monitoring the changes in policy frameworks that may impact the demand for hydrogen in various sectors.

Technological Readiness

Assess the maturity and scalability of hydrogen technologies for each sector. Consider ongoing research and development efforts that may enhance the competitiveness of hydrogen solutions.

Collaboration and Stakeholder Engagement:

Engage with industry stakeholders, research institutions, and government agencies to gather insights and foster collaboration in promoting hydrogen adoption.

By systematically analysing, stakeholders can better understand the potential and demand for hydrogen in different sectors and make informed decisions regarding investments, technology deployment, and market strategies.

A DPR is a very comprehensive outline of a project in which all the essential information such as the resources and tasks to be carried out are included. It is prepared after the planning and the designing part of a project are completed. It can also be referred to as a final blueprint of a project after which the implementation and operational process take place. In this extensive project report, the roles and responsibilities are highlighted along with the safety measures.

The following key factors play an essential role in deciding whether a project will turn into a success:

- Project completion within the specified period
- Delivery of quality products after the completion of the project or customer satisfaction
- Cost-effectiveness

A DPR depicts a practical viewpoint for the project implementation. Effective measures must also be included for an efficient execution of the project. The DPR acts as a ladder toward success and supports the implementation of the project.

A DPR must include the information as shown in the following figure:



Fig 3.1.4: Info to be included in a DPR

Advantages of DPR

Following are the advantages of DPR

- **Budget Management** - DPR helps in planning and managing the budget in such a manner that the cost does not go over the set budget.
- **Minimising Risks** - Every so often, despite giving great attention to details, issues arise during the execution of the project. Hence, it is crucial to recognise and report the risks to the project manager before the implementation. This step makes room for improvement.
- **Project Progress Follow-up** - DPR has control of the project progress. It facilitates tracking the schedule of the project and eliminating any problems.
- **Holdover the Project** - Project reporting maintains hold of the higher authority over the project so that they can keep a check on progress and eliminate factors that cause a halt in the progress of the project. The performance of the team members and the quality of work is also checked.

A detailed project report has countless benefits to make a project successful. Hence, it is important to involve an experienced person or a firm in DPR preparation. A person with relevant expertise, who can handle the project tasks, is also needed in the team.

Creating a Bankable DPR

Developing a bankable Detailed Project Report (DPR) is crucial for obtaining financing and implementing a green hydrogen plant project in India. The following are the various task to be performed to develop a bankable DPR:

- **Executive Summary:** Provide an overview of the project, including its objectives, scope, and expected outcomes. Highlight the key features and benefits of the green hydrogen plant, emphasizing its economic and environmental advantages.
- **Project Description:** Provide detailed information about the green hydrogen plant, including its capacity, design, and technology components. Describe the storage requirements, types of produce to be stored, and the anticipated demand for the facility.
- **Technical Specifications:** Present the technical details of the solar PV system, refrigeration system, insulation, and other relevant components. Specify the capacity and performance parameters of each component, including solar panel capacity, battery storage, refrigeration unit specifications, insulation materials, and thermal energy storage systems.
- **Site Analysis:** Conduct a thorough analysis of the proposed site for the green hydrogen plant facility. Include details such as land availability, location suitability, solar resource potential, infrastructure accessibility, and any environmental considerations. Provide site plans, maps, and relevant photographs.
- **Financial Analysis:** Perform a comprehensive financial analysis of the project to assess its economic viability. Include capital investment costs, operational expenses, revenue projections, and expected returns. Consider factors such as equipment costs, installation costs, maintenance expenses, electricity savings, revenue from storage fees, and any available financial incentives or subsidies.
- **Project Timeline:** Develop a project timeline that outlines the various stages of implementation, including site preparation, equipment procurement, installation, testing, and commissioning. Provide a detailed schedule with milestones and timelines for each stage.
- **Risk Assessment and Mitigation:** Identify potential risks and challenges associated with the project and propose mitigation strategies. Address technical, financial, regulatory, and operational risks. Include contingency plans for unforeseen events that may impact the project's timeline or financial viability.

- **Environmental and Social Impact:** Assess the environmental and social impact of the project. Identify any potential environmental risks and propose mitigation measures to minimize the project's carbon footprint and ensure compliance with environmental regulations. Consider social aspects such as employment opportunities and community benefits.
- **Legal and Regulatory Compliance:** Ensure compliance with relevant laws, regulations, and permits. Include details of required approvals, licenses, and compliance with electrical, building, and environmental regulations.
- **Project Team and Partners:** Highlight the project team's qualifications and expertise, including technical consultants, engineers, and contractors involved in the project. Provide details of any strategic partnerships or collaborations.
- **Conclusion and Recommendations:** Summarize the key findings, conclusions, and recommendations based on the analysis conducted. Emphasize the financial viability, technical feasibility, and sustainability of the green hydrogen plant project.

Developing a bankable DPR requires a comprehensive analysis of technical, financial, environmental, and regulatory aspects of the green hydrogen plant project. It should demonstrate the project's viability, its potential for economic and social benefits, and compliance with relevant regulations and standards. Seek expert advice and engage with relevant stakeholders to ensure the accuracy and credibility of the DPR.

Note:

To know more about Detailed Project Report, you may refer to the following link:

https://www.youtube.com/watch?v=Ke_jwp_lskg

Title of Video: How to Prepare a Bankable Project Report

By: Business Pathshala

Duration: 52 m 39sec

UNIT 3.2: Cost Estimation

Unit Objectives

At the end of this unit, you will be able to:

- Discuss various parameters for calculating lifecycle cost of hydrogen production, storage system, along with cash flow, annual savings, payback period, IRR, inflation, increase in fossil fuel cost, sensitivity analysis by varying costs of renewable energy, water, man power, and such other factors etc. and how to include those in the detailed project report (DPR)
- Discuss key aspects to successfully install, operate, safely store and market Hydrogen as Energy source.
- Discuss O&M related cost for the system. If required, possibility of Annual Maintenance Contract (AMC) of electrolyser from supplier/ manufacturers, and other systems may be explored.

3.2.1 Lifecycle Cost

The lifecycle cost of hydrogen production refers to the total cost incurred over the entire lifespan of a hydrogen production project, from its initial setup and construction to its operation, maintenance, and eventual decommissioning or disposal. It encompasses all relevant expenses and revenues associated with the project, both upfront and ongoing, and is typically expressed in monetary terms.

Following are the key parameters and factors that play a role in this calculation:

- **Initial Investment (CAPEX):** This includes the costs associated with setting up the hydrogen production and storage infrastructure, such as equipment, land acquisition, construction, and installation expenses.
- **Operating and Maintenance Costs (OPEX):** These are the recurring expenses incurred to operate and maintain the hydrogen production and storage systems over their lifespan. OPEX includes labor, materials, energy, regular maintenance, and other operational costs.
- **Hydrogen Production Costs:** This category encompasses the expenses related to hydrogen production, including the cost of renewable energy, water, and any other raw materials used in the production process.
- **Storage Costs:** If the project involves hydrogen storage, these costs include the expenses related to storage infrastructure, such as tanks, reservoirs, or compression facilities.
- **Financing Costs:** These costs involve interest payments on loans or financing used to fund the project. The cost of capital and the interest rate applied to loans are important factors in financing costs.
- **Cash Flow Analysis:** A comprehensive cash flow projection covering the project's entire lifespan is essential. This projection includes annual revenues, costs, savings, and any government incentives or subsidies.
- **Annual Savings and Benefits:** Determine the annual savings and benefits generated by using hydrogen in place of conventional fuels or processes. These may include reduced fuel costs, emissions reductions, and operational efficiency gains.
- **Payback Period:** The payback period is the time it takes for the cumulative net cash flow (revenue minus expenses) to equal the initial investment. A shorter payback period indicates a quicker return on investment.
- **Internal Rate of Return (IRR):** IRR represents the project's annualized rate of return. It measures the project's profitability, with a higher IRR indicating a more financially attractive project.
- **Inflation and Escalation:** Account for inflation and cost escalations over the project's lifespan. These factors impact both costs and revenues.
- **Increase in Fossil Fuel Costs:** Consider the expected increase in fossil fuel costs over time. This can enhance the economic case for hydrogen as an alternative energy source.
- **Sensitivity Analysis:** Conduct sensitivity analyses by varying key parameters like the cost of renewable energy, water, labor, and other critical inputs. Assess how changes in these variables impact the project's financial metrics and viability.

These parameters collectively provide a comprehensive view of the lifecycle cost and financial performance of hydrogen production and storage systems. They help project owners, investors, and decision-makers evaluate the project's economic feasibility and make informed investment

decisions. Additionally, they allow for scenario analysis to understand the impact of changing variables on project outcomes.

The formula used to calculate the lifecycle cost (LCC) of a green hydrogen plant involves summing all relevant costs and benefits over the project's entire lifespan. The general formula for LCC can be expressed as:

- **LCC = Initial Investment + Operating and Maintenance Costs - Annual Savings + Salvage Value**

Where,

- Initial Investment (CAPEX): This is the upfront capital expenditure required to establish the green hydrogen plant. It includes costs for land acquisition, construction, equipment, and infrastructure.
- Operating and Maintenance Costs (OPEX): These are the ongoing expenses incurred to operate and maintain the plant throughout its operational life. OPEX includes labor, materials, energy, maintenance, and other operational costs.
- Annual Savings: These represent the annual benefits and savings generated by the green hydrogen plant compared to alternative energy sources or processes. Savings may include reduced fuel costs, emissions reductions, and operational efficiency gains.
- Salvage Value: The salvage value is the estimated residual value of the plant's assets at the end of their useful life. It accounts for potential resale or reuse value. Salvage value reduces the total cost of the project.

Note: The result of this formula provides the net lifecycle cost of the green hydrogen plant. If the result is positive, it indicates that the project's benefits and savings outweigh its costs, making it financially viable. Conversely, a negative net LCC suggests that the project may not be economically feasible.

3.2.2 Key Aspects to Successfully Install,

Successfully installing, operating, safely storing, and marketing hydrogen as an energy source requires attention to the following key aspects to ensure efficiency, safety, and market acceptance:

- **Production Technology:** Choose the appropriate hydrogen production technology (e.g., electrolysis, steam methane reforming) based on factors like energy source availability, scalability, and environmental impact.
- **Storage Infrastructure:** Invest in safe and efficient hydrogen storage solutions, such as high-pressure tanks, underground caverns, or liquid hydrogen storage.
- **Distribution Network:** Develop a robust distribution network to transport hydrogen to end-users, including pipelines, trailers, or hydrogen refuelling stations.
- **Hydrogen Handling Safety:** Implement strict safety protocols for hydrogen handling, storage, and transportation to mitigate potential risks.
- **Safety Training:** Ensure that all personnel involved in hydrogen production, operation, and distribution receive comprehensive safety training.

- **Regulatory Compliance:** Adhere to national and international safety standards and regulations related to hydrogen handling and storage.
- **Quality Assurance:** Maintain strict quality control measures to ensure the purity and consistency of hydrogen produced.
- **Monitoring and Testing:** Regularly monitor hydrogen quality and conduct testing to identify any impurities or degradation.
- **Environmental Sustainability:**
 - **Renewable Energy Sources:** Whenever possible, use renewable energy sources (solar, wind) for hydrogen production to reduce carbon emissions.
 - **Carbon Capture and Utilization:** Explore carbon capture and utilization technologies to minimize the environmental impact of hydrogen production.
- **Market Development and Promotion:**
 - **Market Research:** Conduct market research to identify potential customers, industries, and regions with high demand for hydrogen.
 - **Marketing Strategy:** Develop a comprehensive marketing strategy to promote hydrogen as a clean energy source to potential users and investors.
- **Partnerships:** Collaborate with industry players, government agencies, and research institutions to expand market reach and credibility.
- **Licensing and Permits:** Obtain all necessary licenses, permits, and approvals for hydrogen production, storage, and distribution.
- **Environmental Clearances:** Ensure compliance with environmental regulations and obtain clearances for projects with potential environmental impacts.
- **Cost Management:** Implement efficient cost management practices to optimize production and operation expenses.
- **Financing:** Secure appropriate financing options, such as loans, grants, or partnerships, to fund the project.
- **Safety Communication:** Public Awareness: Educate the public about the safety and benefits of hydrogen as an energy source through awareness campaigns.
- **Emergency Response Plans:** Develop comprehensive emergency response plans and communicate them to relevant authorities and communities.
- **Continuous Improvement:** Research and Development: Invest in research and development to improve hydrogen production efficiency, storage technologies, and safety measures.
- **Feedback Loop:** Establish a feedback loop for continuous improvement by learning from operational experiences and customer feedback.
- **Scaling Up:** Develop strategies for scaling up hydrogen production and distribution in response to increasing demand.
- **Risk Management:** Identify potential risks, including market volatility, supply chain disruptions, and safety concerns, and develop risk mitigation strategies.
- **International Standards:** Stay informed about international standards and best practices related to hydrogen production, storage, and distribution.

3.2.3 O&M Related Cost for The System

Efficient O&M practices are essential for maintaining the reliability, safety, and cost-effectiveness of a hydrogen production system. A well-designed O&M strategy, which may include supplier or manufacturer maintenance contracts, helps prevent downtime, prolongs equipment life, and ensures the ongoing production of high-quality hydrogen.

Thus, while considering the operation and maintenance (O&M) costs for a hydrogen production system, it's essential to account for various components and explore options for managing these costs efficiently such as:

- **Routine Maintenance:** Routine maintenance activities involve regular inspections, cleaning, and minor repairs to ensure the efficient operation of the hydrogen production system. These tasks typically include checking for leaks, replacing worn components, and verifying the condition of the electrolyzer.
- **Preventive Maintenance:** Preventive maintenance focuses on scheduled inspections and servicing to prevent potential issues or breakdowns. This can include replacing consumables, updating software, and conducting equipment calibrations.
- **Equipment Replacement:** Over time, some components of the hydrogen production system may wear out or become less efficient. Budget for the periodic replacement of critical components, such as membranes in the electrolyzer or other high-wear parts.
- **Supplier/Manufacturer Maintenance Contracts:** Explore the possibility of entering into Annual Maintenance Contracts (AMCs) with equipment suppliers or manufacturers, particularly for the electrolyzer. These contracts can provide peace of mind by ensuring timely maintenance, spare parts availability, and technical support.
- **Spare Parts Inventory:** Maintain a well-stocked inventory of essential spare parts to minimize downtime in case of unexpected equipment failures. Spare parts should be readily available to support rapid repairs.
- **Labor Costs:** Include labour costs associated with O&M activities, such as salaries or wages for maintenance personnel and technicians. Consider whether specialized training is needed for personnel handling hydrogen equipment.
- **Monitoring and Control Systems:** Invest in robust monitoring and control systems that can detect issues early, optimize system performance, and provide data for predictive maintenance. Such systems can help reduce maintenance costs by identifying problems before they escalate.
- **Hydrogen Quality Assurance:** Implement quality control measures to ensure the hydrogen produced meets the required purity standards. This may involve periodic testing and calibration of sensors and analyzers.
- **Asset Management:** Establish an asset management system to track the condition, maintenance history, and replacement schedules of equipment. This ensures a proactive approach to O&M.

Annual Maintenance Contract (AMC)

It is a service agreement or contract between a service provider (such as a manufacturer, supplier, or service company) and a customer. The purpose of an AMC is to ensure the ongoing maintenance, repair, and support of equipment, machinery, or systems over a specific duration, typically one year.

Under an AMC, the service provider commits to providing regular maintenance, inspections, and repair services for the covered equipment during the contract period. These services are often pre-scheduled and may include tasks like preventive maintenance, routine inspections, software updates, repairs, and access to spare parts.

AMCs are commonly used in various industries, including manufacturing, healthcare, information technology, and facility management. They offer several advantages, including proactive maintenance to prevent breakdowns, timely repairs to minimize downtime, access to expert technicians, and extended equipment lifespan.

The terms and conditions of an AMC, including the scope of services, cost, duration, response times, and service-level agreements (SLAs), are typically outlined in a formal contract. Customers pay a fee for the contract, which varies based on the type and complexity of the equipment covered and the level of service provided.

UNIT 3.3: Taxation and Related Compliances

Unit Objectives

At the end of this unit, you will be able to:

1. Explain different taxation and related compliances to setting up and operations
2. To understand the basic concepts and types of taxes applicable to individuals and organizations.

3.3.1 Taxation and Related Compliances in India

Setting up and operating a green hydrogen plant in India involves various taxation and related compliances that businesses need to adhere to. Following are the various taxation relevant to green hydrogen projects:

- **Goods and Services Tax (GST):**
 - GST is applicable to the supply of goods and services in India, including equipment, materials, and services related to the construction and operation of a green hydrogen plant.
 - Businesses need to register for GST, file periodic GST returns, and pay applicable GST on their transactions.
- **Income Tax:**
 - Companies operating green hydrogen plants are subject to corporate income tax in India. The current corporate tax rate varies depending on the type of business entity and annual turnover.
 - Businesses need to file annual income tax returns and comply with transfer pricing regulations for international transactions.
- **Customs Duties and Import Taxes:**
 - Importing equipment, machinery, and components for the green hydrogen plant may incur customs duties and import taxes.
 - Businesses should understand the customs duties applicable to specific imports and comply with customs regulations.
- **State-Specific Taxes:**
 - Different states in India may have their own state-level taxes, such as state GST (SGST) and state-specific incentives or concessions.
 - Businesses should be aware of state-level tax implications and incentives when selecting the plant's location.
- **Pricing Regulations:** For businesses involved in international transactions, transfer pricing regulations must be followed to ensure that transfer prices for goods and services are at arm's length.

Always engage tax experts and legal advisors to plan and ensure compliance with all relevant tax laws and regulations, including exemptions and deductions.

Following are the various compliances relevant to green hydrogen projects:

- **Environmental Compliance:** Green hydrogen plants must comply with environmental regulations and obtain necessary clearances, which may involve environmental impact assessments and mitigation measures.
- **Renewable Energy Certificates (RECs):** Green hydrogen production using renewable energy sources may qualify for RECs, which can be traded and used to demonstrate the use of renewable energy.
- **Labor Laws and Compliance:** Compliance with labor laws, including employment contracts, wages, and social security contributions, is essential for employee management.
- **Environmental Impact Reporting:** Ongoing monitoring and reporting of the environmental impact of hydrogen production and related activities may be required to meet regulatory requirements.
- **Financial Reporting and Compliance:** Proper financial accounting and reporting practices, including compliance with the Companies Act, are necessary for financial transparency and legal compliance.
- **Permitting and Licensing:** Obtain the necessary permits and licenses, including those related to safety, environmental impact, and local regulations, before commencing plant operations.
- **Corporate Social Responsibility (CSR):** Large companies may have CSR obligations and reporting requirements under the Companies Act, which should be fulfilled.
- **Export and Import Regulations:** If hydrogen or related products are intended for export, businesses should comply with export regulations, and if any imports are involved, import regulations must be followed.

In addition to the taxation and compliances, the Indian government and state governments often offer incentives and subsidies to promote renewable energy projects, including green hydrogen. Businesses should explore available schemes and ensure compliance with eligibility criteria.

UNIT 3.4: Policies and Corresponding Incentives/Subsidies

Unit Objectives



At the end of this unit, you will be able to:

1. Discuss National Green Hydrogen policy and corresponding incentives/subsidies available and procedures for availing the same
2. Explain how evolving start-ups and new ventures boost-up hydrogen economy.

3.4.1 Nation Green Hydrogen

The approval of the National Green Hydrogen Mission by the Union Cabinet on January 4, 2023, marks a significant milestone in India's journey toward sustainable and clean energy solutions. With an allocated budget of ₹19,744 crore spanning from fiscal year 2023-24 to 2029-30, this mission underscores the government's commitment to advancing green hydrogen production and utilization. The mission is expected to provide a structured framework encompassing various aspects, including the promotion of renewable energy sources for hydrogen generation, research and development initiatives, infrastructure development, incentives, and regulatory support. This substantial investment reflects India's determination to play a pivotal role in the global transition to green hydrogen as a clean and sustainable energy carrier, contributing to a greener and more energy-independent future.

In February 2022, Ministry of Power announced the Green Hydrogen/Green Ammonia Policy, marking a significant milestone towards India's climate objectives and facilitating the successful execution of the National Hydrogen Mission. This policy also sets out ambitions to establish India as a leading export hub for Green Hydrogen and Green Ammonia. Salient features of the policy are listed below

- Power exchange and developing renewable energy capacity: Manufacturers of Green Hydrogen/Ammonia may purchase renewable power from the power exchange or establish renewable energy capacity themselves or through any other developer, anywhere.
- Open access: Open access will be given within 15 days of receipt of application.
- Unconsumed renewable power: The Green Hydrogen / Ammonia manufacturers can bank their unconsumed renewable power, up to 30 days, with a distribution company and take it back when required.
- Distribution licensees: Distribution licensees can also purchase and supply Renewable Energy to the manufacturers of Green Hydrogen / Green Ammonia in their States at concessional prices which will only include the cost of procurement, wheeling charges and a small margin as determined by the State Commission.
- Waiver of inter-state transmission charges: The manufacturers of Green Hydrogen and Green Ammonia will be allowed a waiver of inter-state transmission charges for a period of 25 years for the projects commissioned before 30th June 2025.

Note that this waiver has been extended to all the projects which are commissioned before 31st Dec 2030.

- **Connectivity to the grid:** To avoid any procedural delays, the manufacturers of Green Hydrogen / Ammonia and the renewable energy plant will be given connectivity to the grid on priority basis.
- **Renewable Purchase Obligation (RPO):** The benefit of Renewable Purchase Obligation (RPO) will be given to the hydrogen/Ammonia manufacturer and the Distribution licensee for consumption of renewable power.
- **A single portal:** To ensure ease of doing business, the MNRE will establish a single portal to carry out all the activities including statutory clearances in a time bound manner.
- **Connectivity:** Connectivity (at the generation end and the Green Hydrogen / Green Ammonia manufacturing end), to the ISTS for Renewable Energy capacity set up for the purpose of manufacturing Green Hydrogen/Green Ammonia will be granted on priority.
- **Storage of green hydrogen/green ammonia:** Green Hydrogen/Green Ammonia manufacturers will be permitted to set up bunkers near Ports for storage of Green Ammonia for export/use by shipping. Moreover, the land for the storage will be provided by the respective Port Authorities at applicable charges.

Now, a number of states like Maharashtra, Odisha, Karnataka, UP etc are following suit, offering incentives on the production of the green gas and renewable energy which is needed to make hydrogen 'green'; subsidies on transmission within their region; SGST reimbursement; government land allotments; and so on. Whatever the states offer in terms of subsidies or incentives under their own green hydrogen policies will be in addition to the schemes rolled out by the central government.

Further here are some key aspects relevant for setting up Green Hydrogen projects

Subsidies for Renewable Energy: Green hydrogen projects that use renewable energy sources like solar and wind power for hydrogen production may be eligible for subsidies and incentives provided by the Ministry of New and Renewable Energy (MNRE) and state-level renewable energy agencies. These subsidies can help offset the cost of renewable energy installations and hydrogen production equipment.

- **Procedure:** Interested parties typically need to apply to the relevant government agency, such as the MNRE or the state's nodal agency for renewable energy. The application process involves submitting project proposals, feasibility studies, and compliance with eligibility criteria.
- **Financial Assistance from Public Financial Institutions:** Public financial institutions, such as the Indian Renewable Energy Development Agency (IREDA), may offer low-interest loans, credit lines, or financial assistance to green hydrogen projects as part of their renewable energy financing programs.
 - **Procedure:** Interested project developers can approach these institutions directly to explore financing options. The application process typically involves submitting project details, financial projections, and compliance with lending criteria.
- **Innovative Financing:** Explore innovative financing mechanisms, including partnerships with private investors, venture capital firms, and international organizations, to secure funding for green hydrogen projects.

- **Procedure:** Developing partnerships and securing financing often involves negotiations and agreements between project developers and investors. Legal and financial advisors can assist in this process.
- **Government Grants:** Some government programs and schemes may offer grants to support the development and deployment of green hydrogen technologies.
 - Procedure: To access government grants, project developers typically need to apply through a competitive application process, which may involve submitting project proposals, budgets, and compliance with specific grant requirements.
- **Renewable Energy Certificates (RECs):** Green hydrogen production using renewable energy sources may qualify for RECs, which can be traded and used to demonstrate the use of renewable energy.
 - Procedure: Project developers can participate in REC trading platforms and follow the standard procedures for trading and certification.

It is important to note that the availability of incentives and subsidies may change over time, and new schemes may be introduced by the government. Therefore, it is advisable to stay updated with the latest government announcements, consult with relevant government agencies and financial experts, and carefully follow the specific procedures outlined for each incentive or subsidy program. Additionally, the recent introduction of the National Green Hydrogen Mission may bring new opportunities for incentives and subsidies related to green hydrogen projects in India.

3.4.2 Evolving Start-Ups and New Ventures

Start-ups and emerging businesses play a vital role in driving the growth of the hydrogen economy in India. These dynamic and innovative companies bring fresh ideas and energy to the world of hydrogen. They focus on finding smarter ways to produce, store, and use hydrogen, making it a cleaner and more practical energy source compared to traditional options like coal or oil.

New ventures and start-ups play a pivotal role in the green hydrogen economy by diversifying markets and applications, fostering an entrepreneurial spirit, creating jobs, attracting investments, promoting research and collaboration, raising market awareness, and expanding their global reach. Their ability to identify niche markets strengthens the green hydrogen economy, reducing reliance on a single industry. Their entrepreneurial spirit drives creativity and innovation, leading to the discovery of groundbreaking solutions. As they grow, they generate employment opportunities, contribute to economic growth, and attract talent. Investors are increasingly drawn to their sustainable initiatives, fuelling growth. Collaboration with research institutions accelerates technological advancements, while active promotion and media attention raise awareness about green hydrogen's potential. Their global perspective contributes to the worldwide expansion of the green hydrogen economy, making them indispensable catalysts for change and sustainability.



4. Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production



UNIT 4.1 Types of Electrolyzers, Their Technology Maturity and Technical Specifications

UNIT 4.2 Tools and Equipment for Installation

UNIT 4.3 Inputs/Outputs of an Electrolyzer System and Calculate Losses and Equipment Efficiency

UNIT 4.4 Input Renewable Power from various sources and its Integration with Electrolyzer

UNIT 4.5 Assembly/Installation of Parts and Components of Electrolyzer

UNIT 4.6 Challenges Associated with Hydrogen in Storage, Handling and Transportation

UNIT 4.7 Hydrogen Compression System

UNIT 4.8: O&M Requirements for Electrolyzer

Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the Electrolyser Type (PEM, AE, SOEC)
2. Explain the technical specification of Electrolyser (PEM, AE and SOEC)
3. Explain the Major components of Electrolyser
4. Discuss the Operation of PEM and AE and SOEC electrolyser.
5. Discuss Comparison between above electrolyzers with reference to the life, cost, efficiency, electricity consumption etc.
6. Discuss the Tools required for Installation, dismantling, removal of components of PEM and AE /Alkaline water electrolyzers.
7. Explain the inputs / outputs of Electrolyzer
8. Discuss in detail the input renewable power from various sources and its integration with electrolyser.
9. Discuss how to oversee the Installation of Parts and Components of Electrolyser
10. Discuss step by step process for Installation of Electrolyser
11. Explain the Challenges associated with Hydrogen in storage, handling and transportation
12. Discuss how to select and install hydrogen compression and storage system
13. Discuss the precautions required to compress and store hydrogen
14. Discuss the safety guidelines to be followed as per industry standard

UNIT 4.1: Types of Electrolyzers, their technology maturity and Technical Specifications

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the Electrolyser Type (PEM, AE, SOEC)
2. Explain the technical specification of Electrolyser (PEM, AE and SOEC)
3. Explain the Major components of Electrolyser
4. Discuss the Operation of PEM and AE and SOEC electrolyser.
5. Discuss Comparison between above electrolyzers with reference to the life, cost, efficiency, electricity consumption etc.

4.1.1. Types and Technological Status of Electrolyzers

We have already learnt about the types of electrolyzers in previous modules (module 1 and 2). In this module we will learn about the technological status of electrolyzers.

Each technology has its own challenges, from critical materials to performance, durability and maturity; there is no clear winner across all applications, which leaves the door open for competition and innovation driving costs down.

Every technology has its own challenges such as performance, materials needed, durability, and how developed they are. Till date there is no one technology that can solve all the purpose or we can say is best for everything. Thus, this encourages competition and innovation and resulting cost reduction.

Now, let us learn about the technology status of the following types of electrolyzers.

- Alkaline electrolyzer
- Polymer Electrolyte Membrane (PEM) electrolyzer
- Solid oxide electrolyzers (SOEC)
- Anion Exchange Membranes (AEM) electrolyzer

Alkaline Electrolyzer

Alkaline Electrolyzer technology is characterized by its easy stack and system design that facilitates relatively easy manufacturing. Currently, they are equipped with electrode areas upto 3m². They operate with high concentrate KOH (typically 57 moles of solute per litre of solution, mol/litre) as electrolyte using robust ZrO₂ based diaphragms and nickel coated stainless-steel for the electrodes.

In these electrolyzers, The hydroxyl ion OH^- acts as the ionic charge carrier. KOH and water move through the diaphragm's porous structure. This permeation enables the diaphragm to facilitate the electrochemical reaction and allows the intermixing of the hydrogen (H_2) and oxygen (O_2) gases that are dissolved in the electrolyte, limiting lower power-operating range and the ability to operate at higher pressure levels.

To prevent this, thicker (0.252 mm) diaphragms are used, but this creates a higher resistance and lower efficiencies. Spacers are sometimes included by some manufacturers between electrodes and diaphragms to further avoid the intermixing of gases. These thick diaphragms and added spacers result into high Ohmic resistances across the two electrodes which drastically reduces the current density at a given voltage. The advanced designs use zero gap electrodes, thinner diaphragms and different electro-catalysts concepts to increase current density have improved their performance in comparison to PEM technology. Alkaline electrolyzers are classic, sturdy, very reliable with a long life.

Polymer Electrolyte Membrane (PEM) Electrolyzer

These PEM electrolyzers use a thin (0.2 mm) Perfluoro Sulfonic Acid (PFSA) membrane which is characterized by high mechanical stability, excellent chemical inertness, good thermal stability, and high proton conductivity. The electrodes with advanced architecture reduce resistance (between anode and cathode) that results in higher efficiencies. The PFSA membrane is also chemically and mechanically robust, which allows for high pressure differentials. Thus, the PEM cells can operate up to 70 bars while the oxygen side at atmospheric pressure.

The acidic environment provided by the PFSA membrane, high voltages, and oxygen evolution at the anode creates a harsh oxidative environment demanding the use of materials that can withstand these conditions. Titanium-based materials, noble metal catalysts and protective coatings are necessary, not only to provide long-term stability to cell components, but also to provide optimal electron conductivity and cell efficiency. These requirements have caused PEM stacks to be more expensive than alkaline electrolyzers. PEMs have one of the most compact but simplest system designs. However, they are sensitive to water impurities due to which iron, copper, chromium and sodium can suffer from calcination (heating). The electrode areas are around 2000 cm^2 which will increase further to suit MW size single stack units. The reliability and lifetime characteristics of MW scale PEM stacks are yet to be validated.

Solid oxide electrolyzers (SOEC)

The operation at high temperatures results in following advantages:

- Using cost-effective nickel electrodes
- Decreasing electricity requirements
- Utilizing heat for separation, potentially resulting in apparent efficiencies surpassing 100% based on electricity;
- Exploring reversibility as it functions both as a fuel cell and electrolyzer;
- Coelectrolyzing CO_2 and water to generate syngas (a basic building block for the chemical industry)

However, there are also some disadvantages, as given below:

- Thermo-chemical cycling, particularly during shutdown or ramping periods, accelerates degradation and reduces lifespans.
 - Stack degradation encompasses:
 - Challenges with respect to sealing under increased differential pressure
 - Electrode contamination due to silica which is used as sealant
 - Additional contaminant sources originating from piping, interconnects, and sealing.
- These are presently used at the kW-scale, although some current demonstration projects have already reached 1 MW.

Anion Exchange Membranes (AEM) electrolyzer

The potential of AEM electrolyzer lies in the combination of a less harsh environment from alkaline electrolyzers with the simplicity and efficiency. Under differential pressure, it allows the use of non-noble catalysts and titanium-free components.

However, the AEM membrane has chemical and mechanical stability problems that leads to unstable lifetime profiles. Moreover, its performance is not yet as good as expected, mostly due to low AEM conductivity, poor electrode architectures and slow catalyst kinetics.

The enhanced performance can only be achieved by tuning membrane conductivity properties, or by adding a supporting electrolyte (e.g. KOH, or sodium bicarbonate NaHCO_3). Such tuning could lead to decreased durability. The OH^- ion is intrinsically three-fold slower (lower conductivity) than protons (H^+ ions) within PEM, which forces AEM developers to make either thinner membranes, or ones with higher charge density. This is the latest technology with only a few companies commercialising it, with limited deployment.

Activity



- Illustrate different types of electrolyzers and their key specifications and technical parameters through Pictures, videos, product data sheet etc.
- Product data sheet of PEM electrolyzer is available in Annexure.
 - Link: file:///C:/Users/Lenovo/Desktop/0070332.pdf
- Explore your resources (e.g. internet) and prepare a product data sheet of other types of electrolyzer.
- Compare technical data/features of PEM electrolyzer with other types of electrolyzers.
- Outline differences in PEM, AE, AEM and SOEC Electrolyzers and illustrate their schematics.
- Dismantle different types of electrolyzers in your Lab./workshop and study.
- Make a tabular statement of the differences you observe in electrolyzers. (For example, the alkaline electrolyzer uses liquid electrolyte whereas solid oxide electrolyzer uses solid electrolyte).

4.1.1 Key Technical Specifications of various Types of Electrolyzer

The Four main electrolyzers; PEM, AE AEM, SOEC are at different stages of their development and commercialization. The following table represents the technical specifications of each electrolyzer.

	Alkaline	PEM	AEM	Solid Oxide
Operating temperature	70-90°C	50-80 °C	40-60 °C	700-850 °C
Operating pressure	1-30 bar	< 70 bar	< 35 bar	1 bar
Electrolyte	Potassium hydroxide (KOH) 5-7 mol/L	PFSA membranes	DVB polymer support with KOH or NaHCO ₃ 1-2 mol/L	Yttria-stabilized Zirconia (YSZ)
Separator	ZrO ₂ stabilized with PPS mesh	Solid electrolyte (above)	Solid electrolyte (above)	Solid electrolyte (above)
Electrode / catalyst (oxygen side)	Nickel coated perforated stainless steel	Iridium oxide	High surface area Nickel or NiFeCo alloys	Perovskite-type (e.g. LSCF, LSM)
Electrode / catalyst (hydrogen side)	Nickel coated perforated stainless steel	Platinum nanoparticles on carbon black	High surface area nickel	Ni/YSZ
Porous transport layer anode	Nickel mesh (not always present)	Platinum coated sintered porous titanium	Nickel foam	Coarse Nickel-mesh or foam
Porous transport layer cathode	Nickel mesh	Sintered porous titanium or carbon cloth	Nickel foam or carbon Cloth	None
Bipolar plate anode	Nickel-coated stainless steel	Platinum-coated titanium	Nickel-coated stainless steel	None
Bipolar plate cathode	Nickel-coated stainless steel	Gold-coated titanium	Nickel-coated Stainless steel	Cobalt-coated stainless steel
Frames and sealing	PSU, PTFE, EPDM	PTFE, PSU, ETFE	PTFE	Ceramic glass

Note: Coloured cells represent conditions or components with significant variation among different companies. PFSA = Perfluoroacidsulfonic; PTFE = Polytetrafluoroethylene; ETFE = Ethylene Tetrafluoroethylene; PSF = poly (bisphenol-A sulfone); PSU = Polysulfone; YSZ = yttria-stabilized zirconia; DVB = divinylbenzene; PPS = Polyphenylene sulphide; LSCF = $\text{La}_{0.58}\text{Sr}_{0.4}\text{Co}_{0.2}\text{Fe}_{0.8}\text{O}_{3-\delta}$; LSM = $(\text{La}_{1-x}\text{Sr}_x)_1\text{MnO}_3$; § = Crofer22APU with co-containing protective coating.

Based on IRENA analysis.

Table 4.1.1: Technical specifications of the electrolyzers

Source: IRENA. 2020. Green hydrogen cost reduction

Sizing of Electrolyzers

The requirement of hydrogen needed may be a starting point. Let us understand sizing of electrolyzers with the help of an example. We know that:

- 1 kg of hydrogen is equivalent to 500 moles.
- 1 mole of H_2 is equal to ~ 22.4 liter under standard temperature and pressure conditions (STP).

An electrolyzer will require 2×96485 Ampere-second or 192970 Coulombs of charge (assuming 100% efficiency), and approximately 2.2 Volt (assuming overpotential of ~1 V for splitting water) to release 1 mole (2 grams of hydrogen).

Here, 96485 is the Faraday's constant ($F = 96485 \text{ C/mol}$) and 2 is the number of electrons required for reduction of hydrogen.

Once the requirement of hydrogen is decided, the need of inputs, mainly land, renewable energy and water can be estimated.

It would be necessary to understand the specifications of a commercially available (i.e. standard capacities/ratings) electrolyzers for economy and also proven technology. Calculate the number of electrolyzers required.

Note:

- **Alkaline:** MW range commercially available. Its technology and performance are well proven.
- **PEM:** Commercially available in kW range. MW capacity is likely to be commercialized in near future.
- **Solid oxide:** It is also available mostly in kW range. Its economy becomes more viable if waste heat of some other industrial process is utilized by the electrolyzer.
- **Anion Exchange electrolyzer:** Its technology has yet to be commercialized.

Activity



- Outline differences in PEM, AE, AEM and SOEC Electrolyzer and illustrate their schematics.

4.1.2 Comparing Various Electrolyzers

The following table list the comparisons between Alkaline: MW and PEM electrolyzers.

Alkaline: MW	PEM
- It is the most proven, cheap, and reliable in operations with a life of over 3 decades. - Alkaline electrolyzers are classic and sturdy. They are very reliable with a long life of above 30 years.	- As compared to alkaline electrolyzers, PEM electrolyzers are new and yet to be proven for larger capacity of MW range. - Further there are some drawbacks which are yet to be resolved e.g. PEM stacks are costlier than alkaline electrolyzers. - PEM is sensitive to water impurities such as iron, copper, chromium and sodium and can suffer from calcination.

Table 4.1.2: Comparisons Between Alkaline: MW and PEM Electrolyzers

The following figure further represents the comparisons of parameters between Alkaline and PEM electrolyzers.

Table 4.3.1

Electrolyzers			
S.	Description	Alkaline	PEM
1	Electrolyte	30% KOH+ DM water	Solid Polymer Membrane +DMW
2	H ₂ purity	99.98%	99.99%
3	Cathode	Ni-based	Platinum nanoparticles on carbon
4	Anode	Ni-based	Ti mesh
5	Diaphragm	Zirfon /ZrO ₂	Nafion (DuPont make)
6	Efficiency	63-70%	56-60%
7	Operating pressure	0.1-30 bar	0.1-30 bar
8	Operating temperature	80-90 °C	50-80 °C
9	Technical Minimum Load	20%	10%
10	Ramp Rate	10% per min	15% per second
11	Footprint without accessories	140 m ² / MW	110 m ² / MW
12	DC Current density (A/cm ²)	≤ 0.5	1.0 ≤ 2.0
13	Size, weight	3.5 ton/MW	1/3 rd – 2/3 rd of Alkaline type
14	Capex ₹Cr/ MW	5.0	6.9
15	Opex ₹Cr/MW/20 yr	10	7
16	Life of Electrolyser Stack	≤ 80,000 Hrs	40,000 - 80,000 Hrs
17	Cold Startup time	~ 30 min	≤ 10 min

Fig 4.1.1. Comparison of parameters of Alkaline and PEM electrolyzers

Source: Panda. DMR. 2023. Key component, Technology selection. Training program on Techno-economic Considerations for the Design of Green Hydrogen Projects, New Delhi.

The following table represents the comparison of technical performance parameters of Alkaline, PEM, and solid oxide electrolyzers.

Parameter	PEM	ALK	SOEC
Technological Readiness (TRL)	9	10	7
Lowest operating power* (%)	5-10	10-15	10 - 30
Space footprint (m ² /MW)	10 - 15	25 - 40	70 - 85
Energy requirement (kWh/kg)	49 - 55	55 - 60	37 - 48
Stack sizes (MW)	1 - 10	1 - 20	1 - 10
Startup time (min)	30 sec warm / < 5 min cold	-	> 2 hours
Output pressure (bars)	30 - 40	Upto 200	
Stack lifetime** (hours, years)	80000 h, 10 years	-	> 5 years
Operating temperature (°C)	60 - 80	50 - 80	700
Ramp rates (%/min)	> 20% /Min	> 20% / Min	-

Table 4.1.3 Comparison of technical performance parameters of electrolyzers

Source: USAID. 2023. System design of a green hydrogen plant. Training program on Techno-economic Considerations for the Design of Green Hydrogen Projects, New Delhi.

Note: We have already about major component of electrolyzers in Module 2.

4.1.4 Operating principles of PEM, AE and SOEC

Operating principle is the ionisation of electrolyte and separation of hydrogen by DC electric power. We have already learnt about the operating principles lets recap those with the help of following points:

Proton Exchange Membrane (PEM) Electrolyzer:

- Utilizes a solid polymer electrolyte membrane (PEM).
- Requires pure water and operates at low temperatures.
- Hydrogen and oxygen form at the cathode and anode respectively.
- Fast response, suitable for intermittent power sources.
- High efficiency but sensitive to impurities.

Alkaline Electrolyzer (AE):

- Uses a liquid electrolyte, often potassium hydroxide (KOH).
- Requires a conductive separator between electrodes.
- Operates at higher temperatures compared to PEM.
- Hydroxide ions migrate, forming hydrogen at the cathode and oxygen at the anode.
- Robust but limited flexibility in operation.

Solid Oxide Electrolysis Cell (SOEC):

- Operates at high temperatures, typically above 600°C.
- Employs solid oxide electrolyte that conducts oxygen ions.

- Oxygen ions migrate from the cathode to anode, where they react with hydrogen to form water.
- Requires external heat, usually from high-temperature sources.
- Well-suited for integration with high-temperature industrial processes.

VIDEO

Topic: PEM Electrolyzer
By: HySA Infrastructure (2016)
Link: <https://www.youtube.com/watch?v=IX93qZbaiuo>
Duration: 1 m 52 sec

The basic technical specifications of electrolyzers of each cell are as follows:

- Cell Type: Specify the type of electrolyzer cell, e.g., PEM, Alkaline, SOEC.
- Cell Voltage: The voltage required for the cell's electrochemical reaction.
- Current Density: The current flowing through the cell per unit area.
- Operating Temperature: The temperature range at which the cell operates efficiently.
- Electrolyte: Mention the electrolyte used in the cell (e.g., KOH, PEM).
- Electrode Materials: Detail the materials used for the anode and cathode.
- Efficiency: The efficiency of converting electrical energy into hydrogen production.
- Pressure Range: The operating pressure range of the cell.
- Hydrogen Production Rate: Amount of hydrogen produced per unit time.
- Cell Dimensions: Physical dimensions of the cell, including length, width, and thickness.
- Stack Configuration: Specify if the cell is designed for standalone use or within a stack.
- Lifetime: The expected operational lifespan of the cell.
- Catalyst: Mention any catalyst used to facilitate the electrochemical reaction.
- Gases Involved: Indicate the gases (hydrogen, oxygen, etc.) involved in the reaction.
- Safety Measures: Outline any safety features or precautions incorporated in the cell design.

UNIT 4.2: Tools and Equipment for Installation

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the Tools required for Installation, dismantling, removal of components of PEM and AE /Alkaline water electrolyzers.
2. Understand safety guidelines and applicable standards for proper execution of work.

4.2.1. Installation of Electrolyzers

To install electrolyzer, you should have to following:

Tools/Tackles	Description
Drawings from Suppliers	The drawings from Supplier includes the following: 1. Single line/block diagram, 2. General layout of the plant. Observe the orientation of each equipment. Check civil drawings for their matching with general layout drawings. 3. Layout of piping and electrical cables 4. Individual assembly diagrams 5. Drawings of fasteners, gaskets, seals, 'O' rings etc. 6. Civil foundation and embedment drawings 7. Earthing pits (electrical and electronic) etc.
Installation Manual from Supplier	The installation manual from supplier include the specific instructions, drawing, safety standards etc.
General tools	General installation tools such which are easily available in market such as spanners, screw drivers etc.
Special tools & tackles	Listed in drawings/manuals along with the instruction for their use. Generally, in the supplier's scope of supply.

Table 4.2.1: Tools and Tackles

As long as hydrogen is not filled or produced, the tools required are similar to other works. The special tools are necessary when work has to be done in the hydrogen environment. The equipment supplier provides all information, manuals, special tools and instructions on their use. If required, the plant contract may include training (limited to ordered plant) of installations, operation, and maintenance of customer's technicians.

Here, the purpose is to give general idea of the activities involved, skills required etc. However, as per the contract obligations concerned to warranty/guarantee clauses, actual work will have to be strictly executed as specified in plant supplier's manuals which may have variations from plant to plant and supplier to supplier.

In installation of hydrogen production plant, workers such as technicians should be able to identify each component, its fasteners (nuts, bolts, washers etc.), quantity as per bill of materials, and individual assembly drawings which are quite simple and could be learnt from seniors. Major and most important work is to tally bill of materials and materials received, their condition and how to store them as specified, e.g. temperature, humidity, shelf life etc.

Sealing of various joints is very important. The sealing rings, gaskets and other sealing materials supplied with the equipment are to be used, as required. They are to be handled carefully to avoid looseness or cracking due to excessive tightening. Gaskets, sealing rings etc. must be checked to ensure that they are in proper shape and free from cracks, folds, cuts etc. These should not come in contact with dust, particles etc.

Procedure to tight electrolyzer cover: As specified by the supplier. Generally, the nuts and bolts are tightened in specified sequence e.g. diagonally opposite in 3- 4 rounds, not to be tightened fully in the first round.

The pipe fittings will require appropriate tapes, e.g. Teflon tape for 100% leak proof sealing.

During installation of hydrogen production plant, the special tools required will be supplied by the equipment supplier along with the instruction manual for their use. Generally, torque wrenches are used to ensure accurate and uniform tightening.

Initially, the sealing will be tested by pressurised helium or mixture of 5% hydrogen and 95% nitrogen for specified duration. It may be 50 to 100% higher than the actual hydrogen pressure developed during the operation of plant.

It is to be remembered that development of any fault or wrong operations always results in the rise in temperature, noise, or vibrations. During daily monitoring, a remote temperature sensing device (most common these days), portable vibrations meter, special wooden sticks can be used to detect change in these basic things.

4.2.3 Technical data sheet

Following figure represents the technical data sheet of solid oxide electrolyzer of M/s Boom Energy, USA.

Sheet 1 of 2

Bloom Electrolyzer™

39.6 kWh/kg

The world's most efficient hydrogen electrolyzer.

The Bloom Electrolyzer utilizes solid oxide technology's high efficiency to produce green hydrogen, leveraging decades of solid oxide experience.

- **Robust and established supply chain base with over 2GW of electrolyzer production capacity**
- **Modular design that allows for continuous operations and concurrent maintenance**
- **No oxygen in the hydrogen stream** (eliminates the need for deoxygenation units)



Efficient. Robust. Economical.

- 

High temperature electrolysis can split steam molecules with less energy than low-temperature electrolysis.
- 

Bloom has over 1GW of solid oxide technology already deployed. The Bloom Electrolyzer uses the same base platform and all the learnings from the past decade.
- 

The Bloom Electrolyzer produces cost-effective hydrogen by virtue of its efficiency, modularity and low reliance on rare earth metals.

Activa

Specifications (sheet 2 of 2)

Efficiency		Modular Bloom Electrolyzer Key Data			
Stack	39.6 kWh/kg	H ₂ output (mt*/day)	H ₂ output (kg/hr)	H ₂ output (Nm ³ /hr)	Power consumption
System	41.9 kWh/kg	1.0	43	478	1.8MW
Warm Start-up Time		4.1	172	1914	7.2MW
	~ 10 minutes	6.2	258	2870	10.8MW
H ₂ Output		14.4	602	6697	25.2MW
Pressure	Atmospheric	28.9	1203	13394	50MW
Temperature	100-180°C	57.8	2407	26789	100MW
Composition	85% H ₂ , 15% H ₂ O mol	115.5	4814	53577	200MW
Steam Input		300	12550	139684	525MW
Pressure	4.1-5.2 bar(g)	573.5**	23987	265973	1GW
Temperature	150-200°C				
Volume	10.5L of H ₂ O/kg of H ₂				
Electrical Input					
Voltage	800VDC				
H ₂ Input (For Start-up)					
Purity	99.9%				

*mt = metric tonnes
 **No cap on the size of the system in increments of 1.8MW
 + Specifications subject to change

Fig. 4..2.1 Technical data sheet of solid oxide electrolyzers of M/s Bloom Energy, USA

Source: Data sheet: Electrolyzers, Bloom Energy. 2023

file:///C:/Users/Lenovo/Desktop/Manuals/DataSheet_Electrolyzer.pdf

Activity



- Show how to identify the tools and equipment to perform the installation of electrolyzer as per concerned technical sheets and relevant safety and technical standards
- Study and understand following manual and drawings of electrolyzer:
 - Installation manual
 - Installation drawings of the electrolyzer
- Write down installation sequence and list out the special tools required.
- Fill-up the following Table.

Activity	Drawing/manual reference	Specifications of the tool required (size, range, digital etc.)	Remarks/your observations etc.

VIDEO

Topic: Gigawatt green hydrogen plant: The advanced design
By: Institute for Sustainable Process Technology (2022)
Link: <https://www.youtube.com/watch?v=fw5zPUbPo1Q>
Duration: 4 m 43 sec

UNIT 4.3: Inputs/Outputs of an Electrolyzer System and Calculate Losses and Equipment Efficiency

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the key material and energy inputs and outputs of an electrolyzer system.
2. Explain the major sources of losses in electrolyzer operation.
3. Calculate system losses using basic energy and mass balance principles.
4. Determine the efficiency of electrolyzer equipment using standard performance metrics.

4.3.1 Inputs/Outputs of an Electrolyzer system

Inputs

The following figure represents the inputs to an electrolyzer.

DC electric power supply (from renewable sources of energy)

Water

Electrolyte (in case of alkaline electrolyser)

Fig 4.3.1. Inputs to an Electrolyzer

The electric connections have to be done for each pair of electrodes with specific size of cables to avoid overheating, sparking, etc. For connections at electrodes, proper lugs with uniform and firm soldering with cable conductor. Dry soldering should be avoided. The lugs should be connected with electrodes with specified bronze nut bolts, washers and heat sink, as specified. In any case, the loose connections must be avoided to prevent overheating of the terminals as well as to avoid possibility (due to high magnitude of current) of sparking at terminals.

Laying and fixing of cables should be such that there are no mechanical stresses and twisting of cables. To avoid this use of cable holder, mild steel wire or strips is made to transfer such load on cable holder/mild steel wire/strip. There should not be any leakage/seepage of electrolyte from drain valves, fittings etc. i.e. it must be leak-proof for which they should first be pressure tested as prescribed in supplier's manuals.

All these points should be taken care of for all the electrical equipment or their components.

Outputs from electrolyzers

Hydrogen is the output and oxygen is the by-product from electrolyzers. Oxygen after separation from water droplets or vapour can be let out in the atmosphere or stored for other uses. Working with oxygen is quite safe similar to working with inert gases. However, care has to be taken to avoid its mixing with hydrogen. The physical layout takes care by keeping oxygen outlet quite apart from hydrogen outlet.

Hydrogen needs to be handled very carefully right from its outlet from electrolyzer to high pressure storage tanks. In between, hydrogen passes through water separator, hydrogen gas holder, hydrogen drier and finally inlet of high-pressure compressor system. Strict monitoring is required at all such places or terminals

4.3.2 Calculate Losses and Equipment

The type tests including efficiency are performed on specific model of electrolyzer in the authorized laboratory. After passing the tests, the manufacture is included in the list of approved manufacturers for a specific model of the electrolyzer.

The tests are performed first time to prove the design and copies of test certificates duly signed with seal are attached with the offers. Customers normally place the order as per such certificates.

For customer's plant, routine tests are performed at manufacturer's works on all the electrolyzers and may be witnessed by the Customer's representatives.

Generally, efficiency, pressure test, temperature withstand test, as applicable, are performed on individual equipment.

After assembly of plant, test is done mainly noting gross input energy, DM water used, and hydrogen collected after high/medium pressure hydrogen compressors. Specific arrangements are to be made for weighing of hydrogen getting stored in high/medium pressure storage.

Calculations are to be done taking into account efficiencies of each equipment recorded during their individual efficiency tests.

In broad sense,

- Plant efficiency = Gross input energy (with due corrections for meters, CT/PT errors and calibrations and permissible tolerances) - Energy consumed by each equipment e.g. dryer, compressor (again with due corrections)
 - = Energy required for production of dry hydrogen.
- Hydrogen produced = Weight of hydrogen filled cylinder – weight of empty cylinder with errors corrections allowed as per the applicable standards.

Above will give energy required for hydrogen production by the electrolyzer.

All these calculations should be essentially performed before acceptance of plant. It avoids future disputes to great extent.

Electrolyzer Efficiency

Water electrolysis is the reverse of the fuel cell reaction. In fact, many fuel cells based on PEM and solid-oxide technology can work both as a fuel cell or a water electrolysis cell, depending on the direction of the electrical current.

The following figure represents the equation for the water electrolysis reaction that is the reverse of the fuel cell equation.



Equation For the Water Electrolysis Reaction

The efficiency calculations, therefore, can be inverted as well. The efficiency of an electrolysis system, can be calculated as the heating value of the hydrogen produced divided by the electrical energy input, shown in the following equation:

- Electrical efficiency (HHV) = $\frac{\text{HHV of H}_2 \text{ produced}}{\text{Electricity used}}$

where,

- HHV is the higher heating value.

HHV is defined as:

The amount of heat released by the unit mass or volume of fuel (initially at 25° C) once it is combusted and the products have returned to a temperature of 25° C. It includes the latent heat of vaporization of water.

HHV

The standard potential of 1.23 Volts corresponds to the higher heating value HHV of hydrogen.

- HHV of hydrogen/kg = 141.9 MJ = 39.4 kWh
- LHV of hydrogen/kg = 120.1 MJ = 33.3 kWh
- Electrical efficiency (LHV) = $\frac{\text{LHV of H}_2 \text{ produced}}{\text{Electricity used}}$

(Source: Harrison K.W., Remick R., and Martin G.W., and Hoskin A. 2010. Hydrogen Production: Fundamentals and Case Study Summaries. NREL)

Other efficiency measures often are listed on the product brochures of electrolyzer manufacturers, the most frequently used being kilowatt hours per normal cubic meter of dry hydrogen produced (kWh/Nm³).

Computation of Efficiency by Simulation

The efficiency of PEM electrolyzer plant can be computed for which the input data has to be carefully worked out with accuracy. The following figure represents the schematic diagram that shows the components of the PEM electrolyzer plant.

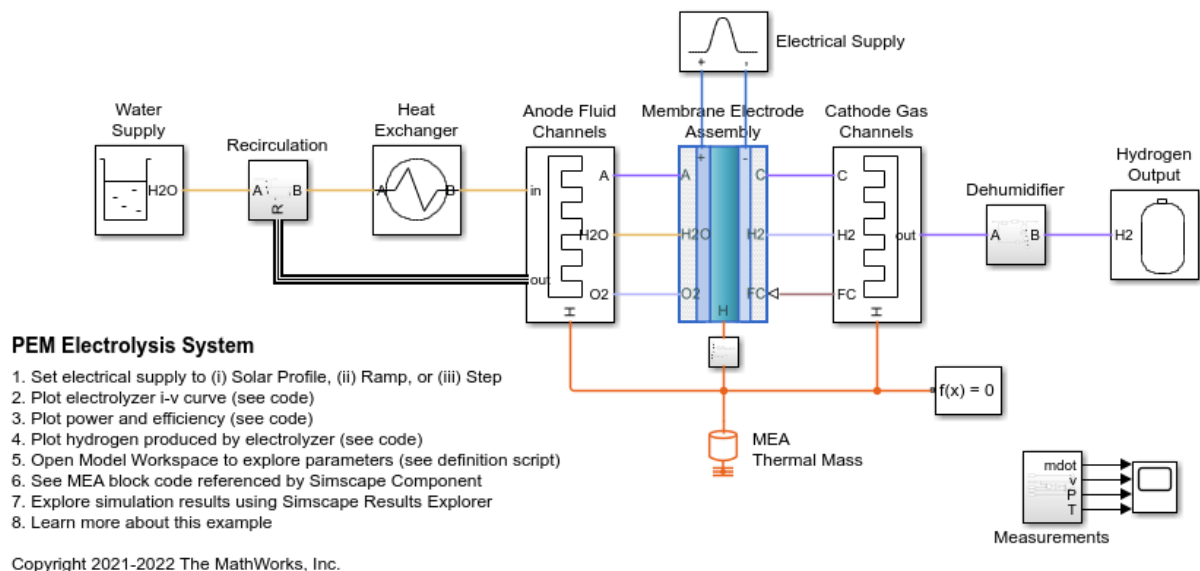


Fig 4.3.2.: Mathematical model of PEM electrolyzer plant.

(Source: MathWorks. PEM Electrolysis System. <https://in.mathworks.com/help/simscape/ug/pem-electrolysis-system.html>)

Activity



- Show how to identify the tools and equipment to perform the installation of electrolyzer as per concerned technical sheets and relevant safety and technical standards
- **TIPS:**
- Special tools & tackles are mentioned in the drawings/manuals along with the instructions for their use. These are generally, in the supplier's scope of supply.
- Drawings from Supplier:
 - Single line/block diagram,
 - General layout of the plant. Observe the orientation of each equipment. Check civil drawings for their matching with general layout drawings.
 - Layout of piping and electrical cables
 - Individual assembly diagrams
 - Drawings of fasteners, gaskets, seals, 'O' rings etc.
 - Civil foundation and embedment drawings
 - Earthing pits (electrical and electronic) etc.
- Installation/O&M manuals of supplier:
 - Specific instructions for the use of special tools/tackles are given in drawings/manuals.

Activity



- Illustrate how to perform calculation for estimating losses and equipment efficiency
- Calculate HHV and LHV electrical efficiency for hydrogen production of 360 kg/day. The input renewable energy may be taken as 55 kWh/kg of hydrogen production.

UNIT 4.4: Input Renewable Power from various sources and its Integration with Electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the various renewable sources of energy and their limitations
2. Explain the concept of integration of incoming renewable power for supply to the electrolyzer

4.4.1 Renewable Sources of Energy and Their Limitations

Solar PV, wind and hydro are the main source of renewable energy in India.



Fig. 4.4.1 Wind, solar PV and hydro power

Source: Resource Library. The Right Balance: Mixing Energy Resources. Society

Solar PV Energy

India is rich in solar energy with around 300 sunny days in a year almost throughout the country. It's main limitation is that solar electrical power can generated for 5 to 6 hours during a day and hence this source is of variable nature.

Wind energy

Globally, India stands fourth in installed wind power capacity. The wind resource is mainly available during night and early morning hours. Following are the two major limitations of wind energy:

- Wind resource is available at some locations and its assessment takes ideally 3 years.
- Variability of wind resource makes its accurate forecasting challenging.

In India, the high wind season is from the month of April to September and low wind season is from the month of October to March.

Wind-solar hybrid energy

Solar power generation typically occurs for approximately 5 hours during midday and afternoon when the sun is at its peak. On the other hand, wind resources tend to be stronger during the nighttime and early morning. Thus, to some extent, wind and solar resources are complementary to each other. However, reduced variability still prevails.

The current trend focuses on a hybrid power system that combines solar PV and wind energy, complemented by energy storage. This system addresses the combined variability of solar and wind power outputs, ensuring a stable energy supply. By harnessing the strengths of both sources and utilizing energy storage to balance fluctuations, this approach delivers reliable power generation, even when individual sources face limitations due to changing weather conditions.

Hydroelectric energy

This is an environment friendly source of reliable energy but its only limitation is that it is subject to the availability of water stored in the dam-reservoir. The cost of electrical energy from hydro power projects has been traditionally very low. There can be some hybrid power combinations with hydro power such as wind-hydro or solar-hydro or wind-solar-hydro hybrid. The provision of energy storage may also be considered, if required. These hybrid power combinations may prove to be very reliable and also offer low-cost energy in medium and long run.

4.4.2 Integration of Renewable Energy

For integration of renewable energy, the requirements and available alternatives are as follows:

- **Requirement:** The electrolyzers operate on DC at low voltage and high current.
- **Available alternatives:**
 - (i) **DC electric power from Solar PV:** D.C. electric power is fed to DC-DC converter to convert DC voltage to match with the DC voltage of electrolyzer.
 - (ii) **AC electric power from wind and hydro:** AC electric power is converted to DC electric power by rectifier. A DC-DC converter changes DC voltage to match with the DC voltage of electrolyzer.

- (iii) **Hybrid e.g. DC power from solar PV and AC power of wind or hydro:** The AC electric power output of wind and/or hydroelectric power projects is converted to DC by rectifier. Next, DC-DC converter changes DC voltage to match with the DC voltage of electrolyzer.

The following schematic diagram shows solar PV DC and utility grid AC power are integrated and fed to DC-DC converter to change the voltage to match with those of PEM electrolyzers.

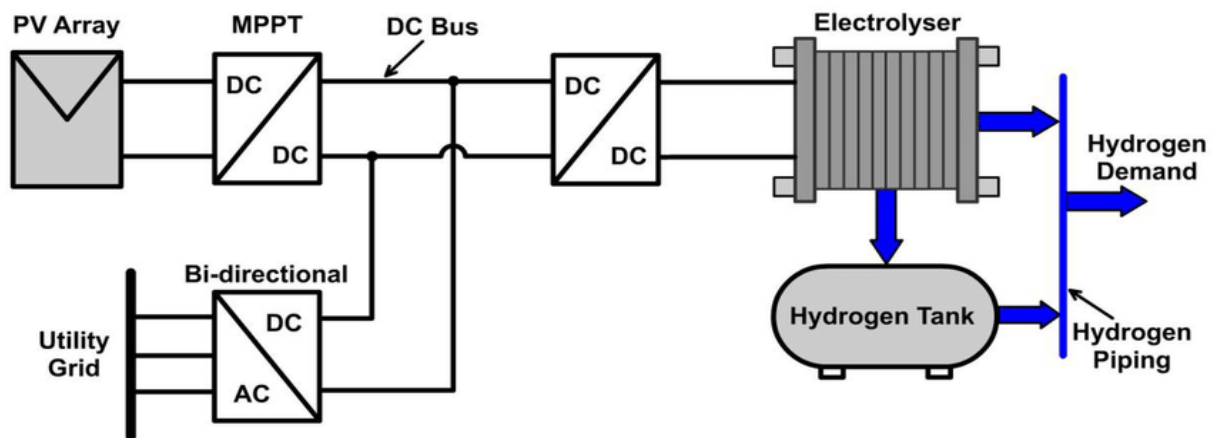


Fig. 4.4.2 Integration of solar PV power for electrolyzer

Source: Tebibel H. and R. Medjebour R. 2018. Comparative performance analysis of a grid connected PV system for hydrogen production using PEM water, methanol and hybrid sulfur electrolysis. International Journal of Hydrogen Energy

Wind power system

Due to wind's intermittent nature, wind electric power fluctuates throughout their operational. AC electric power wind electric generators and AC grid power are converted to DC electric power by rectifiers. A DC-DC converter changes DC voltage to match with the DC voltage of electrolyzer. The following figure shows schematic of wind electric power for green hydrogen production and usage system.

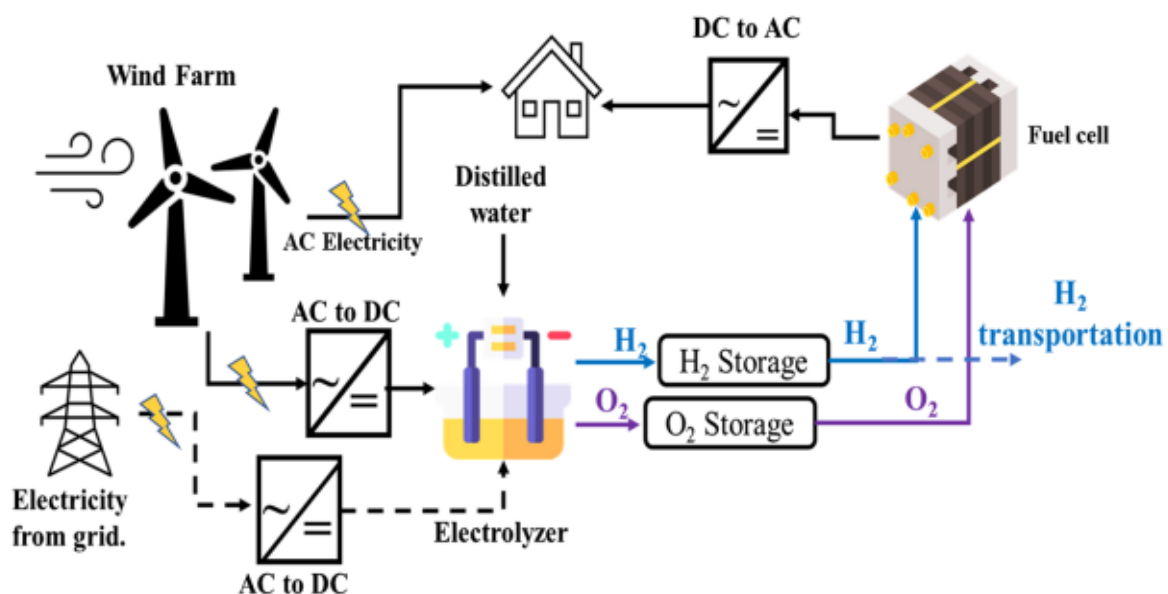


Fig. 4.4.3 Schematic of wind electric power for green hydrogen production and usage

Source: Nasser M., Megahed T.F., Ookawara S., and Hassan H. 2022. A review of water electrolysis-based systems for hydrogen production using hybrid/solar/wind energy systems. *Environmental Science and Pollution Research*

Wind-solar PV hybrid to H₂ system

The hydrogen production from solar and wind hybrid energy is drawing attention. The hybrid of wind and solar PV energy for hydrogen production might be a low-cost solution. Moreover, hybrid power system supplies electric power than individual solar PV or wind.

The basic concept of the wind-solar PV hybrid system is shown in the following figure.

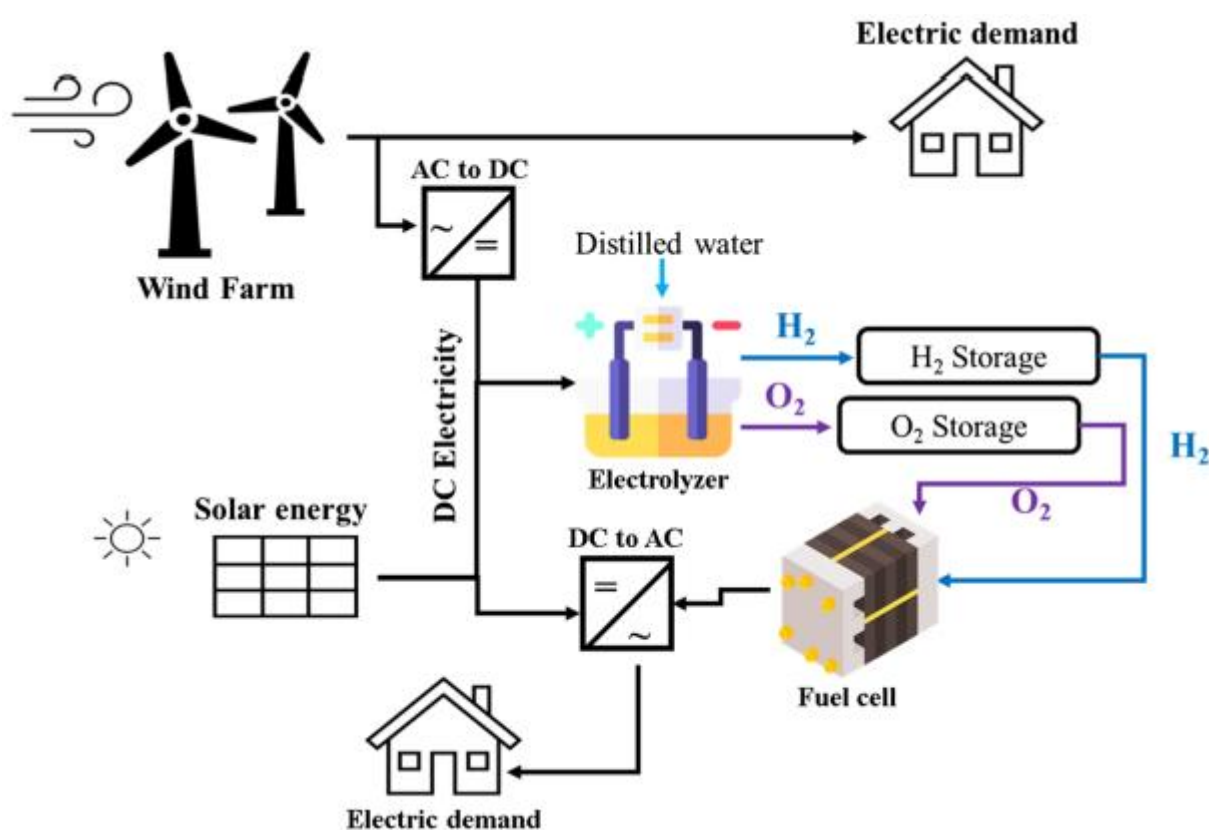


Fig. 4.4.4 Schematic of wind-solar hybrid electric power to green hydrogen production and usage system

AC electric power wind electric generators is converted to DC electric power by rectifiers. A DC-DC converter changes DC voltage to match with the DC voltage of electrolyzer. The wind turbine contribution may be more than PV panels because it produces any time whereas PV panels generate during adequate sunshine hours only.

UNIT 4.5: Assembly/Installation of Parts and Components of Electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Understand electrolyzer components and layout
2. Assemble electrolyzer parts as per specifications
3. Follow standard installation procedures
4. Ensure correct mechanical, electrical, and fluid connections
5. Apply safety and quality control practices
6. Prepare the system for testing and commissioning

4.5.1 Assembly of Electrolyzer

Information available online indicates that typical electrolyzers are commonly manufactured in capacities ranging from 1 Nm³/hour to 100 Nm³/hour and are supplied pre-assembled from factories. The common approach likely involves obtaining pre-assembled electrolyzers for the intended production capacity, including an additional unit for backup purposes.

Assembling an electrolyzer stack requires specialized expertise, dedicated tools, handling equipment, and meticulous testing. These facilities are typically feasible for larger hydrogen plants. Conversely, for small to medium-scale plants, it's more cost-effective and preferable to acquire factory-assembled electrolyzers.

The equipment, such as gas holders, water separators, pump motors, valves, and compressors, can be procured separately. The assembly of these equipment follows standard procedures and doesn't demand specialized skills, except for high-level sealing in the case of high-pressure compressors. Electrolyzer assembly is primarily carried out in controlled factory environments to ensure proper sealing, membrane integrity, and precise oxygen and hydrogen outlets.

Some manufacturers highlight that hydrogen produced by an electrolyzer, after suitable drying, can be directly used for various processes, including applications in the food industry, eliminating the need for separate storage.

The following figure represents the stack assembly of a PEM electrolyzer and parts of a single cell.

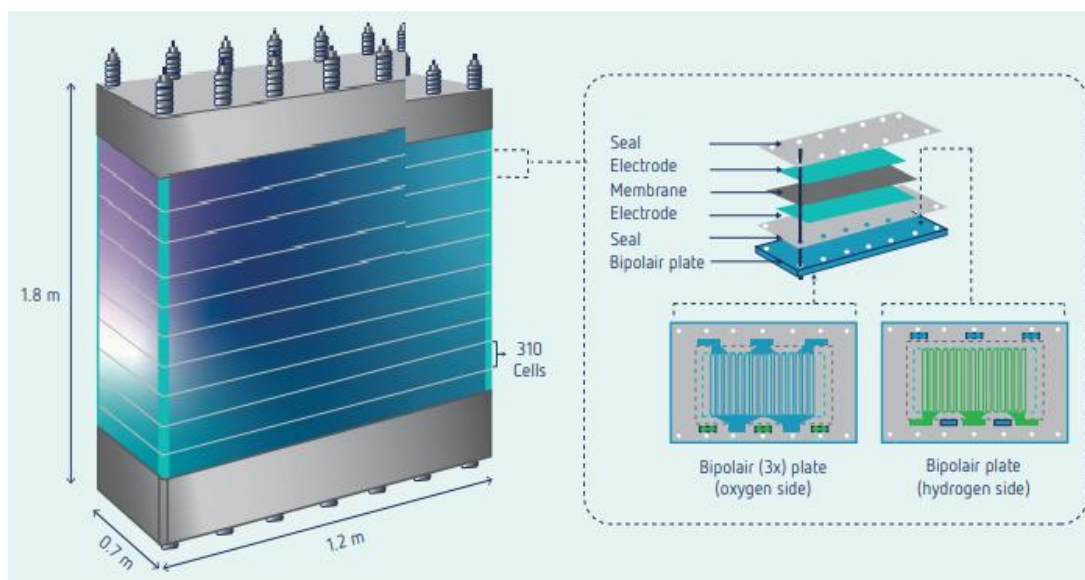


Figure 4.5.1 Schematic illustration of PEM electrolyzer stack and a cell

Source: Noordende H. and Ripson P. 2022. A One-GigaWatt Green-Hydrogen Plant. A Public Report

The factory assembled electrolyzer may be similar to a sort of sealed compressor of house hold fridge. The photographs of 2 kg hydrogen per hour capacity PEM and Alkaline electrolyzers are shown in the following figure. It is in the context of NTPC's green hydrogen project in Leh district, Ladakh.



Alkaline 2 kg/hr

Fig. 4.5.2 PEM and Alkaline electrolyzer assemblies

Source: Panda DMR. 2023. Key component. Technology selection. Training program on Techno-economic Considerations for the Design of Green Hydrogen Projects. New Delhi

Activity



- Show how to handle different tools, equipment as per concerned standard and industry practices.
- TIPS:
- Use all tools and write the activity done by you. (e.g. tightening, loosening of items etc.)
- (From safety point of view insulated or non-metallic tools are to be used as per suppliers' instruction manuals. In hydrogen work area, it is necessary to avoid electrical and mechanical sparking during O&M.)

VIDEO

Topic: Installation of a Sunfire Pressurized Alkaline Electrolyzer
 By: Sunfire (2022)
 Link: <https://www.youtube.com/watch?v=qsygJibJ7eQ>
 Duration: 0 m 53 sec

4.5.3 PEM Electrolyzer/Fuel Cell

The PEM electrolyzer can work as fuel cell and vice-versa. Here is the step-by-step assembly of PEM electrolyzer/fuel cell is given.

Steps

Step 1: Place end plate (aluminium) on flat surface. Assemble bolts vertically, as shown in the following figure.



Fig. 4.5.3: Assembly of PEM fuel cell/electrolyzer

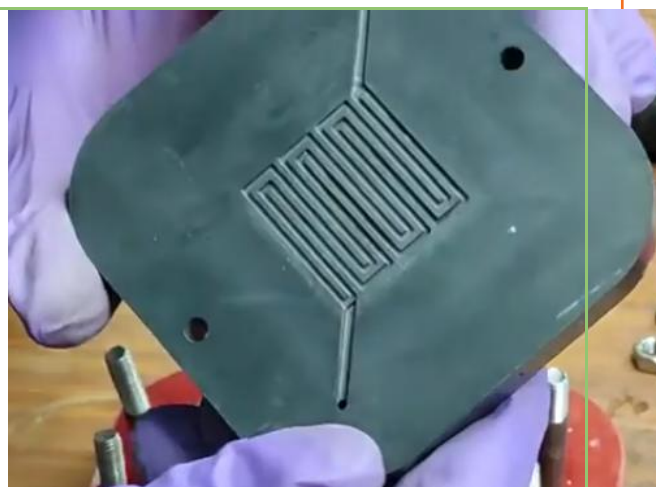


Fig. 4.5.4 Gas flow channel

Source: Video link given at the end of this unit

Step 2: Place Teflon cloth insulation to prevent short circuit of current collector with end shield

Step 3: Place current collector of copper (it has integral terminal)

Step 4: Assemble silicon 'O' seal rings to prevent gas leakage.

Step 5: Place cathode electrode.

Step 6: Assemble membrane with silicon gaskets on its both to stop leakage

Step 6A: Assemble anode electrode.

Step 7: Assemble bottom flow channel (graphite, size 2.8 cm x 2.8 cm).

Step 8: Assemble current collector

Step 9: Assemble silicon 'O' ring seals to stop gas leakage

Step 10: Teflon insulating cloth to prevent short circuit between current collector and end plate.

Step 11: Assemble end plate.

NOTE: Check that there is no gap between various layers of the cell assembly.

Step 12: Assemble Teflon washer before assembling the nuts to protect against electrical short

Step 13: Tighten the nuts hand.

NOTE: A 25 W heater is fitted on the end plate to heat the fuel cell to improve its performance.

Step 14: Tighten the bolts with the concept of differential tightening i.e. tighten opposite bolts.

Step 15: Finally tighten the nuts using torque wrench uniformly by applying torque of 2 Newton metre.

Test: Electrolyzer mode:

- Check the production of hydrogen and oxygen by water bubbles.

Fuel cell mode:

- Input: Hydrogen and air. Check voltage & current at no load and load using electronic load. (fuel cell mode).

NOTE: See the following videos carefully:

1. Fuel cell assembling (PEM)
2. Fuel cell and dismantling (PEM)
3. Methanol fuel cell testing

VIDEO

Topic: Fuel cell assembling (PEM)

By: Dr. Aristatil Ganesan (2021)

Link: https://www.youtube.com/watch?v=bfReGLZwHY&ab_channel=AristatilGanesan

Duration: 13 m 00 sec

VIDEO

Topic: Fuel cell and dismantling (PEM)

By: Dr. Aristatil Ganesan (2021)

Link: https://www.youtube.com/watch?v=ykDiIH7Gtv0&ab_channel=AristatilGanesan

Duration: 4 m 49 sec

VIDEO

Topic: Methanol fuel cell testing

By: Dr. Aristatil Ganesan (2021)

Link: https://www.youtube.com/watch?v=NEp97qGZfeg&ab_channel=AristatilGanesan

Duration: 4 m 34 sec

NOTE: Link for Videos of Mr. Ganesan: <https://www.youtube.com/@aristatilganesan6443>

UNIT 4.6: Challenges Associated with Hydrogen in Storage, Handling and Transportation

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the Challenges associated with Hydrogen in storage, handling and transportation

4.6.1 Need for Hydrogen Storage

The lightest element that exists on Earth is hydrogen. When compared to hydrocarbon fuels that are discovered or used in liquid or gaseous forms. In comparison to hydrocarbon fuels found or utilized in liquid or gaseous states, hydrogen exhibits both the highest energy density (by weight) and the lowest energy density (by volume). The main obstacle to making hydrogen economically competitive with hydrocarbon fuels in terms of cost (including the cost of mobile storage) is its lack of mobile storage.

Additionally, the global strategy is to manufacture as much green hydrogen as possible by electrolysis with renewable electrical energy. Renewable energy resources, namely wind and solar, are unpredictable in terms of time and space. As a result, hydrogen needs to be created as and when renewable energy sources are available.

The consumption of hydrogen during its creation is not guaranteed. As a result, hydrogen storage becomes essential.

This storage is expected to be a fixed local storage at a hydrogen production plant. The following figure shows hydrogen storage and distribution:

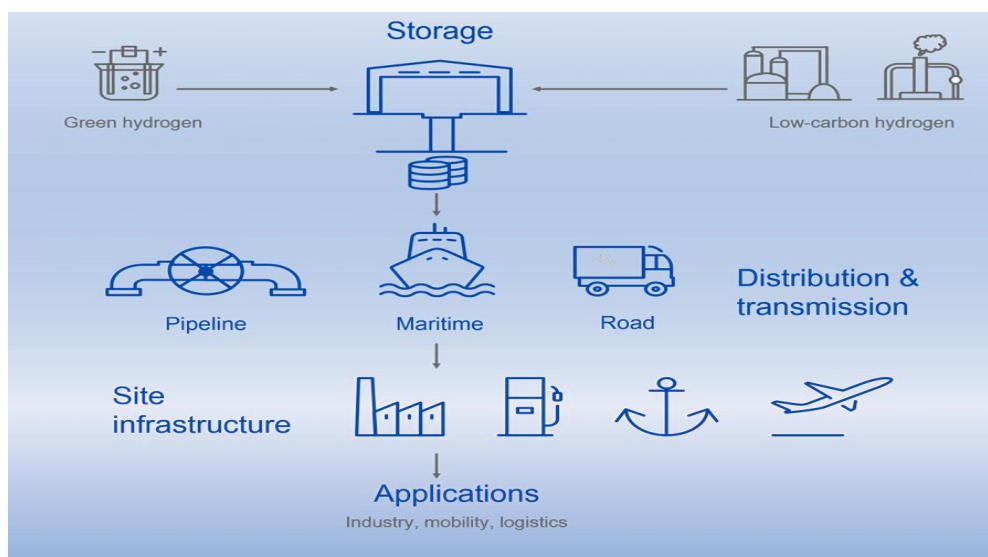
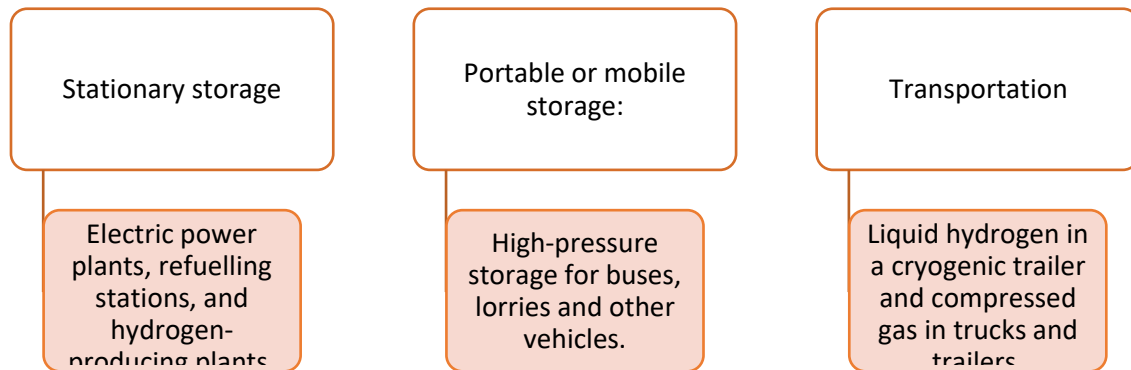


Fig. 4.6.1 Hydrogen storage and distribution

Image Source: TUV. Hydrogen enables the connection of clean energy and applications

Hydrogen storage is a crucial enabling technology for the development of hydrogen and fuel cell technologies. There are different types of hydrogen storage available.

The different types of hydrogen storage, along with some examples, are:



Since hydrogen gas is the foundation of the hydrogen economy, where it replaces fossil fuel in many combustion applications, hydrogen energy storage is of significance.

The market for hydrogen energy storage was valued at USD 14.72 billion in 2021 and is projected to reach USD 26.94 billion by 2030, growing at a 7.1% CAGR between 2022 and 2030.

4.6.2 Challenges in Storing Hydrogen

Since hydrogen is a fairly light gas, storing it is a major challenge. To store it at low temperatures or high pressure, further research is required. It is normal to store compressed hydrogen gas in metal cylinders. However, in these instances, hydrogen embrittlement (brittleness formed in hydrogen resulting in easy breakage) presents a problem that must be addressed.

Safety concerns arise from embrittlement because it causes the metal cylinders used to contain hydrogen gas to degrade. Modern methods, such as magnetic rack detectors, digital cameras, etc., are needed to determine the impact of embrittlement, which results in deterioration or loss of strength, in the case of metallic hydrogen storage. Based on this, it is necessary to decide whether it is possible to fix the frequency of inspection of hydrogen storage. More research is being done on hydrogen storage to remove its challenges and increase the possibility of replacement or strengthening.

For stationary and mobile applications as well as for transportation applications, high-density hydrogen storage is still a considerable hurdle. The majority of the storage methods that are now available require large-volume systems that store hydrogen as a gas. For stationary applications, where the footprint of compressed gas tanks may not be as important, this is less of a problem.

The hydrogen needed for the fuel-cell vehicles must be sufficient to provide them a 500 km or more driving range with quick and simple refuelling. While there are a few light-duty hydrogen FCEVs (fuel cell electric vehicles) on the market that can travel this distance, they rely on compressed gas onboard storage using large-volume, high-pressure composite containers. Large

storage volumes may have less of an impact on larger vehicles, but it is still difficult to provide enough hydrogen storage across all light-duty platforms.

In case 120 MJ/kg of hydrogen is compared to 44 MJ/kg for gasoline, hydrogen has nearly three times the energy of gasoline on a bulk basis. However, the situation is flipped when measured in terms of volume; liquid hydrogen has a density of 8 MJ/L whereas gasoline has a density of 32 MJ/L. The driving range for all light-duty vehicle platforms will require an onboard hydrogen storage capacity of between 5 and 13 kg of hydrogen. For such requirements, the weight of the storage tank should be in the range of 100 kg for 5 kg of hydrogen and 50 kg for 13 kg of hydrogen, and that too at a high pressure of 500 to 700 bars, which increases the risk factor.

The following figure shows the effect of pressure on the volume of hydrogen:

Storage volume for 1kg of hydrogen

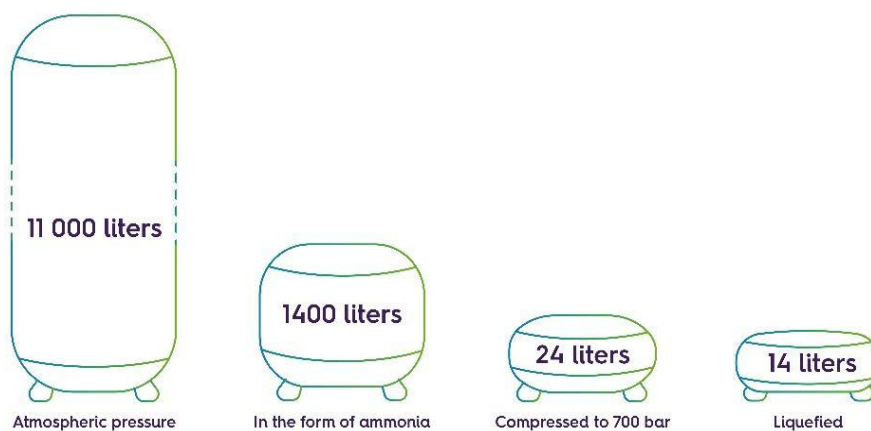


Fig. 4.6.2 Storage volume of 1 kg hydrogen at NTP

Image Source: Terega. Hydrogen Storage: a challenge for the development of the industry

The challenges in handling and transporting hydrogen are:

Challenges in H₂ handling

As hydrogen has a low volumetric energy density, it has to be highly compressed at high pressures (from 350 to 700 bar) to be packed into a tank in sufficient quantities to power a car. 5/6 kg of hydrogen are required to cover about 600 km.

5/6 kg of hydrogen is required to cover about 600 km.

Challenges in H₂ transportation

Reducing cost, increasing energy efficiency, maintaining hydrogen purity, and minimizing hydrogen leakage.

For most applications in India, hydrogen transportation through pipelines could be the most cost-effective solution. However, challenges in obtaining huge investments for building pipelines and technical issues would need to be addressed. Shipping could remain an unviable option for India for domestic transport till 2050 due to the high costs of liquefaction, refrigeration and regasification.

Table 4.6.1 gives a comparison of different modes of transportation with respect to the distance and quantity of hydrogen to be transported.

Technology	Year	Cost (Rs/kg)	Remarks
Transportation pipeline	2020	2.2	Distance 50 km, 1000 t/day
	2050	1.5	Distance 50 km, 1000 t/day
Distribution pipeline	2020	6.7	Distance 50 km, 100 t/day
	2050	5.2	Distance 50 km, 100 t/day
Truck	2020	11.8	Distance 50 km, 700 kg capacity
	2020	40.0	Distance 50 km, 200 kg capacity

Table 4.6.1 Costs of Hydrogen Transportation

The greatest barrier to a successful hydrogen economy is the lack of adequate storage options. But once an achievable solution is established, it's prepared to take off. The key challenges of the available options are shown in the following figure:

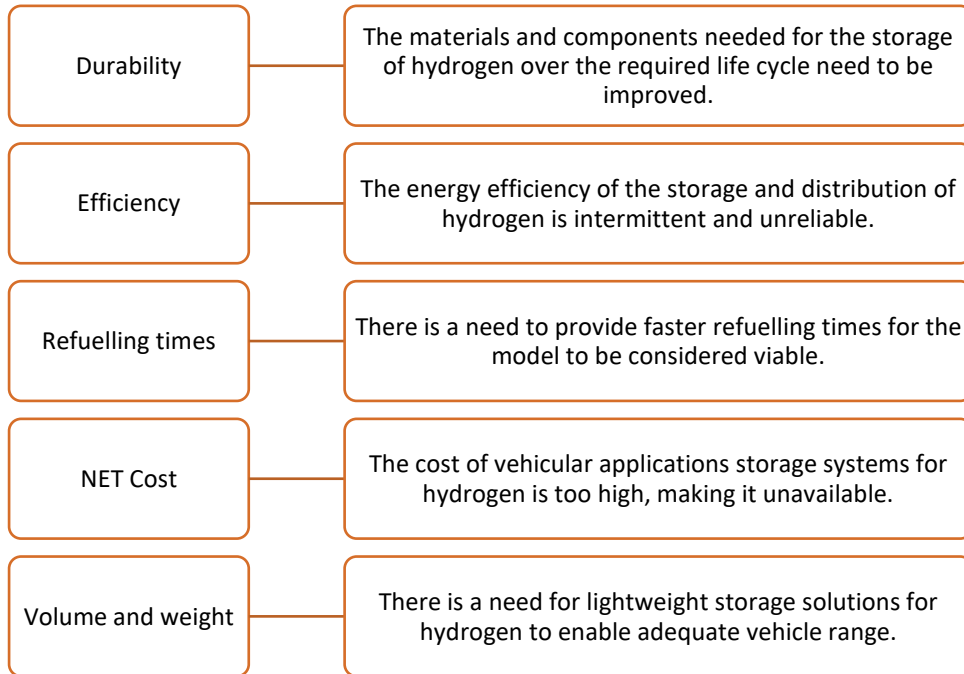


Fig. 4.6.3 Challenges in storage

Source: How to store and transport hydrogen: The ultimate guide. Advanced Structural Technologies, Inc.

UNIT 4.7: Hydrogen Compression System

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss how to select and install hydrogen compression and storage system
2. Discuss the precautions required to compress and store hydrogen
3. Discuss the safety guidelines to be followed as per industry standard

4.7.1 Selection of Compression

Several technical considerations that need to be considered in the selection of a hydrogen compressor are described in the following image:

Pressure range

It is important to select a compressor with a pressure range required for specific application. The pressure range may be from 100 bars to 900 bars.

Flow rate

It is important to select a compressor with a flow rate suitable for specific application. Flow rate of hydrogen is 1 m³/hour to 11 m³/hour for each electrolyser stack.

Efficiency

The efficiency of a compressor for hydrogen refers to the energy required to compress the hydrogen gas. The compressor needs to be selected with high efficiency to minimize energy consumption and cost. The compression efficiency is about 70%.

Reliability

It can affect the overall performance and safety of the system. The compressor needs to be selected with proven track Durability record of reliability.

Durability

It affects long-term performance and life of compressor. The compressor should be strong enough to withstand the tough operating conditions while operating in a hydrogen system.

Size and weight

It is important to select a compressor which can be accommodated in a available space and weight. Typical values: Capacity up to 200 m³/hour.

Discharge pressure > 400 bars for lubricated compressors up to 225 bars for non-lubricated compressors.

Cost

The cost is an important consideration. It is desirable to evaluate trade-off between cost and performance while in the selection of a compressor.

Fig 4.7.1: Technical consideration for selecting compressor.

IMPORTANT: In overall planning of compression system, the standard and proven models of compressors should always be preferred. The manufacturers charge very high cost for other than standard models due to development of new design, drawings, tools etc. in addition to the high cost of type tests to be performed on first piece.

The criteria for selection of compression system mainly depends on end users' requirements, variation in their requirements, quantity to be supplied daily by different modes of transport, transport limitations etc.

Depending on pressure and quantum of hydrogen, compressors of different ratings and types may have to be selected and installed. However, observations of overall economy is equally important. Key points to be considered for the selection and installation of hydrogen compression system are:

1. Extremely good ventilation, no mechanical (frictional) or electrical sparking
2. Compressor HP delivery pipe output valves should be under hydrogen leakage detection. If required, upward ventilation near valves to carry away leaked hydrogen very fast upwardly.
3. Hydrogen storage in separate room with proper leakage detection device
4. The compressor capacity depends on daily requirement of cylinders of hydrogen under pressure. It may have two compressors of 50 to 75% capacity to maintain continuity of compression of hydrogen.
5. The compressors are almost similar to other gaseous compressors except seal rings and their number of levels, non-return valves, safety valves along with arrangements for fast upward diversion of hydrogen, high factor of safety.
6. The dryness of hydrogen is also one of the important requirements for its compression.
7. Availability of nitrogen or carbon dioxide cylinders is to be ensured for purging in case of any exigencies.
8. Device to check purity of hydrogen after compression to detect mixing of oil vapours during compression process.

Reference data

Following table gives data on compressors in NTPC projects for reference.

Project	Hydrogen/ammonia production	Compressors
GH2 mobility, Leh	80 kg/day, 11 hours	2 x 22 kW
GH2 Simhadri, AP	50 kg/day, 8 hours	2 x 37 kW
GH2, Mobility, Delhi	260 kg/day, 10 hours	2 x 110 kW
Green ammonia	9,100 kg/day, 24 hours	1 x 433 kW

Table 4.7.1 Compressors in Green hydrogen and green ammonia projects of NTPC

Source: Panda D.M.R. 2023. Key components, Technology selection. NTPC Online training program on "Green Hydrogen Technology organized by SCGJ

Activity A4.1

Show how to perform selection and sizing of hydrogen tank

Activity A4.1: TIPS

Note down the specifications of the air compressor (for activity purpose) and storage cylinder inside volume in litres. Pressure actuating tripping device and safety valve are to be mounted on storage tank. Check the pressure of the compressor and storage tank before starting the compressor. Start the compressor and note down the pressure at every 10% increase in pressure up to 90% of the rated pressure.

S.No.	Pressure at compressor	Time required, seconds	Pressure at storage tank	Remarks/observations
1				
2				
3				
4				
5				
6				
7				

Table 4.7.2 Study the development of pressure

NOTE: Repeat the activity with different capacities of cylinders.

4.7.2 Precautions

Normally permit card (issued by operating personnel) system is followed to carry out any maintenance work.

A compressor is a precision high-speed mechanical piece of equipment. They require caution in operation to minimize hazards to themselves and their property. The following image lists the safety precautions that should always be observed:

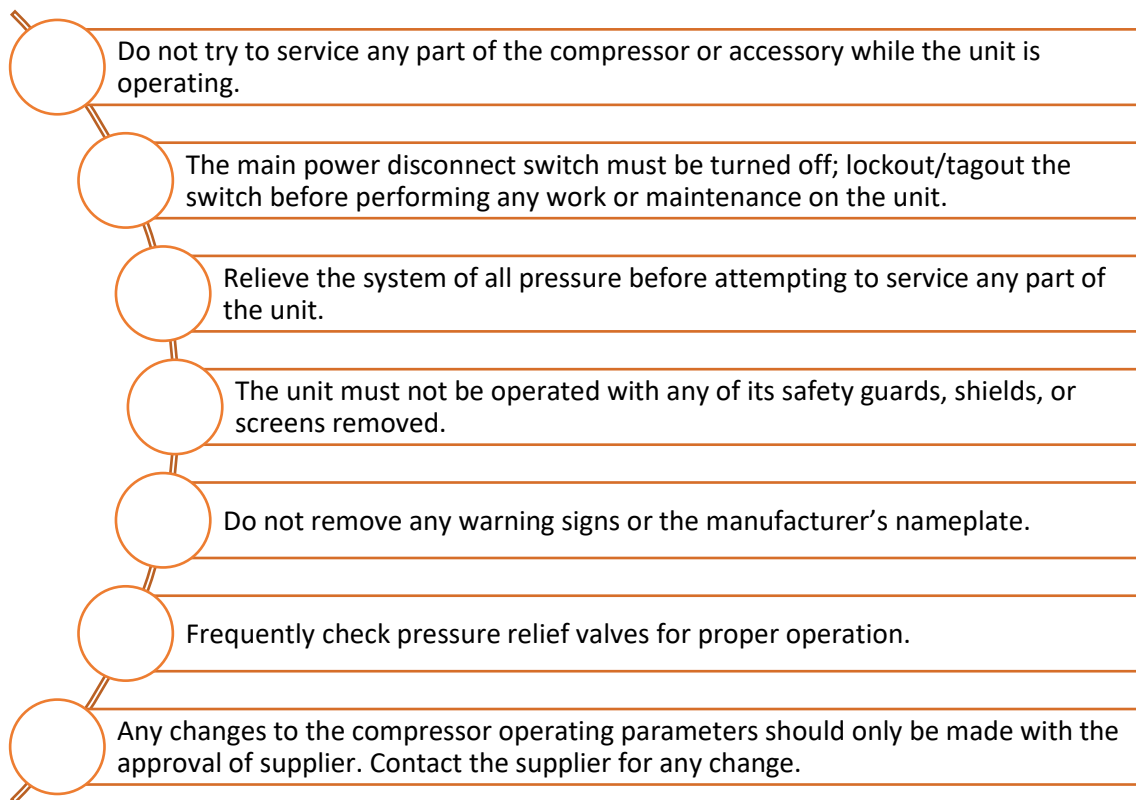


Fig 4.7.3: List of safety precautions

Activity A4.2

Show how compression system comply and follow all the applicable safety and regulatory standards.

Activity A4.2: TIPS

Study the works test certificates of compressor and compare the results. Check the operation of safety valves, non-return valves, and pressure tripping device.

4.7.3 Key components/parameters

Key components/parameters to be considered are:

- The Factor of safety more than 2 i.e., all the components should withstand double of storage pressure.
- Change of seals after certain hours of storage or operation under pressure.
- Valves are the heart of a compressor. The performance of a reciprocating air compressor mostly depends on a suction filter, valves, and piston rings. Hence due care is required to be taken. The valves require periodic inspection and maintenance.

1. Inlet or suction valve

It permits the flow into the cylinder but not back out. Suction valve failures are the most common component failure in a reciprocating compressor. Unloaders may be part of the suction valve assembly.

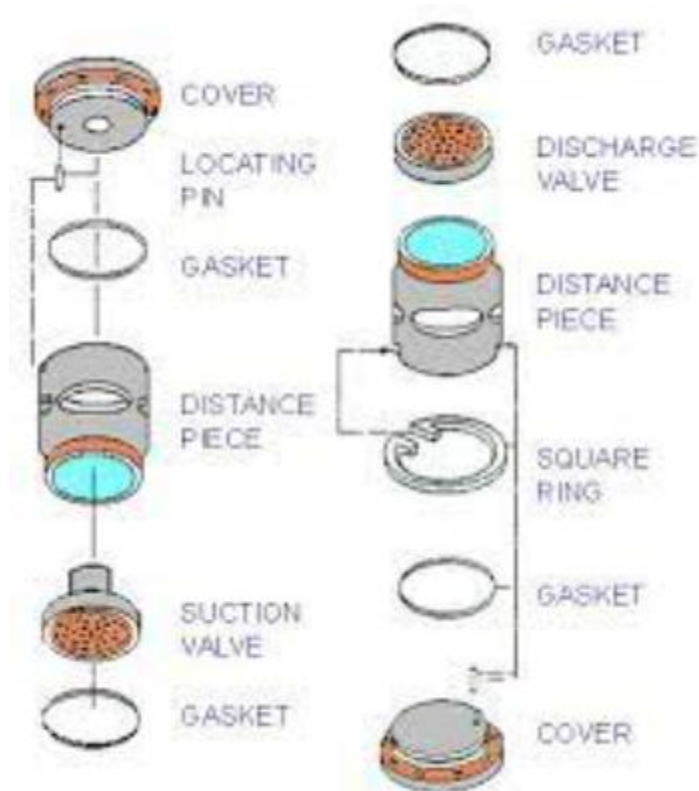


Fig. 4.7.4 Suction (inlet) and discharge (outlet) valves of reciprocating compressor
Source: Keepwin Technology Hebei Co. Ltd.

2. Outlet or discharge valve

It permits the flow to exit the cylinder but not return in. Suction and discharge valve failures are the most common component failures in a reciprocating compressor. Unloaders may be part of the suction valve assembly.

3. Pressure relief valve

A pressure relief valve does not allow the preset pressure to exceed. When the pressure is exceeded, the valve opens, and the excess pressure is released by discharging the hydrogen.

The pressure relief valve is actuated by the hydrogen pressure and no electrical actuation is required.

As soon as the normal working pressure is restored, the pressure relief valve closes, and further outflow of hydrogen is stopped.



Fig. 4.7.5 Hydrogen pressure relief valve

Source: Hyfindr. Hydrogen pressure relief valve

The Leak proofness of the safety valve needs to be checked and ensured regularly. The description given above is indicative. It may vary for hydrogen owing to its extreme lightness.

4. Tank size

The tank size depends on the pressure and quantity of hydrogen to be stored. The volume of 1 kg of hydrogen at atmospheric pressure is 11 m³. The volume to be stored is inversely proportional to the pressure maintained in the storage tank.

$$P_1V_1 = P_2V_2 \quad \text{OR} \quad P_1/P_2 = V_2/V_1$$

On this basis, storage tank pressure and capacity are decided.

The pressure gauge mounted on the storage tank can be used to monitor the leakage of hydrogen over a long duration. The reduction in pressure will indicate leakage of volume/weight of hydrogen.

- **Placement of tank**

The storage tank depending on the availability of space and size of the tank, may be in the open or inside a room with proper guarding. Adequate spacing must be maintained between adjacent storage tanks with minimum high-pressure piping. The provision of hydrogen pressure monitoring, firefighting systems, CCTV monitoring etc. needs to be made.

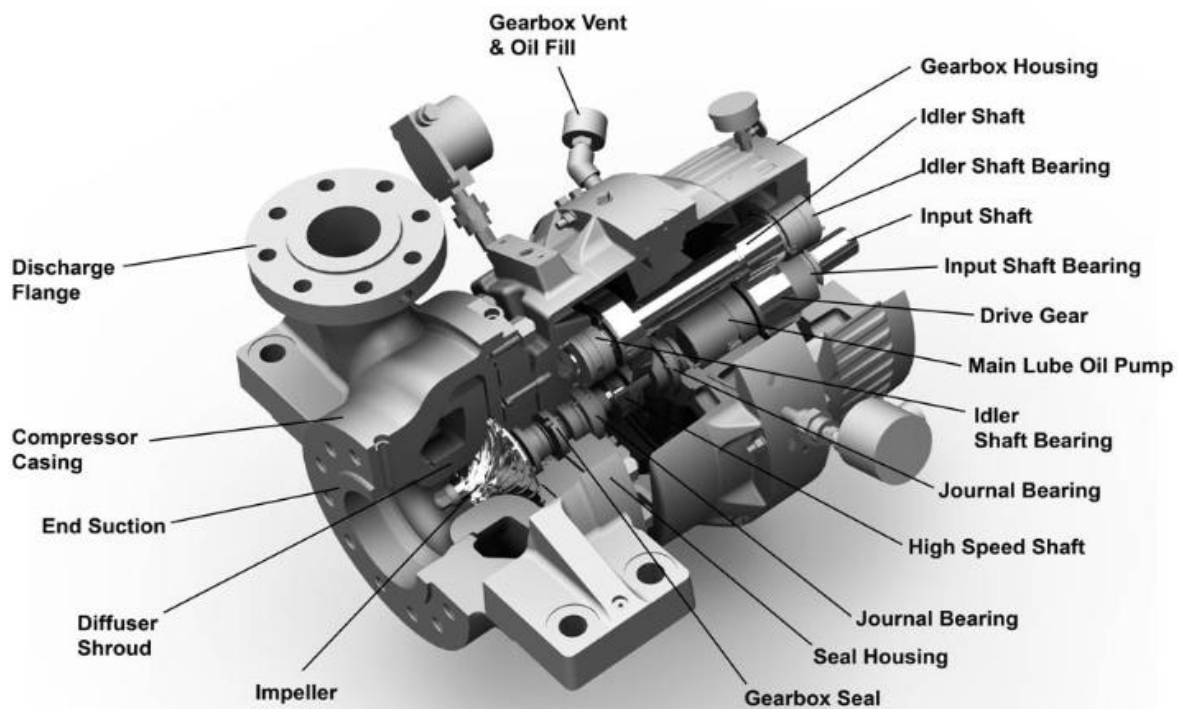


Fig. 4.7.5 Parts of a centrifugal compressor

Source: Sundyne. 2021. The basics: Centrifugal gas compressor

<https://www.sundyne.com/the-basics-centrifugal-gas-compressors-101/>

VIDEO

Topic: How to maintain the suction and discharge valves in a ...

By: Corken (2018)

Link: <https://www.youtube.com/watch?v=knFiyPZqM1M>

Duration: 9 m 14 sec

Activity A4.3

Show how to perform the required safety procedures as per industry standard

Activity A4.3: TIPS

Study the industry safety standards. Perform the checks/tests at less than rated capacity by changing the settings of safety devices.

IMPORTANT: While pressure checks/tests, maintain the safe distance, use of safety appliances.

UNIT 4.8 O&M Requirements for Electrolyzer

Unit Objectives



At the end of this unit, you will be able to:

1. Understand the fundamental operation and performance parameters of electrolyzer systems.
2. Identify routine operation and maintenance (O&M) activities required for safe and efficient functioning.
3. Recognize common operational issues, faults, and preventive maintenance practices.
4. Ensure compliance with safety, environmental, and regulatory standards during electrolyzer operation.
5. Support reliable hydrogen production through optimized maintenance planning and monitoring.

4.8.1 Operation and Maintenance

The main operation activities include the following tasks:

- To monitor the temperatures of electrolyte, electrolyzer terminals, cables etc. with remote temperature sensing devices.
- Further, noise levels, vibrations of rotating equipment such as compressors are to be monitored by portable vibration meters, noise levels by special wooden sticks.
- To monitor the leakages of electrolyte and its level, hydrogen/oxygen leakages, condition of electrolyte filters. Monitor the gas holder for any leakages of hydrogen.
- To monitor hydrogen temperatures and pressure at the inlet and outlet of high-pressure compressor.

The maintenance activities include the following tasks:

- Before any maintenance, purging of hydrogen is to be done by nitrogen or carbon dioxide.
- As per the observations, recorded by the operation staff, check the tightness of the electrolyzer terminals, seal cover, electrode seals, membrane, valve seals.

Operation Safety

A water electrolysis cell stack comprises multiple electrochemical cells linked in series. After stopping, these stacks can retain considerable electrical charge due to leftover hydrogen and oxygen in cells. This charge takes hours to dissipate naturally. Extreme care is needed for maintenance or replacement right after operation. A metal tool bridging current terminals could cause damage or injury through a large current or electrical arc. Lack of proper insulating gear poses risks to personnel. Following are the Best Practices for Handling Water Electrolysis Cell Stacks:

Pre-Check	• Maintenance and service personnel should ensure no significant electric charge remains in the cell stack before proceeding.
Safety Measures	• Before removing safety guards and electrical connections, confirm that the cell stack is free from substantial electrical charge.
Voltage Verification	• Personnel are recommended to check the voltage of the cell stack to guarantee discharge.
Additional Safeguard	• In certain situations, service personnel can use a purpose-designed service tool—a high-current shorting resistor—across the discharged cell stack for added protection.

Fig 4.8.1. Best Practices for Handling Water Electrolysis Cell Stacks

Cleaning of Electrolyzer

Cleaning of electrolyzer includes the following tasks:

- Cleaning of filter mesh by raw water followed by DM water.
- Compressor maintenance: Change of filters, checking of bearings, filing, greasing etc. Note that all these compressor maintenances should be carried out after purging of hydrogen.
- Maintenance of dryer: Cleaning, change of moisture absorbing chemicals e.g. silica gel. To check the valve operation and leakage test.
- Recharging of hydrogen as specified in the manual. Normally it is done with purging of nitrogen or carbon dioxide. After checking the pressure withstand, it is filled with hydrogen.

Checking Instruments

It includes the following tasks:

- Zero position of instruments must be checked in every six months.
- Instruments must be calibrated every year from authorized laboratories only.
- When checking the dismantled instruments, parts contacting medium shall be oil free before they are installed.

Note: These are indicative and instruments may vary from one manufacturer to another.

Start-up Sequence:

Following is the Start-up Sequence:

- Turn on general power switch and instrument switch.
- Switch-on air source
- Start DM feed pump till the DM water reaches required level in separators, around 50% or specified.
- Start electrolyte circulation pump.
 - Rotate the current regulating potentiometer of rectifier panel to zero position. Supply power to control circuit, relevant indicator shall glow. Push 'run' button, 'run' indicator shall glow. Slowly rotate clockwise current regulating potentiometer. Now electrolysis process begins. Liquid level in separator and cell temperature shall rise. Increase current up to its rated value. Turn current regulating potentiometer clockwise to regulate current to its rated value.
- When cell temperature reaches 50°C (or as specified by the manufacturer), set the temperature set point. Regulate the set point (75~85°C) or as specified by the manufacturer.
- After the plant has worked for few hours and gas is vented before catalyst column, then purity of H₂ gas can be analysed. If gas purity is up to the standard, close the vent valve and take the hydrogen gas to catalyst column.

Routine Operation and Maintenance

After the plant is put into operation, operators shall observe the operation situation carefully and operate the plant according to given parameters as specified by the equipment supplier. Readings should be taken periodically/hourly.

Watch process values carefully and deal with abnormal situation well in time. It could be detected with the remote sensing thermometer, vibration meter in the form of excessive noise or change in noise etc.

When adjusting DC output of rectifier panel, do not turn potentiometer knob violently. And selection switch must not be turned if current regulating potentiometer is not in '0' position.

Clean the filters, if the quantity of the circulating electrolyte drops abruptly.

Filter Cleaning

Whenever there is a flow drop or an abrupt change in the flow of electrolyte, the time arrives for the cleaning of the filters. However, before refilling of the electrolyte also, the filters are to be cleaned.

4.8.2 Safety precautions

While operating or performing maintenance tasks, you should adhere to the following safety precautions.

- If the plant shuts down for a long time, for example, half a year, electrolyte solution in electrolyzer must be discharged and electrolyzer must be filled with DM water.
- Since oxygen and hydrogen are generated with pressure, the operators shall strictly follow operating regulations.
- Explosion-proof lights/and fittings shall be installed in electrolyzer room. And the room

shall be well ventilated and protected by a gas detector.

- While the plant is running, repair shall not be carried out.
- Before welding in electrolyzer room, the hydrogen concentration must be measured to find out if it is lower than explosion limit value. To the extent possible, it may be avoided and in exceptional case, it should be done under strict supervision.
- At the same time, the plant and pipelines shall be purged to remove hydrogen and oxygen.
- Smoking is prohibited in electrolyzer room. Shoes with nail, synthetic fibre and woollen clothes are not suitable for operators. Iron wares or tools collided to each other is prohibited which may cause electric or friction spark.
- Fire-proof equipment such as extinguisher, sand and asbestos clothes, etc., are recommended for electrolyzer plant.
- Protective glasses and 2% boric acid shall be available in electrolyzer room. Rubber gloves shall be worn when operators prepare KOH solution.
- The hydrogen and oxygen are not be vented abruptly at high rate to avoid burning and explosion may occur. When oxygen is vented violently, spark may be produced by oxidization layer of pipes. Thus, the pressure in the system should be reduced slowly.
- It is not permitted to drop grease or oil onto the parts, which may contact oxygen. When the plant is operated and maintained, operator's hands and cloths shall be oil free.
- Alkaline solution is not allowed to drop into gaps between tightening bolts and electrode plates which may cause short circuit and lower the current efficiency. For avoiding short circuit of electrode plates, it is strictly forbidden for any conductors to drop on electrolyzer. Keep the surface of electrolyzer clean.
- In case of any accident such as serious leakage of gas or electrolyte, cut-off main circuit power supply and power supply of rectifier panel immediately. Slowly release the pressure in the plant. Analyse the cause for accident and ensure that remedial measures are taken.
- Inflammable and explosive material must not be stored in electrolyzer room. Persons unconcerned are not to be permitted to enter the room.

Exercise

1. Write the names of four types of electrolyzer in the reverse order of their technological status. (Start with the least matured technology i.e. at s.no. 1).
2. Which electrolyzer operates at the highest temperature?
3. Which electrolyzer has the oldest technology?
4. Which electrolyzers are available in the highest capacity range (MW range)?
5. Name the electrolyzers which use (a) acidic electrolyte (b) alkaline electrolyte.
6. What is earthing and why it is necessary?
7. 2. How the healthiness of earthing is checked?
8. 3. Why separate earthlings are required for electrical and electronic equipment?
9. What are the inputs required for production of green hydrogen by electrolysis?
10. If the quantity of water required by alkaline and PEM electrolyzers is the same?
11. What is the output and by-products of an electrolyzer?
12. What do you understand by operation?
13. What is the importance of maintenance?
14. What losses occur in an electrolyzer?



5. Micro-entrepreneurship opportunities in Green Hydrogen



- Unit 5.1– Micro-entrepreneurship Opportunities for Water Supply, Installation and Commissioning of Electrolyzer
- Unit 5.2– Micro-entrepreneurship Opportunities for Renewable Energy Power Supply
- Unit 5.3– Micro-entrepreneurship Opportunities for Storage, Transportation, and Installation
- Unit 5.4– Micro-entrepreneurship for Safety, Operation and Maintenance of Green Hydrogen Plant
- Unit 5.5– Micro-entrepreneurship for Leakage Monitoring



Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss micro-entrepreneurship opportunities for
2. Uninterrupted quality water supply for electrolyzer, water quality check
3. Installation and commissioning of electrolyzer
4. Renewable energy power supply arrangement plant for green hydrogen plant
5. Storage of green hydrogen
6. Installation of pipelines in green hydrogen plant/industry
7. Safe transportation of green hydrogen to consumers
8. Handling of salt enriched water from water feed input system, if required, depending on the type of electrolyser
9. Managing safety of the plant
10. Operation and Maintenance of green hydrogen plant
11. Leakage monitoring, purging of hydrogen by CO₂/NH₃ before start of work.
Recharging of hydrogen by avoiding air contact by using again CO₂/NH₃ and then filling of hydrogen

UNIT 5.1: Micro-entrepreneurship opportunities for water supply, installation and commissioning of electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for Uninterrupted quality water supply for electrolyzer, water quality check
2. Discuss micro-entrepreneurship opportunities for Handling of salt enriched water from water feed input system, if required, depending on the type of electrolyser
3. Discuss micro-entrepreneurship opportunities for Installation and commissioning of electrolyzer

5.1.1 Micro-Entrepreneurship

Micro-entrepreneurship opportunities related to ensuring uninterrupted, high-quality water supply for electrolyzers and conducting water quality checks in India are emerging in the context of the growing green hydrogen sector.

Following are the Micro-entrepreneurship opportunities for the same:

Water Supply Solutions

Water Quality Monitoring

Water Treatment Solutions

Mobile Water Testing Labs

Water Supply Chain Management

Consultancy and Training

Remote Sensing and IoT Solutions

Water Recycling and Circular Solutions

Fig 5.1.1. Micro-entrepreneurship opportunities for Quality Water Supply

- **Water Supply Solutions:** Micro-entrepreneurs can establish small-scale water supply systems or water delivery services specifically designed to meet the demands of hydrogen production facilities, particularly electrolyzers. These systems would ensure a consistent and reliable water supply, which is crucial for the efficient operation of electrolysis.
- **Water Quality Monitoring:** Micro-entrepreneurs can offer water quality testing and monitoring services to hydrogen production plants. Regular water quality checks are essential to ensure that the water used in the electrolysis process meets the required standards. Entrepreneurs can provide on-site testing, remote monitoring, and data analysis services.
- **Water Treatment Solutions:** Entrepreneurs can develop and offer small-scale water treatment solutions tailored to the needs of hydrogen production facilities. These solutions could include filtration, purification, and desalination systems to ensure the water input meets the necessary purity levels for electrolysis.
- **Mobile Water Testing Labs:** Micro-entrepreneurs can establish mobile water testing laboratories equipped with the necessary equipment and technology for on-the-spot water quality analysis. These labs can provide quick and efficient testing services to hydrogen production plants.
- **Water Supply Chain Management:** Entrepreneurs can create digital platforms or services for efficient water supply chain management. This includes real-time monitoring of water sources, scheduling deliveries, and optimizing water usage to minimize waste and costs for hydrogen production facilities.
- **Consultancy and Training:** Micro-entrepreneurs can offer consultancy and training services to hydrogen producers on water supply management, quality control, and regulatory compliance. This expertise can help hydrogen facilities operate efficiently and in compliance with environmental regulations.
- **Remote Sensing and IoT Solutions:** Entrepreneurs can develop remote sensing and Internet of Things (IoT) solutions to monitor water sources and quality remotely. These technologies can provide real-time data and alerts to hydrogen production facilities, ensuring immediate action in case of any water quality issues.
- **Water Recycling and Circular Solutions:** Entrepreneurs can explore opportunities for water recycling and circular economy solutions for hydrogen production plants. Designing and implementing systems to reuse and recycle water can help reduce water consumption and minimize environmental impact.

These micro-entrepreneurship opportunities align with India's increasing focus on green hydrogen production and the critical role that uninterrupted, high-quality water supply plays in this process. As the green hydrogen sector continues to grow, entrepreneurs who address these water-related challenges can contribute to its sustainability and success while building viable businesses.

5.1.2 Handling of salt enriched water from water feed input system

Micro-entrepreneurship opportunities related to handling salt-enriched water from water feed input systems in India are emerging as an essential aspect of the green hydrogen production process, especially considering the diverse types of electrolyzers used.

Following are some of the key opportunities:

- **Water Treatment Solutions:** Micro-entrepreneurs can develop and provide specialized water treatment solutions capable of removing salts and impurities from water feed input systems. These systems can be tailored to the specific requirements of different types of electrolyzers, ensuring the purity of the water supply.
- **Desalination Services:** Entrepreneurs can establish desalination services that focus on removing salts from brackish or seawater sources. Such services are critical for regions with limited access to freshwater sources, and they can cater to the needs of green hydrogen plants utilizing electrolyzers.
- **Water Recycling and Resource Recovery:** Micro-entrepreneurs can explore opportunities for water recycling and resource recovery. By implementing innovative technologies and processes, they can help hydrogen production facilities recover valuable resources from salt-enriched water, reducing waste and operating costs.
- **Mobile Desalination Units:** Entrepreneurs can develop and operate mobile desalination units that can be deployed to green hydrogen production sites as needed. These units can provide on-site desalination services, ensuring a continuous supply of high-quality water for electrolysis.
- **Consultancy and Testing Services:** Entrepreneurs with expertise in water chemistry and desalination technologies can offer consultancy and testing services to green hydrogen producers. They can assess water quality, recommend suitable treatment methods, and help ensure compliance with water quality standards.
- **Brine Management Solutions:** Brine is a byproduct of desalination and salt removal processes. Micro-entrepreneurs can explore opportunities to develop brine management solutions, such as concentrating and disposing of brine in an environmentally responsible manner.
- **IoT and Automation Solutions:** Entrepreneurs can develop IoT-based monitoring and automation solutions for salt-enriched water treatment. These technologies can enable real-time monitoring, remote operation, and predictive maintenance of desalination and water treatment systems.
- **Training and Skill Development:** Entrepreneurs can offer training and skill development programs for operators and technicians involved in the handling and treatment of salt-enriched water. This can help ensure the safe and efficient operation of water treatment systems.
- **Research and Development:** Entrepreneurs can engage in research and development activities to create more efficient and cost-effective desalination and water treatment technologies, catering to the specific needs of various types of electrolyzers.

These micro-entrepreneurship opportunities are essential in addressing the challenges associated with salt-enriched water handling in green hydrogen production. As India's green hydrogen sector continues to grow, entrepreneurs who provide innovative and reliable solutions for salt removal and water treatment will contribute to the sector's sustainability and success while building profitable businesses.

5.1.3 Installation and commissioning of electrolyzer

Micro-entrepreneurship opportunities related to the installation and commissioning of electrolyzers in India are on the rise as the green hydrogen sector expands. These opportunities are crucial for ensuring the efficient and safe operation of electrolysis equipment.

Following are some key opportunities for micro-entrepreneurs in this aspect:

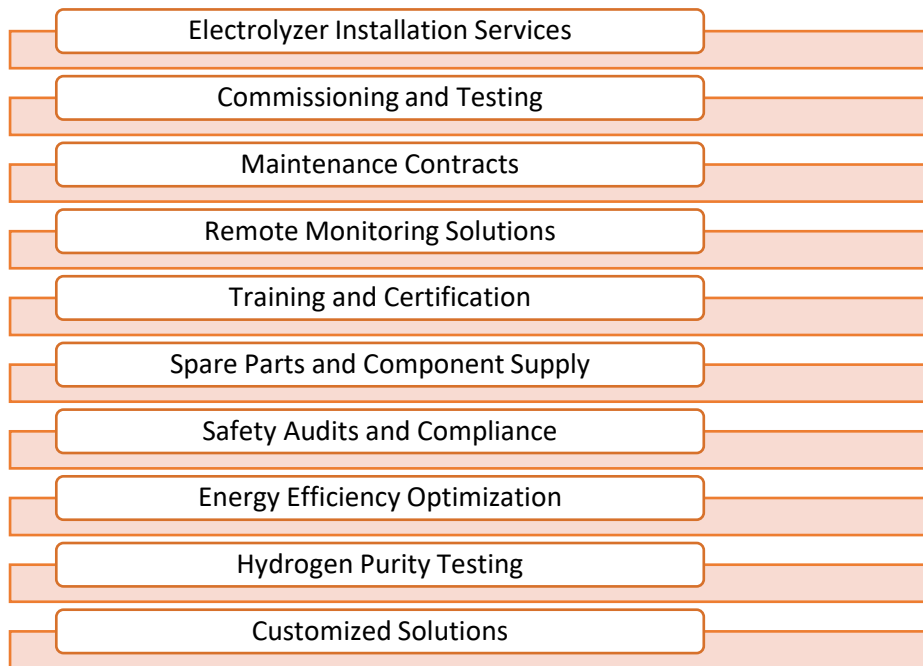


Fig 5.1.2. Micro-Entrepreneurship Opportunities for Installation and commissioning of electrolyzer

- **Electrolyzer Installation Services:** Micro-entrepreneurs can establish businesses that specialize in the installation of electrolyzers. This includes site preparation, equipment installation, and the integration of electrolysis systems into existing infrastructure. Ensuring correct installation is vital for the overall performance of green hydrogen plants.
- **Commissioning and Testing:** Entrepreneurs can offer commissioning and testing services to ensure that electrolyzers operate optimally from the outset. This involves verifying that all components are functioning correctly, performing safety checks, and conducting performance tests.
- **Maintenance Contracts:** Entrepreneurs can enter into maintenance contracts with hydrogen production facilities. These contracts can include routine maintenance, troubleshooting, and repairs to keep electrolyzers in excellent working condition.
- **Remote Monitoring Solutions:** Micro-entrepreneurs can develop and provide remote monitoring solutions for electrolyzer performance. These solutions use IoT technology to collect data in real-time, allowing for proactive maintenance and reducing downtime.
- **Training and Certification:** Entrepreneurs can establish training programs and certification services for technicians and engineers involved in electrolyzer installation and maintenance. Properly trained personnel are essential for safe and efficient operations.

- **Spare Parts and Component Supply:** Entrepreneurs can set up businesses that supply spare parts, components, and consumables for electrolyzers. Ensuring a reliable supply of critical components is vital for minimizing downtime.
- **Safety Audits and Compliance:** Micro-entrepreneurs with expertise in safety standards and regulations can offer safety audits and compliance services to hydrogen production plants. This helps ensure that electrolyzer installations meet safety requirements.
- **Energy Efficiency Optimization:** Entrepreneurs can focus on optimizing the energy efficiency of electrolyzers. This includes implementing energy-saving solutions and providing consultancy services to reduce energy consumption during hydrogen production.
- **Hydrogen Purity Testing:** Micro-entrepreneurs can offer hydrogen purity testing services to ensure that the produced hydrogen meets industry standards and purity requirements.
- **Customized Solutions:** Entrepreneurs can develop customized solutions for specific types of electrolyzers or hydrogen production processes, catering to the unique needs of different green hydrogen projects.

As the green hydrogen sector gains momentum in India, entrepreneurs and micro-entrepreneurs who specialize in electrolyzer installation, commissioning, and related services will play a crucial role in ensuring the reliability, safety, and efficiency of hydrogen production facilities. These services contribute to the overall success and growth of the green hydrogen economy in the country.

UNIT 5.2 Micro-entrepreneurship Opportunities for Renewable Energy Power Supply

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for Renewable energy power supply arrangement for green hydrogen plant

5.2.1 Renewable Energy Power Supply Arrangement for Green Hydrogen Plant

Micro-entrepreneurship opportunities related to providing renewable energy power supply arrangements for green hydrogen plants in India are becoming increasingly important in the transition to sustainable energy sources. These opportunities align with the need for reliable and green energy sources to power electrolysis processes.

Following are the various opportunities for micro-entrepreneurs in this context:

Solar Power Solutions

Wind Energy Services

Hybrid Energy Systems

Microgrid Development

Energy Storage Solutions

Energy Management and Optimization

Off-Grid Solutions

Maintenance and Repairs

Energy Audits and Consultancy

Financing and Leasing Options

Fig 5.2.1. Micro-Entrepreneurship Opportunities for Renewable Energy Power Supply Arrangement Plant for Green Hydrogen Plant

- **Solar Power Solutions:** Micro-entrepreneurs can set up businesses specializing in solar power installations. They can design, install, and maintain solar photovoltaic (PV) systems to generate electricity for green hydrogen production plants. Solar power is a clean and abundant energy source, making it ideal for powering electrolyzers.
- **Wind Energy Services:** Entrepreneurs can provide wind energy solutions by installing wind turbines near green hydrogen plants. Wind energy is another renewable option that can generate electricity for electrolysis processes.
- **Hybrid Energy Systems:** Micro-entrepreneurs can design and implement hybrid energy systems that combine multiple renewable sources, such as solar, wind, and biomass, to ensure a continuous and stable power supply to green hydrogen facilities.
- **Microgrid Development:** Entrepreneurs can specialize in microgrid development for green hydrogen plants. Microgrids integrate renewable energy sources with energy storage systems to ensure a consistent power supply, even in remote areas.
- **Energy Storage Solutions:** Entrepreneurs can focus on energy storage solutions, such as battery storage systems, to store excess renewable energy generated during periods of high production for use when energy demand is high or during cloudy or windless days.
- **Energy Management and Optimization:** Micro-entrepreneurs can offer energy management and optimization services. They can develop and implement energy-efficient strategies to minimize energy waste and reduce overall energy costs for green hydrogen production.
- **Off-Grid Solutions:** Entrepreneurs can target off-grid green hydrogen production facilities that rely entirely on renewable energy sources. These systems can be designed and installed to operate independently of the main power grid.
- **Maintenance and Repairs:** Micro-entrepreneurs can provide maintenance and repair services for renewable energy systems installed at green hydrogen plants, ensuring their continuous and efficient operation.
- **Energy Audits and Consultancy:** Entrepreneurs can offer energy audits and consultancy services to green hydrogen producers. These services help identify opportunities for energy efficiency improvements and cost savings.
- **Financing and Leasing Options:** Entrepreneurs can explore financing and leasing models for renewable energy systems, making it easier for green hydrogen producers to adopt sustainable energy solutions.

As the demand for green hydrogen production grows in India, micro-entrepreneurs who specialize in renewable energy power supply arrangements will play a vital role in providing reliable and eco-friendly energy solutions. These services contribute to the sustainability and success of the green hydrogen industry while promoting a greener and more sustainable energy landscape in India.

UNIT 5.3 Micro-entrepreneurship Opportunities for Storage, Transportation, and Installation

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for Storage of green hydrogen
2. Discuss micro-entrepreneurship opportunities for Safe transportation of green hydrogen to consumers
3. Discuss micro-entrepreneurship opportunities for Installation of pipelines in green hydrogen plant/industry

5.3.1 Micro-Entrepreneurship Opportunities for Storage of Green Hydrogen

Micro-entrepreneurship opportunities related to the storage of green hydrogen in India are emerging as an essential component of the green hydrogen value chain. Adequate and efficient hydrogen storage is critical for the successful integration and utilization of hydrogen as an energy carrier.

Following are some key micro-entrepreneurship opportunities in this domain:

- **Hydrogen Storage Solutions:** Micro-entrepreneurs can specialize in providing hydrogen storage solutions, including compressed gas storage, liquid hydrogen storage, and advanced solid-state storage technologies. They can offer scalable and cost-effective storage options tailored to the specific needs of green hydrogen producers.
- **Hydrogen Transport and Distribution:** Entrepreneurs can establish businesses focused on hydrogen transport and distribution. This includes designing and operating pipelines, tanks, and transportation methods for the safe and efficient delivery of hydrogen to end-users or industrial facilities.
- **Hydrogen Refuelling Stations:** Micro-entrepreneurs can set up hydrogen refuelling stations to support the growing demand for fuel cell vehicles (FCVs). These stations provide a convenient and accessible way for FCV owners to refuel, contributing to the adoption of hydrogen as a clean transportation fuel.
- **Hydrogen Cylinder Services:** Micro-entrepreneurs can offer hydrogen cylinder leasing, maintenance, and exchange services to various industries and applications, including industrial processes, laboratories, and fuel cell applications.
- **Hydrogen Infrastructure Consultancy:** Entrepreneurs with expertise in hydrogen storage and distribution can provide consultancy services to green hydrogen projects. They can assist in the design, optimization, and maintenance of hydrogen storage infrastructure.

- **Hydrogen Storage Safety Solutions:** Micro-entrepreneurs can focus on hydrogen storage safety solutions, including safety inspections, risk assessments, and the implementation of safety protocols and technologies to ensure the safe storage of hydrogen.
- **Hydrogen Storage Materials and Technologies:** Entrepreneurs can engage in research and development activities to create innovative hydrogen storage materials and technologies that enhance storage capacity, efficiency, and safety.
- **Hydrogen Storage Facilities:** Micro-entrepreneurs can invest in and operate hydrogen storage facilities that provide long-term storage solutions for green hydrogen producers, helping them manage fluctuations in supply and demand.
- **Hydrogen Storage Software and Analytics:** Entrepreneurs can develop software solutions and analytics platforms for optimizing hydrogen storage systems, enabling real-time monitoring, predictive maintenance, and data-driven decision-making.

5.3.2 Micro-Entrepreneurship Opportunities for Safe Transportation of Green Hydrogen to Consumers

Micro-entrepreneurship opportunities related to the safe transportation of green hydrogen to consumers in India are integral to the growth of the green hydrogen sector. As hydrogen gains prominence as an energy carrier, ensuring its safe and efficient delivery is paramount.

Following are some key micro-entrepreneurship opportunities in this context:

- **Hydrogen Transportation Services:** Micro-entrepreneurs can establish hydrogen transportation companies equipped with specialized vehicles and containers designed for the safe delivery of hydrogen to consumers. These services can cater to industries, fuel cell vehicle refuelling stations, and other hydrogen end-users.
- **Hydrogen Tanker Trucks and Trailers:** Entrepreneurs can focus on manufacturing, leasing, or operating hydrogen tanker trucks and trailers designed to transport large quantities of green hydrogen. These vehicles should adhere to safety standards and regulations.
- **Hydrogen Cylinder Delivery Services:** Micro-entrepreneurs can offer cylinder delivery services to supply hydrogen to small and medium-scale consumers. This includes industries, laboratories, and other applications that rely on hydrogen cylinders.
- **Hydrogen Pipeline Infrastructure:** Entrepreneurs can invest in the development of hydrogen pipeline networks to transport hydrogen over longer distances efficiently. These pipelines can connect green hydrogen production facilities to various end-users.
- **Hydrogen Transport Safety Consultancy:** Micro-entrepreneurs with expertise in hydrogen transportation safety can provide consultancy services. They can assist in risk assessments, safety protocol development, and compliance with transportation regulations.
- **Hydrogen Safety Training:** Entrepreneurs can establish training programs for drivers and personnel involved in hydrogen transportation. These programs can cover safety procedures, emergency response, and handling practices.

- **Hydrogen Distribution Center:** Micro-entrepreneurs can set up hydrogen distribution centers strategically located to serve multiple consumers. These centers can receive, store, and distribute hydrogen to various industries and users in a safe and efficient manner.
- **Hydrogen Transport Monitoring Solutions:** Entrepreneurs can develop monitoring and tracking solutions for hydrogen transportation. These technologies can provide real-time data on the condition and safety of hydrogen during transit.
- **Hydrogen Transport Container Leasing:** Micro-entrepreneurs can focus on leasing specialized hydrogen transport containers suitable for various modes of transportation, including road, rail, and maritime.
- **Hydrogen Transport Infrastructure Integration:** Entrepreneurs can specialize in integrating hydrogen transport infrastructure with existing logistics and supply chain networks, streamlining the delivery process.

5.3.3 Micro-Entrepreneurship Opportunities for installation of pipelines in Green Hydrogen

Micro-entrepreneurship opportunities related to the installation of pipelines in green hydrogen plants and industries in India are essential for building the infrastructure required to support the growing green hydrogen sector. Efficient and safe transportation of hydrogen through pipelines is crucial for the success of hydrogen-based applications.

Following are some key micro-entrepreneurship opportunities in this domain:

- **Pipeline Installation Services:** Micro-entrepreneurs can establish businesses specializing in the installation of hydrogen pipelines within green hydrogen production facilities, industrial complexes, and distribution networks. This includes site assessment, design, construction, and testing of pipelines.
- **Pipeline Inspection and Maintenance:** Entrepreneurs can offer pipeline inspection and maintenance services to ensure the integrity and safety of hydrogen pipelines. Regular inspections, corrosion prevention, and repair work are essential for pipeline longevity.
- **Pipeline Material Supply:** Micro-entrepreneurs can focus on supplying high-quality materials, including pipes, fittings, and connectors, for hydrogen pipeline installations. Providing reliable materials is critical for the durability of the pipeline system.
- **Pipeline Safety Consultancy:** Entrepreneurs with expertise in pipeline safety can offer consultancy services to green hydrogen projects. They can assist in developing safety protocols, risk assessments, and compliance with safety standards and regulations.
- **Pipeline Leak Detection Solutions:** Micro-entrepreneurs can develop and provide leak detection technologies and solutions designed specifically for hydrogen pipelines. Quick detection and response to leaks are crucial for safety.
- **Pipeline Expansion and Modification:** Entrepreneurs can specialize in expanding and modifying existing hydrogen pipeline networks to accommodate increased demand or new applications. This includes designing and implementing pipeline extensions.

- **Hydrogen Compression and Decompression Stations:** Micro-entrepreneurs can establish hydrogen compression and decompression stations along pipeline routes to ensure the efficient transportation of hydrogen at varying pressures.
- **Pipeline Route Optimization:** Entrepreneurs can offer services for optimizing pipeline routes to minimize construction costs and environmental impact. This involves careful planning and assessment of potential routes.
- **Pipeline Corrosion Protection:** Micro-entrepreneurs can provide corrosion protection solutions, such as coatings and cathodic protection systems, to extend the lifespan of hydrogen pipelines.

Note:

To know more about Business Opportunities in Green Hydrogen Mission of India, you may refer to the following link:

<https://www.youtube.com/watch?v=rHo7rJSvVfA>

Title of Video: New Business Opportunities Under the National Green Hydrogen Mission of India

By: Enterclimate

Duration: 9 m 35 sec

UNIT 5.4 Micro-entrepreneurship for Safety, Operation and Maintenance of Green Hydrogen Plant

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for Managing safety of the plant
2. Discuss micro-entrepreneurship opportunities for Operation and Maintenance of green hydrogen plant

5.4.1 Micro-entrepreneurship Opportunities for Managing Safety of the Plant

Micro-entrepreneurship opportunities for managing the safety of green hydrogen plants in India are crucial to ensure the secure operation of these facilities and minimize risks associated with hydrogen production. Safety is paramount in the hydrogen industry, and entrepreneurs can play a significant role in providing safety-related services and solutions.

Following are key micro-entrepreneurship opportunities in this domain:

- **Safety Training and Certification:** Entrepreneurs can establish safety training centers that offer specialized courses and certifications related to hydrogen safety. These programs can cover safe handling, emergency response, and risk mitigation.
- **Safety Audits and Inspections:** Micro-entrepreneurs can provide safety audit and inspection services for green hydrogen plants. They can assess compliance with safety regulations, identify potential hazards, and recommend corrective measures.
- **Safety Equipment Supply:** Entrepreneurs can focus on supplying safety equipment and systems specifically designed for hydrogen facilities. This includes gas detectors, safety alarms, emergency shutdown systems, and protective gear.
- **Safety Consultancy Services:** Micro-entrepreneurs with expertise in hydrogen safety can offer consultancy services to green hydrogen projects. They can assist in developing safety protocols, risk assessments, and compliance with safety standards and regulations.
- **Emergency Response Services:** Entrepreneurs can specialize in creating customized emergency response plans for hydrogen plants. These plans outline procedures to follow in the event of accidents or incidents, ensuring a swift and coordinated response.
- **Safety Signage and Labels:** Micro-entrepreneurs can provide safety signage, labels, and instructional materials tailored to hydrogen production facilities. Clear and standardized signage is essential for guiding personnel and responders.

- **Safety Software Solutions:** Entrepreneurs can develop software solutions for safety management and monitoring in hydrogen plants. These platforms can help track safety-related data, incidents, and compliance.
- **Safety Equipment Maintenance:** Micro-entrepreneurs can offer maintenance and servicing of safety equipment installed in green hydrogen plants. Regular inspections and upkeep are vital for equipment reliability.
- **Safety Risk Assessment:** Entrepreneurs can conduct comprehensive safety risk assessments for hydrogen production facilities, identifying potential hazards and recommending risk mitigation strategies.
- **Safety Awareness Campaigns:** Micro-entrepreneurs can organize safety awareness campaigns and workshops targeting plant personnel, first responders, and nearby communities to educate them about hydrogen safety.

5.4.2 Micro-entrepreneurship Opportunities for Operation and Maintenance of Green Hydrogen Plant

Micro-entrepreneurship opportunities for the operation and maintenance of green hydrogen plants in India are critical to ensure the efficient and reliable production of clean hydrogen. Proper operation and maintenance are key factors in the success of these facilities.

Following are key micro-entrepreneurship opportunities in this domain:

- **Plant Operation Services:** Entrepreneurs can offer plant operation services, including the operation of hydrogen production equipment, monitoring of processes, and ensuring continuous production.
- **Maintenance and Repairs:** Micro-entrepreneurs can specialize in maintenance and repair services for hydrogen production equipment, ensuring minimal downtime and optimal performance.
- **Condition Monitoring Solutions:** Entrepreneurs can develop and provide condition monitoring solutions that use sensors and predictive maintenance techniques to detect potential equipment failures before they occur.
- **Plant Optimization Services:** Micro-entrepreneurs can focus on optimizing plant operations by improving energy efficiency, reducing waste, and enhancing overall performance.
- **Remote Monitoring and Control:** Entrepreneurs can offer remote monitoring and control systems that allow plant operators to manage equipment and processes from a centralized location, improving efficiency and safety.
- **Hydrogen Quality Assurance:** Micro-entrepreneurs can provide services for monitoring and ensuring the quality of the produced hydrogen, including purity levels and moisture content.
- **Plant Safety Inspections:** Entrepreneurs can offer safety inspection services to assess and enhance the safety protocols and practices within the green hydrogen plant.

UNIT 5.5 Micro-entrepreneurship for Leakage Monitoring

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship Opportunities for Leakage Monitoring
2. Discuss micro-entrepreneurship opportunities for purging of hydrogen by CO₂/NH₃ before start of work.
3. Discuss micro-entrepreneurship opportunities for Recharging of hydrogen by avoiding air contact by using again CO₂/NH₃ and then filling of hydrogen.

5.5.1 Micro-Entrepreneurship Opportunities for Leakage Monitoring

In a green hydrogen plant, which produces hydrogen through electrolysis powered by renewable energy sources, micro-entrepreneurship opportunities related to leakage monitoring are crucial for ensuring safety and efficient operations. Here are some potential avenues:

1. **Leak Detection Technology Services:** Develop or provide advanced leak detection technologies specifically tailored for green hydrogen plants. This could include sensor-based systems, drones, or IoT devices designed to detect leaks in electrolyzers, pipelines, storage tanks, or other relevant components.
2. **Maintenance and Inspection Services:** Offer regular maintenance and inspection services focusing on leak prevention. This could involve scheduled check-ups, calibration of equipment, and proactive measures to identify potential leakage points before they become problematic.
3. **Safety Consulting and Training:** Provide consulting services to green hydrogen plant operators on safety protocols, risk assessment, and staff training for leak detection and emergency response. This could include creating customized training modules or conducting workshops on safety procedures.
4. **Data Analysis and Monitoring Solutions:** Develop software or tools for real-time monitoring and analysis of data from various sensors deployed in the plant. Offering services that interpret this data and provide actionable insights regarding potential leakages or weak points can be valuable.
5. **Remote Monitoring Systems:** Offer remote monitoring solutions that allow plant operators to oversee operations and detect leaks from a distance. Providing a service that enables constant vigilance and immediate response to any issues detected can be highly beneficial.
6. **Supply Chain for Leak Detection Equipment:** Establish a micro-enterprise focused on supplying specialized equipment for leak detection, such as sensors, detectors, or testing kits. Ensuring a steady supply of high-quality, reliable equipment is essential for plant safety.
7. **Integration of AI and Machine Learning:** Develop AI-driven solutions that can predict potential leakages or weak points based on historical data and operational patterns. Offering predictive maintenance services can help prevent leaks before they occur.

8. **Collaboration for R&D:** Collaborate with research institutions or existing companies working on green hydrogen technology to enhance leak detection and prevention methods. Joint ventures for developing innovative solutions can create significant opportunities.
9. **Regulatory Compliance Assistance:** Help green hydrogen plant operators navigate and comply with safety regulations and standards related to leak detection and prevention. Offering services to ensure adherence to safety protocols and legal requirements can be valuable.
10. **Environmental Impact Assessment Services:** Provide services to assess and mitigate the environmental impact of potential leaks in green hydrogen plants. This could involve evaluating potential consequences and proposing solutions to minimize environmental damage.

5.5.2 Micro-Entrepreneurship Opportunities for purging of hydrogen by CO₂/NH₃ before start of work.

The purging of hydrogen by CO₂/NH₃ before commencing work offers several micro-entrepreneurial opportunities:

1. **Purging System Development and Sales:** Create or sell purging systems specifically designed for hydrogen using CO₂/NH₃. Develop portable or fixed purging equipment tailored for different scales of operations, such as small-scale applications or larger industrial settings.
2. **Service Provision for Purging:** Offer purging services to businesses or industries that handle hydrogen. This involves providing skilled personnel and equipment to conduct purging activities, ensuring safety protocols are followed before work starts.
3. **Consulting and Training Services:** Provide consultancy services on purging techniques using CO₂/NH₃. Offer training programs for personnel involved in hydrogen handling, emphasizing the importance of purging for safety and compliance.
4. **Customized Purging Solutions:** Develop customized purging solutions based on the specific needs of different industries or applications. This could involve tailoring purging methods, equipment, or protocols to suit varying requirements.
5. **Safety Equipment Distribution:** Supply safety equipment necessary for purging activities, such as specialized valves, connectors, or sensors. Act as a distributor or reseller for these essential components.
6. **On-Demand Purging Services:** Create a service that provides on-demand purging solutions for smaller businesses or intermittent hydrogen users who might not have the resources or expertise for in-house purging procedures.
7. **Technology Integration and Automation:** Develop or integrate automation technology into purging systems to enhance efficiency and safety. Automation could involve sensors, controls, or IoT solutions for remote monitoring and control of purging processes.
8. **Environmental Impact Mitigation Services:** Offer services focusing on minimizing the environmental impact of purging. Develop methods or systems that capture, store, or neutralize the by-products resulting from the purging process.

9. **Regulatory Compliance Assistance:** Assist businesses in adhering to safety regulations and standards related to purging before commencing work with hydrogen. Ensure that purging procedures comply with industry standards and legal requirements.
10. **Collaboration for Research and Development:** Collaborate with research institutions or companies working on hydrogen technology to improve purging methods. Jointly invest in R&D to develop more efficient, cost-effective, and environmentally friendly purging techniques.

Entrepreneurs entering this space need to be well-versed in hydrogen safety protocols, understand the specific needs of various industries or applications, and stay updated on advancements in purging technologies to offer innovative and effective solutions.

5.5.3 Micro-Entrepreneurship opportunities for recharging of hydrogen by avoiding air contact by using again CO₂/NH₃ and then filling of hydrogen

The recharging of hydrogen while avoiding air contact using CO₂/NH₃ offers several micro-entrepreneurial opportunities:

1. **Closed-Loop Refuelling Systems:** Develop or provide closed-loop refuelling systems that utilize CO₂/NH₃ to ensure hydrogen recharging occurs without exposure to air. This could involve creating specialized equipment for refuelling stations or vehicle applications.
2. **Customized Recharging Solutions:** Offer customized recharging solutions tailored to different industries or applications. Develop systems specific to particular needs, such as those in transportation, energy storage, or industrial processes.
3. **Recharging Station Development:** Establish or contribute to the development of recharging stations that utilize CO₂/NH₃ for hydrogen refuelling. This may involve designing, building, or retrofitting existing stations to accommodate these technologies.
4. **Mobile Recharging Services:** Provide mobile hydrogen recharging services for businesses or applications requiring on-site refueling. Offer convenient solutions for clients who cannot access fixed recharging stations.
5. **Supply Chain Management for Recharging Components:** Create a micro-enterprise focused on managing the supply chain for components needed in CO₂/NH₃-based hydrogen recharging systems. This includes sourcing, distributing, and managing the supply of these materials to support the industry.
6. **Consulting and Implementation Services:** Offer consultancy services to businesses looking to implement CO₂/NH₃-based recharging solutions. Provide expertise in system integration, infrastructure setup, and operational guidance.
7. **Safety Equipment Distribution:** Supply safety equipment required for handling hydrogen during the recharging process. This could involve providing valves, connectors, sensors, and protective gear tailored for these specific applications.
8. **Education and Training Programs:** Conduct educational programs and training sessions on the safe and efficient use of CO₂/NH₃ in hydrogen recharging. Focus on educating technicians, operators, and businesses on best practices and safety protocols.

9. Environmental Impact Assessment and Mitigation: Provide services aimed at assessing and mitigating the environmental impact of hydrogen recharging processes. Develop solutions to manage by-products or emissions resulting from recharging activities.
10. Partnerships for Research and Development: Collaborate with research institutions or companies engaged in hydrogen technology to innovate and improve recharging methods. Partner in R&D efforts to enhance efficiency, safety, and cost-effectiveness of CO₂/NH₃-based recharging.

Entering this space as an entrepreneur requires a deep understanding of hydrogen refuelling technologies, safety standards, market demands, and regulatory compliance. Staying updated on advancements in hydrogen recharging methods will be crucial to provide cutting-edge solutions in this field.

Exercise



1. What role can micro-entrepreneurs play in ensuring uninterrupted quality water supply for electrolyzers in green hydrogen plants?
2. Describe the micro-entrepreneurship opportunities associated with the installation and commissioning of electrolyzers in green hydrogen plants.
3. What skills and services can entrepreneurs offer to support the setup and operation of electrolyzer systems?
4. How can micro-entrepreneurs facilitate the arrangement of renewable energy power supply for green hydrogen plants?
5. How can micro-entrepreneurs contribute to ensuring the safe and efficient storage of green hydrogen?
6. Explain the significance of micro-entrepreneurship in the installation of pipelines within green hydrogen plants and industries.
7. What micro-entrepreneurship opportunities exist for ensuring the safe transportation of green hydrogen to consumers?
8. How can micro-entrepreneurs contribute to the operation and maintenance of green hydrogen plants? What services can they provide?





6. Perform Health and safety measures for installing and operating green hydrogen systems

- UNIT 6.1: Occupational Health & Safety Standards and Regulations
- UNIT 6.2: Emergency Situations in a Hydrogen System
- UNIT 6.3: Detectors and Safety Tools
- UNIT 6.4: Material Safety
- UNIT 6.5: Maintain Ideal Temperature and Humidity Level
- UNIT 6.6: Smart Hazardous Chemical Storage Monitoring
- UNIT 6.7 Administering First Aid
- UNIT 6.8: Personal Protective Equipment
- UNIT 6.9: Hazards Associated with Hydrogen Production System
- UNIT 6.10: Work Safety Procedures and Instructions
- UNIT 6.11: Fire Protection
- UNIT 6.12: Housekeeping and Infection Control Integration

Key Learning Outcomes

At the end of this module, you will be able to:

1. Explain the requirements for safe work area at hydrogen generation project site.
2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems
3. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.
4. Discuss importance of different detectors and safety tools
5. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures
6. Explain need to maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders.
7. Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing dangerous chemicals.
8. Explain the importance of administering first aid.
9. Identify the personal protective equipment used for the specific purpose.
10. Identify the hazards associated with hydrogen generation system;
11. Identify work safety procedures and instructions for working at hydrogen generation plant complying with applicable safety regulations
12. Discuss all applicable statutory requirements along with safety regulations in terms of fire protection.
13. Incorporate good housekeeping practices and infection control guidelines.
14. Dos and Don'ts during normal operation and maintenance as well as in the event of an accident.

UNIT 6.1: Occupational Health & Safety Standards and regulations

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the requirements for safe work area at hydrogen generation project site.
2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems

6.1.1 Introduction

As learnt in previous modules, hydrogen is categorized under high-hazard materials. Its production area should be chosen such that hydrogen production does not get affected by nearby activities and at the same time does not affect other nearby activities.

All fuels have some degree of hazard. The safe use of any fuel focuses on preventing the combination of the following 3 combustion factors:

- ignition source (spark or heat),
- oxidant (air), and
- fuel

The following image is showing the three combustion factors:

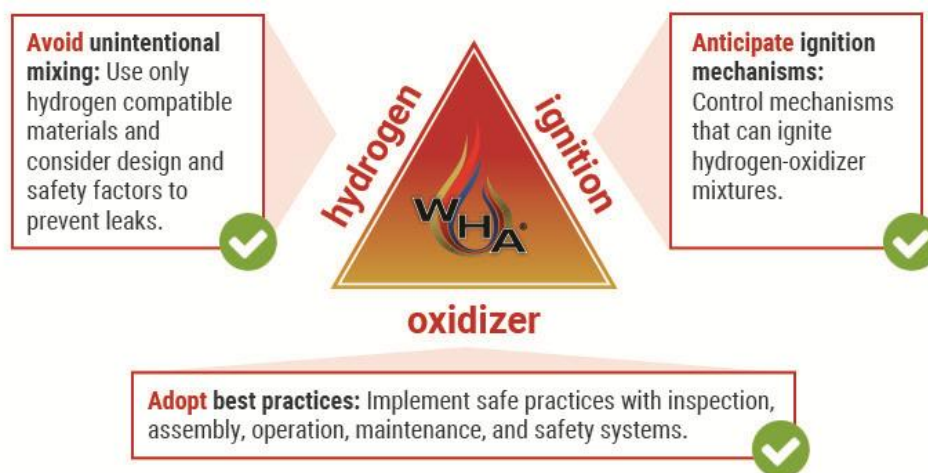


Fig. 6.1.1 Hydrogen Fire Risk

Source: WHA.2019. WHA's Hydrogen Fire Risk Management Philosophy

With a thorough understanding of fuel properties, we can design fuel systems with appropriate engineering controls and establish guidelines to enable the safe handling and use of fuel.

Favourable Properties of Hydrogen

1. Hydrogen is non-toxic.
2. Lighter than air, it dissipates rapidly and moves up in the air in case of leakage.

Risks With Hydrogen

1. A wide range of flammable concentrations in air.
2. Lower ignition energy than petrol or natural gas, which means it can ignite more easily.

Caution

Adequate ventilation and leak detection are necessary. The production area should have adequate horizontal spacing from other machinery and adjacent industries. Hydrogen burns with a nearly invisible flame and hence special flame detectors; and thermo-vision cameras are required.

As far as possible, the hydrogen pressure storage area should be open. The location should be such that in case of an accident, help from other agencies should be available in a short time.

Embrittlement	The hydrogen reacts with some metals and creates voids/cracks. Hence, the selection of appropriate materials is important while designing the hydrogen systems.
Testing of hydrogen systems	Tank leak tests, garage leak simulations, and hydrogen tank drop tests will help to produce, store, and dispense hydrogen safely.
Training	Safe hydrogen handling practices are a key element for ensuring the safe use of hydrogen.

Occupational health is an area of work in public health to promote and maintain the highest degree of physical, mental, and social well-being of workers in all occupations. The objectives are:

- To reduce the risk of accidents or injuries by identifying and mitigating hazards.
- Improved efficiency and productivity due to fewer employees missing work due to illness or injury.
- Improved employee relations and morale (a safer work environment is less stressful work)
- Train and improve awareness regarding adverse effects and precautions to be taken etc. by the working personnel.

6.1.1. Health Effects of Hydrogen

Hydrogen is not toxic, but it harms in the following ways:

- Hydrogen displaces the oxygen and occupies upper space in the closed workplace having no or poor ventilation. If the closed workspace does not contain enough oxygen, then inhalation can cause dizziness, unconsciousness or even death. This phenomenon is called asphyxiation.
- Direct contact of body tissue with liquid hydrogen can cause cold burns or frost bite. The possibility of such burns is much less but one should know about the care to be taken.

Hydrogen as an Asphyxiant

Hydrogen is an asphyxiant. It can create an oxygen-deficient atmosphere by displacing oxygen. Symptoms of an oxygen-deficient atmosphere may be headache, dizziness, breathing difficulty, nausea, and/or unconsciousness.

Proper ventilation is a must! One should not be trapped in an oxygen-deficient atmosphere. However, emergency oxygen cylinders may be stocked at a safe distance. The hydrogen moves upwards and gets evacuated automatically due to its lightness, subject to proper ventilation. Normally hydrogen does not exist at the breathing level if leakage occurs.

6.1.3 OSHA

Occupational Safety and Health Administration (OSHA) deals with all aspects of health and safety in the workplace and has a strong focus on the primary prevention of hazards. Its goal is to prevent accidents and harm to the people from work-related activities.

The scope of OSHA activities is shown in Figure 6.1.2.



Fig. 6.1.2 Scope of OSHA activities

Source: KEN. OSHA Standards for the safety of workers & workplace

6.1.4 Standards

OSHA standards describe the methods that employers must use to protect their employees from hazards. OSHA regulations set out uniform national standards for workplace safety and health practices throughout the country.

The International Organization for Standardization (ISO) is an international standard development organization composed of representatives from the national standards organizations of member countries.

ISO 45001 is an International Standard for health and safety at work developed by national and international standards committees independent of Government.

The ISO 45001 certification for Occupational Health and Safety Management Systems (OHSMS) helps organisations prevent work-related injuries and illnesses and provides safe and healthy workplaces.

ISO 18001 (2007) Occupational health and safety management system — Requirements with guidance for use

A first aid treatment dispensary should be located near to the exit gate of plant to facilitate quick entry of ambulance and other outside help. This will help to shift the affected persons to dispensary for initial treatment who may be shifted to dedicated hospital, as required.

VIDEO

Topic: Introduction to OSHA

By: National Safety Compliance (2011)

Link: <https://www.youtube.com/watch?v=H6wRRWi6i0c>

Duration: 16m 19 sec

UNIT 6.2: Emergency Situations in a Hydrogen System

Unit Objectives

At the end of this unit, you will be able to:

1. List the probable emergency situations in a hydrogen system.
2. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.

Introduction

Hydrogen gas is generally supplied at quite high pressure and hence possibility of leakage and explosion is high. The possibility of burn or cut injury due to explosion is there. The possibility of inhaling hydrogen is however much less unless a person is trapped in a closed space.

Further fire gets extinguished if the hydrogen supply is stopped in the shortest possible time by closing the remotely operated shut-off valves provided at the starting end of the pipelines. If there are no other flammable burning materials, then fire can be controlled by an available firefighting system. However, looking at the situation, help from outside agencies needs to be asked or alerted. For sabotage-type activities, CCTV supervision, drone supervision etc. are better options. Natural calamities e.g., lightning strokes, earthquakes, flood etc. are to be taken care of with special protection nowadays in practice to be adopted.

- **Flammability and explosion**

Hydrogen has a very wide flammability range of 4 to 75% concentration in air as compared to other fuels and 4 to 94% in oxygen. Hence, the prevention of mixing of air or oxygen with hydrogen inside a confined space is very important.

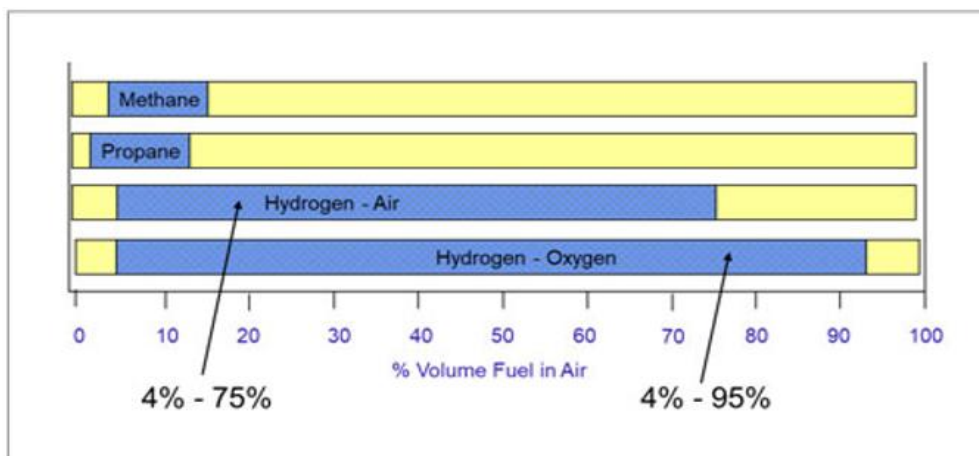


Fig. 6.2.1 Hydrogen flammability range

Source: WHA.2019. WHA's Hydrogen Fire Risk Management Philosophy

Similarly, hydrogen gas forms explosive mixtures with atmospheric oxygen in the range of 18%–59%.

The explosion may be in the hydrogen production plant or storage due to hydrogen leakage or may be caused by a terrorist attack. In view of this, the possibility of underground storage (generally viable for large storage) may be explored which is much safer.

- **Low Ignition Energy**

Small energy is needed to ignite hydrogen as compared to other common fuels as shown in Figure 10.3.2. Consequently, there is a risk of explosion for hydrogen pipelines, storage tanks and containers.

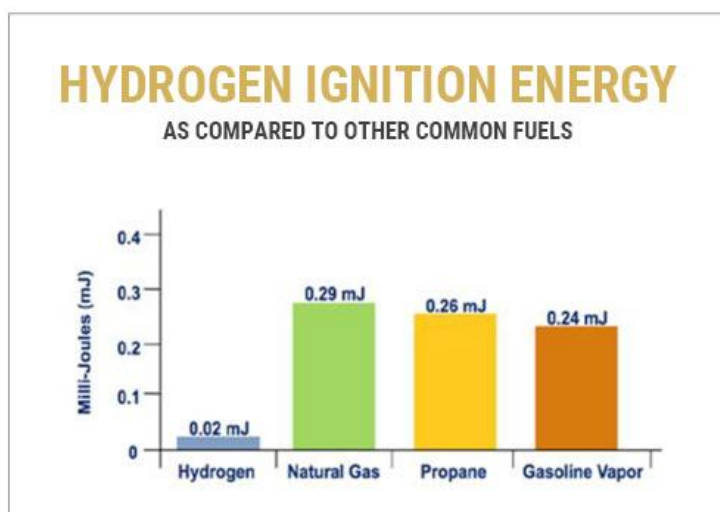


Fig. 6.2.2 Hydrogen ignition energy

Source: WHA.2019. WHA's Hydrogen Fire Risk Management Philosophy

Natural calamity

A *natural calamity* is a disaster caused by natural activities of the Earth. These calamities can cause widespread and serious damage to life and *property*. These include *earthquakes, landslides, volcanic eruptions, floods, hurricanes, tornadoes, tsunamis, cyclones, wildfires, pandemics* etc.

The severity of damage likely to be caused to the hydrogen production plant and area by natural calamity depends on quantum of hydrogen stored at site and rate of hydrogen production per hour. It can be limited by maintaining safe distance all around or by constructing protective wall.

UNIT 6.3: Detectors and Safety Tools

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the importance of detectors and safety tools.
2. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.

6.3.1 Hydrogen Leak Detection

The hydrogen may leak with or without flame and in the form of continuous very small leakage. Its detection is in two parts:

Without flame by bubble detector,
gas detector

With flame by flame detector,
thermo-vision camera

In case of very small leakage, take sample of air near to highest points in plant and check hydrogen percentage by gas detector.

Large quantities of hydrogen are safely used in industries though it is difficult to contain and detect the leaks. Hydrogen would be a much larger industrial commodity with a growing market with its use in hard to abate sectors. The introduction of hydrogen as a consumer fuel has been just started with blending which has raised serious concerns over safety and interest has generated in hydrogen sensors and leak detection. In oil refineries, leakage detectors are provided in the areas prone to hydrogen leakage. The detectors are capable to detect 5% of minimum inflammable hydrogen (i.e. 5% of 4% = 0.2%) which is quite safe. Also, the mobile hydrogen detectors are provided to operation & maintenance personal.

Embrittlement

It is a process caused by the reaction of hydrogen with some metals which results in development of voids and cracks. The early detection of embrittlement gives an opportunity for corrective action and avoid leakage of hydrogen.

6.3.2 Methods of Hydrogen Leak Detection

In Hydrogen leak testing, hydrogen pressure vessel or installation is checked for leaks. The hydrogen is generally used as a tracer gas which flows into the device undergoing testing, with any leaking gas detected by hydrogen sensors. Various test mechanisms are described.

Hydrostatic test

In the hydrostatic test, a vessel is filled with a nearly incompressible liquid – usually water or oil – and checked for leaks or permanent changes in shape. The test pressure is always considerably higher than the operating pressure to give a margin for safety, typically 150% of the operating pressure.

Burst test

In this test, a vessel is filled with a gas and tested for leaks. The test pressure is at least double of the operating pressure.

Helium leak test

In this test, helium is used as a tracer gas and detects it in concentrations as small as 0.1 ppm. The helium is selected primarily because it penetrates small leaks easily and being inert will not react with the test piece while having a naturally low quantity in air makes detection less complicated. It is possible to detect leaks as small as 5×10^{-10} Pa·m³/s in vacuum mode whereas modern digital machines can detect 5×10^{-10} Pa·m³/s in sniffing mode too.

Vacuum test

A vacuum inside the object is generally created with an external pump connected to the instrument. Alternatively, helium can be injected inside the object while the object itself is enclosed in a vacuum chamber connected to the instrument. In this case, burst and leakage tests can be combined in one operation.

Hydrogen sensor test

During the hydrogen sensor test, the object is filled with a mixture of 5% hydrogen and 95% nitrogen. The reason being that hydrogen below 4% as stated above is non-flammable. This is typically called a sniffing test. The hand probe is connected to the microelectronic hydrogen sensors to check the object. An audio signal increases as the leakage detector is moved near to the leakage spot. Detection of leaks go down to 5×10^{-7} cubic centimetres per second. The hydrogen is cheaper than helium and, in this test, there is no need for a vacuum but it is not as sensitive as a helium leak detector and hence it will not find smaller leaks.

Chemo-chromic hydrogen leak detectors can be proactively applied. If there is a hydrogen leakage, the chemo-chromic material changes colour to that indicates the presence of a leak. Chemo-chromic indicators can also be added to silicone tapes for hydrogen detection purposes.

6.3.3 Controls and Protection

Safety of hydrogen production plant are required to be improved by using interlock systems working in unison with hydrogen and fire detectors. The leak and flame detection systems should be integrated into the control systems such that when a hydrogen leak or a fire is detected they:

1. Shut-off and isolate hydrogen sources
2. Shut down/deactivate/de-energize the process equipment to a safe mode
3. Activate audible and visual alarms

The active ventilation should be increased as required. There should be manual reset to restart.

Remote notification of facility management and first responders need also to be considered where appropriate.

An indicative shutdown control actions table is shown in Figure 6.3.1.

Equipment/Alarms	Instrument Air Dispenser	Compressor Control Panel	Compressor Diaphragm H2 Storage	Hydrogen System Entry way 25 ft away from H2 System	Hydrogen Fill	Equipment Enclosure Site Shutdown Scenario
E-Stop	•	•	•	•	•	•
Heat Detection	•	•	•	•	•	•
Flame Detection	•	•	•	•	•	•
Leak Detection	•	•	•	•	•	•
Hose Break	•			•		•
Mechanical Ventilation				•		•
High Pressure		•				•
Low Pressure	•	•				•
High Temperature	•	•				•
Low Temperature	•	•				•
Mechanical Relief Device	•	•	•	•	N/A	N/A

Fig. 6.3.1 Indicative shutdown control actions

Source: Hydrogen tools. Instruments and controls. <https://h2tools.org/bestpractices/instruments-controls>

Warning systems are installed to detect abnormal conditions, malfunctions, and incipient failures. An alarm system (consisting of both audible and visible alarms) with distinctive signals is provided for each type of emergency. Periodic practice drills help to ensure that employees are familiar with the alarms and are trained at regular intervals about how to respond. The various signals are displayed in the operational areas.

Above are indicative for general information and to understand protection/safety features of hydrogen production plant.

6.3.4 Safety Tools

After detection of hydrogen leakage, role of safety tools comes into the picture. The safety tools are like other tools but have special features of non-sparking, no slippage etc. to avoid mechanical sparks etc.

The safety tools are to be carried to immediately attend the minor leakage detected/noticed during monitoring. However, in the case of a major leakage, the back-up isolating valve has to be closed using safety tools.

If the safety valve has operated, diversion of hydrogen to exhaust to be seen to ensure its proper working.

VIDEO

Topic: Hydrogen gas detector (Battery room)

By: Forensics Detectors (2021)

Link: <https://www.forensicsdetectors.com/blogs/articles/hydrogen-gas-monitors-expert-guidance>

Duration: 4 m 18 sec

UNIT 6.4: Material Safety

Unit Objectives

At the end of this unit, you will be able to:

1. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures

6.4.1 Chemical labels

The Globally Harmonized System (GHS) is an international system that *the United Nations created for the unified classification and labelling of chemicals*.

GHS labels are usually required by chemical hazard communication regulations or occupational health and safety regulations. Those regulations are often based on the Globally Harmonized System of Classification and Labelling of Chemicals (GHS). The primary purpose of GHS labels is to communicate chemical hazards to workers or recipients through signal words, pictograms, hazard statements and precautionary statements.

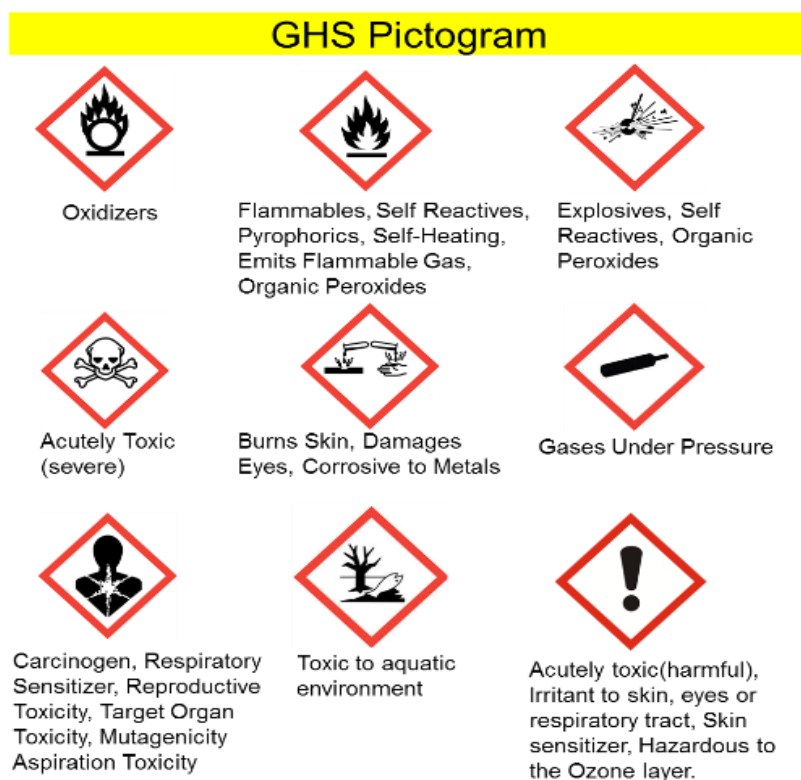


Fig. 6.4.1 GHS safety symbols

Source: ChemSafetyPRO. 2019. Comparison of GHS Labels and Dangerous Goods Labels



Fig. 6.4.2 Symbols indicating nature of chemicals

Source: GA International industry Tag

<https://www.industritag.com/chemical-resistant-labels/>

IMPORTANT

In addition to the above, specific labels conforming to company activities e.g. food industry, semiconductor industry etc. needs to be prepared to caution workers to keep related items/activities away from hydrogen. Such caution needs to be displayed essentially in local languages also.

6.4.2 Use of MSDS for safe handling

Material Safety Data Sheets (MSDSs) are prepared by the manufactures/suppliers of the chemicals which gives information on potential hazards associated with the material and precautions for safe working with these materials. It contains information on the physical and chemical properties of the material. The use of MSDS related to working with hydrogen are summarized below and a sample MSDS is attached as Annexure for reference.

- A. Always check the Material Safety Data Sheet (MSDS) and label for information about the possible hazards and precautions for the product being used. Ask questions if you are in doubt.
- B. Release the gas into the well-ventilated areas.
- C. Always wear personal protection clothing and equipment which includes eye protection (chemical safety goggles) when working with gases under pressure and a face shield.
- D. For gas cylinders:

- i. Inspect all cylinders and valves for damage. Make sure cylinders are not giving off an odour or making a hissing sound. Never open a damaged valve.
- ii. Use the appropriate regulator. Make sure that equipment is compatible with the cylinder pressure and contents. Do not use homemade adaptors or force connections between cylinder valve and gas handling equipment. Never tamper with safety devices in cylinders, valves or equipment.
- iii. Secure cylinders to a wall or rack in an upright position. Leave the cylinder cap in place until the cylinder is secured and ready for use.
- iv. Close all valves when cylinders are not in use.
- v. Do not apply any lubricant, joint compound or tape to cylinder valves, fittings or regulator threads.
- vi. Keep dirt, rust, oil or grease away from all cylinders or fittings.
- vii. Do not drop or bang cylinders against each other. Move cylinders using a hand truck or cart designed for the purpose.
- viii. Avoid direct skin contact with gas escaping from a cylinder.

UNIT 6.5: Maintain Ideal Temperature and Humidity Level

Unit Objectives

At the end of this unit, you will be able to:

1. Identify ideal temperature and humidity ranges
2. Monitor environmental conditions
3. Apply control measures to maintain set levels
4. Ensure safe and efficient system operation

6.5.1 Effect of Temperature and Humidity on Hydrogen Purity

The normal room temperatures are not harmful and can be monitored by simple thermometers. Similarly, humidity can also remain as per general weather conditions if the ventilation is adequate.

The hydrogen purity analyser is used for measuring the purity of hydrogen gas. The purity of hydrogen is influenced by the temperature and humidity. Generally, hydrogen purity measuring instruments have setting to compensate for the effect of temperature and humidity. Therefore, temperature and humidity are measured, and a correction is applied to obtain the correct value of hydrogen purity.

Temperature and its effects

The performance and efficiency of a PEM fuel cell increases with the increase in temperature. The optimum temperature maintains the required humidity.

Humidity and its effects

Humidity indicates the water vapor in the air. If water vapor is more in the air, the humidity will be high and vice-versa.

The humidity is usually expressed as relative humidity. Relative humidity is the amount of water vapor in the air, expressed as a percentage of the maximum amount of water vapor the air can hold at the same temperature.

- **Effect on steel**
 - Under normal relative humidity (about 40%) at 25 °C, the hydrogen concentration in steel is 0.4 ppm. When exposed to higher relative humidity, the hydrogen concentration in steel increases slowly and reaches a stable value, about 0.8 ppm.

- **Effect on fuel cells**

- As expected, the fuel cell performance decreases dramatically with decreasing RH. For example, at a current density of 0.34 A/cm², the cell voltages are 0.675 V (100% inlet RH), 0.642 V (70% RH), 0.556 V (50% RH), 0.414 V (35% RH), and 0.358 V (25% RH). It is shown in Figure 6.5.1

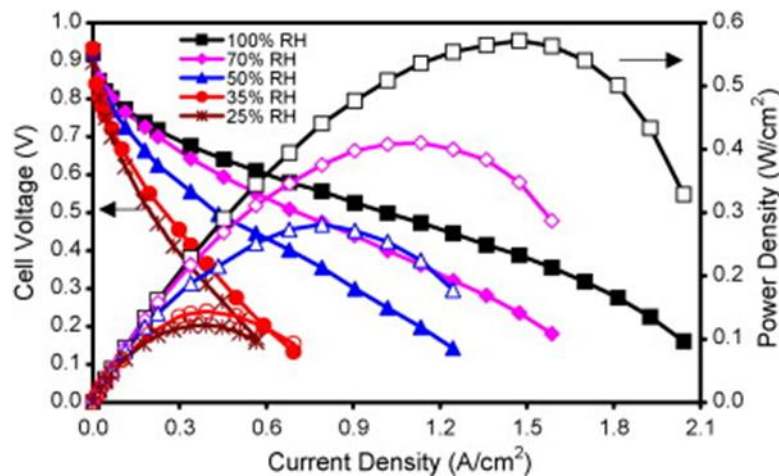


Fig. 6.5.1 Effect of relative humidity on fuel cell voltage

Source: Zhang J., Tang Y., Song C., Xia Z., Li H., Wang H., and Zhang J.. 2008. PEM fuel cell relative humidity (RH) and its effect on performance at high temperatures. *Electrochimica Acta* (Elsevier)

Monitoring temperature and humidity

The temperature and humidity are to be monitored near hydrogen handling equipment conditioning unit to limit them as per the general atmospheric conditions.

The temperature can be monitored by remote temperature sensing device. Humidity also by normally available device. It is important to limit the temperature rise of a particular item/equipment e.g., compressor, electrical connections, DC high current cables. Humidity is harmful to open metallic surfaces water starts condensing cause corrosion, rusting etc. Regular cleaning, hot air blowing etc. needs to be used to control adverse effects of humidity. Daily housekeeping should cover these aspects. If temperature at any part of equipment is found to be high then checking, remedial action needs to be taken.

UNIT 6.6: Smart Hazardous Chemical Storage Monitoring

Unit Objectives

At the end of this unit, you will be able to:

1. Understand principles of safe hazardous chemical storage
2. Identify risks associated with improper chemical storage
3. Monitor storage conditions using smart sensors and systems
4. Ensure compliance with safety standards and regulations

6.6.1 Smart Monitoring

Large hydrogen storage area needs to be monitored continuously and concerned persons for necessary action, as required. Hydrogen storage room needs cleanliness, good ventilation, less humidity, availability of fire protection equipment, emergency exit etc.

Large hydrogen storage room needs to be away from main plant, electric systems,

Ensure proper functioning of ventilation system with a provision of alarm to alert the concerned operational person(s). -

Bubble leakage detection device

Conventional gas detector at higher elevation points where hydrogen concentration is normally high

Humidity measurement/detector

Temperature detector

Specific CCTV supervision facility

Restricted entry ((that too with specified safety appliances, PPE)), strict watch on entries made by regular personnel. It can be monitored by CCTV.

Strict prohibitions on smoking, spark, air tight electric switch boards at lower levels, electrical switch boards preferably in adjacent safe room.

Monitoring by thermo-vision camera with the periodic scanning and recording of the focused equipment/conditioning unit. It will show spots of high temperature of focussed equipment/ conditioning unit. It can be monitored from remote also.

supply systems. It should have:

Fig 6.6.1: Smart hazardous chemical storage monitoring methods

UNIT 6.7 Administering First Aid

Unit Objectives

At the end of this unit, you will be able to:

1. Understand basic principles of first aid
2. Identify common injuries and emergency conditions
3. Provide immediate and appropriate first aid response
4. Ensure safety of the victim until professional help arrives

6.7.1 Health Monitoring

Hydrogen is a hazardous gas. Even with the compliance with the best practices, the possibility of accident/hazard due to manual/machine error/malfunctioning cannot be ruled out. In order to handle such a situation, administering of first aid is a must which is in line with the compliance of mandatory regulation.

Employees should take a bath daily after finishing work. Eating, drinking or storing of food should be prohibited near the place where hydrogen is handled.

Physical Examination

- Proper preplacement medical examinations of a personnel should be carried out to determine the physical fitness before assigning the job.
- Annual medical examination should be carried out as required by laws/regulations (Factories Act/State Factories Rules).

6.7.2 First aid and medical

First aid treatment should be started at once in all cases of contact with hydrogen in any form or injury may result. Refer all injured persons to a physician even when the injury appears to be slight. Give the physician a detailed account of the accident.

First Aid Treatment

Excess inhalation of hydrogen gas causes asphyxiation. Remove the affected person from exposure/contamination area. Place him in the recovery position, as necessary. If breathing has stopped begin rescue breathing. If heart action has stopped begin CPR (Cardiopulmonary Resuscitation). If medical oxygen and trained personnel are available, administer oxygen to the affected person. If unconscious, do not give anything to drink. If conscious make the affected person lie or sit down quietly.

Skin Contact with Liquid Hydrogen

- (A) Treat for frost bite. Soak the affected part in lukewarm water. Seek medical attention.
- (B) Treat the eyes for frost bite.

An indicative first aid kit containing hydrogen peroxide, antiseptic, physiological saline, emergency blanket, triangular bandage etc. is shown in Figure 6.7.1.



Fig. 6.7.1 An indicative first aid kit
Source: Ubuy. First Aid kit Equipped with 100 items

Non-Respiratory Equipment

Personal protecting equipment should include non-respiratory equipment like splash-proof safety goggles and face shield when working with liquid, safety helmets and boots or high top shoes. Where exposure to cold equipment, vapours, or liquid may occur, employees should wear special clothing designed to prevent the freezing of body tissues. Standards/safety codes need to be framed on these points.

Telephone/ mobile numbers of dedicated hospitals, regular doctors, ambulance (in case of plant ambulance failure or more personal affected), oxygen cylinders and their masks etc.

Activity A6.1 Demonstrate how to administer first aid.

Hands-on

Mock exercise under supervision of a supervisor/Doctor. Administer first aid to the workers.

Activity A6.1: TIPS

Refer video details (including link) given below.

VIDEO

Topic: Inhaling hydrogen H₂

By: dchummer Chemistry (2009)

Link: <https://www.youtube.com/watch?v=WeDAZLrk4do>

Duration: 0 m 50 sec

UNIT 6.8: Personal Protective Equipment

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the personal protective equipment used for the specific purpose.

6.8.1 Personal Protective Equipment

As a fuel, hydrogen is highly flammable and hence hydrogen leaks create a risk of fire. When petrol or diesel, leak they pool close to the ground but when hydrogen leaks, it rapidly disperses upwards. This makes ignition less likely, but hydrogen ignites and burns more easily than petrol or diesel. In fact, even a spark of static electricity from a person's finger or clothing is good enough to cause hydrogen explosion. Hydrogen flame is invisible and it tends to burn out quickly.

In view of above personal protection equipment (PPE) are used in hydrogen environment.

The following image shows the list of items in personal protective equipment:

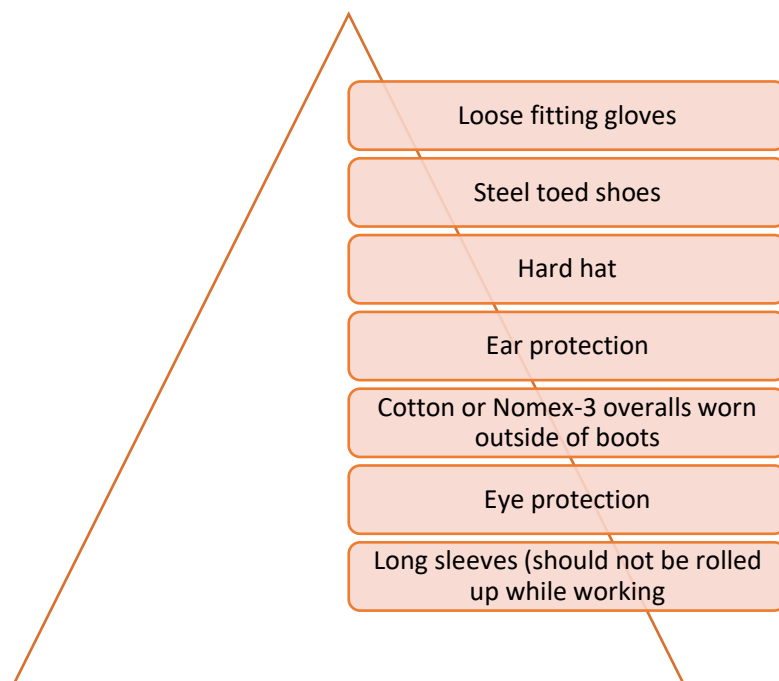


Fig 6.8.1: Items in personal protective equipment

The items of personal protection equipment to be worn in hazardous area are shown in Figure 6.8.1.



Fig. 6.8.2 Items of PPE

A person working in hazardous plant wearing PPE is shown in Figure 6.8.2



Fig. 6.8.2 A person bearing PPE in hazard plant

Source: Bhattacharjee S., Joshi R., Chughtai A.A., and Macintyre C.R. 2019. Wiley Online Library. Graphene Modified Multifunctional Personal Protective Clothing

Note

Personal protective equipment is not an adequate substitute for good, safe working conditions, adequate ventilation, and intelligent conduct on the part of the employees. One should understand that personal protective equipment protects only the person wearing it whereas unprotected workers may be exposed to danger.

Activity A6.2 Demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work.

Hands-on

Wear the personal protection equipment and do the work such as carrying out some pipe fittings, electrical connections, replacement of seals/gaskets, assembly/dismantling etc.

VIDEO

Topic: Personal Protection Equipment

By: National Safety Compliance: OSHA Safety Training (2017)

Link: <https://www.youtube.com/watch?v=YE-63aeECVs>

Duration: 24 m 09 sec

UNIT 6.9: Hazards Associated with Hydrogen Production System

Unit Objectives



At the end of this unit, you will be able to:

1. Know various items of personal protection equipment

6.9.1 Hydrogen Production Equipment Related Hazards

The hazards related to hydrogen production equipment are leakage, fire, explosion, and inhaling.

The following image explains the hydrogen production equipment related hazards:

Electrolyser

- Sudden rise in production rate of hydrogen. It can be controlled by switching off electric supply if production has risen causing pressure increasing mixing of oxygen in hydrogen. It will have to be vented off immediately and stop further process units.

Water separator

- Mainly leakage of water, high percentage of water in hydrogen coming from the hydrogen outlet. Shut-down the plant, Check, replace or repair in case of a minor problem. Check chemicals.

Oxygen separator

- High percentage of oxygen in hydrogen. To check chemicals, catalyst, Stop /repair/replace.

Dryer

- Low temperature, high moisture in hydrogen, check heater electric supply, any leakage of air in dryer. Stop/repair/replace.

Compressor

- High temperature, vibrations, low pressure, leakages. Stop faulty compressor and check thoroughly, repair/replace if standby is available. Safety valve operated shutdown whole plant, forced ventilation, vacate personal.

Storage:

- Small leakage, operation of safety valve. Action same as at s.no. 5 above.

Fig 6.9.1: Hydrogen production equipment related hazards

All persons to alert, firefighting, medical services at high alert, vehicles in ready position. Needs periodic mock exercise.

6.9.2 Spillage and Leakage Hazards

Spillage/Leakages

Frequent inspections of equipment, containers and vessels containing hydrogen should be made to detect or prevent leaks.

- If spills or leaks of hydrogen occur, only properly protected personnel should remain in the area and entry to others should be restricted.
- Every organization handling hydrogen must form an emergency response team, which is available round the clock. Spills or leaks should be handled by the emergency response team specially trained to handle emergency situations.
- If spills or leaks develop, take measures to hasten the dissipation of harmful vapours.
- In case of a major spill or equipment rupture, the exposed area should be evacuated and unprotected personnel must not re-enter the area unless declared safe.
- Proper respiratory protection should be provided for use by personnel entering the area for tests or to repair damaged equipment as high hydrogen concentration may exist.
- Stop flow of gas/shut off leak, if it can be done without risk.
- If source of a leak is a cylinder and the leak cannot be stopped in place, remove the leaking cylinder to a safe place in the open air, and repair leak or allow the cylinder to empty.
- In both gas or liquid spills, remove all ignition sources.

Evacuation

If the release is not contained in an appropriate device or system, all personnel not appropriately protected must evacuate the contaminated spaces. Evacuation of additional areas should be considered as a precaution against the spread of the release or subsequent explosion or fire.

UNIT 6.10: Work Safety Procedures and Instructions

Unit Objectives

At the end of this unit, you will be able to:

1. Understand workplace safety rules and standard operating procedures
2. Identify potential hazards and associated risks
3. Follow safety instructions and use protective equipment correctly
4. Ensure safe and compliant work practices

6.10.1 Introduction

At present, hydrogen is used in some dedicated industries e.g. fertilizer, refinery, steel etc. having their own standards of safety related to the plant and people. However, now onwards, hydrogen is going to be used in private and public transport, electric power generation for local isolated microgrids, cooking etc. Therefore, common men are going to use hydrogen. This has made it necessary to update the standards, regulations, codes to make safe use of hydrogen.

6.10.2 Safety guidelines

The current ANSI/AIAA standard for hydrogen safety guidelines is AIAA G-095-2004, Guide to Safety of Hydrogen and Hydrogen Systems. As NASA has been one of the world's largest users of hydrogen, this evolved from NASA's guidelines, NSS 1740.16 (8719.16). These documents cover both the risks posed by hydrogen in its different forms and how to overcome them. NASA also uses Safety Standard for Hydrogen and Hydrogen Systems and also the Sourcebook for Hydrogen Applications.

Another organization responsible for hydrogen safety guidelines is the Compressed Gas Association (CGA), which has a number of references of their own covering general hydrogen storage, piping, and venting.

6.10.3 Training

All personnel engaged in the operation, maintenance and attending emergency of hydrogen and related systems shall receive specific training for the work on which they are engaged.

Training shall be arranged to cover all the aspects and potential hazards that the operator is likely to encounter. In addition, individuals shall receive specific training in the activities for which they

are employed. Refresher courses should be arranged on a periodic basis. Older employees should be re-educated.

Each employee should know the location and correct operation of gas and fire alarm systems, sprinklers, monitors and other firefighting equipment. They should know the location of safety showers, eye washes and first aid boxes.

6.10.4 Storage, Handling, and Labelling

The storage, handling, and labelling are the main activities before the transportation of hydrogen.

Storage

Storage of hydrogen must avoid contact with heat, flames, sparks and oxygen. Sources of ignition such as smoking and open flames are prohibited where hydrogen is used, handled or stored. Metal containers involving transfer of large quantities of hydrogen and piping should be bonded and grounded. Containers must be equipped with self-closing valves, pressure vacuum bungs and flame arrestors.

Only non-sparking tools and equipment should be used while opening and closing of the hydrogen containers. While handling, manufacturing or storing the hydrogen, approved explosion-proof electrical equipment and fittings should be used.

In storage areas, the healthiness of automatic or manual shutdown processes, sprinklers, gas and heat sensors with early warning audio-visual alarms, flooding, water spraying, extinguishing systems, explosion relief devices should be ensured.

The building shall have adequate natural/forced ventilation, especially at higher elevations.

Indoor Storage

Avoid mechanical damage or overheating of storage tanks and cylinders. The vents from the storage tanks, relief valve must be led to a safe location to avoid any explosions with suitable flame arrestors fitted in it.

Outdoor Storage

Outside storage tanks may be located at least 15 m away from building or adjacent to blank masonry building walls. The location should be away from any flammable liquid storage with adequate approved fire protection system.

Cylinders must be protected from direct sun rays. Provision of shed should be done for outdoor storage of cylinders.

Bulk Storage (Non-refrigerated)

Hydrogen is lighter than air and any leak goes up the atmosphere readily. It shall be stored in gas tight containers either under atmospheric condition or under pressure. Storage area should be protected by fire hydrants and automatic sprinklers for extinguishing a fire.

The storage tanks shall be designed as per applicable standards. Each storage tank shall be equipped with relief valves. Vent pipes from the valves should terminate upward and should have a flame arrestor. Suitable provision should be made to prevent anything from entering the vents. A drain should also be arranged at the bottom of the vent pipe. Any discharge through the relief valves should not enter work area or contact source of ignition.

Handling

Hydrogen should be handled wearing an approved respirator, chemical resistant gloves and safety goggles and other protective clothing. Sources of ignition such as smoking and open flames are prohibited where hydrogen is handled.

Safety recommendations and safe operating procedures should be strictly followed by the persons handling hydrogen.

Handling of Cylinders

The image below explains the ways of handling the cylinders:

Before filling the cylinder with hydrogen, check and confirm for its validity. It shall have valid test certificate from the concerned authorities. Cylinder must be hydraulically tested before filling hydrogen gas. Both seller and consumer must verify that the period of date of testing is still valid.

Cylinders should never be subjected to rough handling or to abnormal mechanical shock such as dropping, bumping, dragging or sliding. They should always be kept in vertical position.

Ropes, slings, hooks, tongs and similar handling devices which may produce sparks or static electricity should not be used during handling and use of hydrogen cylinder.

A suitable hand truck, fork truck or similar device to which the cylinders can be firmly secured should be used for transporting the cylinders.

While handling by crane or derrick, a suitable platform, cradle or boat should be used. Use rack or chain to hold cylinders in place when hooked up for discharging.

Do not place or handle cylinders where they might form part of an electrical circuit.

Do not remove protection cap unless ready to withdraw hydrogen from the cylinder. Appropriate and standard valves must be used. Explosive gas check of valves and fittings should be done before usage.

Use only standard colour code (Red colour without any band) for hydrogen cylinders.

Cylinders for hydrogen, whether full or empty should never be used as rollers for moving heavy or bulky articles.

No one should tamper with numbers, markings, or test dates stamped on cylinders.

When the cylinder is empty, disconnect it, insert the valve plug and replace the cylinder protective cap.

Store empty cylinders separate from filled cylinders and fasten an 'EMPTY' tag on cylinders immediately upon emptying. Close valve, replace plug or nut on valve outlet and secure valve protecting cap snugly.

Fig 6.10.1: Handling cylinders

Labelling

1. Any vessel containing hydrogen should carry an identifying label or stencil.
2. Label should be as per rules and regulations.
3. Each cylinder must be labelled with 'FLAMMABLE GAS' 'CLASS 2'.



Fig. 6.10.2 Label on compressed hydrogen vessel

Source: ACCUFORM. Safety Label: Hydrogen Compressed

Pre-departure Checks for trucks Carrying Hydrogen in Cylinders in Manifold Systems

Prior to and after loading, the drivers must take a complete inspection of the vehicle to ensure that it meets all performance safety requirements and is roadworthy for transporting hydrogen. A few of the numerous items that are checked are the lights, tyres, suspensions, and brakes.

The driver should maintain a log entry book for critical parameters like pressure, etc., during transit.

1. Ensure availability of fire extinguishers (DCP/carbon dioxide) on the vehicle.
2. Ensure availability of emergency kit on the vehicle.
3. Ensure availability of transport emergency (TREM) card and instruction manual during transportation on the vehicle.

6.10.4 Transportation of hydrogen

Labels: Class 2 labels must be displayed on the transport vehicle:

Compressed gas:

UN No. 1049 (United Nations number for hazardous hydrogen)

Refrigerated liquid:

UN No. 1966.

Fig 6.10.3: Labels on transport vehicles

Labels and Markings for transportation

- Product Identification
- Product Hazard Warning
- Safety Venting Decals
- Flammable Gas Warning
- No Smoking
- No Open Flames
- Personal Protection
- Pipeline Labelling
- Authorized Personnel
- Pressure Vessel Identification
- Emergency Information Valve and Component Identification
- Periodic Safety Inspection and routine checking/testing
- Vent Stack Purge Identification
- Static Grounding etc.

Precautions during transportation

No smoking in the cabin. All sources of ignition must be kept away from the truck during transportation. Driver should follow specified route only, maintain speed limit, never park the truck near residential areas, drive truck carefully and observe all rules and signals, avoid overtaking of moving vehicles and do not leave the truck without watch at any time.

No repairs involving gas cutting, welding, etc., should be done during transportation period. If required at all, repairs should be done by authorized and competent personnel of maintenance department/plant who are deputed by manufacturer/supplier. The driver should physically inspect complete mounting periodically and immediately report any abnormality to the supplier.

In case of emergency (for example leaks, fires, accident, etc.) follow the instructions mentioned in the 'Instruction manual during transportation' to control and mitigate an emergency. Immediately contact manufacturer/supplier, police, RTO, fire brigade and other local authorities.

Activity A6.5

Demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at hydrogen production site

Hands-on

Perform shift duty with plant personnel. Note down his activities related to safety and discuss with them the reasons for safety activities involved.

Activity A6.5: TIPS

Go through safety aspects and try to see, discuss them at hydrogen production plan.

UNIT 6.11: Fire Protection

Unit Objectives

At the end of this unit, you will be able to:

1. Understand basic principles of fire safety and prevention
2. Identify fire hazards and potential sources of ignition
3. Use fire protection equipment correctly
4. Respond effectively to fire emergencies to minimize risk and damage

6.11.1 Introduction

Occurrence of fire during handling of hydrogen is a common hazard to be dealt with in all the activities of hydrogen such as hydrogen production, storage, transportation, distribution, uses in various fields.

6.11.2 Statutory Requirements for Fire

The National Fire Protection Association (NFPA), USA issues various standards related to fire hazards related to various industries, power plants etc. Table 10.11.1 gives a list of standards related to hydrogen fire hazards. These cover fundamental safeguards, design of refuelling stations, electrical equipment for hydrogen plant etc.

Name of Standard	Short Title
NFPA 2	Hydrogen Technologies Code: This code provides fundamental safeguards for the generation, installation, storage, piping, use, and handling of hydrogen in compressed gas (GH ₂) form or cryogenic liquid (LH ₂) form
NFPA 30A	Rules for design of refueling stations
NFPA 50A	Standard for gaseous hydrogen systems at consumer sites
NFPA 50B	Standard for liquefied hydrogen systems at consumer sites
NFPA 55	Compressed gases and cryogenic fluids code
NFPA 70	National Electrical Code: It is the benchmark for safe electrical design, installation, and inspection to protect people and property from electrical hazards. These can be used for selection of the electrical fittings in hydrogen handling areas.

Table 6.11.1 Standards on hydrogen

6.11.2 Best practices

The following image describes the best practices for dealing with a gaseous hydrogen fire include:

Do not extinguish the fire until the hydrogen supply has been cut-off, as unburned hydrogen gas may lead to explosion. Spray water on adjacent equipment to cool them.

Do not try to extinguish a hydrogen cylinder fire unless the cylinder is outdoor or in a well-ventilated area free of combustibles and ignition materials.

Do not try to remove a burning cylinder. Keep it and surrounding cylinders cool by spraying with water.

If many cylinders are under fire, the fire fighting should be from a safe distance to protect against the possibility of flying debris.

Fig 6.11.2: Best practices for dealing with a gaseous hydrogen fire

From above points, it could be inferred that in case of hydrogen fire, utmost care is necessary to avoid explosion at the cost of controlled fire of hydrogen till hydrogen supply is stopped or its stored quantity gets exhausted. In other words, the objective is to avoid severe damages at the cost of controlled small damage.

6.11.3 Hydrogen fire fighting

A hydrogen flame is invisible. It causes little damage from radiation. It can burn in a strong wind and be stretched out away from its source a number of feet. Any venting should be done carefully to prevent accumulation in pockets which may cause an explosive mixture. Use of remote sensing thermometer, thermos-vision camera needs to be explored.

Detection

Daylight fires can be detected by heat waves or a broom. Shut off the source and allow the fire to burn out. Very small fires can be controlled with carbon dioxide and water spray. Larger fires may be controlled with steam and/or nitrogen.

If the fire is extinguished before all the gas burns off, watch for pockets which may suddenly re-ignite.

Tankers containing hydrogen should be cooled with water, if near a fire. Hydrogen vent stack fires can be extinguished by shutting off the source and snuffing the fire with nitrogen or helium gas. **WATER SHOULD NEVER BE SPRAYED ON OR NEAR A VENT STACK!!!** Make sure nitrogen or steam line is attached securely.

Activity A6.3: TIPS

Mock test of the fire extinguishing system

Activity A6.4 Show how to comply with all applicable statutory requirements along with safety regulations in terms of Fire protection.

Hands-on

Visit a hydrogen production plant. Study the labels and instructions/manuals/regulations on safety. Form a team of 3, 4 trainees and try to get acquainted on different aspects of safety. Go with the team to each equipment, conditioning units. Make observations, discuss and widen your exposure regarding hydrogen plant working, process. Prepare note and sketches for your future reference.

Activity A6.4: TIPS

Go through safety aspects and try to see, discuss them at hydrogen production plant.

UNIT 6.12: Housekeeping and Infection Control

Unit Objectives

At the end of this unit, you will be able to:

1. Understand the importance of cleanliness and hygiene in the workplace
2. Follow standard housekeeping procedures
3. Apply infection prevention and control measures
4. Maintain a safe and healthy working environment

6.12.1 Introduction

Housekeeping and cleanliness at the workplace are closely linked to the industrial safety. The degree, to which these activities are effectively managed, is an indicator of the safety culture of the organization.

Housekeeping and cleanliness not only make the organization a safer place to work in but also provide a big boost to the image of the organization. These activities also (i) improve efficiency and productivity, (ii) helps in maintaining good control over the processes, and (iii) assist in maintaining the quality of the product.

6.12.2 Housekeeping

Housekeeping is the routine support activities such as tidying, cleaning, routine maintenance etc. of running an organized physical institution occupied or used by people. For example, a house, school, hospital, factory etc.

Importance of Housekeeping

The start/beginning of any abnormal event indicates its presence in the form of rise in temperature, vibrations, noise, abnormal sound, which may appear all together or single or few together depending on severity of abnormal event.

Two rounds of whole plant minute watching of each equipment, components, joints, valves, temperature, pressure, noise level, vibrations level, abnormal sound etc. is the key to open lock of good housekeeping.

By promoting good hand-washing practices, identifying high-touch surfaces and regularly cleaning and disinfecting, housekeeping managers can help keep all areas clean and hygienic. This, in turn, can help curb outbreaks and illnesses in your properties.

Good Housekeeping Practices

The management concept of a good housekeeping practice in workplaces includes five complementary principles of:

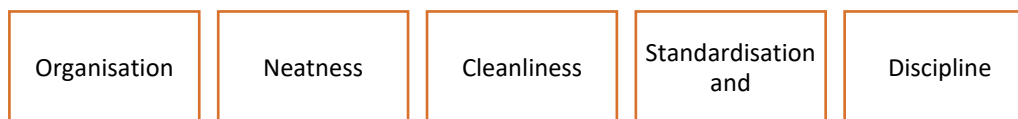


Fig 6.12.1: Five principles of good housekeeping practice

Above 5 principles are known as “5S” in Japan. A typical example of good housekeeping and cleanliness of a green hydrogen production plant is shown in Figure 6.12.1.



Fig. 6.12.2 Typical green hydrogen production plant: Housekeeping

Source: Energy. 2020. In-depth Q&A: Does the world need hydrogen to solve climate change? Carbon Brief

The benefits of housekeeping and cleanliness are shown in Figure 6.12.2.



Fig. 6.12.3 Housekeeping and cleanliness

Source: Satyendra. 2017. *Importance of Housekeeping and Cleanliness at Workplace*. Ispatguru

6.12.3 Infection Control

Infection prevention & control is the discipline concerned with preventing healthcare-related infections. It basically comprises of:

- Hand Hygiene. Hand hygiene is the most important measure to prevent the spread of infections among patients and DHCP
- Respiratory Hygiene/Cough Etiquette
- Sharps Safety
- Safe Injection Practices
- Sterilization and Disinfection of Patient-Care Items and Devices

The precautions to be taken for protection against infectious diseases is shown in Figure 6.12.3.

Infectious Disease Precautions

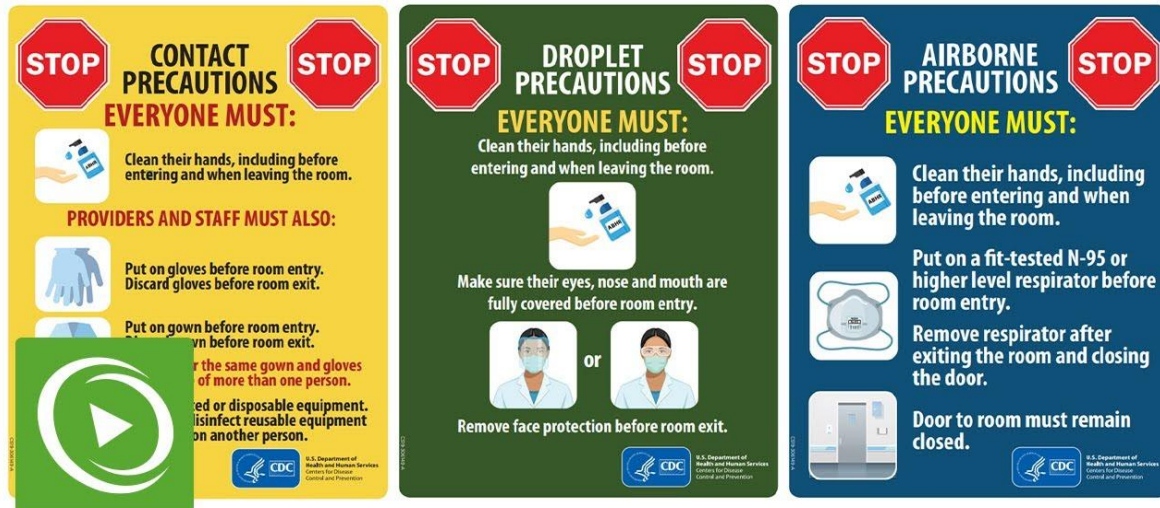


Fig. 6.12.3 Infectious disease precautions

Source: Physiopedia. Infection Prevention and Control

Activity A6.6 Demonstrate good housekeeping and infection control & prevention practices.

Hands-on

Clean the plant area, equipment, component without disturbing the settings of valves, electrical settings. Note readings and observations such as any leakage, cracks, abnormal sound etc.

Activity A6.6: TIPS

Show your observations to your seniors. Seek necessary clarification and clear your doubts.

6.12.4 Dos and Don'ts During Normal Operation and Maintenance

Operating and maintaining a green hydrogen production plant safely is important for the environment, worker safety, and the overall success of the facility. Following are some dos and don'ts during normal operation and maintenance, as well as in the event of an accident:

Dos During Normal Operation and Maintenance:

- **Follow Safety Protocols:** Always adhere to safety protocols and guidelines established for the plant.
- **Regular Inspections:** Conduct regular inspections and maintenance checks to ensure all equipment and systems are in good working condition.
- **Training:** Provide comprehensive training to all personnel involved in the operation and maintenance of the plant.
- **Emergency Response Plan:** Have a well-defined emergency response plan in place, and ensure all employees are familiar with it.
- **Monitoring:** Continuously monitor hydrogen production processes, pressure levels, and temperature to identify any deviations or anomalies.
- **Ventilation:** Ensure proper ventilation systems are in place to prevent the buildup of hydrogen gas.
- **Hydrogen Leak Detection:** Install hydrogen leak detection systems and regularly check for leaks in pipes, valves, and equipment.
- **Fire Safety:** Implement fire safety measures, including fire extinguishers and fire suppression systems.
- **Proper Labeling:** Clearly label all equipment and areas with appropriate safety warnings and instructions.
- **Personal Protective Equipment (PPE):** Require employees to wear appropriate PPE, including safety goggles, gloves, and flame-resistant clothing.

Don'ts During Normal Operation and Maintenance:

- **Neglect Safety Procedures:** Never compromise safety procedures, even for the sake of production efficiency.
- **Unauthorized Access:** Restrict unauthorized access to critical areas and equipment.
- **Ignoring Alarms:** Never ignore alarms or warning signals; investigate and address any issues immediately.
- **Improper Handling:** Do not mishandle or misuse equipment, especially those involving hydrogen storage and handling.
- **Overlook Maintenance:** Avoid neglecting routine maintenance and inspections.
- **Exceed Capacity:** Do not operate the plant beyond its designed capacity or under conditions that may compromise safety.

6.12.5 Dos and Don'ts During Event of an Accident

Dos in the Event of an Accident

Following are Do's for the event of an accident:

- **Safety Shutdown:** Immediately initiate safety shutdown procedures if an accident or abnormal situation occurs.
- **Alert Authorities:** Notify appropriate authorities, including emergency services, in the event of a major accident or hydrogen leak.
- **Evacuation:** Evacuate personnel from the affected area to a safe distance and assembly point.
- **Isolate Sources:** Isolate the source of the accident, if possible, and shut down equipment to prevent further incidents.
- **First Aid:** Administer first aid to injured personnel and provide medical assistance as needed.

Don'ts in the Event of an Accident:

Following are Don'ts for the event of an accident:

- **Panicking:** Avoid panicking; follow emergency procedures calmly and efficiently.
- **Delay in Reporting:** Do not delay reporting an accident or incident to the appropriate authorities and management.
- **Ignoring Safety Protocols:** Do not disregard safety protocols or attempt to fix major issues without proper training and equipment.
- **Re-Entry Without Clearance:** Do not re-enter the affected area until it has been declared safe by qualified personnel.
- **Neglecting Documentation:** Do not neglect documenting the accident, including the sequence of events, actions taken, and lessons learned for future prevention.

Exercise

1. Why safety requirements are more stringent in the case of hydrogen work area?

2. What do you understand by occupational health?

3. What do you understand by natural calamities?

Exercise

4. Discuss the importance of different detectors and safety tools.

5. What is the use of MSDS and who prepares it?

6. What is the importance of temperature and pressure in hydrogen storage?

Exercise

7. What are the ideal conditions for storing the hazardous chemicals?

8. What is first aid and when it is administered?

9. What is the necessity of personal protection equipment?

Exercise

10. Name the hazards associated with the hydrogen production system.

11. What do you understand by safety procedures?

12. What do you understand by the statutory requirements and safety regulations in terms of fire protection?

Exercise

13. What is housekeeping?





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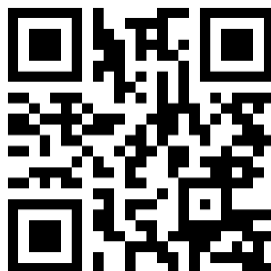
Employability Skills



It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed:

[Employability skills workbook](#)



7. Annexures



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Module No.	Unit No.	Topic Name	Page no.	URL	QR Code(s)
Module 1 - Introduction to Green Hydrogen	Unit 1.1	What is Green hydrogen	5	https://www.youtube.com/watch?v=GqVxE2cVGk	
Module 1 - Introduction to Green Hydrogen	Unit 1.3	How to produce hydrogen gas at home	16	https://www.youtube.com/watch?v=RAbjgk1Zlwc	
Module 1 - Introduction to Green Hydrogen	Unit 1.6	Hydrogen Electrolyzer	28	https://www.youtube.com/watch?v=WfkNf7kMZPA	
Module 1 - Introduction to Green Hydrogen	Unit 1.6	What is Hydrogen Fuel with Full Information	28	https://www.youtube.com/watch?v=Bn6_fQGqiJQ	
Module 1 - Introduction to Green Hydrogen	Unit 1.7	Renewable Energy Sources	32	https://www.youtube.com/watch?v=wI6eKPZ_qkc	
Module 2 - Components of Green Hydrogen Plant and its Layout	Unit 2.1	Demineralization process	47	https://www.youtube.com/watch?v=gCMxFmGkG8s	
Module 3 - Key technical and entrepreneurial aspects for supporting growth and business development green hydrogen production	Unit 3.1	How to Prepare a Bankable Project Report	113	https://www.youtube.com/watch?v=Ke_jwp_lskg	

Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.1	PEM Electrolyzer	135	https://www.youtube.com/watch?v=IX93qZbaiuo	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.2	Gigawatt green hydrogen plant	140	https://www.youtube.com/watch?v=fw5zPUbPo1Q	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.5	Installation of a Sunfire Pressurized Alkaline Electrolyzer	152	https://www.youtube.com/watch?v=qsygJibJ7eQ	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.5	Fuel cell and dismantling	154	https://www.youtube.com/watch?v=ykDiIH7Gtv0	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.7	How to maintain the suction and discharge valves	166	https://www.youtube.com/watch?v=knFiyPZqM1M	
Module 5 - Micro- entrepreneurship opportunities in Green Hydrogen	Unit 5.3	New Business Opportunities Under the National Green Hydrogen Mission	185	https://www.youtube.com/watch?v=rHo7rJSvVfA	
Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.1	Occupational Health & Safety Standards and Regulations	200	https://www.youtube.com/watch?v=H6wRRWi6i0c	

Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.3	Best Hydrogen Detector	206	https://www.forensicsdetectors.com/blogs/articles/hydrogen-gas-monitors-expert-guidance	
Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.7	ling hydrogen H2	215	https://www.youtube.com/watch?v=WeDAZLrk4do	
Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.8	National Safety Compliance: OSHA Safety Training (2017)	218	https://www.youtube.com/watch?v=YE-63aeECVs	





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